



Semi-Conductor
Laboratory

Name: Based on SCL SL18

Revision: 0

Type: Design Rule

Pages: 178

NOTE: This document is comprehensive and gives coverage to variety of devices but this version should be strictly limited and to be used for the legal device list as defined in Annexure-1



Design Rules for SL18 Standard Logic Process for SCL

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Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

1 Purpose and Scope

1.1 Purpose

This document contains Layout Design rules and electrical specification parameters for core Logic CMOS and Analog Devices only, supported by SCL SL18 process

This document is based on SCL SL18 DRM, provided to SCL as part of PDK2020 deliverables

1.2 Scope and Platform Description

SCL 0.18 Micron foundry compatible process includes Core CMOS devices of 1.8V, with several Vt options for leakage reduction / performances enhancement and I/O of 3.3V or real 5V.

For Core CMOS:

- 1.8V transistors ($T_{ox}=40\text{\AA}$, CV measured in inversion), with several Vt options: SVt for general-purpose applications, Low Leakage (high Vt) for low power applications. Low Vt option for high performance.
- Native devices are also supported.

For I/O: one of the two options:

- 3.3V transistors ($T_{ox}=70\text{\AA}$, CV measured in inversion), SVt
- 5V transistors ($T_{ox}=109\text{\AA}$, CV measured in inversion),
- 3.3V and 5V Native devices are also supported.
- Local Cobalt Salicidation for lower delay time, and design with soft CS allows butted straps with single CS, for lower design area.
- Tungsten Contacts, Tungsten Vias, with AlCu metals and USG dielectrics ($k=4.2$) or FSG dielectrics ($k=3.7$). 4 metals and up to 6 metals
 - Tight line and space M1 and internal (0.53 μm Al), for routing and high density design,
 - Top metal layer (ML, 0.94 μm physical thickness Al) for I/O,
 - Thick Top metal (MT, 2 μm Al, or MT3, 2.8 μm Al) for power busses
- Analog Modules includes:
 - Salicided and un-salicided AA and Poly resistors (including HIPO resistor), well resistors
 - MOS varactors and junction varactors.
 - Metal Fringe Capacitor (MFC, Interdigitized capacitors),



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

- Metal-Insulator-Metal (MIM) capacitors, single or double stack, up to 5.6fF/um².
- LV and HV diodes, Deep WN,
- Silicon-based BJTs: PNP and NPN

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1.3 General X-section

The figure below shows the 0.18 platform cross section.

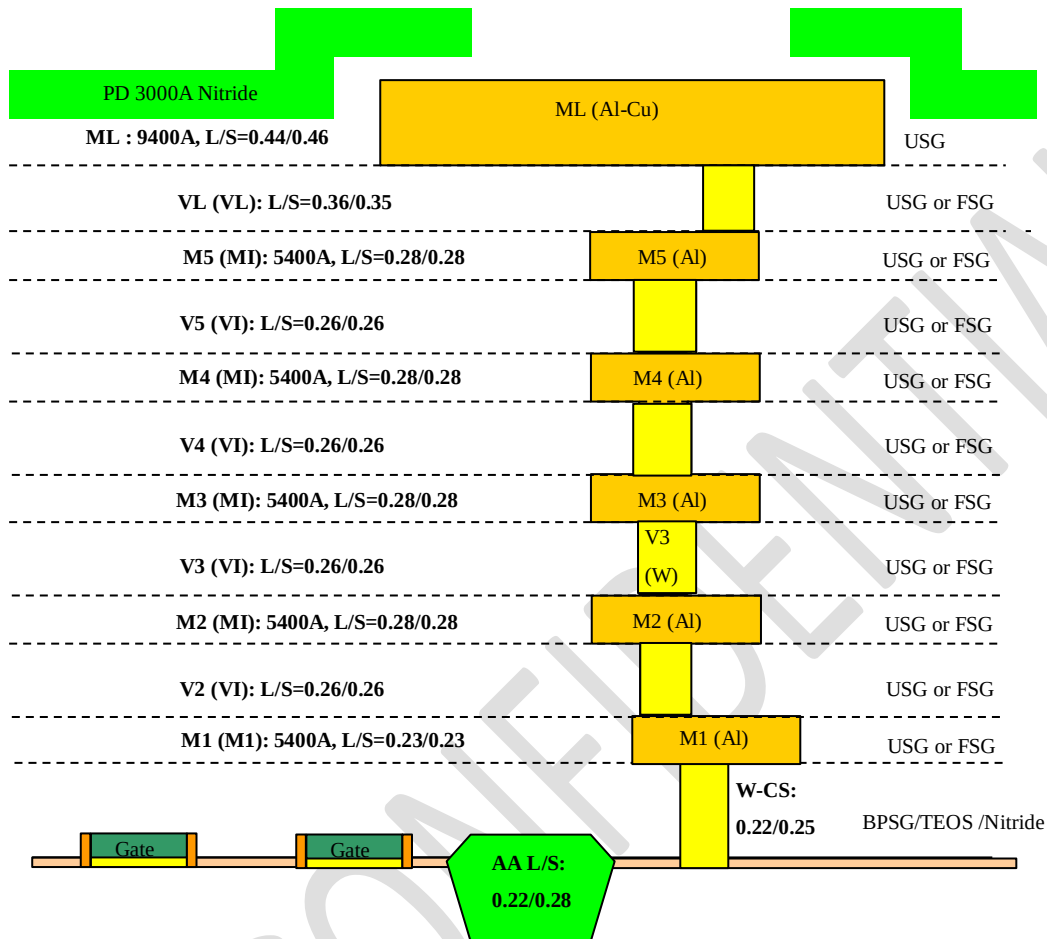
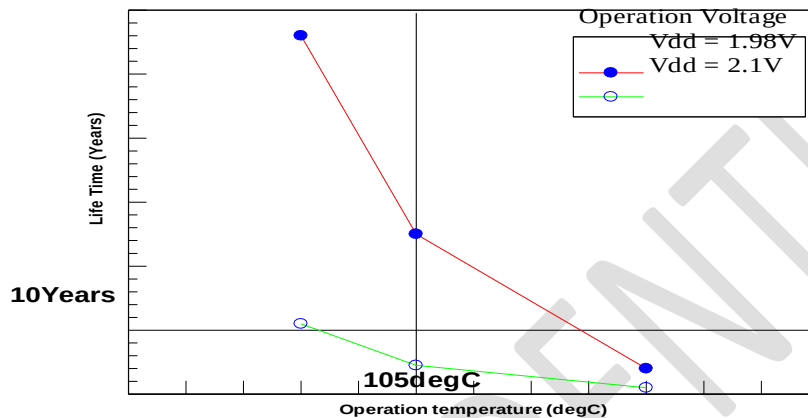


Figure 1-1: 0.18um platform cross section, for this example :6M1L

1.4 Voltage Supply and Operational Temperature Range

1.4.1 Temperature and Voltage Operation Range

Ambient temperature operation range for devices in the platform is Min= -40C; Max= +125C. The operation conditions (Temp, Voltage), the device dimensions, and the reliability failure criteria have effect the device lifetime (LT). The figure below shows, an example, the effect of the temperature and operation voltage on the device LT.



Voltage range of each device is specified in table below.

Legal Device Type	Voltage Range
1.8V Transistors	Vgs : 0~1.8V, Vds : 0~1.8V, Vbs : 0~1.8V
3.3V Transistors	Vgs : 0~3.3V, Vds : 0~3.3V, Vbs : 0~3.3V
5V Transistors	Vgs : 0~5V, Vds : 0~5V, Vbs : 0~5V
Poly, Metal Resistors	-5V to 5V
N+ Diffusion, WN Resistors	0V to 5V
P+ Diffusion Resistors	-5V to 0V
Diodes	See SPICE Document (per specific device)
1.8V MOS (Acc.) Capacitors	0V to 1.8V
3.3V MOS (Acc.) Capacitors	0V to 3.3V
5V MOS (acc.) capacitors	0V to 5V
Metal Fringe Capacitor	-5V to 5V
MIM Capacitors	For 1 or 1.7 or 3.4fF/um2: -5V to 5V For 2.8 or 5.6fF/um2: -3.3V to 3.3V
BJT's	See SPICE Document (per specific device)



1.4.2 Maximum Storage Temperature

Different package types may have different limitations on the storage temperatures. The customer needs to consult with the assembly house.

	Temperature
Storage Temperature (Plastic)	-40C* to + 150C

- Definitions for Vdd, Vdd_max and Absolute Max Rating:

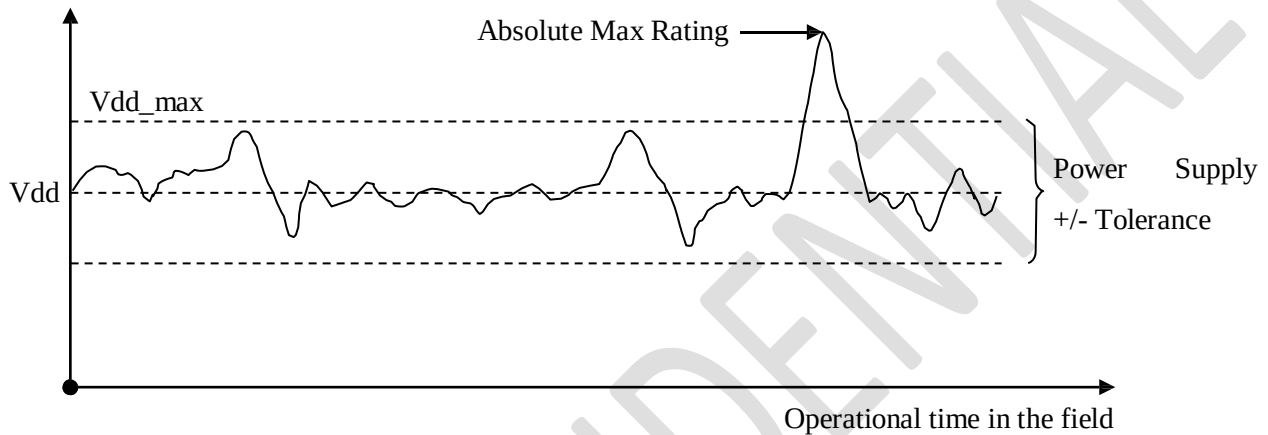


Figure 1-2: General illustration platform voltage (Vdd), Maximum operational Voltage (Vdd_max) and absolute maximum rating voltage.

1.4.3 Max Operational Voltage

The following table summarizes max operation voltage for LV/HV devices. The voltage is measured between any two terminals of the device. Naturally, these limits are ONLY for transient stresses, and NOT recommended for DC conditions.

Vdd	Max operational Voltage
1.8V	+/-10% <1.98V
3.3V	+/-10% <3.63V
5V	+/-10% <5.5V



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

1.4.4 Absolute Max Rating

Vdd	Absolute Max Rating
1.8V	2.2V
3.3V	4.3V
5V	6.9V



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

2 Table of Contents

1	Purpose and Scope	2
1.1	Purpose	2
1.2	Scope and Platform Description	2
1.3	General X-section	4
1.4	Voltage Supply and Operational Temperature Range	5
1.4.1	Temperature and Voltage Operation Range	5
1.4.2	Maximum Storage Temperature	6
1.4.3	Max Operational Voltage	6
1.4.4	Absolute Max Rating	7
2	Table of Contents	8
3	Referenced Documents	9
4	Environmental Protection	10
5	Limitations, Disclaimers And Indemnity	10
6	Layout Terminology	10
6.1	Definition of Layout Terminology	10
6.2	Definition of Layout Design Rule Parameters	12
7	Layout Information	14
7.1	Layout and Grid	14
7.1.1	P (General Pattern Layout and Grid) Rules	14
7.2	Technology Infrastructure – Layers, Metallization, Devices	15
7.2.1	Drawing Layers	15
7.3	Support Layers for MDP	16
7.4	Support Layers for DRC and LVS	18
7.5	Drawing Layer for Labels	20
7.6	Reserved layers for Internal Customers use	20
7.7	Metals Options – the Metal Combination Table	21
7.8	List Masks Needed for Tape Out	24
7.9	List of Legal Devices – Names for PDK, LVS, SPICE Core and I/O Devices	25
7.9.1	Resistor Devices	27
7.9.3	Bipolar Devices	31
7.10	Drawing Instructions for 3-Terminal Marking Layer 77dt86	35
7.11	PDK Limits and Coverage	36
7.12	Drawing and Support Layers TrueTable	40
7.13	LVS Definition for Legal Devices	48
7.14	Rule for Drawing the Support Layer DEV_AREA (lp)	48
7.14.1	DEV_AREA(lp) Support layer Rules	48
8	Topological Design Rule	50
8.1	Rule Coding and General Notes	50
8.1	Mask Information and Re-ordering Masks	50
8.2	Layout Rules	51
8.2.1	AA (Active Area) Rules	51
8.2.2	WN (N-Well) Rule	54
8.2.3	LA (Thick Oxide, AREA2 and AREA6) Rules	56
8.2.4	LA (Thick Oxide, AREA2 and AREA6) Rules	56
8.2.5	GC (Poly) Rules	58
8.2.6	XN (N+ Implant) Rules	60
8.2.7	XP (P+ Implant) Rules	62
8.2.8	NS (Salicide Block) Rules	64
8.2.9	CS (Contact) Rules	66
8.2.10	M1 (Metal 1) Rules	68
8.2.11	VI (I=2~5, Inter Via: V2, V3, V4, V5) Rules	70
8.2.12	MI (I=2~5, Inter Metal: M2, M3, M4, M5) Rules	71
8.2.13	VL (TOP_VIA) Rules	73
8.2.14	ML (TOP_M) Rules	74
8.2.15	MT (20kA Thick Metal Last) Rules	76
8.2.16	MT3 (28kA Thick Al Metal Last) Rules	78
8.2.17	WB (Well Buried, Deep N-Well) Rules	80
8.2.1	VNN (Native NMOS) Devices Rules	84
8.2.2	CNH (High Vt for Low Leakage 1.8V NMOS Devices) Rules	86
8.2.3	CPH (HVt for Low Leakage 1.8V PMOS devices) Rules	88
8.2.4	CNL (1.8V LVt nMOS Devices) Rules	90
8.2.5	CPL (1.8V LVt pMOS Devices) Rules	91
8.2.5.2	MR (HIPO- High Ohmic P-Type for Poly w/o Salicide RESISTOR) Rules	92
8.2.5.3	Medium Ohmic Poly (MOPO) Resistor Rules	97
8.2.6	NAR (N+ AA diffusion w/o Salicided Resistor) Rules	98
8.2.7	PAR (P+ AA diffusion w/o Salicided Resistor) Rules	98
8.2.8	WNSR (N-Well under STI) Resistor Rules	100



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

8.2.8.1	RES (General POLY Resistor) Rules.....	101
8.2.9	RES (General POLY Resistor) Rules	101
8.2.11	NSDN and NSDP (Non salicided 1.8V P+/WN and 1.8V N+/WP diode) Rules	104
8.2.12	Accumulation Capacitor Layout Rules and Recommendations	104
8.2.13	MIM (Metal- Insulator- Metal) Capacitors.....	106
8.2.14	Metal Fringe Capacitor (MFC) Layout Design Rules.....	117
8.2.15	Design Rules for Electrical Poly Fuse (EPF)	119
8.2.16	Silicon Based Vertical Bipolar Junction (VNP and VNP) Transistor Rules	122
8.2.16.1	BJT (Vertical Bipolar Transistor) Rules.....	122
9	Design Guidelines	123
9.1	Antenna Rules	125
9.2	Coverage Guide Lines and Dummy Patterns Insertion / Blocking.....	127
	Restriction of AA dummy's, GC dummy's, Metal Dummy's.....	127
9.2.1	Maximum and Minimum Active Area (AA) Coverage	127
9.2.2	GC/ Metal Coverage.....	130
9.2.3	DGC (Dummy GC) Rules.....	130
9.2.4	DM (Dummy Metal) Rules	131
9.3	Metal Slit (for Metal Coverage Reduction and for Stress Relaxation)	133
9.3.1	MS (Metal Slits for stress relief) Rules	133
10	Bonding Pads and External Structures	135
10.1	Design Rules for Standard Pad Structures (Stacked, Wire-Bond)	135
10.1.2	Common Pad Rules for Both In-Line Pads and Staggered Pads	135
10.1.3	PD (Pad) Rules.....	135
10.1.4	Single In-line Bond Pad Rules	138
10.1.5	Staggered Bond Pad Rules.....	139
10.1.5.1	PD2 (Staggered bond Pad) Rules.....	139
10.1.6	Dummy Pads Rules.....	140
10.2	Die Corner Rules	142
10.2.1	DC (Die Corner) Rules.....	142
10.3	Seal Ring Layout Rules and Scribe Line	144
10.4	Instructions for Including External Structures.....	147
10.4.1	EX (External Structures) Rules.....	147
11	Parasitic Capacitance Model	148
11.1	Cross Section Technology Parameters.....	148
11.1.1	Metal Layers Thicknesses.....	148
11.2	Capacitance Modeling BEOL X-Sections	149
11.2.1	Definition for Generic Structure Type A	153
11.2.2	Definition for Generic Structure Type B.....	153
11.2.3	Examples for Parasitic Capacitance Modeling.....	154
12	Electro Static Discharge (ESD) Protection.....	158
13	Latch Up Rules	158
13.1	LU (Both Internal Circuit and Periphery Circuit Latch Up) Rules.....	158
14	Reliability Specification	159
14.5	Current limits for DC (J_{DC_max}) and Average DC (J_{AC_avgr}).....	162
14.6.1	Root-Mean-Square Current (Irms) limit.....	164
15	Device Tables: Specification of Electrical Parameters	166
16	Appendix A: Mask List	174
17	Revision History	175
18	Annexure-1: Legal Device List as on 16-10-23.....	176

3 Referenced Documents

Table 3-1: Referenced Documents

Document Title	Document or run-set name
SCL 0.18 Micron Generation Runset for Dummy ACTIVE, Dummy Poly and Dummy Metals.	DK2_DRC_18SL_DUMMY_FILL_CALIBRE
External Structures for SL18SL; GDS of pads keys and text	DK2 18SLEXT1



SCL CONFIDENTIAL



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

4 Environmental Protection

All personnel must review their operations with reference to the requirements as noted in SCL's safety and energy conservation policy.

5 Limitations, Disclaimers And Indemnity

Products that use SCL processes are not authorized, designed, or intended for use in, no warranty is made with respect to use of such products in, and Customer shall not permit the use of such products in the design, construction, operation or maintenance of any nuclear equipment, aircraft except with SCL's written approval, or weaponry. Customer shall indemnify, defend and hold SCL (including its employees, agents and affiliates) harmless from and against any and all loss, cost, expense (including, without limitation, attorneys' fees and costs), liabilities or damages associated with any claims, actions or proceedings brought against SCL (1) relating to any such unauthorized use of such products or (2) relating to use of such products in, or use of SCL processes for the manufacturing of critical components in, life support of life critical medical devices or systems, including but not limited to use in any surgical implant into the human body. For these purposes, "life support of life critical medical devices or systems" are devices or systems which (a) are intended for the surgical implant into the human body; or (b) support or sustain human life, and whose failure to perform can be reasonably expected to result in a significant injury to, or death of, the human subject, and a "critical component" is any component of a life support device or system whose failure to perform can be reasonably expected to cause the failure of the life support device or system, or to affect its safety or effectiveness.

6 Layout Terminology

6.1 Definition of Layout Terminology

Table 6-1: Abbreviations and Terms Descriptions

Abbreviations/Terms	Description
CS	Contact
VI	Inetr-Via: V2, V3, V4, V5,
MI	Inetr-Metal: M2, M3, M4, M5,
VL	Via Last (located under ML): TOP_V,
ML	Metal Last (located above VL): TOP_M,
MT	Thick Metal Last (Thickness= 2um)
MT3	Thick Metal Last (Thickness= 2.8um)
Butted diffusion	N+ Active touches P+ Active. The butted diffusion is made when XN touch (just touch but NOT overlap) XP on active. The N+ active should be connected to the same potential of the P+ active potential in order to eliminate non-zero biased Zener diode. N+/P+ butted diffusion is used to connect P+ diffusion of pMOS source to WN pick-up in order to reduce the resistance between the pMOS source and the WN pick-up. It is also used to connect N+ diffusion of nMOS source to the WP pick-up in order to reduce the resistance between the nMOS source and the WP pick-up.
Hot Well	Well that is NOT directly connected to Vdd, which is highest voltage. Hot WN area should interact with support layer 3dt27. For more details, refer to section 8.2.2.1



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Abbreviations/Terms	Description
Well pick-up	CS to AA that having well that is similar to the AA type (CS to SUB). For example, WN pick-up is CS to NSUB. WP pick-up is CS to PSUB.
In WP	In P-Well area, that is outside WN.
IPW	Isolated WP. WP that isolated from the P-Sub by WN (closed ring). Refer to section 8.2.17 For DRC purposes: $IPW = ((NOT\ WN)\ inside\ WB)$.
NSUB	NSUB is AA implanted with N+ over WN, using for WN pick-up. For electrical connection, a CS is needed.
PSUB	PSUB is AA implanted with P+ over WP, using for WP pick-up. For electrical connection, a CS is needed.
GATE	GC over AA, implanted with XN or XP. For DRC purposes: $GATE = GC\ AND\ AA\ NOT\ DEV_AREA(cp)$
NGATE	$NGATE = GATE\ AND\ XN$
PGATE	$PGATE = GATE\ AND\ XP$
LV_NGATE	$LV_NGATE = NGATE\ NOT\ AREA2$
LV_PGATE	$LV_PGATE = PGATE\ NOT\ AREA2$
NDIFF_DIO	NDIFF_DIO gives the exact N+ AA diode area to be used by LVS. For LVS and DRC purposes: $NDIFF_DIO = \{AA\ AND\ XN\ AND\ DEV_AREA(do)\}\ NOT\ (WN\ OR\ GC)$
PDIFF_DIO	PDIFF_DIO gives the exact P+ AA diode area to be used by LVS. For LVS and DRC purposes: $PDIFF_DIO = \{AA\ AND\ XP\ AND\ WN\ AND\ DEV_AREA\}\ NOT\ GC$
LV_WN	WN not covered by AREA2 layer For DRC purposes: $WN_LV = WN\ NOT\ AREA2$
HV_WN	WN covered by AREA2 layer For DRC purposes: $WN_HV = WN\ AND\ AREA2$



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

6.2 Definition of Layout Design Rule Parameters

Table 6-2: Definition of Layout Design Rule Parameters

Code	Parameter	Description
W	Width	Check for the polygon size of a single layer. For example, the width of the AA or the Poly layers.
S	Space	Check the space between two polygons of the same layers. For example, the space between two AA polygons or two poly lines.
E	Enclosure	Check the enclosure of a layer around another layer. For example, the enclosure of under metal around via.
D	Distance	Check the space between two different layers. For example, the distance from GC polygon to AA polygon.
X	Extension	Check the extended "Out" of polygon of layer A which covers some of polygon of layer B. For example, extension of GC beyond active edge, (end cap).
O	Overlap	Check the intersection of polygons of layer A with polygon of layer B.
C	Coverage	Coverage % of a layer at a unit area. For example, the minimum coverage % of M1 area on 200um x 200um area. The coverage step calculation is by step of 100um.
A	Area	Check the area of polygon. For example, the WN area
N	Not allowed	Check for restricted layout violations. For example, CS interact NS is not allowed.
L	Length	Check the length of a polygon edge. For example, length of 45degree metal edge.
R	Recommended	Recommend rule and a value, for high yield and process robustness. This rule is NOT checked by DRC
P	Pattern	Define the layout and pattern rules. For example, the pattern digitized grid.

More definitions and abbreviations are included at the DRC header. Please refer to rule coding as describe in section 8.1.



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

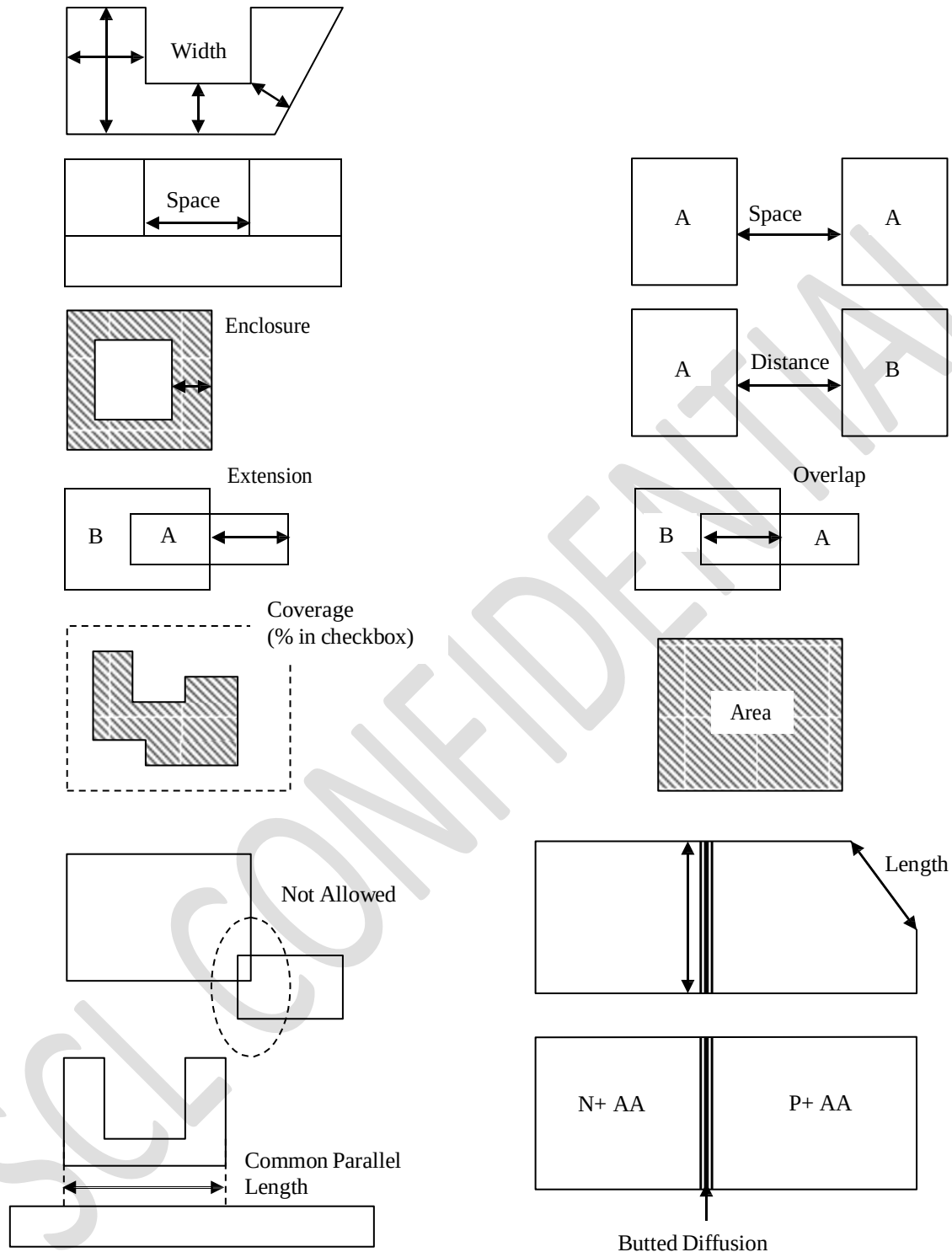


Figure 6-1: Definition of Layout Design Rule



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

7 Layout Information

7.1 Layout and Grid

All patterns should be digitized as per rule P.1. The designer may use a less recommended grid of 0.001um. In that case, the grid definition should be changed in the DRC run set by the customer.

If the designer uses path (routing) data, width of interconnect should be defined with multiple of 0.01um.

7.1.1 P (General Pattern Layout and Grid) Rules		Action	Drawn
P.1	Pattern digitized grid.	min	0.005
P.3	All polygons of all layers should be drawn with only orthogonal or 45deg edges except under layer DEV_AREA(et), and except under layer DEV_AREA(sl).		
P.4	Design should be made using the layers specified in Section 0 ONLY. Using a layer that is not defined in this table is NOT allowed.		
P.5	GDS layer number exceeds 253 is not allowed.		
P.6	Design having both 1.8V LVt layers (CNL, CPL) and 1.8V HVt layers (CNH, CPH, 77dt41) is not allowed. For DRC a purpose, bulk interacts with (CNL OR CPL) interacts (CNH OR CPH OR 77dt41) is not allowed.		
P.7	Design data need to include only metal layers and via layers per the metal combination for manufacturing (see Table 7-6) Design data base having metal/via layers that are not needed is NOT allowed		



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

7.2 Technology Infrastructure – Layers, Metallization, Devices

7.2.1 Drawing Layers

The following table lists the drawing layers which are used for mask generations. Layers marked as optional, are needed if (and only if) the related devices are used.

Table 7-1: Drawing Layers

SCL Layer ID	Technology File For OPUS		GDS		Description
	Layer I.D.	Purpose	Number	Data Type	
Active	AA	logic(lc)	9	0	Active Area
WN	WN	logic(lc)	3	0	Nwell
HVArea	AREA2	logic(lc)	72	0	This layer is area for thick gate oxide for HV transistors, used to generate masks of LA, and more.
GC	GC	logic(lc)	8	0	Gate/Line Poly
XN	XN	logic(lc)	32	0	N+ Implant
XP	XP	logic(lc)	31	0	P+ Implant
Salicide Block	NS	logic(lc)	88	0	Salicide Block
CS	CS	logic(lc)	34	0	Hole to connect M1 to AA or GC
M1	M1	logic(lc)	40	0	1 st Metal
V2	V2	logic(lc)	41	0	Hole to connect M1 to M2
M2	M2	logic(lc)	42	0	2 nd Metal
V3	V3	logic(lc)	43	0	Hole to connect M2 to M3
M3	M3	logic(lc)	44	0	3 rd Metal
V4	V4	logic(lc)	45	0	Hole to connect M3 to M4
M4	M4	logic(lc)	46	0	4 th Metal
V5	V5	logic(lc)	47	0	Hole to connect M4 to M5
M5	M5	logic(lc)	48	0	5 th Metal
TOP_V	TOP_V	logic(lc)	49	0	Top Via (VL)
TOP_M	TOP_M	logic(lc)	50	0	Top Metal (ML)
MT	MT	logic(lc)	59	0	20kÅ Al Thick top Metal
MT3	MT3	logic(lc)	184	0	28kÅ Al Thick top Metal
PAD	PAD	logic(lc)	68	0	PAD opening for Bonding PAD.
Option- WB	WB	logic(lc)	4	0	Deep N-well
Option- CNH	CNH	logic(lc)	10	0	1.8V Low Leakage NMOS (higt Vt WP)
Option- CPH	CPH	logic(lc)	65	0	1.8V Low Leakage PMOS (higt Vt WN)
Option- CNL	CNL	logic(lc)	162	0	1.8V low Vt NMOS Devices
Option- CPL	CPL	logic(lc)	163	0	1.8V low Vt PMOS Devices
VNN	VNN	logic(lc)	17	0	Blocks P-well implant for Native 1.8V, 3.3V and 5V NMOS and for isolation purposes. This drawn layer used to generate WP and other masks, see section 8.2.1
Optional-MR	MR	logic(lc)	87	0	HIPO implant for HIPO (1 kohm/sq) resistor.
Optional-MR2	MR2	logic(lc)	164	0	HIPO implant for 2kohm/sq HIPO Resistor
Option-CE	CE	logic(lc)	99	0	MIM Capacitor Top Electrode
Dummy AA	AA	dummy(dy)	9	8	Dummy AA
Dummy GC	GC	dummy(dy)	8	8	Dummy GC



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

SCL Layer ID	Technology File For OPUS		GDS		Description
	Layer I.D.	Purpose	Number	Data Type	
Dummy M1	M1	dummy(dy)	40	8	Dummy M1
Dummy M2	M2	dummy(dy)	42	8	Dummy M2
Dummy M3	M3	dummy(dy)	44	8	Dummy M3
Dummy M4	M4	dummy(dy)	46	8	Dummy M4
Dummy M5	M5	dummy(dy)	48	8	Dummy M5
Dummy TOP_M	TOP_M	dummy(dy)	50	8	Dummy TOP_M

7.3 Support Layers for MDP

Table 7-2: Support Layers for Masks Generation Preparation

Technology File for OPUS		GDS		In Use for Generation of:
Layer I.D.	Purpose	Number	Data type	
AREA6	logic(lc)	76	0	Area for 5V CMOS transistors for MDP, DRC and LVS support.
ESD5OVT	keep-out(kt)	116	26	Support layer to disable ESD implant mask generation during MDP See EDM2 0018SL
M1	mslits (ms)	40	59	Support layer for DRC of M1 slits Rules and generation slits by MDP (section 1.1)
M2	mslits (ms)	42	59	Support layer for DRC of M2 slits Rules and generation slits by MDP ((section 1.1)
M3	mslits (ms)	44	59	Support layer for DRC of M3 slits Rules and generation slits by MDP (section 1.1)
M4	mslits (ms)	46	59	Support layer for DRC of M4 slits Rules and generation slits by MDP (section 1.1)
M5	mslits (ms)	48	59	Support layer for DRC of M5 slits Rules and generation slits by MDP (section 1.1)
TOP_M	mslits (ms)	50	59	Support layer for DRC of TOP_M slits Rules and generation slits by MDP (section 1.1)
MT	mslits(ms)	59	59	Support layer for DRC of MT slits Rules and generation slits by MDP (section 1.1)
MT3	mslits(ms)	184	59	Support layer for DRC of MT3 slits Rules and generation slits by MDP(section 1.1)
AREA 10	opc (oc)	80	30	Block OPC operation of all layers under during MDP. See Note 2.
WN	opc (oc)	3	30	Blocks OPC operation for N-well only. See Note 2.
AA	opc (oc)	9	30	Blocks OPC operation for AA only. See Note 2.
GC	opc (oc)	8	30	Blocks OPC operation for GC only. See Note 2.
M1	opc (oc)	40	30	Blocks OPC operation for M1 only. See Note 2.
M2	opc (oc)	42	30	Blocks OPC operation for M2 only. See Note 2.
M3	opc (oc)	44	30	Blocks OPC operation for M3 only. See Note 2.
M4	opc (oc)	46	30	Blocks OPC operation for M4 only. See Note 2.
M5	opc (oc)	48	30	Blocks OPC operation for M5 only. See Note 2.
TOP_M	opc(oc)	50	30	Blocks OPC operation for TOP_M only. See Note 2.
MT	opc(oc)	59	30	Blocks OPC operation for MT only. See Note 2.
MT3	opc(oc)	184	30	Blocks OPC operation for MT3 only.
ACTIVE	nodummy (ny)	9	35	Blocks generation of Dummy AA
GC	nodummy (ny)	8	35	Blocks generation of Dummy GC
M1	nodummy (ny)	40	35	Blocks generation of Dummy M1



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Technology File for OPUS		GDS		In Use for Generation of:
Layer I.D.	Purpose	Number	Data type	
M2	nodummy (ny)	42	35	Blocks generation of Dummy M2
M3	nodummy (ny)	44	35	Blocks generation of Dummy M3
M4	nodummy (ny)	46	35	Blocks generation of Dummy M4
M5	nodummy (ny)	48	35	Blocks generation of Dummy M5
TOP_M	nodummy (ny)	50	35	Blocks generation of Dummy TOP_M
MT	nodummy (ny)	59	35	Blocks generation of Dummy MT
MT3	nodummy (ny)	184	35	Blocks generation of Dummy MT3
SPC_AREA	mim(mm)	78	56	Marking layer for stacked mim capacitor area. Used for DRC, LVS and MDP. See section 8.2.13.8
MR2	logic(lc)	164	0	Marks HIPO implant for 2 kohm/sq HIPO resistor. Used for DRC, LVS and MDP.

- NOTE 2: The instructions for placement of OPC blocking layer are specified in the special DR specs for IP library SRAM bitcells only.
For other special cases, drawing of OPC blocking layers is strongly dependent on other drawing layers. For such non standard cases, special working procedures need to be implemented with SCL Engineer ahead of time prior to T/O.
- NOTE 3: Using the keep-out(kt) blocking layer (dt26) is for special cases like test chips, ONLY. It is NOT allowed to use keep-out(kt) blocking layers besides test chips or special cases .



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

7.4 Support Layers for DRC and LVS

Table 7-3: Support Layer for DRC and LVS Table

Technology File for OPUS		GDS		In Use for Generation of:
Layer I.D.	Purpose	Number	Datatype	
WN	drc(dc)	3	27	Hot N-well for DRC purpose. See section 8.2.2.1
WN	dio(do)	3	44	Support layer to mark Hot N-Well modeled with diode to substrate for LVS/SPICE purposes. This model is used as sub-circuit for PMOS placed in this Hot N-Well.
WN	vss(vs)	3	51	For LVS purposes only: In case of separated Grounds, this support layer identifies Vss, which is connected to P-well (Bulk). The support layer should cover the entire area of P+ AA, which is connected to the bulk, physically.
PAD	dummy(dy)	68	8	Marks Dummy PAD for DRC. See section 10.1.6
PAD	drc(dc)	68	27	Identify the inner line of staggered pads for DRC. See section 10.1
DEV_AREA	res(rs)	77	42	Resistors (poly, diffusion or WN) area
DEV_AREA	LL_OP(lp)	77	41	Support layer to convert large area to LL / HVt.
AREA10	SHVT(shv)	80	40	Support layer to convert large area to supper HVt.
DEV_AREA	low_voltage(le)	77	113	Support layer to convert a full chip area to LVt
DEV_AREA	cap(cp)	77	43	Capacitor (accumulation capacitor) for LVS. The layer should cover the entire capacitor gate area.
DEV_AREA	fix(fx)	77	93	Define devices having fix area. See section 8.2.12
DEV_AREA	mim(mm)	77	56	MIM capacitor area, DRC, LVS, MDP. See section 8.2.13
AREA10	mim(hc)	80	56	Define high capacitance MIM. See section 8.2.13
SPC_AREA	cap(cp)	78	43	Define UHD (2.8fF/um ² or 5.6fF/um ²) MIM capacitor. See section 8.2.13
DEV_AREA	dio(do)	77	44	Diode for LVS. See section 8.2.11
DEV_AREA	bjt(bt)	77	47	Bipolar transistors for LVS. See section 8.2.16.1
DEV_AREA	mfc(mc)	77	73	Metal Fringe capacitor (MFC) area of any type for DRC, LVS, MDP and metal tiles blocking. See section 8.2.14
DEV_AREA	seal(sl)	77	45	Defines Seal ring area for DRC. See section 10.3
DEV_AREA	pad(pd)	77	46	Identify the outer line of staggered pads for DRC. See section 10.1.5
DEV_AREA	esd(ed)	77	50	ESD of CMOS buffer for DRC. See section 12
ACTIVE	esd(ed)	9	50	Support layer to identify SOURCE area of ESD CMOS buffer transistors. See section 12
DEV_AREA	cvs(cs)	77	58	ESD NMOS clamp for DRC for Vdd/Vss protection. See section 12
DEV_AREA	scr(sr)	77	123	Marking Layer for SCR. Used for DRC and LVS
DEV_AREA	ext(et)	77	65	Covers external features which are not part of the electrical circuits, like Rev letters etc. It is used for DRC purposes. Section 10.4
SPC_AREA	fuse(fe)	78	105	Marking layer for ePolyFuse (Type-1). Used for DRC, LVS and Dummy fill insertion. See section 8.2.15
M1	res1(r1)	40	55	M1 resistors area
M2	res1(r1)	42	55	M2 resistors area
M3	res1(r1)	44	55	M3 resistors area
M4	res1(r1)	46	55	M4 resistors area
M5	res1(r1)	48	55	M5 resistors area
TOP_M	res1(r1)	50	55	TOP_M resistors area
MT	res1(r1)	59	55	MT resistors area
MT3	res1(r1)	184	55	MT3 resistors area



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Technology File for OPUS		GDS		In Use for Generation of:
Layer I.D.	Purpose	Number	Datatype	
GC	pin(pn)	8	19	For place-and-route tools
M1	pin(pn)	40	19	For place-and-route tools
M2	pin(pn)	42	19	For place-and-route tools
M3	pin(pn)	44	19	For place-and-route tools
M4	pin(pn)	46	19	For place-and-route tools
M5	pin(pn)	48	19	For place-and-route tools
TOP_M	pin(pn)	50	19	For place-and-route tools
MT	pin(pn)	59	19	For place-and-route tools
MT3	pin(pn)	184	19	For place-and-route tools
GC	net(nt)	8	20	For place-and-route tools
M1	net(nt)	40	20	For place-and-route tools
M2	net(nt)	42	20	For place-and-route tools
M3	net(nt)	44	20	For place-and-route tools
M4	net(nt)	46	20	For place-and-route tools
M5	net(nt)	48	20	For place-and-route tools
TOP_M	net(nt)	50	20	For place-and-route tools
MT	net(nt)	59	20	For place-and-route tools
MT3	net(nt)	184	20	For place-and-route tools
GC	pnrblock(pk)	8	84	For place-and-route tools, blocking GC wiring
M1	pnrblock(pk)	40	84	For place-and-route tools, blocking M1 wiring
M2	pnrblock(pk)	42	84	For place-and-route tools, blocking M2 wiring
M3	pnrblock(pk)	44	84	For place-and-route tools, blocking M3 wiring
M4	pnrblock(pk)	46	84	For place-and-route tools, blocking M4 wiring
M5	pnrblock(pk)	48	84	For place-and-route tools, blocking M5 wiring
TOP_M	pnrblock(pk)	50	84	For place-and-route tools, blocking TOP_M wiring
MT	pnrblock(pk)	59	84	For place-and-route tools, blocking MT wiring
MT3	pnrblock(pk)	184	84	For place-and-route tools, blocking MT3 wiring
V2	pnrblock(pk)	41	84	For place-and-route tools, blocking V2 wiring
V3	pnrblock(pk)	43	84	For place-and-route tools, blocking V3 wiring
V4	pnrblock(pk)	45	84	For place-and-route tools, blocking V4 wiring
V5	pnrblock(pk)	47	84	For place-and-route tools, blocking V5 wiring
TOP_V	pnrblock(pk)	49	84	For place-and-route tools, blocking TOP_V wiring
WN	Lvs(ls)	3	28	Support layer LVS/SPICE purposes.
WB	Lvs(ls)	4	28	Marking layer for isolated transistors(transistors over WB)
CS	lvs(ls)	34	28	Marking layer for Metal-option device
V2	lvs(ls)	41	28	Marking layer for Metal-option device
TOP_V	lvs(ls)	49	28	Marking layer for Metal-option device
TOP_M	lvs(ls)	50	28	Marking layer for Metal-option device
MT	lvs (ls)	59	28	Marking layer for Metal-option device
MT 3	lvs (ls)	184	28	Marking layer for MT3 Probe-point devices
M1	vss(vs)	40	51	For M1 LVS identification.
M2	vss(vs)	42	51	For M2 LVS identification.
M3	vss(vs)	44	51	For M3 LVS identification.
AREA 10	a_bitcell (al)	80	60	Covers a_bitcell Single Port IP library cell area
DEV_AREA	a_bitcell (al)	77	60	Covers a_bitcell Dual Port IP library cell area
Prboundary	logic(lc)	235	0	2 nd layer for cell boundary definition.
DEV_AREA	t3(t3)	77	86	Marking layer for 3 terminal devices for LVS. See section7.10

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

7.5 Drawing Layer for Labels

Table 7-4: Drawing Layer for Labels

Technology File for OPUS		GDS		In Use for Generation of:
Layer I.D.	Purpose	Number	Data	
GC	label(ll)	8	7	Label for GC
M1	label(ll)	40	7	Label for M1
M2	label(ll)	42	7	Label for M2
M3	label(ll)	44	7	Label for M3
M4	label(ll)	46	7	Label for M4
M5	label(ll)	48	7	Label for M5
TOP_M	label(ll)	50	7	Label for TOP_M
MT	label(ll)	59	7	Label for MT
MT3	label(ll)	184	7	Label for MT3

7.6 Reserved layers for Internal Customers use

The following layers are reserved at the platform Tech File for INTERNAL customer use. All SCL EDA tools and MDP flow will ignore these layers. Customers are allowed doing tapeout having data at these layers. Note that these data will NOT be available after MDP for mask review.

Table 7-5: Reserved layers for internal customer use Table

TECHNOLOGY FILE FOR OPUS		GDS		DESCRIPTION
SCL LAYER I. D.	PURPOSE	NO	DATA TYPE	
DRCERR	drawing1(d1)	60	241	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing2(d2)	60	242	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing3(d3)	60	243	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing4(d4)	60	244	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing5(d5)	60	245	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing6(d6)	60	246	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing7(d7)	60	247	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing8(d8)	60	248	Reserved layer for INTERNAL customer use, ONLY
DRCERR	drawing9(d9)	60	249	Reserved layer for INTERNAL customer use, ONLY



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

7.7 Metals Options – the Metal Combination Table

SCL SL18 platform offers a large variety of BEOL options. The BEOL is having two dielectric options:

- BEOL with USG (“USG”)
- BEOL with FSG (“FSG”)

The dielectric selection does NOT affect the total number of masks needed for tapout.

There are several types of metal lins:

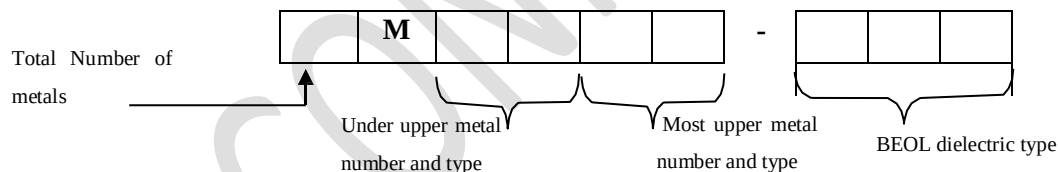
- First metal Al (M1), Line/Space=0.23um/0.23um, Aluminum Thickness=5400A.
- Inter Metal Al (MI). Line/Space=0.28um/0.28um, Aluminum Thickness=5400A.
- TOP_M Al, also called “Last metal” (ML), Line/Space=0.44um/0.46um, physical Thickness=9400A.
- Thick top Al (MT), can be used as top metal once. Line/Space= 1.2/1um, thickness =20kA.
- Thick top Al (MT3), can be used as top metal once. Line/Space= 2.5/2um, thickness =28kA.

Some layout rules refer to more than one metal layer. For example, the MI (inter-metal rules, Section 8.2.12) refer to M2, M3, M4, M5. ML (Metal Last) rules refer to TOP_M.

According to the metal types, there are three different types of vias:

- Inter Via (VI), located under every MI,
- TOP_V, also called “Via Last” (VL) are located under every ML (as VL rules). TOP_V may also be located under MT and MT3.

Metal Combination coding name:





Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

For Example :

- **5M1L-USG**, having M1 , 3 time MI for M2, M3, M4 and (once) ML for M5, using USG dielectric,

MIM capacitors can be integrated into the backend. Single plate or two level (stacked) MIM capacitors are allowed. The bottom capacitor in stacked MIM is generated by MDP after Tape-Out. The MIM capacitors are always located over MI.

MFC (Metal Fringe Capacitor, section 8.2.14) are also supported with Pcells, DRC and SPICE models. Table 7-6 summarizes via and metal name and type for each metallization option, as well as MIM and MFC availability.



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

Table 7-6: Applicable Metal combinations, MIM, MFC offering

MIM: O represent allowed MIM location or upper plate of stacked MIM capacitor , S represents allowed location of lower plate of stacked MIM capacitor, X represents not allowed location for MIM.

Product	M-Comb	USG	FSG	CS/M1	2 nd M	MIM	3 rd M	MIM	4 th M	MIM	5 th M	MIM	6 th M
4LM →	4M1L	Yes	Yes	CS/M1	VI/MI	S	VI/MI	O	VL/ML				
4LM →	4M1T	Yes	Yes	CS/M1	VI/MI	S	VI/MI	O	VL/MT				
4LM →	4M1T3	Yes	Yes	CS/M1	VI/MI	S	VI/MI	O	VL/MT3				
5LM →	5M1L	Yes	Yes	CS/M1	VI/MI	X	VI/MI	S	VI/MI	O	VL/ML		
5LM →	5M1T	Yes	Yes	CS/M1	VI/MI	X	VI/MI	S	VI/MI	O	VL/MT		
5LM →	5M1T3	Yes	Yes	CS/M1	VI/MI	X	VI/MI	S	VI/MI	O	VL/MT3		
6LM →	6M1L	Yes	Yes	CS/M1	VI/MI	X	VI/MI	X	VI/MI	S	VI/MI	O	VL/ML
6LM →	6M1T	Yes	Yes	CS/M1	VI/MI	X	VI/MI	X	VI/MI	S	VI/MI	O	VL/MT
6LM →	6M1T3	Yes	Yes	CS/M1	VI/MI	X	VI/MI	X	VI/MI	S	VI/MI	O	VL/MT3



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

7.8 List Masks Needed for Tape Out

The following table specifies the masks needed for the different technology flavors. Some of the masks included in the process flow are optional. Some of them are generated, and do NOT require any customer drawing. For mask definitions and generation formulas, refer to Appendix A: Mask List.

Note that design fix by correcting one (or more) drawing layers, may require re-order of several masks. For example, correction made to XN drawing layer data, will most likely require re-order of one or more from the flowing masks: XN, XP, LN, HN.

The following table specified the masks needed to be ordered by the customer, for the different technologies and flavors. Note, that some masks are optional, as per the customers' needs.

Table 7-7: Mask Count needed for Tape out

(1.8V)	SVt or LVt or HVt	SVt
Thick oxide	3.3V	5V
Mandatory Masks	AA, WP, WN, GC, XN, XP, NS, CS, M1, (PD) (Total=10masks) CNS(CN), LA, LP, HP, HN, LN (Total=6masks)	
Per Case		CPS(CP)
Optional	WB (for DNW), MR (for 1k Res), MR2 (for 2k Res), CE (for single MIM), BCE+CE (for stacked MIM)	
metals	see section 6.8.	
Overall	22masks for 4LM	23masks for 4LM

Design Rules for SL18 Standard Logic Process
Doc. Name: DR2 0018 SL
Rev. Num: 0
List of Legal Devices – Names for PDK, LVS, SPICE Core and I/O Devices

Core and I/O Devices	Spice Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
nmos_18 (1.8V SVt NMOS Transistor)	n18	N	nmos_18	Drawn: AA, GC, XN. Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
pmos_18 (1.8V SVt PMOS Transistor)	p18	P	pmos_18	Drawn: AA, GC, XP, WN. Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
n18lvt (1.8V Low Vt NMOS transistor)	n18lvt	N18LVT	n18lvt	Drawn: AA, GC, XN, CNL Terminals: S, G, D, B Parameters: w, l, m	Terminals: S, G, D, B Parameters: w,l,m, AD, AS, PD, PS
p18lvt (1.8V Low Vt PMOS transistor)	p18lvt	P18LVT	p18lvt	Drawn:AA,GC, XP, CPL ,WN Terminals: S, G, D, B Parameters: w, l, m	Terminals: S, G, D, B Parameters: w,l,m, AD, AS, PD, PS
n18hvt (1.8V HVt NMOS transistor)	n18hvtb3	N18HVT	n18hvt	Drawn: AA, GC, XN, CNH or DEV_AREA(lp). Terminals: S, G, D, B. Parameters: w, l, m	Terminals:S,G,D,B. Parameters: w, l, m, AD, AS, PD, PS
p18hvt (1.8V HVt PMOS transistor)	p18hvtb3	P18HVT	p18hvt	Drawn: AA, GC, XP, WN,CPH or DEV_AREA(lp). Terminals: S, G, D, B. Parameters: w, l, m	Terminals:S,G,D,B. Parameters: w, l, m, AD, AS, PD, PS
nmos_native (1.8V NMOS Native Transistor) See Figure 8-20	natlv	NA	nmos_native	Drawn: AA, GC, XN, VNN. Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
nmos_18_iso (1.8V SVt NMOS Transistor in IPW)	n18	N_ISO	nmos_18_iso	Drawn: AA, GC, XN,WB, WB(ls) Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
n18hvt_iso (1.8V HVt NMOS transistor in IPW)	n18hvtb3	N18HVT_ISO	n18hvt_iso	Drawn: AA, GC, XN,WB, WB(ls), CNH or DEV_AREA(lp), WB(ls). Terminals: S, G, D, B. Parameters: w, l, m	Terminals:S,G,D,B. Parameters: w, l, m, AD, AS, PD, PS
n18lvt_iso(1.8V Low Vt NMOS transistor in IPW)	n18lvt	N18LVT_ISO	n18lvt_iso	Drawn: AA, GC, XN, CNL,WB, WB(ls) Terminals: S, G, D, B Parameters: w, l, m	Terminals: S, G, D, B Parameters: w,l,m, AD, AS, PD, PS
nmos_33 (3.3V SVt NMOS Transistor)	nhv	NH	nmos_33	Drawn: AA, GC, XN, AREA2. Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
pmos_33 (3.3V SVt PMOS Transistor)	phv	PH	pmos_33	Drawn: AA, GC, XP, WN, AREA2. Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
nmos_33_iso (3.3V SVt NMOS Transistor in IPW)	nhv	NH_ISO	nmos_33_iso	Drawn: AA, GC, XN,WB, WB(ls), AREA2, WB(ls) Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
nmos_native_33 (3.3V NMOS Native Transistor)	nathv	NB	nmos_native_33	Drawn: AA, GC, XN,,AREA2, VNN. Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS
nmos_5V (5V SVt NMOS transistor)	n5v	N5	nmos_5V	Drawn Layers: AA, GC, XN, AREA2, AREA6 Terminals: S, G, D, B Parameters: w, l, m	Terminals: S, G, D, B Parameters: w, l, m, AD, AS, PD, PS, NRD, NSR

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Core and I/O Devices	Spice Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
pmos_5V (5V SVt PMOS transistor)	p5v	P5	pmos_5V	Drawn Layers: AA, WN, GC, XP, AREA2, AREA6 Terminals: S, G, D, B Parameters: w, l, m	Terminals: S, G, D, B Parameters: w, l, m, AD, AS, PD, PS, NRD, NSR
nmos_5V_iso (5V SVt NMOS transistor)	n5v	N5_ISO	nmos_5V_iso	Drawn Layers: AA, GC, XN, WB, WB(l), AREA2, AREA6 Terminals: S, G, D, B Parameters: w, l, m	Terminals: S, G, D, B Parameters: w, l, m, AD, AS, PD, PS, NRD, NSR
nmos_native_5 (5V NMOS Native Transistor)	nath5v	NB5	nmos_native_5	Drawn: AA, GC, XN,, AREA2, AREA6, VNN. Terminals: S, G, D, B. Parameters: w, l, m	Terminals: S, G, D, B. Parameters: w, l, m, AD, AS, PD, PS

Design Rules for SL18 Standard Logic Process
Doc. Name: DR2 0018 SL
Rev. Num: 0
7.9.1 Resistor Devices

Resistor Devices	Spice Model Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
rphpoly2t (HIPO-1k High Ohmic P-type Poly w/o salicide 2 terminal resistor), see Figure 8-31	rphpoly2t	RC	rphpoly2t	Drawn: GC, MR, NS, XP, DEV_AREA(rs). Terminals: Plus, Minus. Parameters: r, w, l or r only	Terminals: Plus, Minus, Parameters: w, l
rphpoly3t (HIPO-1k High Ohmic P-type Poly w/o salicide 3 terminal resistor), see Figure 8-31	rphpoly3t	RC3	rphpoly3t	Drawn: GC,MR,NS,XP, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, (bulk). Parameters: r, w, l or r only	Terminals: Plus, Minus, (Bulk) Parameters: w, l
mr22t (2k High Ohmic P-type Poly w/o salicide 2 terminal resistor). Section 8.2.5.2	mr22t	MR22T	mr22t	Drawn: GC,MR, MR2,NS, XP, DEV_AREA(rs). Terminals: Plus, Minus, (bulk). Parameters: r, w, l or r only	Terminals: Plus, Minus, (Bulk) Parameters: w, l
mr23t (2k High Ohmic P-type Poly w/o salicide 3 terminal resistor).	mr23t	MR23T	mr23t	Drawn: GC,MR,MR2,NS,XP, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, (bulk). Parameters: r, w, l or r only	Terminals: Plus, Minus, (Bulk) Parameters: w, l
rnmpoly2t (NMOPO–N-type Poly w/o salicide 2 terminals resistor), see Figure 8-32	rnmpoly2t	RE	rnmpoly2t	Drawn: GC, XN, NS, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w,l or r only	Terminals: Plus, Minus Parameters: w, l
rnmpoly3t (NMOPO–N-type Poly w/o salicide 3 terminals resistor), see Figure 8-32	rnmpoly3t	RE3	rnmpoly3t	Drawn: GC, XN, NS, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, (Bulk) Parameters: r,w,l or r only	Terminals: Plus, Minus, (Bulk) Parameters: w, l
rpmpoly2t (PMOPO–P-type Poly w/o salicide 2 terminals resistor)See Figure 8-30	rpmpoly2t	RD	rpmpoly2t	Drawn: GC, XP, NS, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w,l or r only	Terminals: Plus, MinusParameters: w, l
rpmpoly3t (PMOPO–P-type Poly w/o salicide 3 terminals resistor), see Figure 8-30	rpmpoly3t	RD3	rpmpoly3t	Drawn: GC, XP, NS, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, (Bulk) Parameters: r,w, l or r only	Terminals: Plus, Minus, (Bulk) Parameters: w, l
rnplus2t (N+ AA Diffusion w/o salicide 2 terminals resistor)	rnplus2t	RB	rnplus2t	Drawn: AA, XN, NS, DEV_AREA(rs). Terminals: Plus, Minus. Parameters: r,w,l or r only	Terminals: Plus, MinusParameters: w, l
rnplus3t (N+ AA Diffusion w/o salicide 3 terminals resistor,)	rnplus3t	RB3	rnplus3t	Drawn: AA, XN, NS, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, P-sub. Parameters: r,w, l or r only	Terminals: Plus, Minus, P- sub. Parameters: w, l
rpplus2t (P+ AA Diffusion w/o salicide 2 terminals resistor)	rpplus2t	RA	rpplus2t	Drawn: AA, WN, XP, NS, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w,l or r only	Terminals: Plus, Minus Parameters: w, l
rpplus3t (P+ AA Diffusion w/o salicide 3 terminals resistor)	rpplus3t	RA3	rpplus3t	Drawn: AA, WN, XP, NS, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, N-sub. Parameters: r,w, l or r only	Terminals: Plus, Minus, N- sub. Parameters: w, l
rnlpoly2t (N-poly salicide 2 terminals resistor)	rnlpoly2t	RH	rnlpoly2t	Drawn: GC, XN, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w, l or r only	Terminals: Plus, Minus, Parameters: w, l

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Resistor Devices	Spice Model Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
rnlpoly3t (N-poly salicide 3 terminals resistor)	rnlpoly3t	RH3	rnlpoly3t	Drawn: GC, XN, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, (Bulk), Parameters: r,w, l or r only	Terminals: Plus, Minus, (Bulk) Parameters: w, l
rpplpoly2t (P-poly salicide 2 terminals resistor)	rpplpoly2t	RG	rpplpoly2t	Drawn: GC, XP, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w, l or r only	Terminals: Plus, Minus, Parameters: w, l
rpplpoly3t (P-poly salicide 3 terminals resistor)	rpplpoly3t	RG3	rpplpoly3t	Drawn: GC, XP, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, (Bulk), Parameters: r,w, l or r only	Terminals: Plus, Minus, (Bulk) Parameters: w, l
rnplus_sal2t (N+ AA Diffusion salicide 2 terminals resistor)	nasr2t	RK	rnplus_sal2t	Drawn: AA, XN, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w,l or r only	Terminals: Plus, Minus, Parameters: w, l
rnplus_sal3t (N+ AA Diffusion salicide 3 terminals resistor)	nasr3t	RK3	rnplus_sal3t	Drawn: AA, XN, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, P-sub. Parameters: r,w, l or r only	Terminals: Plus, Minus, P-sub. Parameters: w, l
rpplus_sal2t (P+ AA Diffusion salicide 2 terminals resistor)	pasr2t	RM	rpplus_sal2t	Drawn: AA, WN, XP, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w,l or r only	Terminals: Plus, Minus, Parameters: w, l
rpplus_sal3t (P+ AA Diffusion salicide 3 terminals resistor)	pasr3t	RM3	rpplus_sal3t	Drawn: AA, WN, XP, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, N-well Tap. Tap Parameters: r,w,l or r only	Terminals: Plus, Minus, N-well Tap. Parameters: w, l
rnwell_aa2t (N-Well under AA 2 terminals resistor)	rnwellaa2t	RF	rnwell_AA2t	Drawn: AA, WN, XN, NS, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w,l or only r	Terminals: Plus, Minus, Parameters: w, l
rnwell_aa3t (N-Well under AA 3 terminals resistor)	rnwellaa3t	RF3	rnwell_AA3t	Drawn: AA, WN, XN, NS, DEV_AREA(rs). DEV_AREA(t3) Terminals: Plus, Minus, P-sub. Parameters: r,w, l or only r	Terminals: Plus, Minus, P-sub. Parameters: w, l
rnwell_sti2t (N-Well under STI 2 terminals resistor)	rnwellsti2t	RW	rnwell_STI2t	Drawn: WN, XN, DEV_AREA(rs). Terminals: Plus, Minus, Parameters: r,w,l or only r	Terminals: Plus, Minus Parameters: w, l
rnwell_sti3t (N-Well under STI 3 terminals resistor)	rnwellsti3t	RW3	rnwell_STI3t	Drawn: WN, XN, DEV_AREA(rs), DEV_AREA(t3) Terminals: Plus, Minus, P-sub. Parameters: r,w, l or only r	Terminals: Plus, Minus, P-sub. Parameters: w, l
rm1 (M1 resistor)	rm1	R1	rm1	Drawn: M1, M1(r1). Terminals: Plus, Minus. Parameters: r, w,l or r only	Terminals: Plus, Minus, Parameters: w, l
rm2 (M2 resistor)	rm2	R2	rm2	Drawn: M2, M2(r1). Terminals: Plus, Minus. Parameters: r, w,l or r only	Terminals: Plus, Minus, Parameters: w, l
rm3 (M3 resistor)	rm3	R3	rm3	Drawn: M3, M3 (r1) Terminals: Plus, Minus. Parameters: r, w,l or r only	Terminals: Plus, Minus, Parameters: w, l

Design Rules for SL18 Standard Logic Process
Doc. Name: DR2 0018 SL
Rev. Num: 0

Resistor Devices	Spice Model Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
rm4 (M4 resistor)	rm4	R4	rm4	Drawn: M4, M4(r1) Terminals: Plus, Minus. Parameters: r, w, l or r only	Terminals: Plus, Minus, Parameters: w, l,
rm5 (M5 resistor)	rm5	R5	rm5	Drawn: M5, M5(r1) Terminals: Plus, Minus. Parameters: r, w, l or r only	Terminals: Plus, Minus, Parameters: w, l,
rml (ML resistor)	rml	RL	rmL	Drawn: TOP_M, TOP_M (r1) Terminals: Plus, Minus. Parameters: r, w, l or r only	Terminals: Plus, Minus, Parameters: w, l,
rmt (MT resistor)	rmt	RT	rmt	Drawn: MT, MT(r1). Terminals: Plus, Minus. Parameters: r, w, l or r only	2 Terminals: Plus, Minus, Parameters: w, l,
rmt3 (MT3 resistor)	rmt3	RT3	rmt3	Drawn: MT3, MT3(r1). Terminals: Plus, Minus. Parameters: r, w, l or r only	2 Terminals: Plus, Minus, Parameters: w, l,

7.9.2 Diode Devices

Diode Devices	Spice Model Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
ndio_sal (1.8V N+ /Pwell Diode,) See Figure 8-39	dn18	DN	ndio_sal	Drawn: AA, XN, XP, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter, m	Terminals: Plus, Minus. Parameters: area, perimeter, m
pdio_sal (1.8V P+ /Nwell Diode) See Figure 8-40	dp18	DP	pdio_sal	Drawn: AA, WN, XP, XN, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter, m	Terminals: Plus, Minus. Parameters: area, perimeter, m
ndio_33 (3.3/5V N+ /Pwell Diode)	dnh	N3	ndio_33	Drawn: AA, XN, XP, AREA2, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter, m	Terminals: Plus, Minus. Parameters: area, perimeter, m
pdio_33 (3.3/5V P+ /Nwell Diode)	dph	P3	pdio_33	Drawn: AA, WN, XP, XN, AREA2, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter, m	Terminals: Plus, Minus. Parameters: area, perimeter, m
ndio_nosal (1.8/3.3/5V N+ w/o salicidation/Pwell diode)	dnh	NS	ndio_nosal	Drawn: AA, XN, NS, XP, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter, m	Terminals: Plus, Minus. Parameters: area, perimeter, m
pdio_nosal (1.8/3.3/5V P+ w/o salicidation/Nwell diode)	dph	PS	pdio_nosal	Drawn: AA, WN, XP, NS, XN, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter, m	Terminals: Plus, Minus. Parameters: area, perimeter, m
nwldio (1.8V Nwell/Psub Diode)	dnwell	DW	nwldio	Drawn: AA, WN, XN, XP, DEV_AREA(do), Terminals: Plus, Minus. Parameter: m	Terminals: Plus, Minus. Parameters: area, perimeter, m

Design Rules for SL18 Standard Logic Process
Doc. Name: DR2 0018 SL

Rev. Num: 0

Diode Devices	Spice Model Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
nwldio_33 (3.3/5V Nwell/Psub Diode)	dnwell33	W3	nwldio_33	Drawn: AA, WN, XN, XP, AREA2, DEV_AREA(do), Terminals: Plus, Minus. Parameters: m	Terminals: Plus, Minus. Parameters: area, perimeter, m
ddwnps18 (1.8/3.3/5V Deep-N-Well/Psub Diode)	ddwnps18	DDWNPS18	ddwnps18	Drawn: AA, WB, WN, XN, XP, Terminals: Plus, Minus. Parameters: m	Terminals: Plus, Minus. Parameters: area, perimeter, m
ddwnpw18 (1.8/3.3/5V Isolated-P-Well / Deep N-Well diode)	ddwnpw18	DDWNPW18	ddwnpw18	Drawn: AA, WB, WN, XN, XP, Terminals: Plus, Minus. Parameters: m	Terminals: Plus, Minus. Parameters: area, perimeter, m
dn50 (N+ in 5V Pwell diode)	dn50	DN50	dn50	Drawn: AA, XN, XP, AREA2, AREA6, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter	Terminals: Plus, Minus. Parameters: area, perimeter, m
dp50 (P+ in 5V Nwell diode)	dp50	DP50	dp50	Drawn: AA, WN, XP, XN, AREA2, AREA6, DEV_AREA(do). Terminals: Plus, Minus. Parameters: area, perimeter	Terminals: Plus, Minus. Parameters: area, perimeter, m

**Design Rules for SL18 Standard Logic Process**

Doc. Name: DR2 0018 SL

Rev. Num: 0

7.9.3 Bipolar Devices

VPNP BJT transistors are not characterized on EPI wafers of High resistance.

Bipolar Devices	Spice Model Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	Spice: Terminals and Parameters
vpnp18_10 (1.8V Vertical PNP 10um x 10um BJT) See 8.2.16	vd10	VD10	vpnp18_10	Drawn: AA, WN, XN, XP, DEV_AREA(bt). Terminals: Emitter, Base, Collector. Parameters: m,	Terminals: Emitter, Base, Collector. Parameters: m,
vpnp18_5 (1.8V Vertical PNP 5um x 5um BJT) See 8.2.16	vd5	VD5	vpnp18_5	Drawn: AA, WN, XN, XP, DEV_AREA(bt). Terminals: Emitter, Base, Collector. Parameters: m,	Terminals: Emitter, Base, Collector. Parameters: m,
vpnp18_2 (1.8V Vertical PNP 2um x 2um BJT) See 8.2.16	vd2	VD	vpnp18_2	Drawn: AA, WN, XN, XP, DEV_AREA(bt). Terminals: Emitter, Base, Collector. Parameters: m,	Terminals: Emitter, Base, Collector. Parameters: m,
vpnp18_5 (1.8/3.3/5V Vertical NPN 5um x 5um BJT)	vdn5	VDN5	vpnp18_5	Drawn: AA, WN, WB, XN, XP, DEV_AREA(bt). Terminals: Emitter, Base, Collector. Parameters: m,	Terminals: Emitter, Base, Collector. Parameters: m,
vpnp33_10 (3.3/5 Vertical PNP 10um x 10um BJT) See 8.2.16	vd10	VH10	vpnp33_10	Drawn: AA, WN, XN, XP, AREA2 DEV_AREA(bt). Terminals: Emitter, Base, Collector. Parameters: m,	Terminals: Emitter, Base, Collector. Parameters: m,
vpnp33_5 (3.3/5 Vertical PNP 5um x 5um BJT) See 8.2.16	vd5	VH5	vpnp33_5	Drawn: AA, WN, XN, XP, AREA2, DEV_AREA(bt). Terminals: Emitter, Base, Collector. Parameters: m,	Terminals: Emitter, Base, Collector. Parameters: m,
vpnp33_2 (3.3/5 Vertical PNP 2um x 2um BJT) See 8.2.16	vd2	VH	vpnp33_2	Drawn: AA, WN, XN, XP, AREA2, DEV_AREA(bt). Terminals: Emitter, Base, Collector. Parameters: m,	Terminals: Emitter, Base, Collector. Parameters: m,
Scalable VPNP 1.8V for LVS purposes ONLY	NA	VPNP18_S	N/A	Drawn: AA, WN, XN, XP, DEV_AREA(bt). DEV_AEWA (cp) Terminals: Emitter, Base, Collector. Parameters: area, perimeter	N/A
Scalable VPNP 3.3V for LVS purposes ONLY	N/A	VPNP33_S	N/A	Drawn: AA, WN, XN, XP, AREA2, DEV_AREA(bt), DEV_AREA(cp) Terminals: Emitter, Base, Collector. Parameters: area, perimeter	N/A
Scalable VNPV 1.8V for LVS purposes ONLY	N/A	VNPV18_S	N/A	Drawn: AA, WN, WB, XN, XP, DEV_AREA(bt), DEV_AREA(cp) Terminals: Emitter, Base, Collector. Parameters: area, perimeter	N/A

7.9.4 Capacitor Devices

**Design Rules for SL18 Standard Logic Process**

Doc. Name: DR2 0018 SL

Rev. Num: 0

Capacitor Devices	Spice Name	LVS Name	PDK Name	LVS: Layers, OPUS Terminals and Parameters	SPICE: Terminals Parameters and Sub circuit
cnwnmos_thn2t (1.8V Scalable Accumulation capacitor N+ Gate/ NW, 2 terminals)	nwcap2t	CL	cnwnmos_thn2t	Drawn: AA, WN, GC, XN. DEV_AREAC(cp). Terminals: Plus, Minus. Parameters: w,l,m or only m	Terminals: Plus, Minus. Parameters: w, l
cnwnmos_thn2t (1.8V Scalable Accumulation capacitor N+ Gate/NW)3 terminals , See Figure 8-41	nwcap3t	CL3	cnwnmos_thn2t	Drawn: AA, WN, GC, XN. DEV_AREAC(cp). DEV_AREA (t3) Terminals: Plus, Minus, PSUB. Parameters: w,l m or only m	Terminals: Plus, Minus, PSUB. Parameters: w, l
cnwnmos_thk2t (3.3V Scalable Accumulation capacitor, N+ Gate/NW,2 terminals)	nwcaph2t	CH	cnwnmos_thk2t	Drawn: AA, WN, GC, XN, AREA2. DEV_AREA(cp). Terminals: Plus, Minus. Parameters: w,l,m or only m	Terminals: Plus, Minus. Parameters: w, l
cnwnmos_thk3t (3.3V Scalable Accumulation capacitor, N+ Gate/NW,3 terminals)	nwcaph3t	CH3	cnwnmos_thk3t	Drawn: AA, WN, GC, XN, AREA2. DEV_AREA(cp). DEV_AREA(t3). Terminals: Plus, Minus, PSUB. Parameters: w,l,m or only m	Terminals: Plus, Minus, PSUB. Parameters: w, l
5V Accumulation capacitor, N+ Gate/NW. See section 8.2.12	ch53t	CH53T	ch53t	Drawn: AA, WN, GC, XN, AREA2. AREA6, DEV_AREA(cp) DEV_AREA(t3). Terminals: Plus, Minus, PSUB. Parameters: w,l,m	Terminals: Plus, Minus, PSUB. Parameters: w, l
cmim_sq (MIM Capacitor, 1fF/um2 capacitor)	cmim_sq	CB	cmim_sq	Drawn: M(top-1), CE Terminals: Plus, Minus. Parameters: w ,l ,c, and m or c and m	Terminals: Plus, Minus. Parameters: w, l and m.
cmim_mt (MIM Capacitor, 1fF/um2 with MT capacitor)	cmim_sq	CB	cmim_mt	Drawn: M(top-1), CE Terminals: Plus, Minus. Parameters: w ,l ,c, and m or c and m	Terminals: Plus, Minus. Parameters: w, l and m.
cmim_hc (High Capacitance single plate 1.7fF/um2 MIM Capacitor) see Figure 8-46	cmim_hc	CMIM_HC	cmim_hc	Drawn: M(top-1), CE, AREA10(mim). Terminals: Plus, Minus. Parameters:w, l, c and m or c and m	Terminals: Plus, Minus. Parameters: w, l and m.
cmim_stk (High Capacitance stacked MIM 3.4fF/um2 capacitore)	cmim_stk	CMIM_STK	cmim_stk	Drawn: TOP_M, TOP_M-1, TOP_M-2, CE, AREA10(mim), SPC_AREA(mim). Terminals: Plus, Minus. Parameters:w, l, c and m or c and m	Terminals: Plus, Minus. Parameters: w, l and m.
Cmim_uhd (Ultra High Density Single plate 2.8fF/um2 MIM) capacitor.	cmim_uhd	CMIM_UHD	cmim_uhd	Drawn: M(top-1), CE, 78dt43 Terminals: Plus, Minus. Parameters:w, l, c and m or c and m	Terminals: Plus, Minus. Parameters: w, l and m.
Cmim_uhdstk (Ultra High Density stacked 5.6fF/um2 MIM) capacitor.	cmim_uhdstk	CMIM_UHDSTK	cmim_uhdstk	Drawn: TOP_M, TOP_M-1, TOP_M-2, CE, 78dt43, SPC_AREA(mim). Terminals: Plus, Minus. Parameters:w, l, c and m or c and m	Terminals: Plus, Minus. Parameters: w, l and m.

**Design Rules for SL18 Standard Logic Process**

Doc. Name: DR2 0018 SL

Rev. Num: 0

Capacitor Devices	Spice Name	LVS Name	PKD Name	LVS: Layers, OPUS Terminals and Parameters	SPICE: Terminals Parameters and Sub circuit
CM52t (M15 FSG 2T MFC) See section 8.2.14	cm52t	CM5	cm52t	M1...M5, V2...V5. DEV_AREA(mc). Terminals: Plus, Minus. Parameters: nf, l	Terminals: Plus, Minus. Parameters: nf, l
CM42t (M14 FSG 2T MFC)	cm42t	CM42T	cm42t	M1...M4, V2...V4. DEV_AREA(mc). Terminals: Plus, Minus. Parameters: nf, l	Terminals: Plus, Minus,. Parameters: nf, l
CM32t (M13 FSG 2T MFC) see section 8.2.14	cm32t	CM32T	cm32t	M1, M2, M3, V2, V3. DEV_AREA(mc). Terminals: Plus, Minus. Parameters: nf, l	Terminals: Plus, Minus,. Parameters: nf, l
CM53t (M15 FSG 3T MFC) See section 8.2.14	cm53t	CM53	cm53t	M1...M5, V2...V5. DEV_AREA (mc) DEV_AREA (t3). Terminals: Plus, Minus, (Bulk). Parameters: nf, l	Terminals: Plus, Minus, (Bulk). Parameters: nf, l
CM43t (M14 FSG 3T MFC) See section 8.2.14	cm43t	CM43T	cm43t	Layers: M1...M4, V2...V4. DEV_AREA (mc) DEV_AREA (t3). Terminals: Plus, Minus, Bulk. Parameters: nf, l	Terminals: Plus, Minus, Bulk. Parameters: nf, lSub circuit: No
CM33t (M13 FSG 3T MFC)	cm33t	CM33T	cm33t	M1, M2, M3, V2, V3. DEV_AREA(mc) DEV_AREA (t3). . Terminals: Plus, Minus, Bulk. Parameters: nf, l	Terminals: Plus, Minus,. Parameters: nf, l
USGCM52T (M15 USG 2T MFC) See section 8.2.14	usgcm52t	usgcm52t	usgcm52t	M1...M5, V2...V5. DEV_AREA(cp), DEV_AREA(mc). Terminals: Plus, Minus. Parameters: nf, l	Terminals: Plus, Minus. Parameters: nf, l
USGCM42T (M14 USG 2T MFC) See section 8.2.14	usgcm42t	usgcm42t	usgcm42t	M1...M4, V2...V4. DEV_AREA(cp), DEV_AREA(mc). Terminals: Plus, Minus. Parameters: nf, l	Terminals: Plus, Minus,. Parameters: nf, l
USGCM32T (M13 USG 2T MFC) See section 8.2.14	usgcm32t	usgcm32t	usgcm32t	M1, M2, M3, V2, V3. DEV_AREA(cp), DEV_AREA(mc). Terminals: Plus, Minus. Parameters: nf, l	Terminals: Plus, Minus,. Parameters: nf, l
USGCM53T (M15 USG 3T MFC) See section 8.2.14	usgcm53t	usgcm53t	usgcm53t	M1...M5, V2...V5. DEV_AREA(cp), DEV_AREA (mc) DEV_AREA (t3). Terminals: Plus, Minus, (Bulk). Parameters: nf, l	Terminals: Plus, Minus, (Bulk). Parameters: nf, l
USGCM43T (M14 USG 3T MFC) See section 8.2.14	usgcm43t	usgcm43t	usgcm43t	Layers: M1...M4, V2...V4. DEV_AREA(cp), DEV_AREA (mc) DEV_AREA (t3). Terminals: Plus, Minus, Bulk. Parameters: nf, l	Terminals: Plus, Minus, Bulk. Parameters: nf, lSub circuit: No
USGCM33T (M13 USG 3T MFC) See section 8.2.14	usgcm33t	usgcm33t	usgcm33t	M1, M2, M3, V2, V3. DEV_AREA(cp), DEV_AREA(mc), DEV_AREA (t3). . Terminals: Plus, Minus, Bulk. Parameters: nf, l	Terminals: Plus, Minus,. Parameters: nf, l

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

7.9.5 Devices for Debugging Purposes

Devices for Debugging Purposes	Spice Name	LVS Name	PDK Name	LVS: Layers, Terminals and Parameters	SPICE: Terminals Parameters and Sub circuit
epf (ePolyFuse, Type-1) See section 8.2.15.2	r	EPF	epf	Drawn: GC, XP, 78dt105. Terminals: Plus, Minus. Parameters: R	Terminals: Plus, Minus. Parameters: R



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

7.10 Drawing Instructions for 3-Terminal Marking Layer 77dt86

The aim of layer DEV_AREA(3t)=77dt86 is, to enable the LVS to handle a mixed design contained both 2-terminal and 3-terminal devices. Basically, the LVS includes several “TERM” flags for terminal definition. Using this flags and the 77dt86 marking layer, designer can set the recognition and the modeling attached:

2TERM	3TERM	Description	2t device	3t device
Off	Off	For having mixed design of 2t and 3t	LVS=2t	LVS=3t
On	Off	For having 2t all over the design	LVS=2t	LVS=2t
Off	On	For having 3t all over the design	LVS=3t	LVS=3t
On	On	N/A. The LVS will error	Error	Error

- Any new design of 3-terminal devices made using SCL PDK, will have layer 77dt86, as part of the generated layout. However, existing GDS that contain 3t devices are still supported, as explained by the above table.
- For 2t/3t mixing design, designer has the possibility to use 2t/3t .
- 3t devices are using the SAME spice model and having the same CDL. However, the symbol will be different.

Drawing guidelines: the marking layers needs to be coincidence with the device recognition layer. For example, in case of 3t MR resistor, layer 77dt86 should be coincidence with the DEV_AREA(rs) marking layer (see

Figure 8-27). For accumulation capacitors, layer 77dt86 should be coincidence with the DEV_AREA(cp) marking layer (see

Figure 8-41).



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

7.11 PDK Limits and Coverage

TS1 platform PDK covers the overall list of the legal devices, as specified in Section 7.9 (Legal Device List). Using the PDK, cells in verity of dimensions can be created. All the devices created in the range of dimensions specified below, will be DRC clean and will have a full SPICE and LVS coverage. Devices with dimensions outside the range specified below may not have a full modeling or DRC coverage. All values are in um.

Table 7-8: PDK Limits and Coverage

Transistors	Lmin	Lmax	Wmin	Wmax	Wdefault	Ldefault
nmos_18 (1.8V SVt NMOS Transistor)	0.18	10	0.22	100	0.42	0.18
pmos_18 (1.8V SVt PMOS Transistor)	0.18	10	0.22	100	0.42	0.18
n18lvt (1.8V Low Vt NMOS transistor)	0.18	10	0.22	100	0.42	0.18
p18lvt (1.8V Low Vt PMOS transistor)	0.18	10	0.22	100	0.42	0.18
n18hvt (1.8V HVt NMOS transistor)	0.18	10	0.22	100	0.42	0.18
p18hvt (1.8V HVt PMOS transistor)	0.18	10	0.22	100	0.42	0.18
nmos_18_iso (1.8V SVt NMOS Transistor in IPW)	0.18	10	0.22	100	0.42	0.18
n18hvt_iso (1.8V HVt NMOS transistor in IPW)	0.18	10	0.22	100	0.42	0.18
n18lvt_iso (1.8V Low Vt NMOS transistor in IPW)	0.18	10	0.22	100	0.42	0.18
nmos_native (1.8V NMOS Native Transistor)	0.5	10	0.42	100	0.42	0.5
nmos_33 (3.3V SVt NMOS Transistor)	0.35	10	0.22	100	0.42	0.35
pmos_33 (3.3V SVt PMOS Transistor)	0.3	10	0.22	100	0.42	0.3
nmos_33_iso (3.3V SVt NMOS Transistor in IPW)	0.35	10	0.22	100	0.42	0.35
nmos_native_33 (3.3V NMOS Native Transistor)	1.2	10	0.42	100	0.42	1.2
nmos_5V (5V SVt NMOS transistor)	0.5	10	0.42	100	0.42	0.6
pmos_5V (5V SVt PMOS transistor)	0.5	10	0.42	100	0.42	0.5
nmos_5V_iso (5V SVt NMOS transistor in IPW)	0.5	10	0.42	100	0.42	0.6
nmos_native_5 (5V NMOS Native Transistor)	1.5	10	0.6	100	0.42	1.5

Resistors	Lmin	Lmax	Wmin	Wmax	Wdefault	Ldefault
rphpoly2t (HIPO-1k High Ohmic P-type Poly w/o salicide 2 terminal resistor)	1	NA	0.18	NA	0.18	1
rphpoly3t (HIPO-1k High Ohmic P-type Poly w/o salicide 3 terminal resistor)	1	NA	0.18	NA	0.18	1
mr22t (2k High Ohmic P-type Poly w/o salicide 2 terminal resistor)	1	NA	0.18	NA	0.18	1
mr23t (2k High Ohmic P-type Poly w/o salicide 3 terminal resistor)	1	NA	0.18	NA	0.18	1
rnmpoly2t (NMOPO-N-type Poly w/o salicide 2 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43
rnmpoly3t (NMOPO-N-type Poly w/o salicide 3 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43
rpmpoly2t (PMOPO-P-type Poly w/o salicide 2 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43
rpmpoly3t (PMOPO-P-type Poly w/o salicide 3 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Resistors	Lmin	Lmax	Wmin	Wmax	Wdefault	Ldefault
resistor)						
rnplus2t (N+ AA Diffusion w/o salicide 2 terminals resistor)	0.43	NA	0.42	NA	0.42	0.43
rnplus3t (N+ AA Diffusion w/o salicide 3 terminals resistor,)	0.43	NA	0.42	NA	0.42	0.43
rpplus2t (P+ AA Diffusion w/o salicide 2 terminals resistor)	0.43	NA	0.42	NA	0.42	0.43
rpplus3t (P+ AA Diffusion w/o salicide 3 terminals resistor)	0.43	NA	0.42	NA	0.42	0.43
rnlpoly2t (N-poly salicide 2 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43
rnlpoly3t (N-poly salicide 3 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43
rpplpoly2t (P-poly salicide 2 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43
rpplpoly3t (P-poly salicide 3 terminals resistor)	0.43	NA	0.18	NA	0.18	0.43
rnplus_sal2t (N+ AA Diffusion salicide 2 terminals resistor)	0.25	NA	0.42	NA	0.42	0.25
rnplus_sal3t (N+ AA Diffusion salicide 3 terminals resistor)	0.25	NA	0.42	NA	0.42	0.25
rpplus_sal2t (P+ AA Diffusion salicide 2 terminals resistor)	0.25	NA	0.42	NA	0.42	0.25
rpplus_sal3t (P+ AA Diffusion salicide 3 terminals resistor)	0.25	NA	0.42	NA	0.42	0.25
RNPLUS2T (N+ AA Diffusion w/o salicidation resistor, 2 terminals)	0.43	NA	0.42	NA	1.42	0.43
RNPLUS3T (N+ AA Diffusion w/o salicidation resistor, 3 terminals)	0.43	NA	0.42	NA	1.42	0.43
RPPLUS2T (P+ AA Diffusion w/o salicidation resistor, 2 terminals)	0.43	NA	0.42	NA	1.42	0.43
RPPLUS3T (P+ AA Diffusion w/o salicidation resistor, 3 terminals)	0.43	NA	0.42	NA	1.42	0.43
rnwell_aa2t (N-Well under AA 2 terminals resistor)	0.44	NA	2.1	NA	2.1	0.44
rnwell_aa3t (N-Well under AA 3 terminals resistor)	0.44	NA	2.1	NA	2.1	0.44
rnwell_sti2t (N-Well under STI 2 terminals resistor)	2.1	NA	2.1	NA	2.1	2.1
rnwell_sti3t (N-Well under STI 3 terminals resistor)	2.1	NA	2.1	NA	2.1	2.1
rm1 (M1 resistor)	0.12	NA	0.23	35	0.23	0.23
rm2 (M2 resistor)	0.12	NA	0.28	35	0.28	0.28
rm3 (M3 resistor)	0.12	NA	0.28	35	0.28	0.28
rm4 (M4 resistor)	0.12	NA	0.28	35	0.28	0.28
rm5 (M5 resistor)	0.12	NA	0.28	35	0.28	0.28
rml (ML resistor)	0.12	NA	0.44	35	0.44	0.44
rmt (MT resistor)	0.12	5000	1.2	35	1.2	1.2
rmt3 (MT3 resistor)	2.5	100	2.5	100	2.5	2.5

Diodes	Lmin	Lmax	Wmin	Wmax	Wdefault	Ldefault
ndio_sal (1.8V N+ /Pwell Diode)	0.45	60066	0.45	60066	1.97	1.97
pdio_sal (1.8V P+ /Nwell Diode)	0.42	60066	0.42	60066	1.97	1.97
ndio_33 (3.3/5V N+ /Pwell Diode)	0.45	60066	0.45	60066	1.97	1.97
pdio_33 (3.3/5V P+ /Nwell Diode)	0.45	60066	0.45	60066	1.97	1.97
ndio_nosal (1.8/3.3/5V N+ w/o salicidation/Pwell diode)	1.15	60066	1.15	60066	1.97	1.97
pdio_nosal (1.8/3.3/5V P+ w/o salicidation/Nwell diode)	1.15	60066	1.15	60066	1.97	1.97



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Diodes	Lmin	Lmax	Wmin	Wmax	Wdefault	Ldefault
nwldio (1.8V Nwell/Psub Diode)	0.86	60066	0.86	60066	2.21	2.21
nwldio_33 (3.3/5V Nwell/Psub Diode)	0.86	60066	0.86	60066	2.21	2.21
ddwnps18 (1.8/3.3/5V Deep-N-Well/Psub Diode)	1.15	100	2	100	2	1.97
ddwnpw18 (1.8/3.3/5V Isolated-P-Well / Deep N-Well diode)	1.15	100	2	100	2	1.97
dn50 (N+ in 5V Pwell diode)	0.45	100	0.45	100	0.6	0.6
dp50 (P+ in 5V Nwell diode)	0.45	100	0.45	100	0.6	0.6

BJT	Lmin	Lmax	Wmin	Wmax	Wdefault	Ldefault
PDK=VPNP18_10 (VPNP 1.8V 10umx10um BJT)	10	10	10	10	10	10
PDK=VPNP18_5 (VPNP 1.8V 5umx5um BJT)	5	5	5	5	5	5
PDK=VPNP18_2 (VPNP 1.8V 2umx2um BJT)	2	2	2	2	2	2
PDK=VPNP33_10 (VPNP 3.3V 10umx10um BJT)	10	10	10	10	10	10
PDK=VPNP33_5 (VPNP 3.3V 5umx5um BJT)	5	5	5	5	5	5
PDK=VPNP33_2 (VPNP 3.3V 2umx2um BJT)	2	2	2	2	2	2
PDK=VNP18_5 (VNP 1.8V 5umx5um BJT)	5	5	5	5	5	5
LVS=VPNP18_S (Scalable VNP 1.8V)	1.5	11	1.5	11	N/A	N/A
LVS=VPNP33_S (Scalable VNP 3.3V)	1.5	11	1.5	11	N/A	N/A
LVS=VNP18_S (Scalable VNP 1.8V)	1.5	11	1.5	11	N/A	N/A

Capacitors	Lmin	Lmax	Wmin	Wmax	Wdefault	Ldefault
NWCAP2T (1.8V Scalable Accumulation N+ Gate/NW 2 terminals)	1	60	1	60	5	5
NWCAP3T (1.8V Scalable Accumulation N+ Gate/NW 3 terminals)	1	60	1	60	5	5
NWCAPH2T (3.3V Scalable Accumulation N+ Gate/NW 2 terminals)	1	60	1	60	5	5
NWCAPH3T (3.3V Scalable Accumulation N+ Gate/NW 3 terminals)	1	60	1	60	5	5
CH53T (5C Accumulation N+ Gate/NW Capacitor)	0.6	60	0.6	60	0.6	0.6
CMIM_HC (MIM High Capacitance single plate capacitor)	4	1000000	4	1000000	4	4
CMIM_STK (MIM High Capacitance stacked capacitor)	4	1000000	4	1000000	4	4
CMIM_UHD (Ultra High Density Single plate 2.8fF/um2 MIM) capacitor.	4	1000000	4	1000000	4	4
CMIM_UHDSTK (Ultra High Density stacked 5.6fF/um2 MIM) capacitor.	4	1000000	4	1000000	4	4
CM52t (M15 FSG 2T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
CM42t (M14 FSG 2T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
CM32t (M13 FSG 2T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
CM53t (M15 FSG 3T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
CM43t (M14 FSG 3T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
CM33t (M13 FSG 3T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
USGCM52T (M15 USG 2T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
USGCM42T (M14 USG 2T MFC)	1.1	1000000	N/A	N/A	N/A	1.1



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

USGCM32T (M13 USG 2T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
USGCM53T (M15 USG 3T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
USGCM43T (M14 USG 3T MFC)	1.1	1000000	N/A	N/A	N/A	1.1
USGCM33T (M13 USG 3T MFC)	1.1	1000000	N/A	N/A	N/A	1.1

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Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

7.12 Drawing and Support Layers TrueTable

Transistors

Description	AA	WN	Thick-Ox	5V devices	Poly	N+ S/D	P+ S/D	No-Sal	DWN	CNL	CPL	CNH	CPH	WB	Native MOS	Res Mask	Cap Mark	Diode Mark	BJT Mark	DEV_ARE A(le)	DEV_ARE A(lp)
Layer No/dt	9dt0	3dt0	72dt0	76dt0	8dt0	32dt0	31dt0	88dt0	22dt0	162dt0	163dt0	10dt0	65dt0	4dt0	17dt0	77dt42	77dt43	77dt44	77dt47	77dt113	77dt41
Device	AA(lc)	WN(lc)	AREA2(lc)	AREA6(lc)	GC(lc)	XN(lc)	XP(lc)	NS(lc)	WB(lc)	CNL(lc)	CPL(lc)	CNH(lc)	CPH(lc)	WB(lc)	VNN(lc)	(rs)	(cp)	(dio)	(bt)	(le)	(lp)
nmos_18	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No
pmos_18	Yes	Yes	No	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No
n18lvt	Yes	No	No	No	Yes	Yes	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No
p18lvt	Yes	Yes	No	No	Yes	No	Yes	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No
n18hvt	Yes	No	No	No	Yes	Yes	No	No	No	No	No	Yes	No	No	No	No	No	No	No	No	Yes
p18hvt	Yes	Yes	No	No	Yes	No	Yes	No	No	No	No	No	Yes	No	No	No	No	No	No	No	Yes
nmos_18_iso	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No
n18hvt_iso	Yes	No	No	No	Yes	Yes	No	No	No	No	No	Yes	No	Yes	No	No	No	No	No	No	Yes
n18lvt_iso	Yes	No	No	No	Yes	Yes	No	No	No	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No
nmos_native	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No
nmos_33	Yes	No	Yes	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No
pmos_33	Yes	Yes	Yes	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No
nmos_33_iso	Yes	No	Yes	No	Yes	Yes	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No
nmos_native_33	Yes	No	Yes	No	Yes	Yes	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No
nmos_5V	Yes	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No
pmos_5V	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No
nmos_5V_iso	Yes	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No
nmos_native_5	Yes	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Resistors

Description	AA	WN	Thick-Ox	Poly	N+ S/D	P+ S/D	No-Sal	CE	MR	MR2	M1	M2	M3	M4	M5	ML	MT	MT3	DEV_A REA(3t)	Res Mask
Layer No/dt	9dt0	3dt0	72dt0	8dt0	32dt0	31dt0	88dt0	99dt0	87dt0	164dt0	40dt0	42dt0	44dt0	46dt0	48dt0	50dt0	59dt0	184dt0	77dt86	77dt42
Device	AA(lc)	WN(lc)	AREA2 (lc)	GC(lc)	XN(lc)	XP(lc)	NS(lc)	CE(lc)	MR(lc)	MR2 (lc)	M1(lc)	M2(lc)	M3(lc)	M4(lc)	M5(lc)	ML(lc)	MT(lc)	MT3(lc)	(3t)	(rs)
rphpoly2t	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rphpoly3t	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
mr22t	No	No	No	Yes	No	Yes	Yes	No	No	Yes	No	No	No	No	No	No	No	No	No	Yes
mr23t	No	No	No	Yes	No	Yes	Yes	No	No	Yes	No	No	No	No	No	No	No	No	Yes	Yes
rnmpoly2t	No	No	No	Yes	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rnmpoly2t	No	No	No	Yes	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rpmpoly2t	No	Yes	No	Yes	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rpmpoly3t	No	Yes	No	Yes	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rnlpoly2t	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rnlpoly3t	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rpplpoly2t	No	Yes	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rpplpoly3t	No	Yes	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rnplus2t	Yes	No	No	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rnplus3t	Yes	No	No	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rpplus2t	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rpplus3t	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rnplus_sal2t	Yes	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rnplus_sal3t	Yes	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rpplus_sal2t	Yes	Yes	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rpplus_sal3t	Yes	Yes	No	No	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rnwell_aa2t	Yes	Yes	No	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rnwell_aa3t	Yes	Yes	No	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
rnwell_sti2t	No	Yes	No	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes
rnwell_sti3t	No	Yes	No	No	Yes	No	Yes	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Description	AA	WN	Thick-Ox	Poly	N+ S/D	P+ S/D	No-Sal	CE	MR	MR2	M1	M2	M3	M4	M5	ML	MT	MT3	DEV_A REA(3t)	Res Mask
Layer No/dt	9dt0	3dt0	72dt0	8dt0	32dt0	31dt0	88dt0	99dt0	87dt0	164dt0	40dt0	42dt0	44dt0	46dt0	48dt0	50dt0	59dt0	184dt0	77dt86	77dt42
Device	AA(lc)	WN(lc)	AREA2 (lc)	GC(lc)	XN(lc)	XP(lc)	NS(lc)	CE(lc)	MR(lc)	MR2 (lc)	M1(lc)	M2(lc)	M3(lc)	M4(lc)	M5(lc)	ML(lc)	MT(lc)	MT3(lc)	(3t)	(rs)
rm1	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No	No	Yes
rm2	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	No	Yes
rm3	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	No	Yes
rm4	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No	No	Yes
rm5	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	No	Yes
rml	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No	No	Yes
rmt	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No	Yes
rmt3	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	Yes

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Diodes

Description	AA	WN	Thick-Ox	5V devices	Poly	N+ S/D	P+ S/D	No-Sal	DWN	HN	CPL	CNH	CPH	MR	Diode Mark
Layer No/dt	9dt0	3dt0	72dt0	76dt0	8dt0	32dt0	31dt0	88dt0	22dt0	23dt0	163dt0	10dt0	65dt0	87dt0	77dt44
Device	AA(lc)	WN(lc)	AREA2(lc)	AREA6(lc)	GC(lc)	XN(lc)	XP(lc)	NS(lc)	WB(lc)	HN(lc)	CPL(lc)	CNH(lc)	CPH(lc)	MR(lc)	(dio)
ndio_sal	Yes	No	No	No	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes
pdio_sal	Yes	Yes	No	No	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes
ndio_33	Yes	No	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	Yes
pdio_33	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	Yes
ndio_nosal	Yes	No	No	No	No	Yes	Yes	No	No	No	No	No	No	No	Yes
pdio_nosal	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	Yes
nwldio	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No
nwldio_33	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	No
ddwnps18	Yes	Yes	No	No	No	Yes	Yes	No	Yes	No	No	No	No	No	Yes
ddwnpw18	Yes	Yes	No	No	No	Yes	Yes	No	Yes	No	No	No	No	No	Yes
dn50	Yes	No	Yes	Yes	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes
dp50	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	No	No	No	No	No	No	Yes

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

BJT's

Description	AA	WN	Thick-Ox	5V devices	Poly	N+ S/D	P+ S/D	No-Sal	DWN	CNL	CPL	CNH	CPH	MR	WPL	DEV_A REA(hg)	DEV_A REA(bt)
Layer No/dt	9dt0	3dt0	72dt0	76dt0	8dt0	32dt0	31dt0	88dt0	22dt0	162dt0	163dt0	10dt0	65dt0	87dt0	13dt0	77dt24	77dt47
Device	AA(lc)	WN(lc)	AREA2 (lc)	AREA6 (lc)	GC(lc)	XN(lc)	XP(lc)	NS(lc)	WB(lc)	CNL (lc)	CPL (lc)	CNH (lc)	CPH (lc)	MR (lc)	WPL (lc)	(hg)	(bt)
vpnp18_10	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes
vpnp18_5	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes
vpnp18_2	Yes	Yes	No	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes
vpnp33_10	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes
vpnp33_5	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes
vpnp33_2	Yes	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes
vnnp18_5	Yes	Yes	No	No	No	Yes	Yes	No	Yes	No	No	No	No	No	No	No	Yes

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Capacitors

Description	AA	WN	Thick-Ox	5V devices	Poly	N+ S/D	P+ S/D	DWN	CE	M1	M2	M3	M4	M5	SPC_ AREA (mm)	DEV_ AREA (3t)	AREA10 (mm)	DEV_ AREA (mc)	DEV_ AREA (cp)	DEV_ AREA (mm)	DEV_ AREA (mm)
Layer No/dt	9dt0	3dt0	72dt0	76dt0	8dt0	32dt0	31dt0	22dt0	99dt0	40dt0	42dt0	44dt0	46dt0	48dt0	78dt56	77dt86	80dt56	77dt73	77dt43	77dt57	77dt56
Device	AA(lc)	WN(lc)	AREA2 (lc)	AREA6 (lc)	GC(lc)	XN(lc)	XP(lc)	WB (lc)	CE (lc)	M1 (lc)	M2 (lc)	M3 (lc)	M4 (lc)	M5 (lc)	(mm)	(3t)	(mm)	(cp)	(cp)	(mm)	(mm)
cnwnmos_thn2t	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
cnwnmos_thn3t	Yes	Yes	No	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes	No	No	Yes	No	No
cnwnmos_thk2t	Yes	Yes	Yes	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes	No	No
cnwnmos_thk3t	Yes	Yes	Yes	No	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes	No	No	Yes	No	No
ch53t	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	No	No	No

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Description	CE	M1	V2	M2	V3	M3	V4	M4	V5	M5	ML	SPC_ AREA (mm)	DEV_ AREA (3t)	AREA10 (mm)	SPC_AREA (cp)	DEV_ AREA (mc)	DEV_ AREA (mm)	DEV_ AREA (mm)
Layer No/dt	99dt0	40dt0	41dt0	42dt0	43dt0	44dt0	45dt0	46dt0	47dt0	48dt0	50dt0	78dt56	77dt86	80dt56	78dt43	77dt73	77dt57	77dt56
Device	CE (lc)	M1 (lc)	V2 (lc)	M2 (lc)	V3 (lc)	M3 (lc)	V4 (lc)	M4 (lc)	V5 (lc)	M5 (lc)	ML (lc)	(mm)	(3t)	(mm)	(mm)	(mc)	(nm)	(mm)
cmim_sq	Yes	No	No	Depend to metal combination refer to Table 7-6								No	No	No	No	No	No	No
cmim_hc	Yes	No	No	Depend to metal combination refer to Table 7-6								No	No	Yes	No	No	No	No
cmim_stk	Yes	No	No	Depend to metal combination refer to Table 7-6								Yes	No	No	No	No	No	No
cmim_uhd	Yes	No	No	Depend to metal combination refer to Table 7-6								No	No	No	Yes	No	No	No
Cmim_uhdstk	Yes	No	No	Depend to metal combination refer to Table 7-6								Yes	No	No	Yes	No	No	No
cm52t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	No	No
cm42t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	Yes	No	No
cm32t	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes	No	No
cm53t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No	Yes	No	No
cm43t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	Yes	No	No	Yes	No	No
cm33t	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	Yes	No	No	Yes	No	No
usgcm52t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	No	No
usgcm42t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	Yes	No	No
usgcm32t	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	No	Yes	No	No
usgcm43t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No	Yes	No	No
usgcm33t	No	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	Yes	No	No	Yes	No	No
usgcm53t	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No	Yes	No	No

**ePF**

Description	Poly	XP	M1	M2	SPC_AREA (fe)
Layer No/dt	8dt0	31dt0	40dt0	42dt0	78dt105
Device	GC(lc)	XP(lc)	M1(lc)	M2(lc)	(fe)
epf (any type)	Yes	Yes	No	No	Yes

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Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

7.13 LVS Definition for Legal Devices

The general LVS definition for every legal device is listed at the relevant LVS file.

7.14 Rule for Drawing the Support Layer DEV_AREA(lp)

This marking layer is used for LVS, Modeling and masks generation purposes:

To define entire chip to run at 1.8V LL (HVt) flavor: For example, for 1.8HVt/3.3V. Designer should cover the overall chip area, corner-to-corner (which also includes pads, seal ring, IP's, I/O, thin-oxide devices, thick-oxide devices, etc) by the support layer DEV_AREA(lp). Do not cover external structures.

Make sure, that at any case, CNH/CPH layers do not exist under DEV_AREA(lp), because it will cause MDP problems. See rule LLSUP.N.2 , and Rule BAD_EXT.N.1, see section 10.7.

7.14.1 DEV_AREA(lp) Support layer Rules		Action	Drawn
LLSUP.W.1	Width of DEV_AREA(lp)	min	0.44
LLSUP.S.1	Space of DEV_AREA(lp) layer	min	0.44
LLSUP.D.1	Distance from DEV_AREA(lp) layer to SVt AA (AA that is NOT LL). Overlap is NOT allowed.	min	0.44
LLSUP.D.2	Distance between DEV_AREA(lp) to AREA2. DEV_AREA(lp) polygon is not allowed to cut AREA2	min	0.86
LLSUP.D.3	Distance from DEV_AREA(lp) layer to WN(lc). DEV_AREA(lp) is not allowed to cut WN	min	0.86
LLSUP.E.1	Enclosure of DEV_AREA(lp) around LL AA. AA “cutting” is not allowed	min	0.18
LLSUP.X.2	Extension of DEV_AREA(lp) beyond LL AA along GC gate. This extension rule must extend 0.32um perpendicular to GC end cap	min	0.35
LLSUP.N.1	DEV_AREA(lp) having 45deg shapes is not allowed. Layer must be rectangular		
LLSUP.N.2	DEV_AREA(lp) interact with CNH or interact with CPH is NOT allowed		
LLSUP.A.1	Area of 77dt41	min	4um2



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

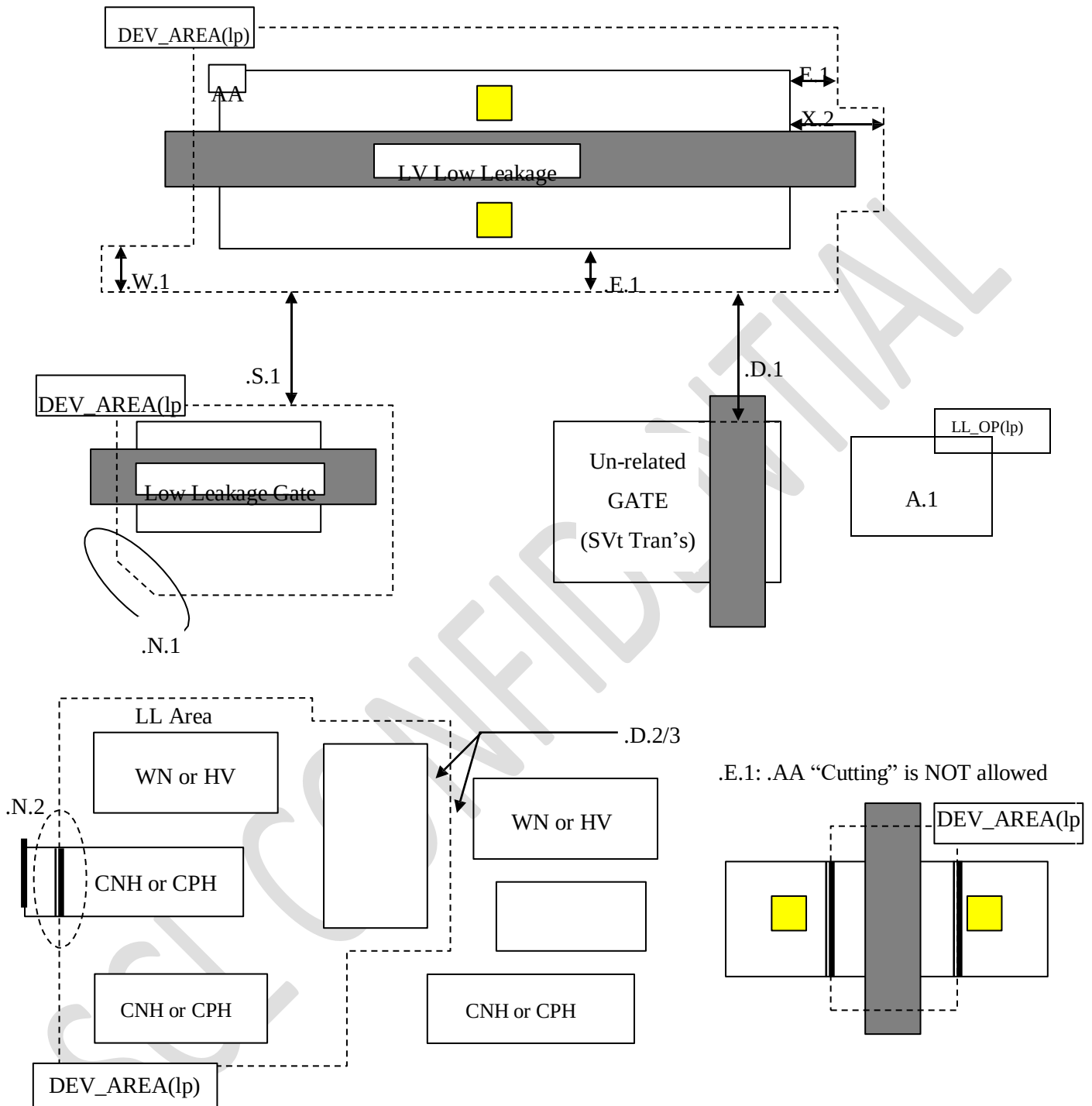


Figure 7-1: Rules of DEV_AREA (lp) Support Layer

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

8 Topological Design Rule

8.1 Rule Coding and General Notes

Rule formats: (Layer or topic).(Parameter).(Code). For example, DR of AA space is AA.S.1.

Before using the DRC, some “flags” should be activated. For example, 5V or 3.3V.

Some of the layout rules are “related”. For example, anytime rule GC.N.4 is violated, rule XN.N.1 will also flagged.

There are two types of recommended rules:

- Rules that specify guidelines, notes etc, are marked: (*.R.*). These rules are NOT checked by DRC.
- Recommended TDR values for specific rules. These rules start with R. For example, the recommended value for rule AA.S.1 is RAA.S.1. The sign [R] at the rule description indicates that this rule has a recommendation related.

We are targeting that all layout rules will be covered by DRC. However, on some cases, the DRC cannot be fully reliable or under development. These rules are noted accordingly (“This rule is NOT checked by DRC”).

8.1 Mask Information and Re-ordering Masks

Some of the masks included in the process flow, are optional. Please refer to section 7.8 for the mandatory and optional masks available for this technology.

Some of the masks included in the process flow are generated, and do NOT require any customer drawing.

However, design fix by correcting one (or more) drawing layer, may require order of several masks. For example, correcting XN layer will probably require re-ordering the flowing masks: XN, XP, LN, HN. It is recommended to consult with SCL PE.

Native transistors (LV as well as HV) do not require an additional mask. However, a dedicated native layer, 17dt0 is needed for native transistor definition.



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2 Layout Rules

8.2.1 AA (Active Area) Rules		Action	Drawn
AA.W.1	Width of AA.	min	0.22
AA.W.2	Width of AA.	max	60
AA.W.3	For 5V: Width of AA. For DRC purposes, ignore AA of Nsub and Psub	min	0.42
AA.S.1	Space between two AA's over the same well for 1.8V or 3.3V. This rule refers to space between two N+ Diff or between two P+ Diff or between Nsub to P+ Diff or between Psub to N+ Diff.	min	0.28
AA.S.3	Space between Nsub to Psub, having common parallel length >0. This rule refers to 1.8V or 3.3V Wells	min	0.36
AA.S.4	For 5V. Space between Nsub to Psub, having common parallel length >0. This rule refers to the case where at least one of the wells is 5V	min	0.28
AA.S.5	For 5V. Space between two AA's over the same well for 5V. This rule refers to space between two N+ Diff or between two P+ Diff or between Nsub to P+ Diff or between Psub to N+ Diff. The rule is also applicable to any case of space between 1.8V to 5V	min	0.28
AA.E.2	Enclosure of N-Well around NSUB, for WN pick-up. For DRC purposes, this rule is NOT applicable for Nwell resistors	min	0.12
AA.E.25	For 5V. Enclosure of 5V N-Well around NSUB, for WN pick-up. For DRC purposes, this rule is NOT applicable for Nwell resistors, vnp33_2, vnp33_5, vnp33_10	min	0.18
AA.E.3	Enclosure of N-Well around P+ AA in 1.8 or 3.3V N-Well	min	0.43
AA.E.35	For 5V. Enclosure of N-Well around P+ AA in 5V N-Well	min	0.65
AA.D.2	Distance between 1.8V or 3.3V WN to PSUB, for WP pick-up (WP enclose PSUB)	min	0.12
AA.D.25	For 5V. Distance between 5V WN to PSUB, for WP pick-up (WP enclose PSUB)	min	0.18
AA.D.3	Distance between 1.8V or 3.3V WN to N+ AA in P-Well.	min	0.43
AA.D.35	For 5V. Distance between 5V WN to N+ AA in P-Well.	min	0.65
AA.C.1	AA coverage on any square of 200um x 200um step 100um. For DRC purposes, the AA coverage is calculated as: {(AA(lc) OR AA(dy) Sized 0.469um/side) Sized -0.469um/side}	min	15%
AA.C.3	AA coverage on any square of 100um x 100um step 100um. For DRC purposes, the AA coverage is calculated as: {(AA(lc) OR AA(dy) Sized 0.469um/side) Sized -0.469um/side}. Ignore windows having all AA spaces <8um.	min	20%
AA.C.5	AA coverage on any square of 100um x 100um step 100um. For DRC purposes, the AA coverage is calculated as: {(AA(lc) OR AA(dy) Sized 0.469um/side) Sized -0.469um/side}. Ignore windows having all AA spaces <20um.	min	30%
AA.C.8	AA coverage on any square of 64um x 64um step 32um. .	max	95%
AA.C.10	AA coverage on any square of 200um x 200um step 100um. For DRC purposes, the AA coverage is calculated as:	max	98%



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.1 AA (Active Area) Rules		Action	Drawn
	{{(AA(lc) OR AA(dy) Sized 0.469um/side) Sized -0.469um/side}}		
AA.A.1	AA island area, that do NOT enclose any STI (field oxide)	min	0.202um ²
AA.A.2	Area that enclosed by AA (Area of STI)	min	0.28
AA.A.3	For butted diffusion. Area of any butted diffusion	min	0.176um ²
AA.N.4	AA must be implanted with XN or XP (but not both). Un-implanted AA or inter-mixing implanted AA is NOT allowed. See also rule XN.N.1, XP.N.1		
AA.L.1	For butted diffusion. Length of N+/P+ butted diffusion edge. For N+/P+ butted having jogs, at least one edge of the butted area must satisfy this rule. Please also refer to rules XN.A.1 and XP.A.1.	min	0.42
Notes: 5V AA. is defined as AA enclosed by {AREA2 AND AREA6}. AA.C.* rules are not checked under embedded IPs areas			



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

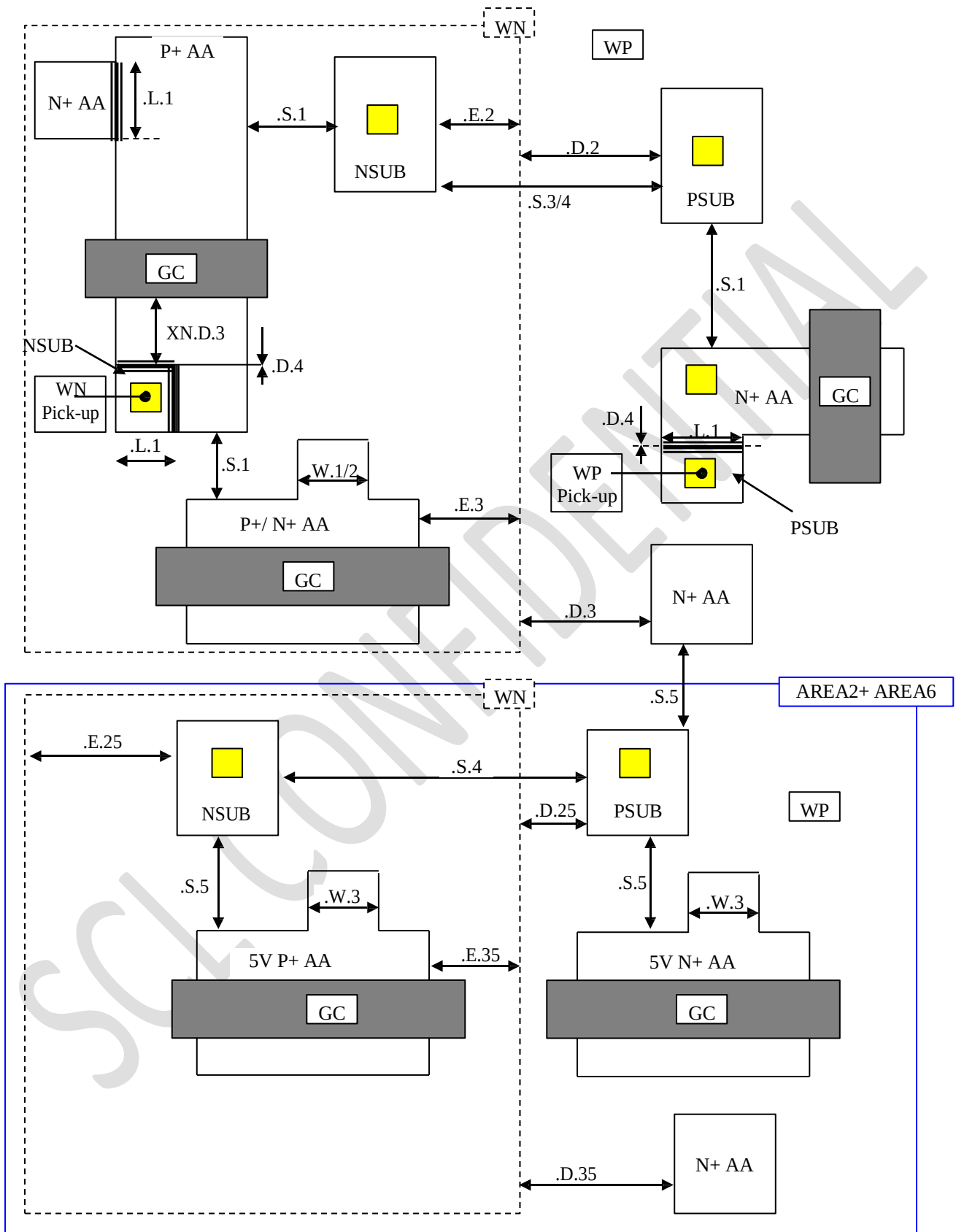


Figure 8-1: AA (Active Area) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.2 WN (N-Well) Rule		Action	Drawn
WN.W.1	Width of 1.8V or 3.3V WN	min	0.86
WN.W.2	For 5V. Width of 5V WN	min	1.3
WN.W.3	Width of WN, were the WN is NOT connected to Vdd. (width of Hot WN). Refer to 1.8V, 3.3V, 5V well	min	2.1
WN.S.1	Space between two WN's having the same potential. That is: 1.8V ~ 1.8V, 3.3V ~ 3.3V, 1.8V ~ 3.3V, Merge if space under this value	min	0.6
WN.S.2	For 5V. Space between two WN's having the same potential. That is: 5V ~ 1.8/3.3V, 5V ~ 5V. Merge if space under this value	min	0.9
WN.S.3	Space between two WN's having different potentials. That is: 1.8V ~ 1.8V, 3.3V ~ 3.3V, 1.8V ~ 3.3V.	min	1.4
WN.S.4	For 5V. Space between two WN's having different potentials. That is: 1.8V ~ 5V or 5V ~ 5V.	min	2
Notes: 5V nwell is defined as nwell enclosing 5V P+ AA			

8.2.2.1 Definition and Instructions for Drawing Hot WN Support Layer

Use **Hot WN- 3dt27** to define WN that is NOT connected to any kind of Vdd Power voltage, trough NSUB.

WN resistor, WN of MOSCAP and the WN of P+/WN Junction Varactor are also hot WN's.

It is the designer's responsibility to draw the Hot WN layer. The layer should co-inside with WN.

For DRC purposes, Hot WN is defined s as follows: Hot WN = WN Interact 3dt27 OR {WN Interact 77dt42}.



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

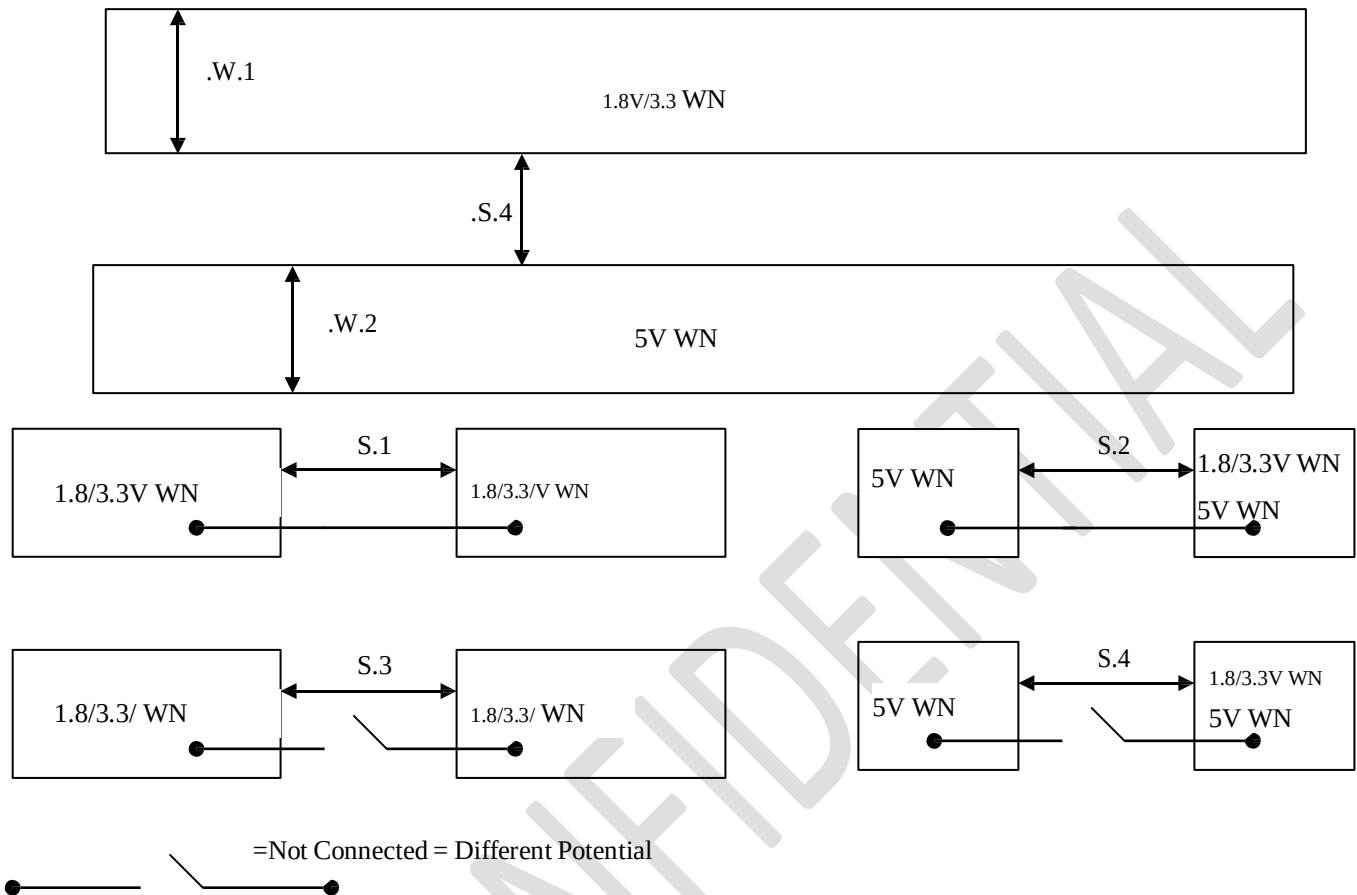


Figure 8-2: WN (N-Well) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.3 LA (Thick Oxide, AREA2 and AREA6) Rules

Two layers are used to define thick gate oxide 5V for mask generation and device recognition:

AREA2 defines areas with thick gate oxide: either 3.3V. This layer is used for LA mask generation, as well as for implant masks generation.

AREA6 marks thick gate oxide as 5V for DRC, LVS, and MDP purposes.

NOTE: Using 3.3V and 5V devices on the same design is not allowed.
It is enforced by DRC rule LA.N.1/2

8.2.4 LA (Thick Oxide, AREA2 and AREA6) Rules		Action	Drawn
LA.W.1	Width of AREA2 (HV Area)	min	0.88
LA.S.1	For 3.3V: Space between two AREA2 areas. Merge if space is under this value	min	0.45
LA.S.2	For 5V: Space between two AREA2 areas. Merge if space is under this value	min	0.8
LA.E.1	For 3.3V: Enclosure of AREA2 around 3.3V N+AA in WP or 3.3V P+ AA in WN. The rule does not check dummy AA, nsub, Psub, or AA interacting LV gate extending beyond AREA2.	min	0.32
LA.E.2	For 5V: Enclosure of AREA2 around 5V N+AA in WP or 5V P+ AA in WN. The rule does not check dummy AA, Nsub, Psub, or AA interacting LV gate extending beyond AREA2.	min.	0.5
LA.D.1	Distance between AREA2 to AA located outside AREA2. (Not include dummy AA or N+ AA in WN or P+ AA outside WN). This rule is applicable for 3.3V and for 5V designs	min.	0.32
LA.D.2	Distance between AREA2 and LV GATE located outside AREA2 The rule refers to source/ drain edges of the LV GATE and not to the STI edges. This rule is also applicable for 5V designs	min.	0.4
LA.X.3	For 3.3V: Extension of AREA2 beyond 3.3V gate, perpendicular to the transistor length (along gate edge)	min.	0.4
LA.X.4	For 5V: Extension of AREA2 beyond 5V gate, perpendicular to the transistor length (along gate edge)	min.	0.6
LA.X.5	For 5V: Extension of AREA2 beyond 5V gate of a common 1.8/5V AA, perpendicular to the transistor length (along gate edge).	min.	1.2
LA.N.1	For 5V: AREA6 AND AREA2 should completely cover any 5V devices. For DRC purposes, check that: (1) AREA2 is fully covered by AREA6. (2) AREA6, that is not interact with AREA2 is NOT allowed. AREA6 should be drawn all over the full chip area, corner to corner.		

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

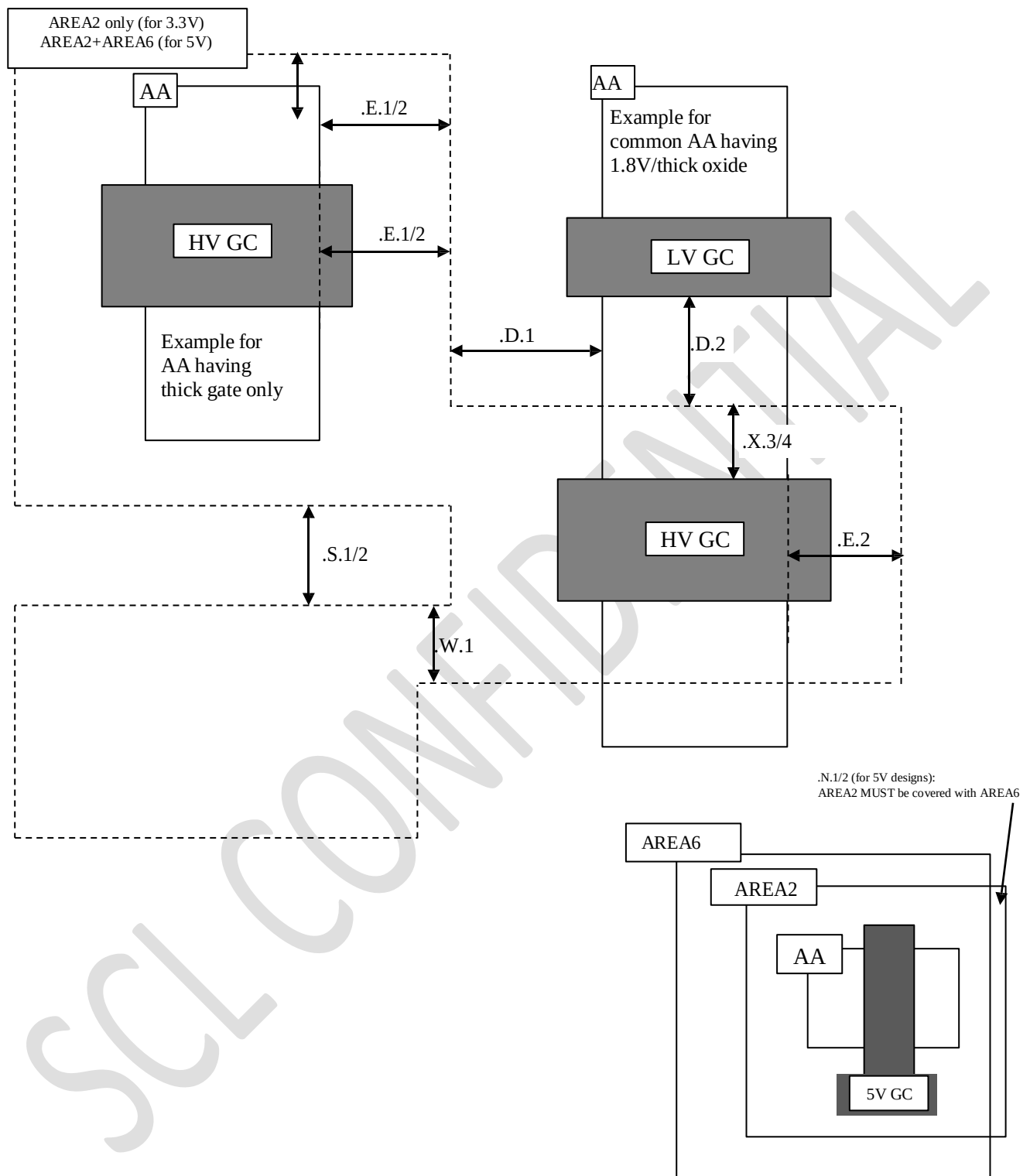


Figure 8-3: LA (Thick oxide, AREA2 and AREA6) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.5 GC (Poly) Rules		Action	Drawn
GC.W.1	Gate length of 1.8V nMOS transistor	min	0.18
GC.W.2	Gate length of 1.8V pMOS transistor	min	0.18
GC.W.3	Gate length of 3.3V nMOS transistor	min	0.35
GC.W.4	Gate length of 3.3V pMOS transistor	min	0.3
GC.W.5	Gate length of 5V nMOS transistor	min	0.5
GC.W.6	Gate length of 5V pMOS transistor	min	0.5
GC.W.7	For 1.8V: Gate length of 45 degrees 1.8V transistor	min	0.21
GC.W.8	Width of GC for interconnects (GC width over field oxide)	min.	0.18
GC.S.1	Space between two GC gate's on AA without any contacts in between gates	min	0.25
GC.S.2	Space between two GC gate's on AA, having CS between gates	min	0.375
GC.S.3	Space between two GC lines located on STI (field oxide)	min	0.25
GC.D.1	Distance from GC on STI (field oxide) to AA edge of 1.8 or 3.3	min	0.1
GC.D.2	For 5V: Distance from GC on STI (field oxide) to AA edge of 5V	min	0.1
GC.X.1	Extension of AA beyond GC edge located on AA	min	0.32
GC.X.2	Extension of GC beyond AA edge (end cap)	min	0.22
GC.C.1	Coverage of GC area on overall chip area. See section 9.2.2 for guidelines on achieving minimum coverage. For DRC purposes, check coverage of: GC(lc) OR GC(dy)	min	10%
GC.C.2	Coverage of GC area on overall chip area. For DRC purposes, check coverage of GC(lc) or GC(dy)	max	40%
GC.C.4	Coverage of (Poly OR WN resistor) over STI, measured in widow of 200um x 200um step 100um. This rule helps to meet rules the minimum AA coverage. For DRC purposes, check the coverage as: (((GC not AA) Sized up-down 1.45) or ((WN and DEV_AREA;res not AA) sized up-down 0.85)) not (AA sized up-down 0.469)	max	70%
GC.C.6	GC coverage on any square of 1000um x 1000um step 500um. For DRC purposes, the GC coverage is calculated as: {GC(lc) OR GC(dy)} Sized 0.3um/side	max	95%
GC.A.1	GC Island area	min	0.203 um ²
GC.N.2	45 degrees GC gate crossing AA edge is not allowed		
GC.N.3	GC gate bend at 90degree is not allowed		
GC.N.4	Non-salicided GC must be implanted with XN or XP or MR. Un- implanted GC, or inter-mixing implanted GC (both XN and XP or both XN and MR) is not allowed. See also rules XN.N.1 and XP.N.1 For DRC purposes, check implant over Poly that is non-salicided.		
GC.L.1	Length of narrow (W<0.24um) Salicided GC on STI	max	50
Notes: GC.C.* rules are not checked under embedded IPs areas.			



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

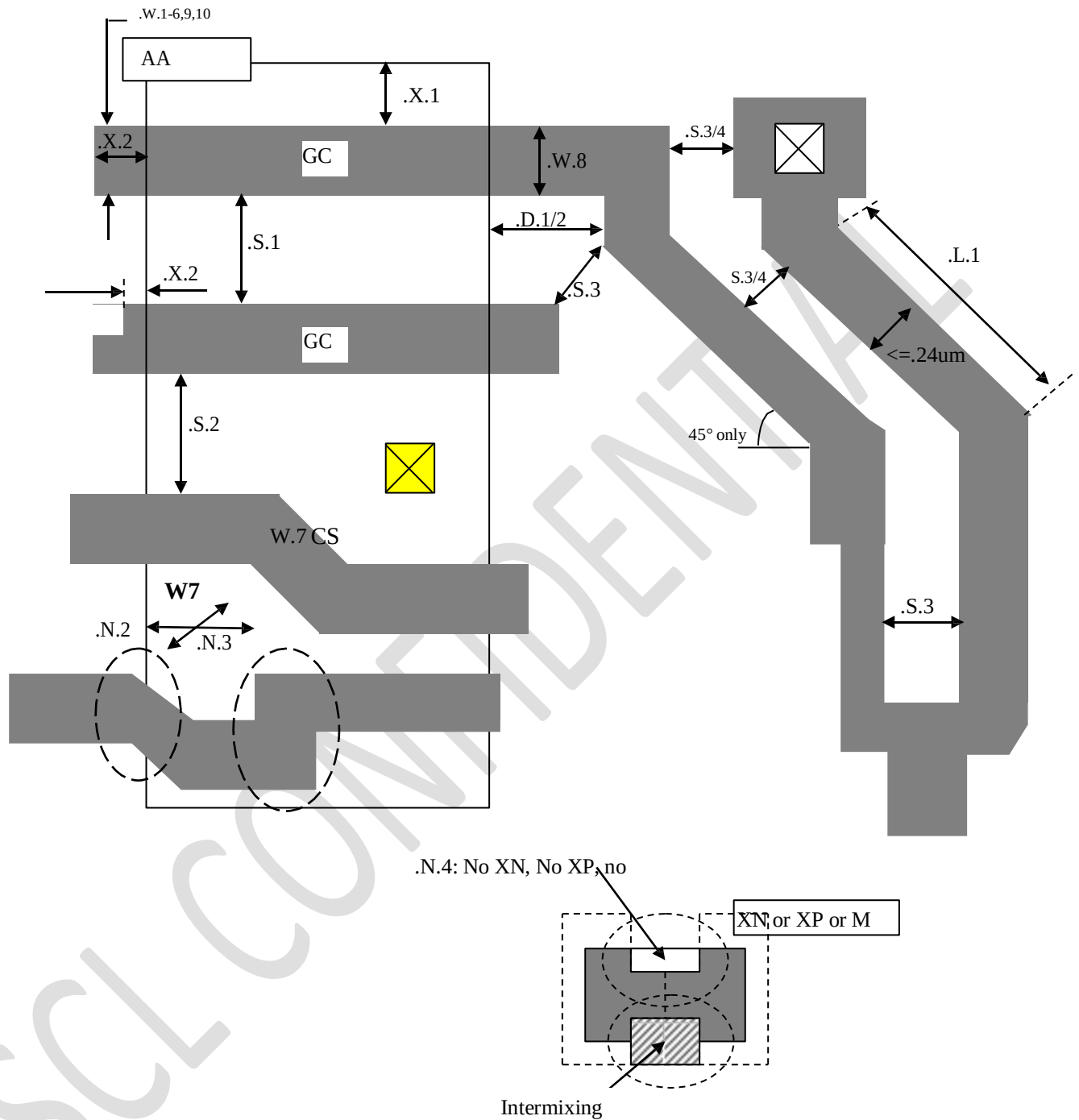


Figure 8-4: GC (Poly) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

8.2.6 XN (N+ Implant) Rules		Action	Drawn
XN.W.1	Width of XN (for photolithography purposes)	min	0.44
XN.S.1	Space between two XN areas. Merge if space is under this value	min	0.44
XN.RS.2	For butted diffusion. Space between XN butted to XP on the same AA over WP.	exact	0
XN.RS.3	For butted diffusion. Space between XN butted to XP on the same AA over WN. Please refer also to rule XN.N.1.	exact	0
XN.E.1	Enclosure of XN around N+ AA, on WP	min	0.18
XN.E.2	Enclosure of XN around N+ AA, on WN (on WN pick-up).	min	0.02
XN.E.9	Enclosure of XN around N+ AA, on WN (on WN pick-up). This rule is checked if enclosure of WN on the WN pickup is <0.43um	min	0.18
XN.D.1	Distance from un-related XN to P+ AA on WN.	min	0.26
XN.D.2	Distance from un-related XN to P+ AA on WP (WP pick-up) This rule is checked if distance of WN to the WP pickup is <0.43um	min	0.18
XN.D.9	Distance from un-related XN to P+ AA on WP (WP pick-up) This rule is checked if distance of WN to the WP pickup is >0.43um	min	0.1
XN.D.3	For butted diffusion. Where the N+/P+ butted diffusion edge is parallel to the GC transistor gate, the minimum distance over AA from pMOS GC gate edge to the XN of the WN pick-up (w/ or w/o CS). Please refer also to rule AA.D.2 and XN.D.5	min	0.32
XN.D.5	For 5V butted diffusion. Where the N+/P+ butted diffusion edge is parallel to the GC transistor gate, the minimum distance over AA from pMOS GC gate edge to the XN of the WN pick-up. Please refer also to rule AA.D.2 and XN.D.3	min	0.32
XN.X.1	For butted diffusion. Where the N+/P+ butted diffusion edge is parallel to the GC transistor gate, the minimum extension of XN beyond nMOS GC gate toward (over AA) the N+/P+ butted diffusion (w/ or w/o CS) edge. Please refer also to rule AA.D.2 This rule is checked by rule XP.D.3	min	0.32
XN.O.1	Overlap of XN on (N+ AA over WP) to have butted diffusion. The distance is measured between the N+ AA edge to the N+/P+ butted diffusion edge.	min	0.23
XN.A.1	XN island area. Also refer to XN implant for WN pick-up of butted diffusion. Please refer also to rule AA.L.1	min	0.3844um ²
XN.N.1	Overlap of XN and XP is NOT allowed.		
XN.R.1	Recommended extension of XN beyond N+ AA along GC. This extension rule must extend 0.32um perpendicular to GC end cap.	min	0.35



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

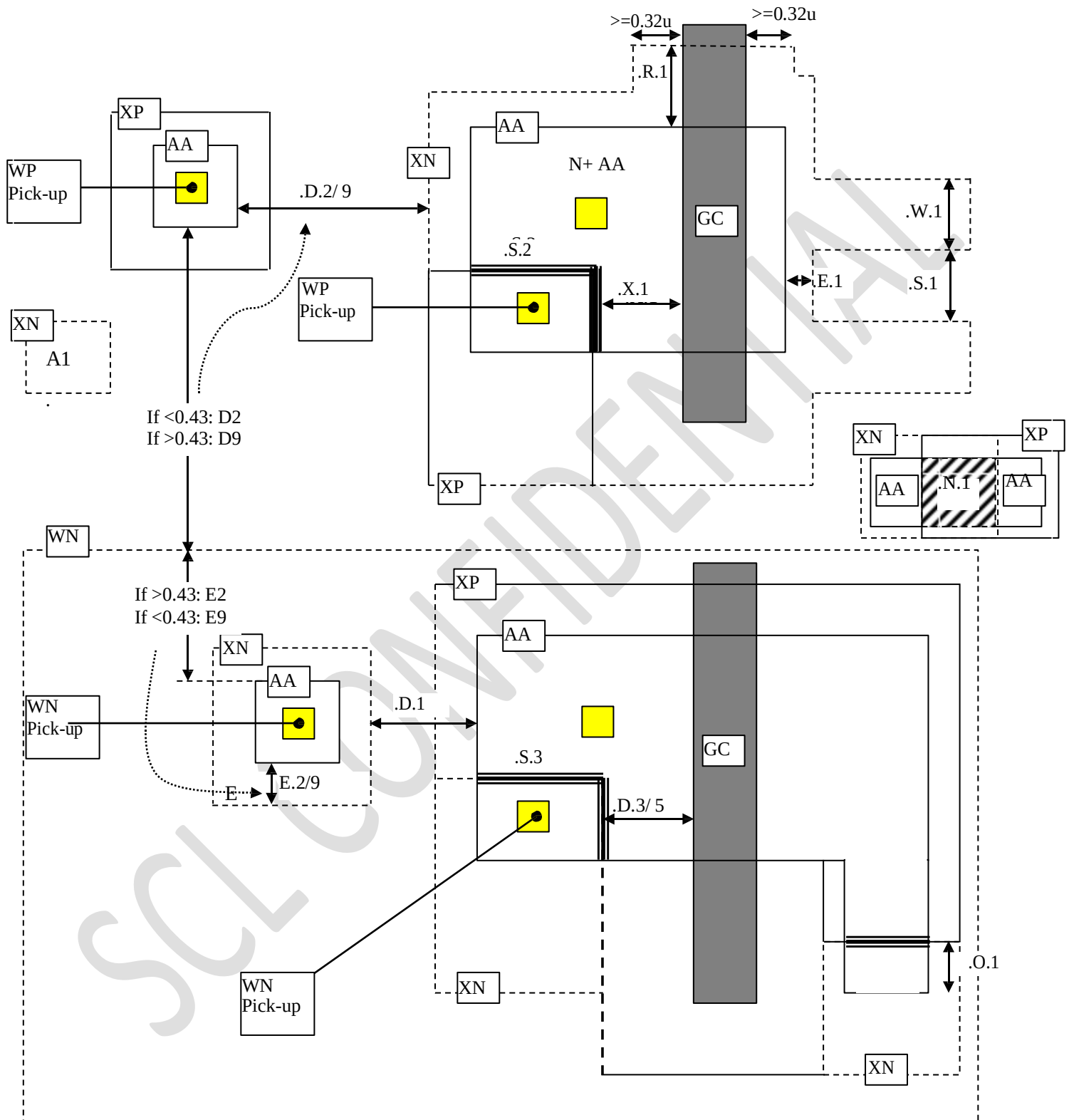


Figure 8-5: XN (N+ Implant) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.7 XP (P+ Implant) Rules		Action	Drawn
XP.W.1	Width of XP (for photolithography purposes)	min	0.44
XP.S.1	Space between two XP areas. Merge if space is under this value	min	0.44
XP.RS.2	For butted diffusion. Space between XP butted to XN on the same AA over WN .	exact	0
XP.RS.3	For butted diffusion. Space between XP butted to XN on the same AA over WP. Please refer also to rule XP.N.1.	exact	0
XP.E.1	Enclosure of XP around P+ AA, on WN	min	0.18
XP.E.2	Enclosure of XP around P+ AA, outside WN. (on WP pick-up).	min	0.02
XP.E.9	Enclosure of XP around P+ AA, outside WN. (on WP pick-up). This rule is checked if distance of WN to the WP pickup is <0.43um	min	0.18
XP.D.1	Distance from un-related XP to N+ AA on WP.	min	0.26
XP.D.2	Distance from un-related XP to N+ AA on WN (WN pick-up) . This rule is checked if enclosure of WN around WN pickup is <0.43um	min	0.18
XP.D.3	For butted diffusion. Where the P+/N+ butted diffusion edge is parallel to the GC transistor gate, the minimum distance over AA from nMOS GC gate edge to the XP of the WP pick-up (w/ or w/o CS). Please refer also to rule AA.D.2 and to rule XP.D.5	min	0.32
XP.D.5	For 5V butted diffusion. Where the P+/N+ butted diffusion edge is parallel to the GC transistor gate, the minimum distance over AA from nMOS GC gate edge to the XP of the WP pick-up. Please refer also to rule AA.D.2 and to rule XP.D.3	min	0.32
XP.D.9	Distance from un-related XP to N+ AA on WN (WN pick-up) . This rule is checked if enclosure of WN around WN pickup is >0.43um	min	0.1
XP.X.1	For butted diffusion. Where the P+/N+ butted diffusion edge is parallel to the GC transistor gate, the minimum extension of XP beyond pMOS GC gate toward (over AA) the P+/N+ butted diffusion (w/ or w/o CS) edge. Please refer also to rule AA.D.2 This rule checked by rule XN.D.3	min	0.32
XP.O.1	Overlap of XP on (P+ AA over WN) to have butted diffusion. The distance is measured between the P+ AA edge to the P+/N+ butted diffusion edge	min	0.23
XP.A.1	XP island area. Also refer to XP implant for WN pick-up of butted diffusion. Please refer also to rule AA.L.1	min	0.3844um ²
XP.N.1	Overlap of XP and XN is NOT allowed. This rule is checked by XN.N.1		
XP.R.1	Recommended extension of XP beyond P+ AA along GC. This extension rule must extend 0.32um perpendicular to GC end cap.	min	0.35

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

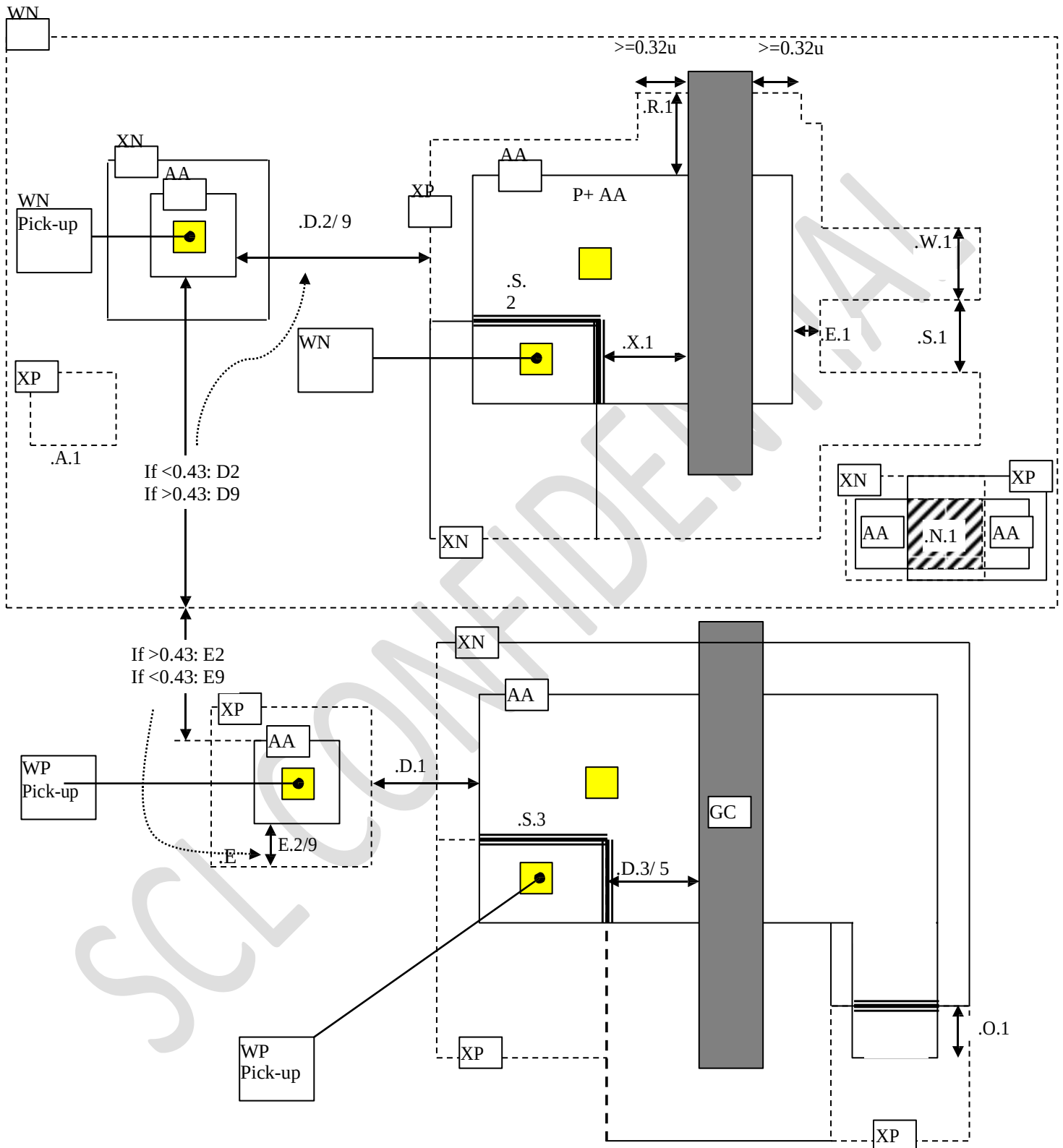


Figure 8-6: XP (P+ Implant) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.8 NS (Salicide Block) Rules		Action	Drawn
NS.W.1	Width of NS (for photolithography purposes)	min	0.43
NS.S.1	Space between two NS areas where AA or GC with Contact does not exist in the space. SCL will merge space if less than 0.66um during MDP. Space for photolithography purposes is 0.66um	min	0.72
NS.S.1MDP	Space between two NS areas where AA or GC with Contact exists in the space SCL will merge space if less than 0.66um without Contact during MDP.	min	0.66
NS.D.1	Distance from NS to salicided AA	min	0.22
NS.D.2	Distance from NS to CS	min	0.22
NS.D.4	Distance from NS to GC gate	min	0.45
NS.D.5	Distance from NS to GC over STI	min	0.3
NS.X.1	Extension of NS beyond AA.	min	0.22
NS.X.2	Extension of AA beyond NS over the same AA. The distance is measured between the NS over AA edge to the diffusion edge.	min	0.22
NS.X.3	Extension of NS beyond GC on STI	min	0.22
NS.A.1	NS island area	min	2um ²
NS.A.2	Area that enclosed by NS	min	0.66um ²



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

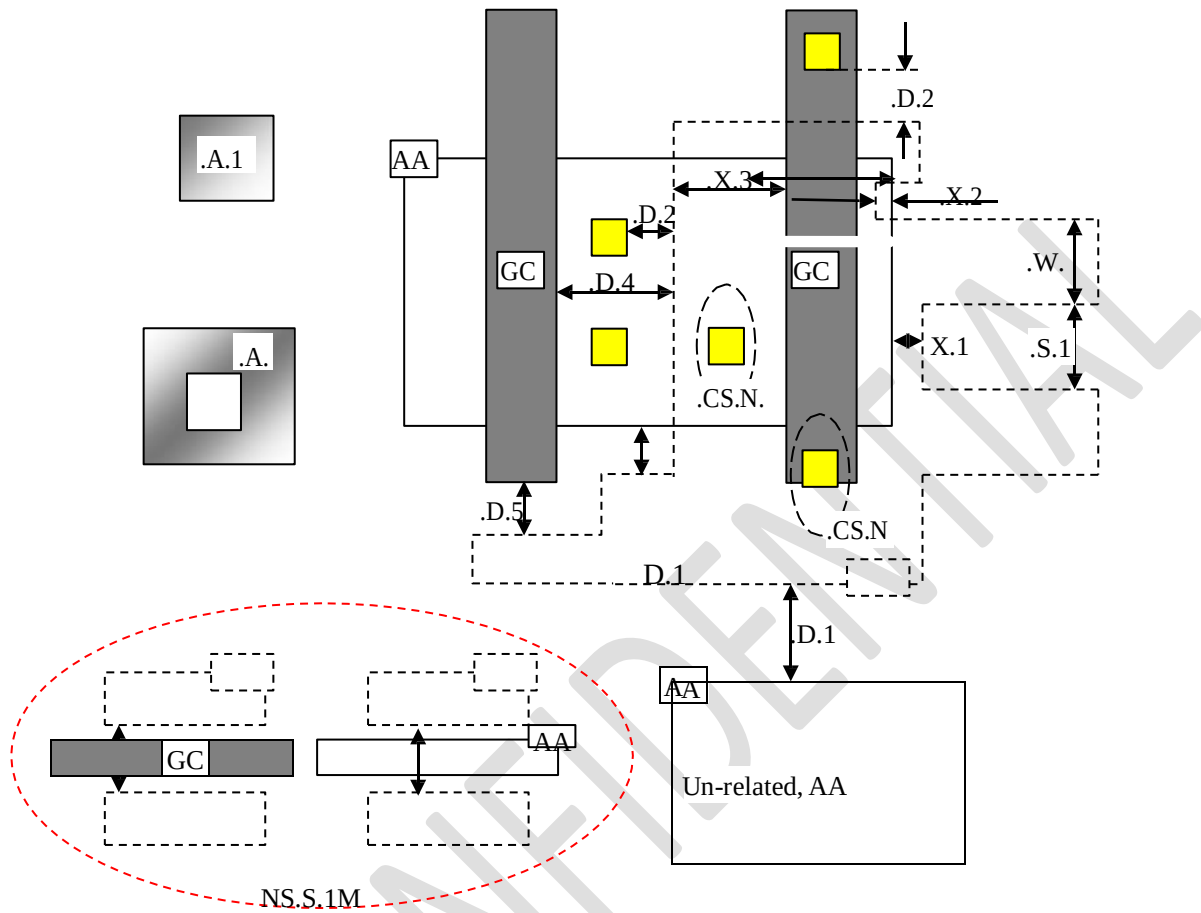


Figure 8-7: NS ("No Sal", Local Salicide Block) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.9 CS (Contact) Rules		Action	Drawn
CS.W.1	Width of CS	exact	0.22
CS.S.1	Space between two CS	min	0.25
CS.S.2	Space between two CS that are a part of CS array. Array of CS is defined as an area includes min 4 x min 4 CS, having a maximum space inside the array of 0.29um. Two CS with space larger than 0.29um must be “related” to two different CS arrays.	min	0.28
CS.E.1	Enclosure of AA around CS	min	0.1
CS.E.2	Enclosure of GC around CS	min	0.1
CS.D.1	Distance from CS over AA to GATE	min	0.16
CS.D.3	Distance from CS over GC to AA	min	0.2
CS.X.1	For butted diffusion. Extension of XP beyond CS	min	0.11
CS.X.2	For butted diffusion. Extension of XN beyond CS	min	0.11
CS.N.1	CS to GATE is not allowed. (CS to GC must be over field oxide)		
CS.N.2	CS on N+/P+ butted diffusion edge is not allowed		
CS.N.3	CS (to AA or to GC) that covered with NS is NOT allowed. CS MUST be salicided.		
CS.N.5	CS must be inside AA or GC. CS to STI (field oxide) is NOT allowed.		

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

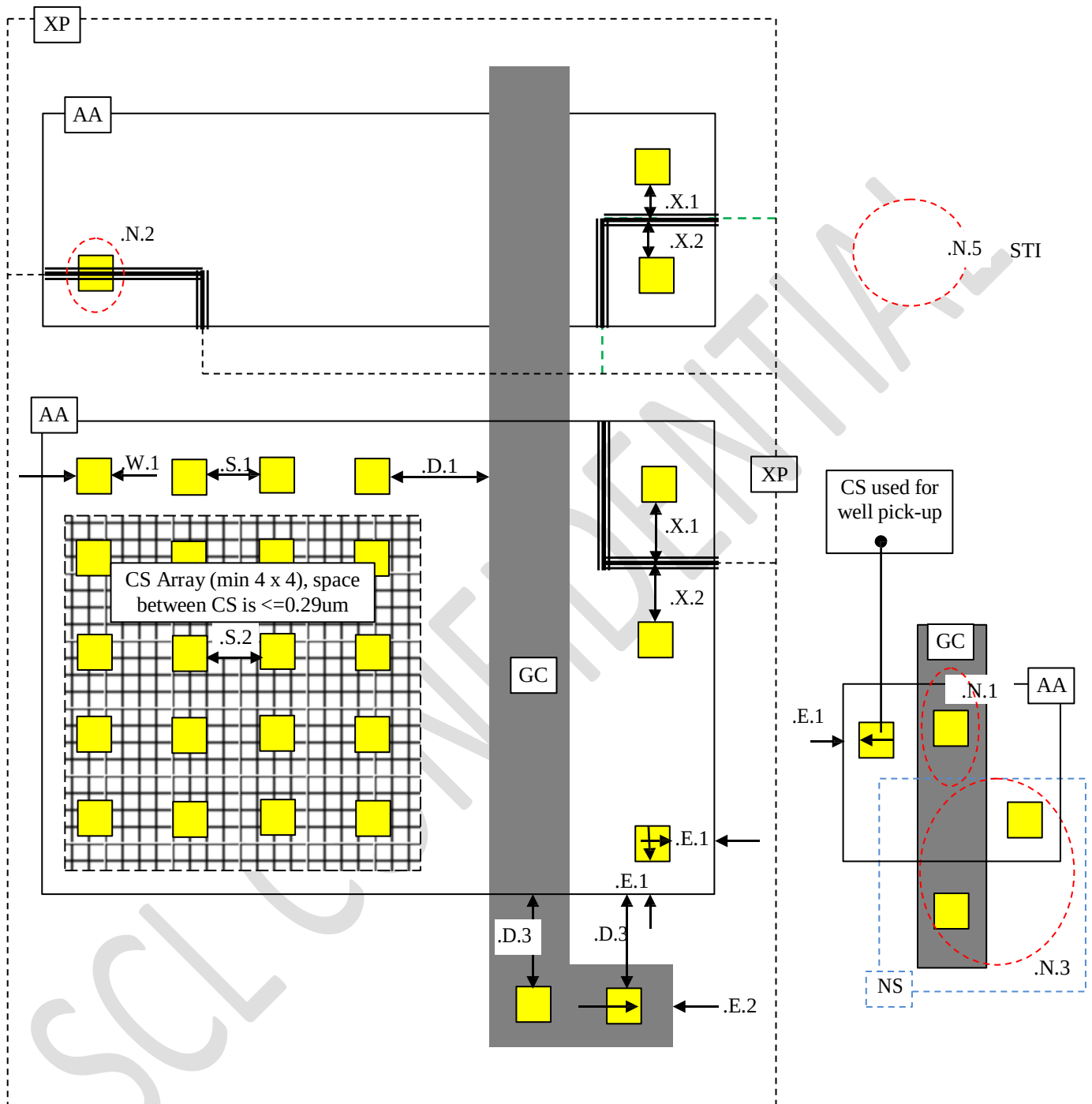


Figure 8-8: CS (Contact) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.10 M1 (Metal 1) Rules		Action	Drawn
M1.W.1	Width of M1	min	0.23
M1.W.2	Width of M1 line connected to a wide M1 (>35um in both width and length). For DRC purposes, refer to the first 1um length of the connected line to the wide metal.	min	1
M1.S.1	Space between two M1	min	0.23
M1.S.2	Space between two M1 lines where at least one M1 line (width AND length) is > 10um. This rule is applicable only where the parallel edges projected length of this M1 spacing is >= 0.28um.	min	0.6
M1.E.1	Enclosure of M1 on CS (not in end of line)	min	0.005
M1.E.2	Enclosure of M1 around CS in end of line DRC checks that at least one CS grow direction (X or Y) by 0.06um is covered by M1	min	0.06
M1.E.3	Enclosure of angled M1 line end around CS DRC checks that at least one CS GROW direction (X or Y direction) by 0.02um is covered by M1	min	0.02
M1.C.1	Coverage of M1 area on overall chip area. For DRC purposes, coverage is calculated by $[(M1(lc)+M1(dy)-M1(ms))/(chip\ area)]$ This rule is not checked under embedded IP area	min	20%
M1.C.2	M1 Coverage on any square of 750um x 750um step of 100um	max	88.5%
M1.C.3	M1 Coverage on any square of 900um x 900um step of 100um	max	83%
M1.C.5	M1 Coverage on any square of 1050um x 1050um step of 100um	max	80%
M1.C.4	Coverage of M1 area on overall chip area. For DRC purposes, coverage is calculated by $[(M1(lc)+M1(dy)-M1(ms))/(chip\ area)]$ This rule is not checked under embedded IP area	max	60%
M1.A.1	M1 island area	min	0.202um ²



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

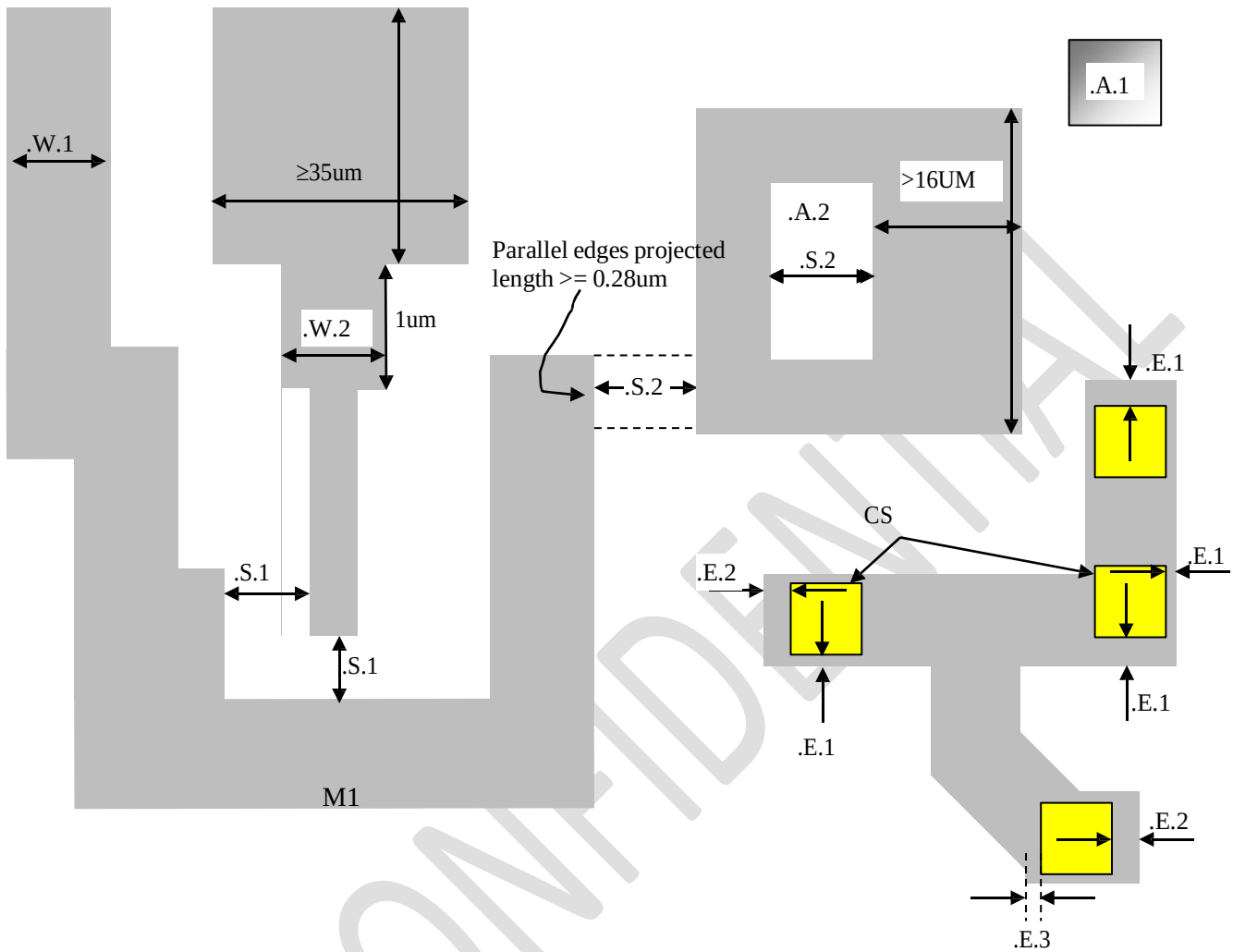


Figure 8-9: M1 (Metal 1) Rules

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.11 VI (I=2~5, Inter Via: V2, V3, V4, V5) Rules		Action	Drawn
VI.W.1	Width of VI (V2, V3, V4, V5)	exact	0.26
VI.S.1	Space between two VI	min	0.26
VI.E.1	Enclosure of M(I-1) around VI (not in end of line)	min	0.01
VI.E.2	Enclosure of M(I-1) around VI above (not in M(I-1) end-of-line) DRC checks that at least one VI GROW direction (X direction or Y direction) by min 0.06um is covered by M(I-1).	min	0.06
VI.E.3	Enclosure of angled M(I-1) line end around VI above DRC checks that at least one VI GROW direction (X or Y direction) by min 0.02um is covered by M(I-1)	min	0.02

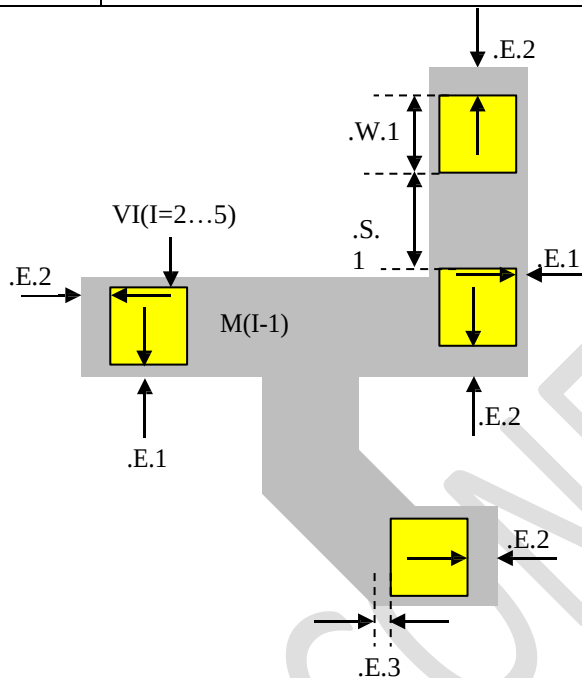


Figure 8-10: VI (I=2~5, Inter Via: V2, V3, V4, V5) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.12 MI (I=2~5, Inter Metal: M2, M3, M4, M5) Rules		Action	Drawn
MI.W.1	width of MI (M2, M3, M4, M5)	min	0.28
MI.W.2	width of MI line connected to a wide MI (>35um in both width and length). For DRC purposes, refer to the first 1um length of the connected line to the wide metal.	min	1
MI.S.1	Space between two MI	min	0.28
MI.S.2	Space between two MI lines where at least one MI line (width AND length) is > 10um. This rule is applicable only where the parallel edges projected length of this MI spacing is >= 0.28um.	min	0.6
MI.E.1	Enclosure of MI around VI (not in end-of-line)	min	0.01
MI.E.2	Enclosure of MI around VI in end-of-line DRC checks that at least one VI GROW direction (X direction or Y direction) by min 0.06um is covered by MI.	min	0.06
MI.E.3	Enclosure of angled MI line end around VI DRC checks that at least one VI GROW direction (X or Y direction) by min 0.02um is covered by MI	min	0.02
MI.C.1	Coverage of MI area on overall chip area. For DRC purposes, coverage is calculated by $[(MI(lc)+MI(dy)-MI(ms))/(chip\ area)]$	min	20%
MI.C.2	MI Coverage on any square of 900um x 900um step of 100um	max	96%
MI.C.3	MI Coverage on any square of 1050um x 1050um step of 100um	max	80%
MI.C.4	Coverage of MI area on overall chip area. For DRC purposes, coverage is calculated by $[(MI(lc)+MI(dy)-MI(ms))/(chip\ area)]$	max	60%
MI.A.1	MI island area	min	0.202um ²

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

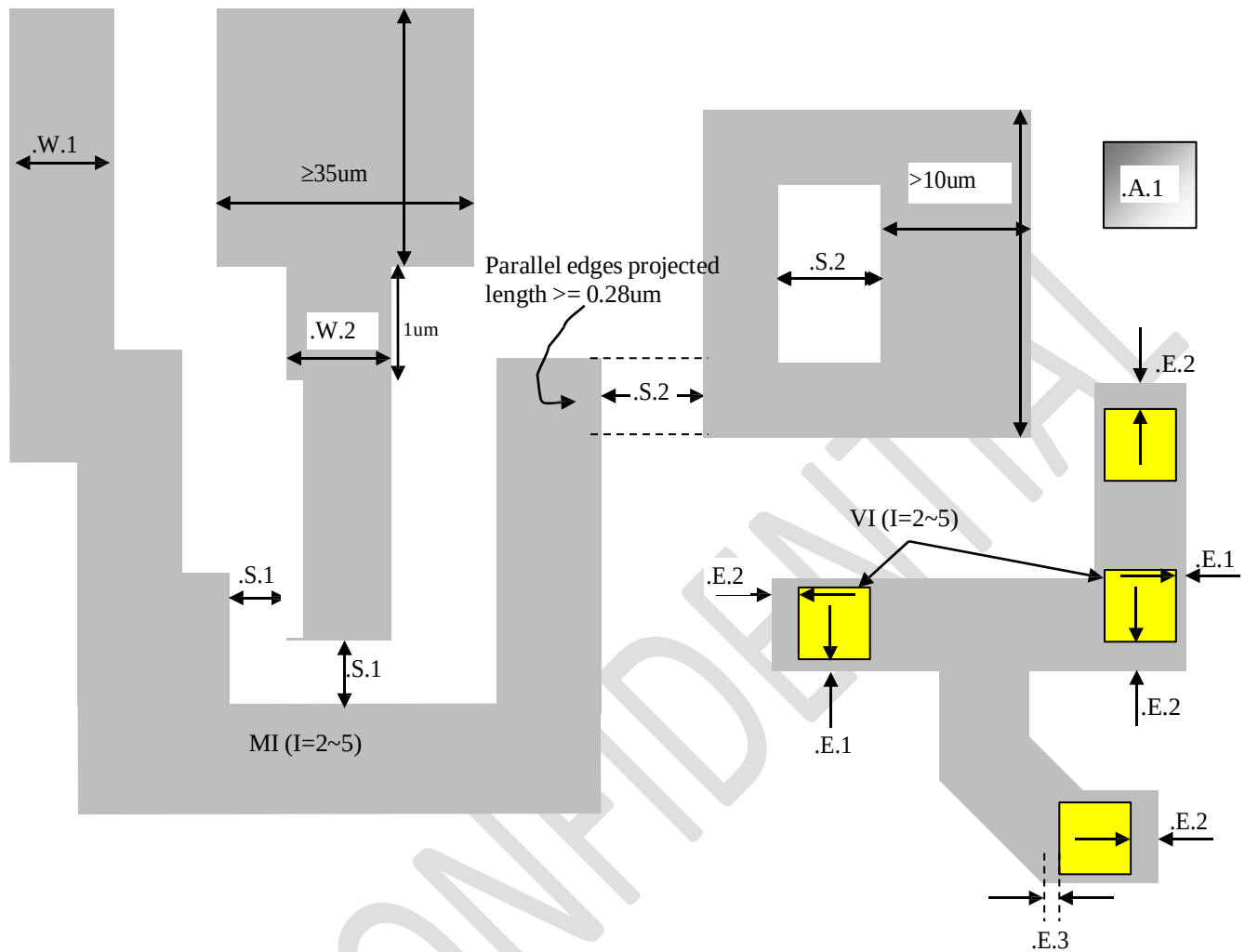


Figure 8-11: MI (I=2~5, Inter Metal: M2, M3, M4, M5) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.13 VL (TOP_VIA) Rules		Action	Drawn
VL.W.1	Width of VL (TOP_V)	exact	0.36
VL.S.1	Space between two VL	min	0.35
VL.E.1	Enclosure of ML or MI or M1 around VL (not in end of line) For example, M3 under TOP_V in 4M1L design.	min	0.01
VL.E.2	Enclosure of M1/ MI/ ML around VL above (in end of line). DRC checks that at least one VL GROW direction (X direction or Y direction) by min 0.06um is covered by ML/ MI/ M1.	min	0.06
VL.E.3	Enclosure of angled ML/ MI/ M1 line end around VL above DRC checks that at least one VL GROW direction (X or Y direction) by min 0.02um is covered by ML/ MI/M1	min	0.02

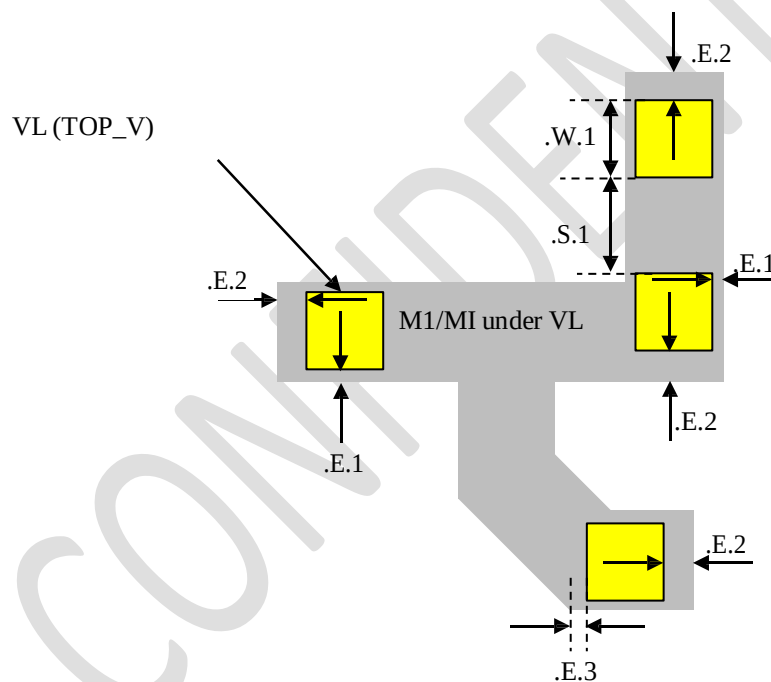


Figure 8-12: VL (TOP_VIA) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.14 ML (TOP_M) Rules		Action	Drawn
ML.W.1	width of ML (TOP_M)	min	0.44
ML.W.2	width of ML line connected to a wide ML (>35um in both width and length). For DRC purposes, refer to the first 1um length of the connected line to the wide metal.	min	1
ML.S.1	Space between two ML	min	0.46
ML.S.2	Space between two ML lines where at least one ML line (width AND length) is > 10um. This rule is applicable only where the parallel edges projected length of this ML spacing is >= 0.44um.	min	0.6
ML.E.1	Enclosure of ML around VL below	min	0.09
ML.C.1	Coverage of ML area on overall chip area. For DRC purposes, check coverage of {ML OR MLdt8}. This rule is not checked under embedded IP area	min	20%
ML.C.2	Coverage of ML area on overall chip area in the case where the chip contains one or more places where spacing between two ML is <0.55um. For DRC purposes, check coverage of {ML OR MLdt8}.	max	58%
ML.C.4	Coverage of ML area on overall chip area in the case where all spacing between two ML >0.55um. For DRC purposes, check coverage of {ML OR MLdt8}.	max	73%
ML.A.1	ML island area	min	0.562um ²



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

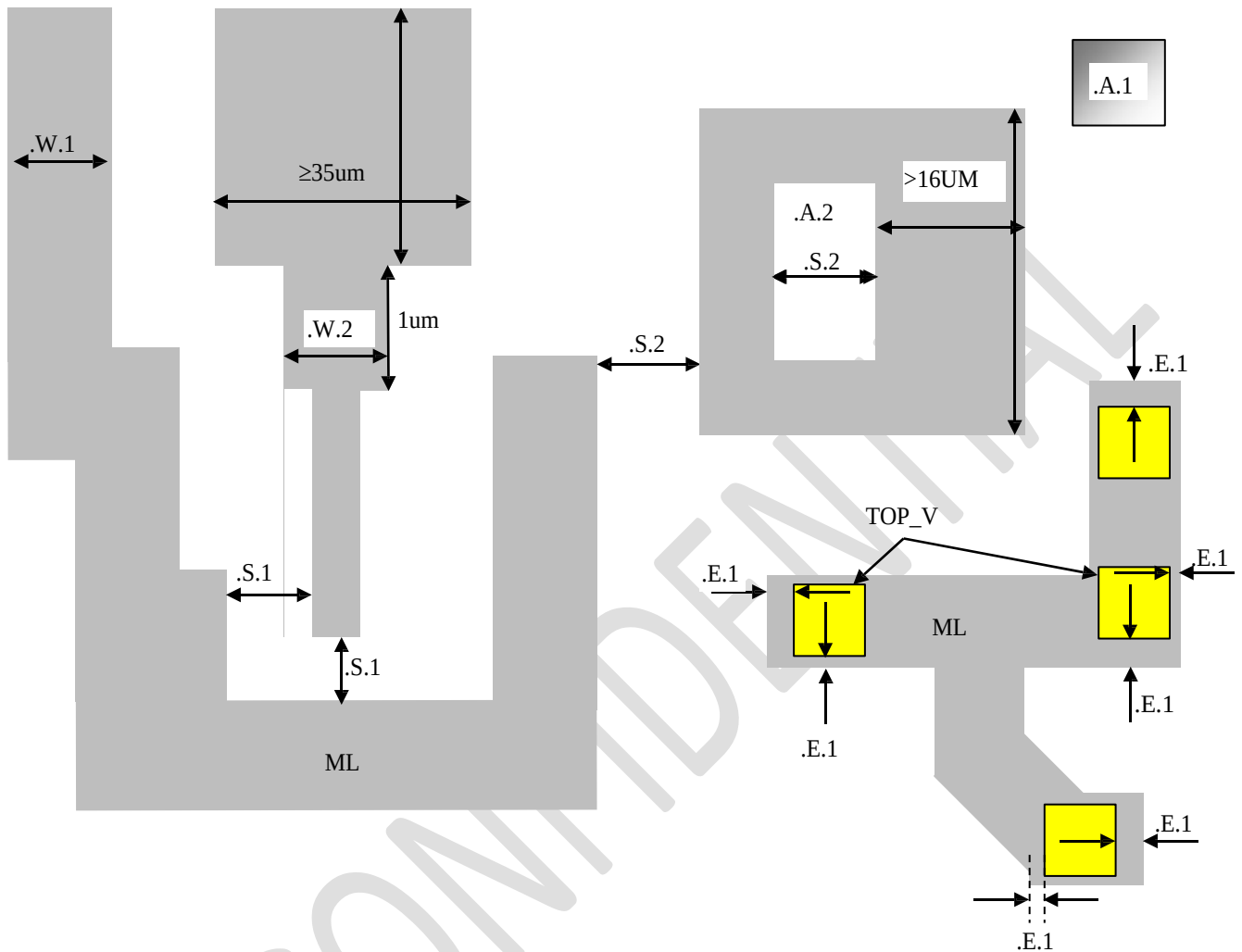


Figure 8-13: ML (TOP_M) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.15 MT (20kA Thick Metal Last) Rules		Action	Drawn
MT.W.1	Width of MT	min	1.2
MT.W.2	width of MT line connected to a wide MT (>35um in both width and length). For DRC purposes, refer to the first 1um length of the connected line to the wide metal.	min	2
MT.S.1	Space between two MT	min	1
MT.S.2	Space between two MT lines that at least one of the MT lines is wider than 16um.	min	1.5
MT.E.1	Enclosure of MT around TOP_V (not in end-of-line)	min	0.3
MT.E.2	Enclosure of MT around TOP_V in end-of-line	min	0.4
MT.E.3	Enclosure of only angled MT line- end beyond TOP_V. DRC checks that at least one TOP_V GROW direction (X or Y direction) by mean 0.40um is covered by MT.	min	0.4
MT.C.1	Coverage of MT area on overall chip area. For DRC purposes, coverage is calculated by $[(MT(lc)+MT(dy)-MT(ms))/(chip\ area)]$	min	25%
MT.C.4	Coverage of MT area on overall chip area. For DRC purposes, coverage is calculated by $[(MT(lc)+MT(dy)-MT(ms))/(chip\ area)]$	max	60%
MT.A.1	MT island area	min	2.25um ²
MT.A.2	Area that enclosed by (hole area of) MT	min	2.25um ²



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

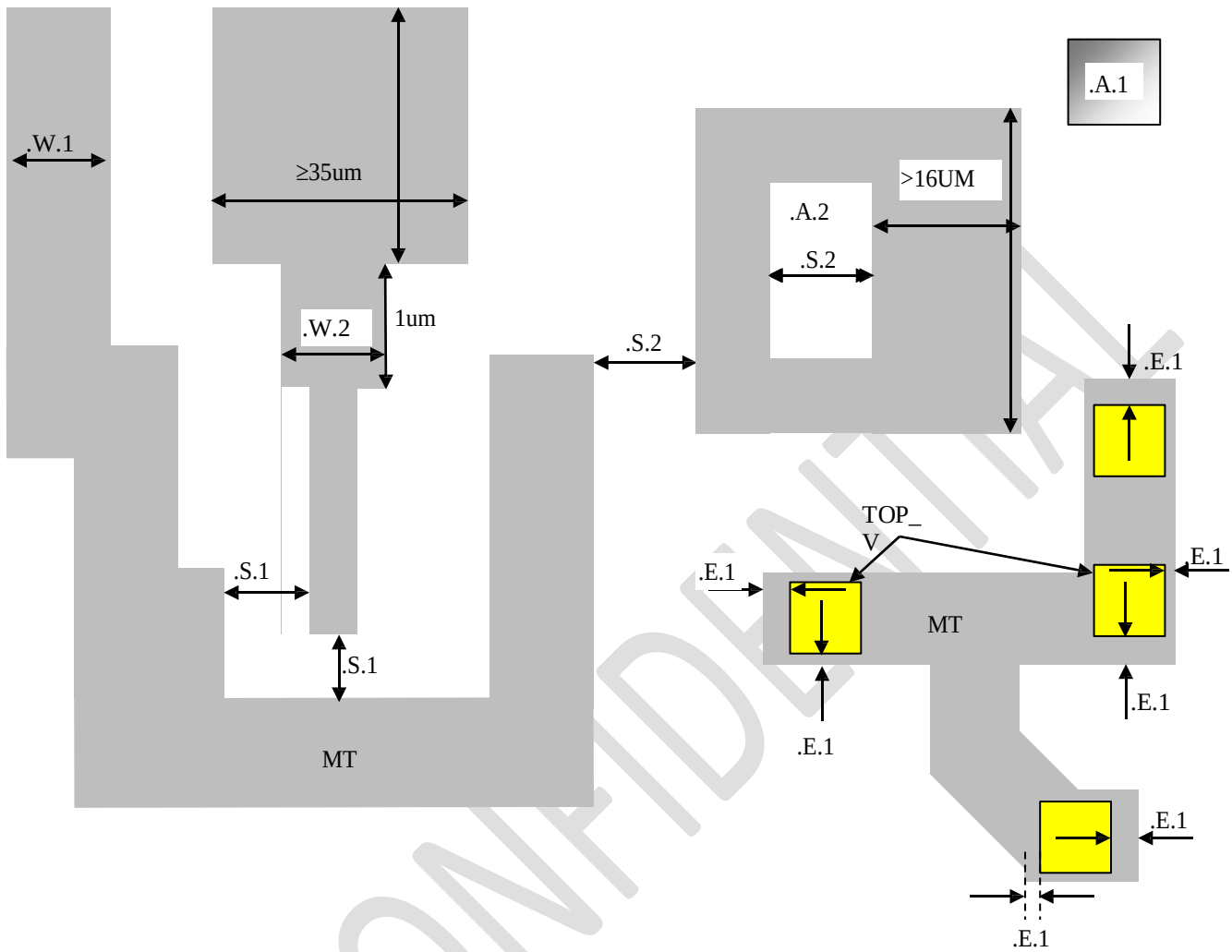


Figure 8-14: MT (20kA Thick Metal Last) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.16 MT3 (28kA Thick Al Metal Last) Rules		Action	Drawn
MT3.W.1	Width of MT3	min	2.5
MT3.S.1	Space between two MT3	min	2
MT3.S.2	Space between two MT3 lines that at least one of the MT3 lines is wider than 16um.	min	2.5
MT3.E.1	Enclosure of MT3 around TOP_V (not in end-of-line)	min	0.35
MT3.E.2	Enclosure of MT3 around TOP_V in end-of-line	min	0.4
MT3.C.1	Coverage of MT3 area on overall chip area. For DRC purposes, coverage is calculated by $[(MT3(lc)+MT3(dy)-MT3(ms))/(chip\ area)]$	min	25%
MT3.C.4	Coverage of MT3 area on overall chip area. For DRC purposes, coverage is calculated by $[(MT3(lc)+MT3(dy)-MT3(ms))/(chip\ area)]$	max	70%
MT3.A.1	MT3 island area	min	6.25um ²
MT3.A.2	Area that enclosed by (hole area of) MT3	min	6.25um ²



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

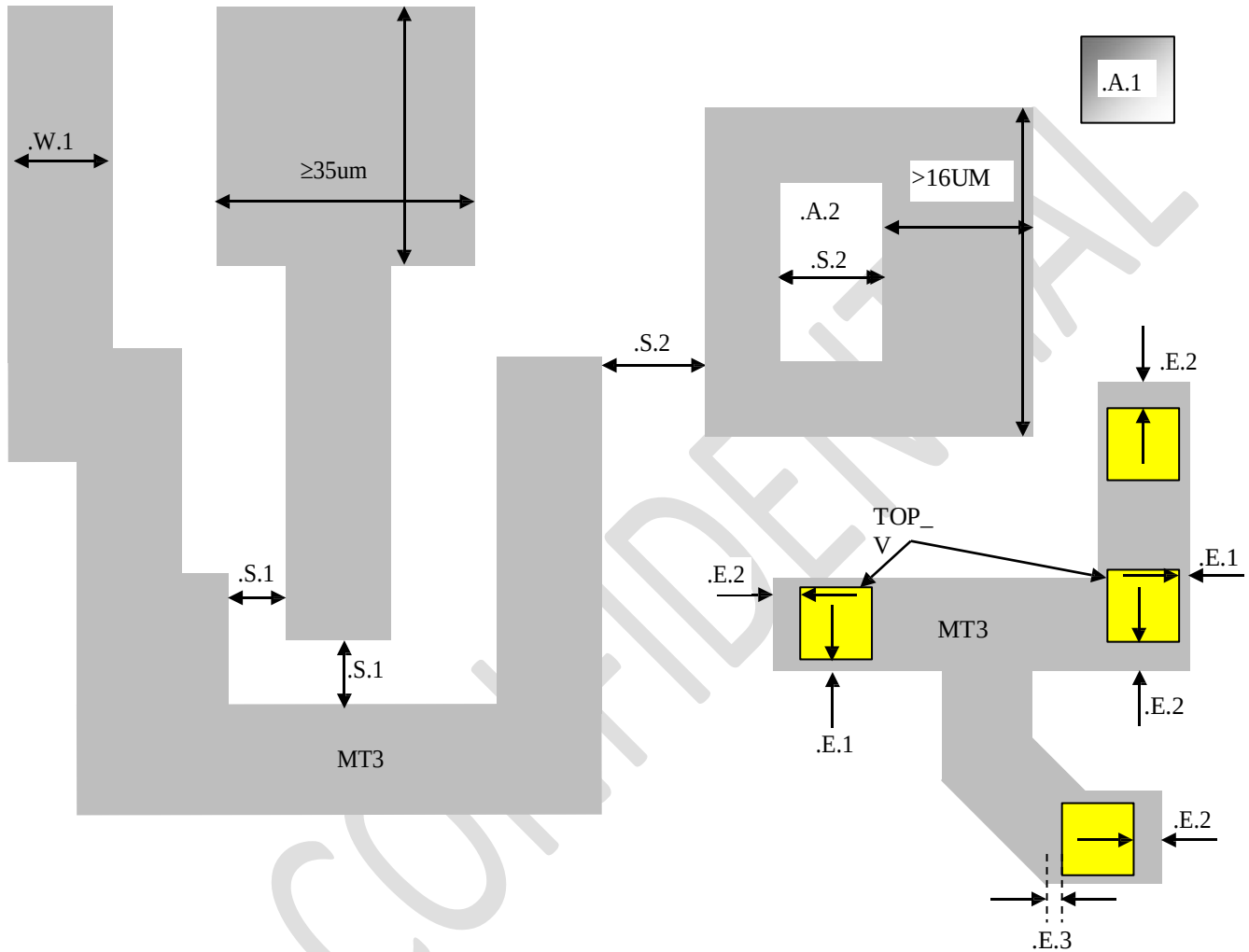


Figure 8-15: MT3 (28kA Thick Al Metal Last) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.17 WB (Well Buried, Deep N-Well) Rules			
Rule No.	Feature	Action	Drawn
WB.W.1	WB Width	min	3.0
WB.S.1	Space between two WB	min	5.0
WB.E.1	Extension of WN beyond WB	min	0
WB.O.1	Overlap of WN with WB This rule is applicable only to WB periphery [WB periphery is defined as (WB “NOT” A), A = WB size (-) 2um/per side], see Figure 8-16.	min	2.0
WB.D.1	Distance from WB to WN enclosed another WB	min	3.5
WB.D.2	Distance from WB region to un-related WN	min	3.5
WB.D.3	Distance from N+ AA to WN, where both N+ AA and WN are inside WB. (Enclosure of isolated P-well on N+ AA). This rule is not checked under marking layer of (*_bitcell) of SRAM arrays	min	0.43
WB.D.4	Distance from WB to un-related N+ AA in WP	min	3.0
WB.N.1	Only the entire WB periphery should be overlapped with WN.		
WB.RN.2	WN tap to WB that is not in reverse bias is not allowed. The WN tap to WB MUST be in reverse bias.		
WB.RN.3	Tap to WB without WN is not allowed		
WB.RN.8	WB resistor is not allowed.		
WB.N.9	WN Resistor made of WN interact WB is not allowed This rule is checked for WN resistors marked by DEV_AREA;res marking layer.		
WB.RN.10	All WN regions interacting with the same WB polygon must be in the same potential		
WB.N.11	VNN interacting with WB is not allowed.		
WB.N.12	VPNP devices over WB are not allowed. VNP devices WILL use WB layer. For DRC purposes, (77dt47 AND P+ AA) interacts {WN interact WB} is NOT allowed		



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

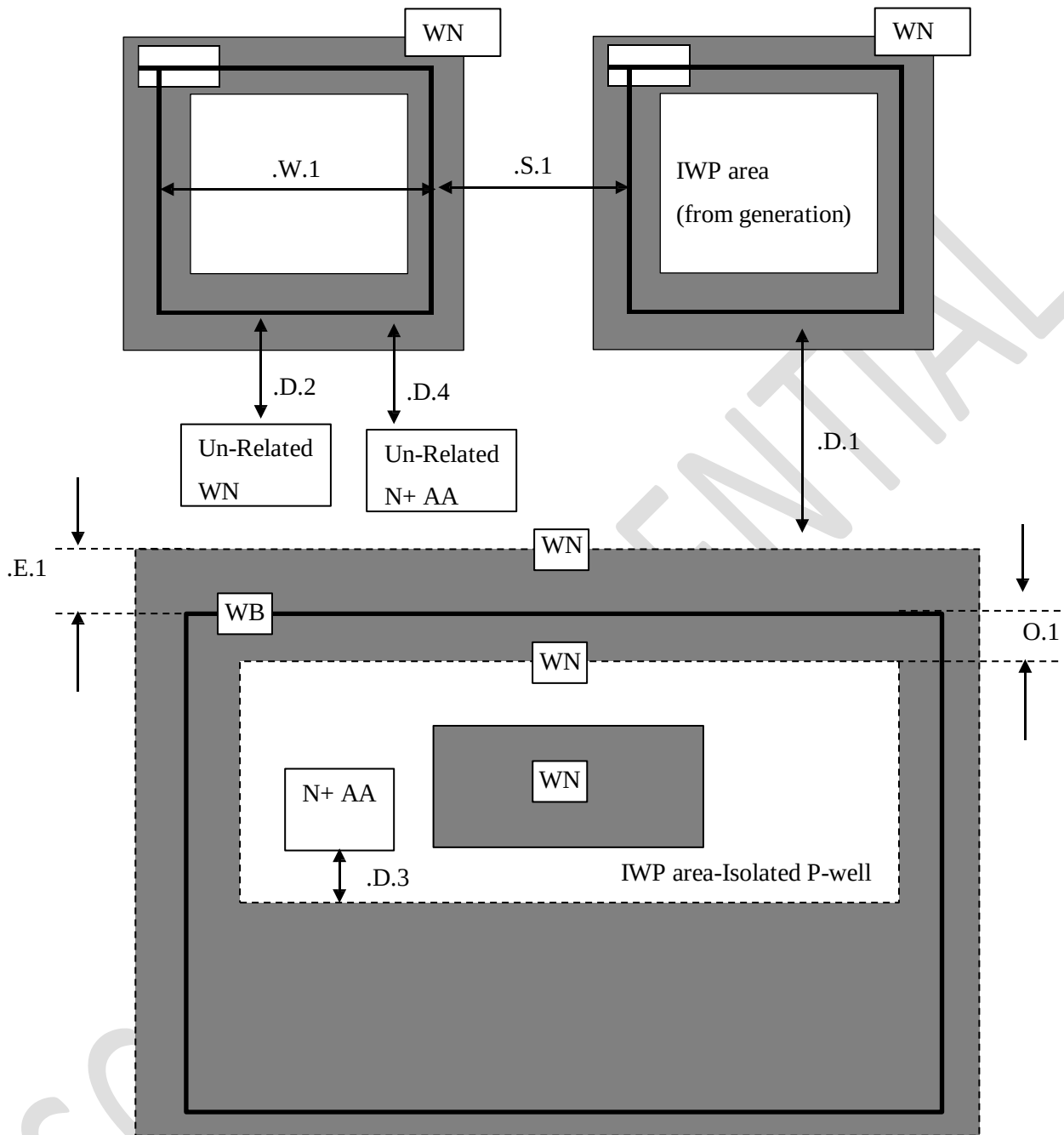


Figure 8-16: WB (Well Buried, Deep N-Well) Rules (1/2)



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

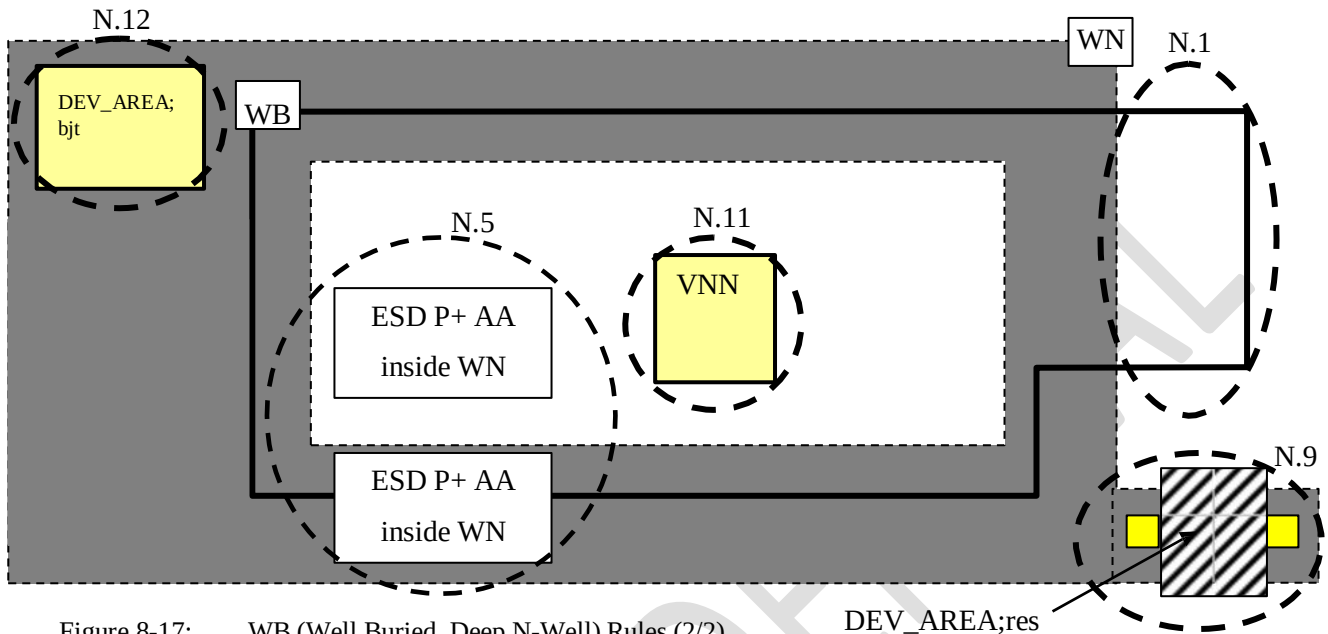


Figure 8-17: WB (Well Buried, Deep N-Well) Rules (2/2)

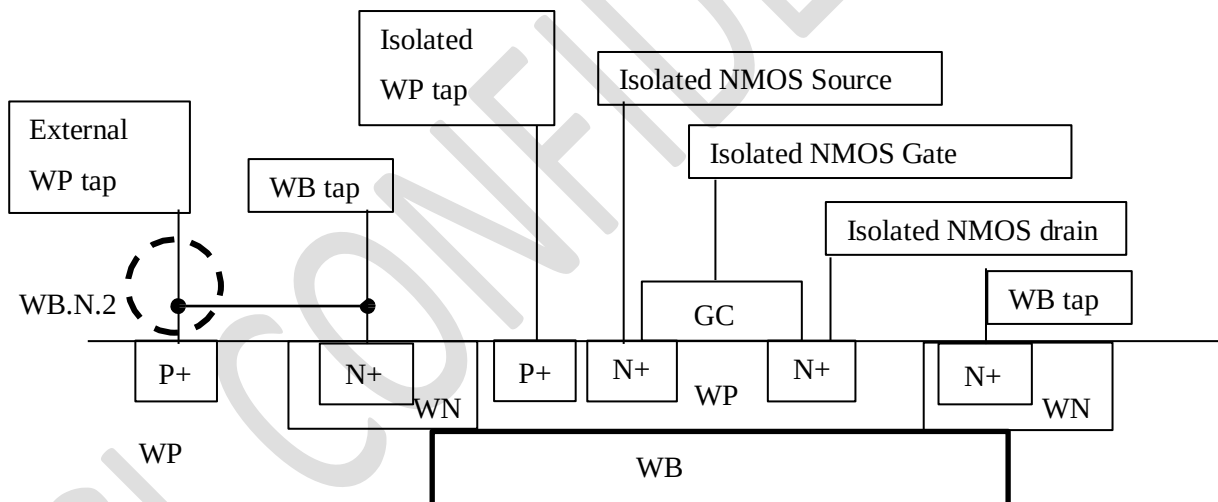
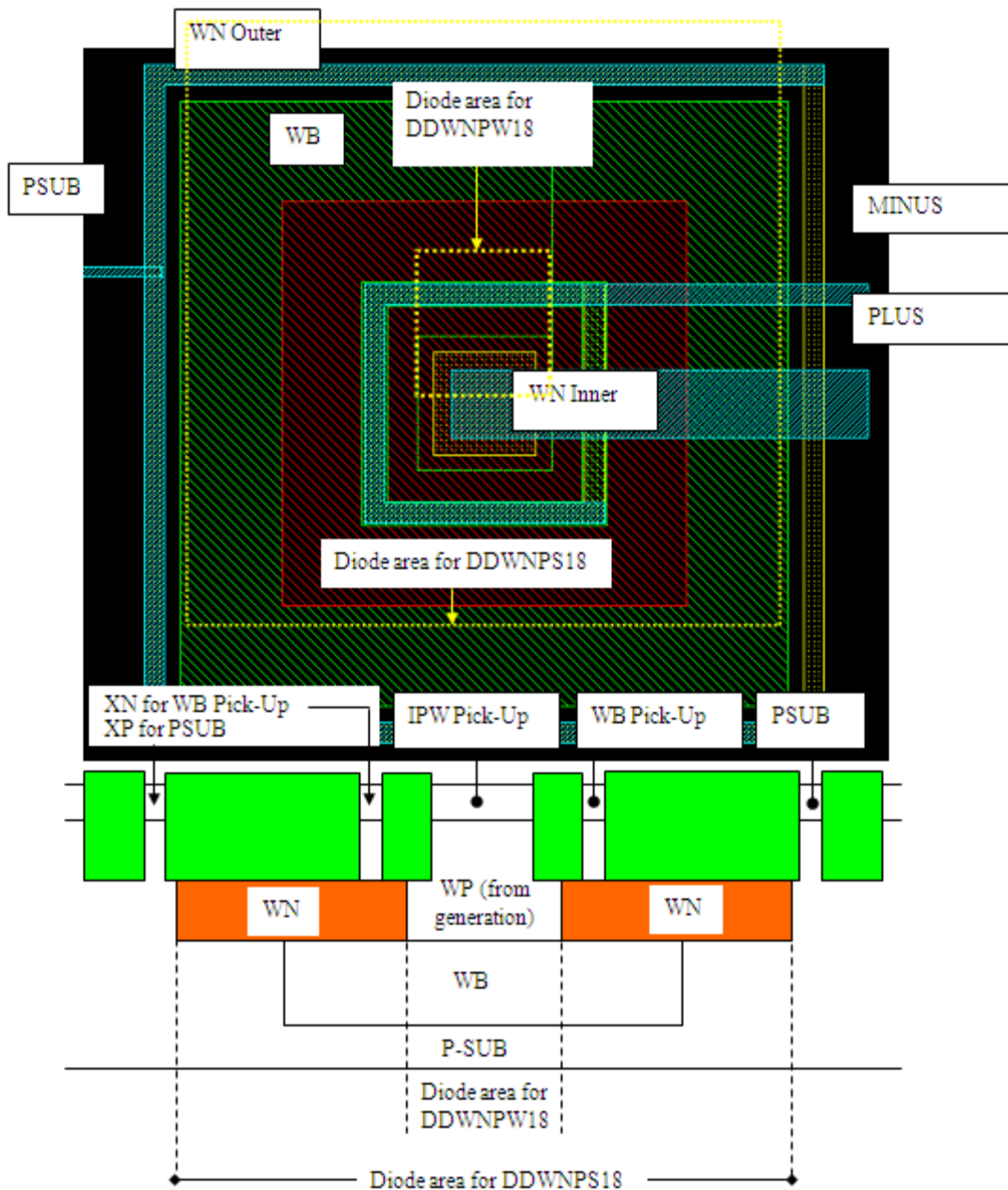


Figure 8-18: Cross Section of WB Region with the Electrical Connections

Note that by placing nMOS transistor over WB (in “isolated WP”), two additional diodes are created TOGETHER : Isolated P-Well to WB (DDWNPW18) and WB to substrate (DDWNPS18). See the related SPICE document for a schematic circuit. Connection of these two diodes is “head-to-head” between the transistor body to the substrate. By keeping the WB potential so the two diodes are in reverse-bias, the deep N-Well can isolate the P-well above it from the substrate.



The area of the Isolated-P-Well to WB (DDWNPW18) diode, defined by the WP area (the “hole” of the WN “donut”).
The area of WB to substrate (DDWNPS18) diode includes both the WB area, and the enclosure of WN around it.

Figure 8-19: DDWNPS18 (1.8V WB/Psub Diode) and DDWNPW18 (1.8V WP/WB Diode)

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.1 (Native NMOS) Devices Rules

Native NMOS transistor is defined in the area where all substrate implants, including P-well implant are blocked.

VNN layer defines the area where P-well implants are blocked; see generation of substrate implant mask for Native Transistors in Appendix A.

8.2.1.1 VNN (Native NMOS) Devices Rules

Rule No.	Feature	Action	Drawn
VNN.W.1	Minimum width	min	0.86
VNN.S.1	Minimum Space. Merge if space is less than.	min	0.86
VNN.W.2	GC gate width for Native 1.8V NMOS devices	min	0.50
VNN.W.3	GC gate width for Native 3.3V NMOS devices	min	1.20
VNN.W.4	GC gate width for Native 5V NMOS devices	min	1.5
VNN.E.1	Enclosure of VNN around N+ diffusion outside WN	exact	0.26
VNN.D.1	Distance from VNN region to N+ diffusion outside VNN (un-related device)	min	0.52
VNN.D.2	Distance from VNN (outside WN) to WN edge	min	1.66
VNN.X.1	Extension of Gate beyond field Ox.(End Cap)	min	0.35
VNN.N.1	VNN inside or cross over WN is not allowed (VNN interact WN is not allowed)		
VNN.N.2	XP inside VNN region is not allowed		
VNN.N.3	Each VNN area may include ONE AA region, only		
Note 1:	If Native Transistor is used (VNN is drawn) then CN mask and LN mask are not identical. See Appendix A.		

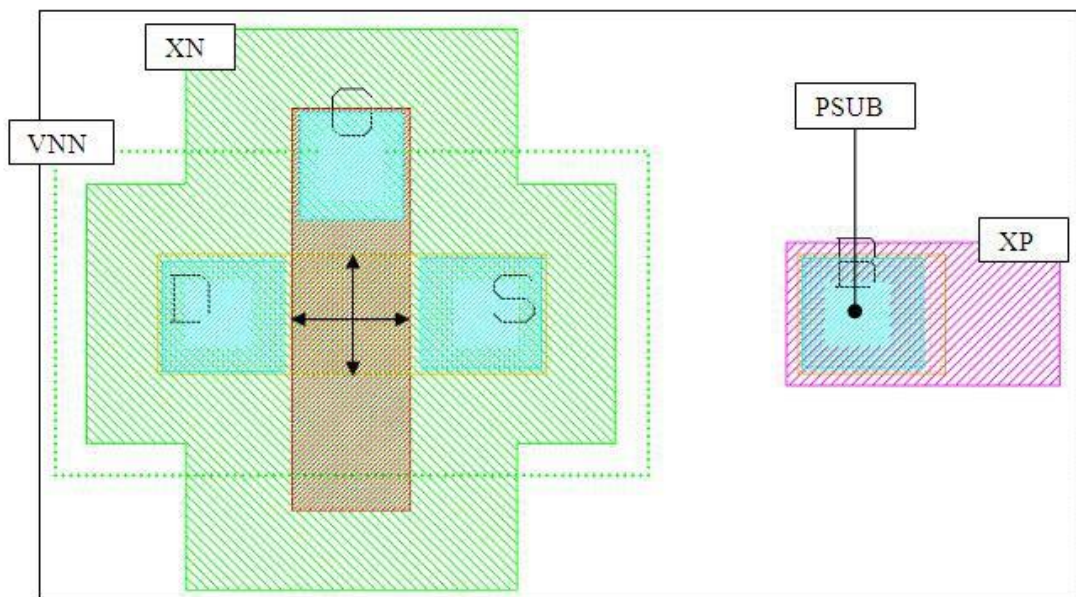


Figure 8-20: NATLV (1.8V Native Vt Transistor) SPICE, LVS Parameters: w-AA width, l-Gate length

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

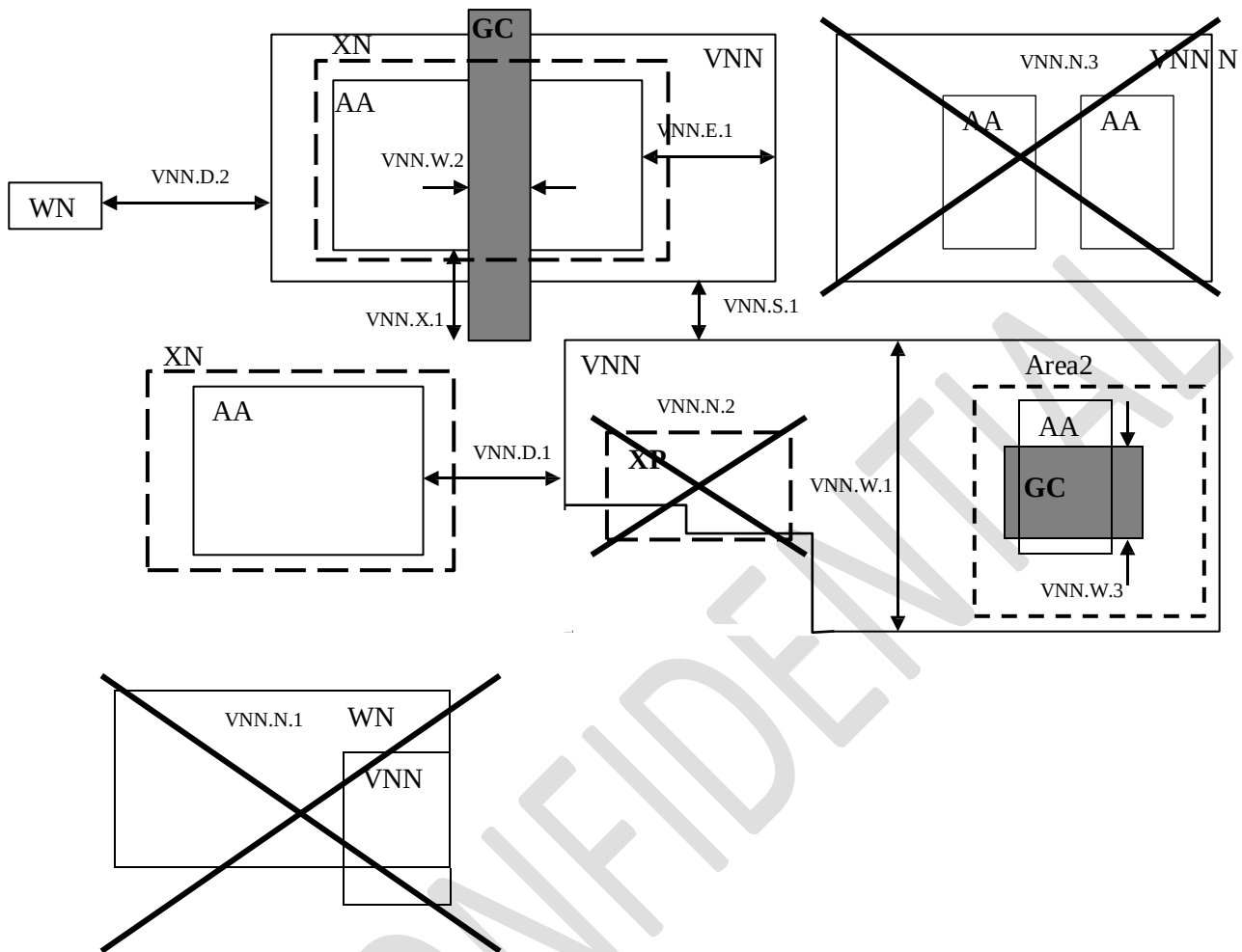


Figure 8-21: VNN (Native NMOS) Devices Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.2 NH (High Vt for Low Leakage 1.8V NMOS Devices) Rules

This layer is used for HVt design. The layer should cover the HVt device, only. Designs with HVt devices only, are not required to be covered with this layer.

8.2.2.1 CNH (High Vt for Low Leakage 1.8V NMOS Devices) Rules		Action	Drawn
CNH.W.1	Width of CNH	min	0.44
CNH.S.1	Space of CNH layer. Merge if space is under this value	min	0.44
CNH.W.2	Width of LV Low Leakage nMOS GC gate	min	0.18
CNH.X.1	Extension of CNH beyond XN AA region, perpendicular to the transistor width (along the GC end cap). This extension rule must extend 0.32um perpendicular to GC end-cap	min	0.35
CNH.D.1	Distance from CNH to GC over AA (that is not low leakage device), along the S/D direction. AA can be common to CNH transistor or have a separated AA	min	0.32
CNH.D.2	Distance from CNH to un-related GATE	min	0.44
CNH.X.2	Extension of CNH beyond LV low leakage gate, perpendicular to the transistor length (along the gate edge).	min	0.32
CNH.N.1	CNH overlap with WN is not allowed		
CNH.N.2	CNH interact with VNN is not allowed		



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

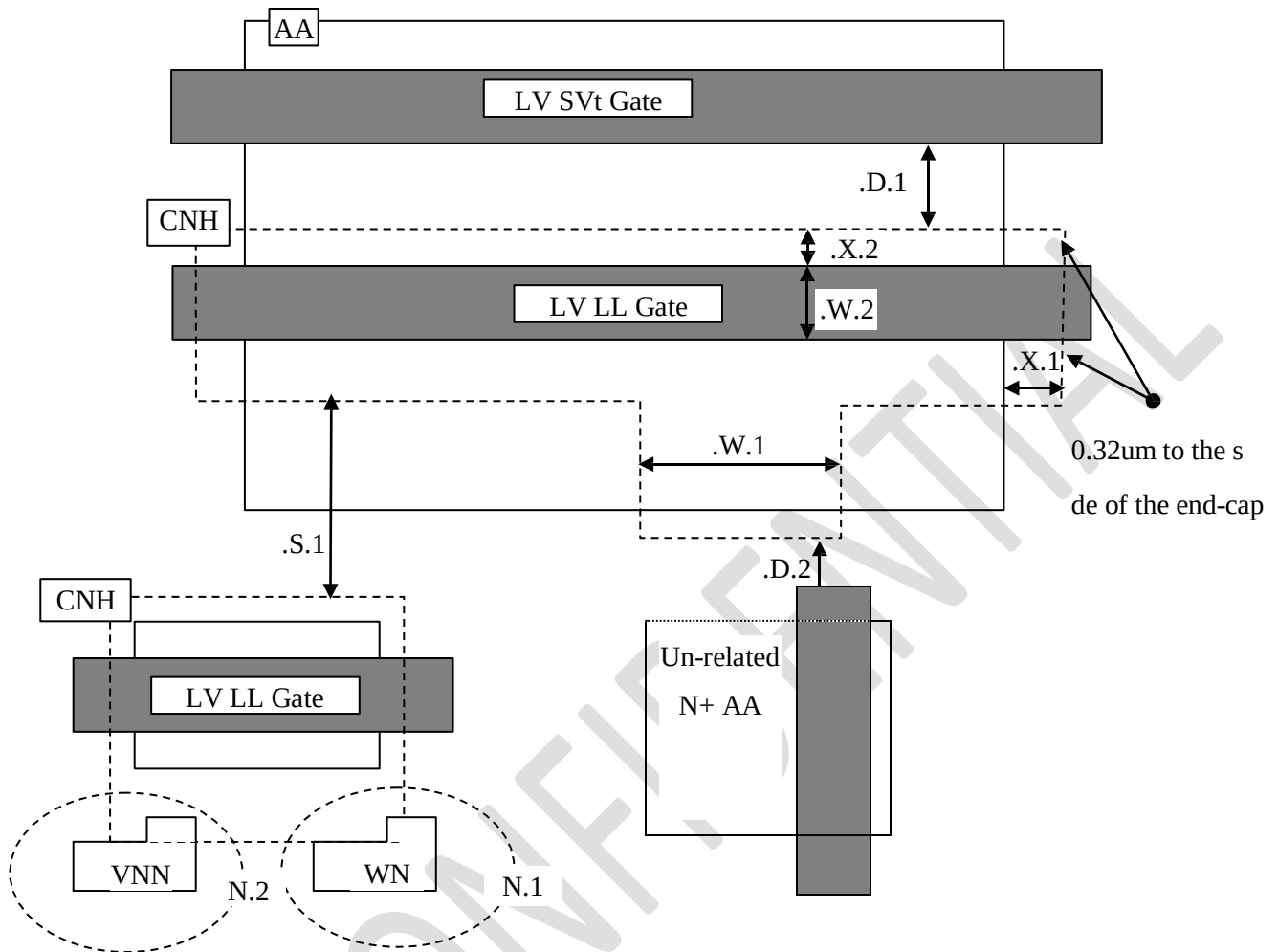


Figure 8-22: CNH (HVt Low Leakage 1.8V NMOS Devices) Rules

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.3 (HVt for Low Leakage 1.8V PMOS devices) Rules

This layer is used for HVt design. The layer should cover the HVt device, only. Designs with HVt devices only, are not required to be covered with this layer.

8.2.3.1 CPH (HVt for Low Leakage 1.8V PMOS devices) Rules		Action	Drawn
CPH.W.1	Width of CPH	min	0.44
CPH.S.1	Space of CPH layer. Merge if space is under this value	min	0.44
CPH.W.2	Width of LV Low Leakage PMOS GC gate	min	0.18
CPH.X.1	Extension of CPH beyond XP AA region, perpendicular to the transistor width (along the GC end cap). This extension rule must extend 0.32um perpendicular to GC end-cap	min	0.35
CPH.D.1	Distance from CPH to GC over AA (that is not low Leakage device), along the S/D direction. AA can be common to CPH transistor or have a separated AA	min	0.32
CPH.D.2	Distance from CPH to un-related GATE	min	0.44
CPH.X.2	Extension of CPH beyond LV low leakage gate, perpendicular to the transistor length (along the gate edge).	min	0.32
CPH.N.1	CPH overlap with WP is not allowed. CPH MUST be inside WN		
CPH.N.2	CPH interact with VNN is not allowed		

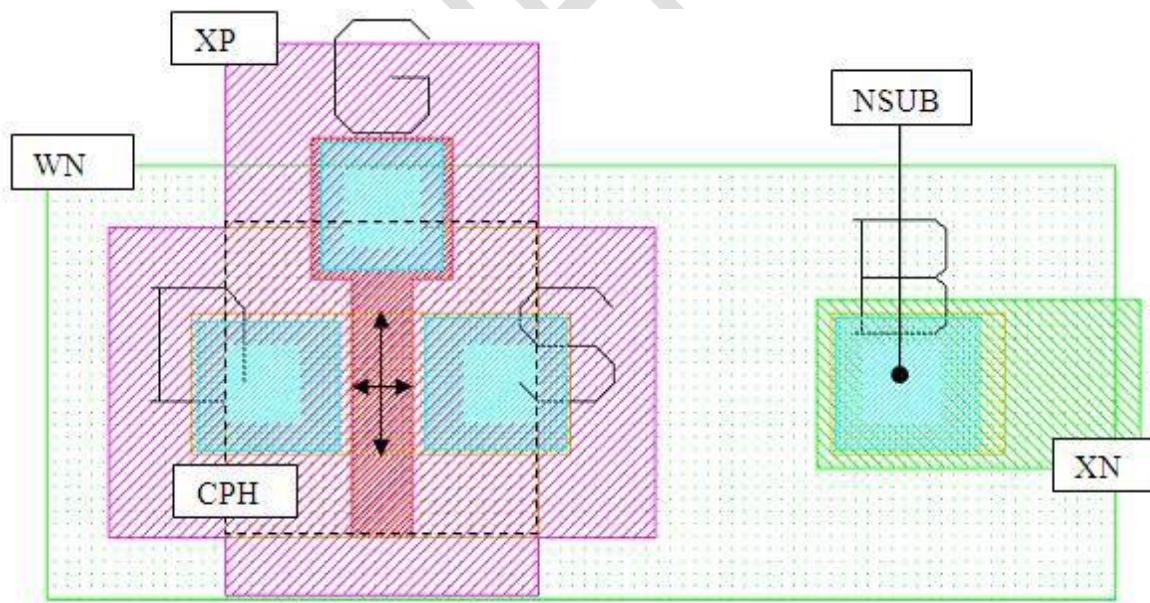


Figure 8-23: P18HVt (1.8V PMOS Low Leakage Transistor) in Dual LL. Note that CPH was drawn for minimum DR (but it is very much recommended to enclosure the AA from all sides)

SPICE, LVS Parameters: w-AA width, l-Gate length

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

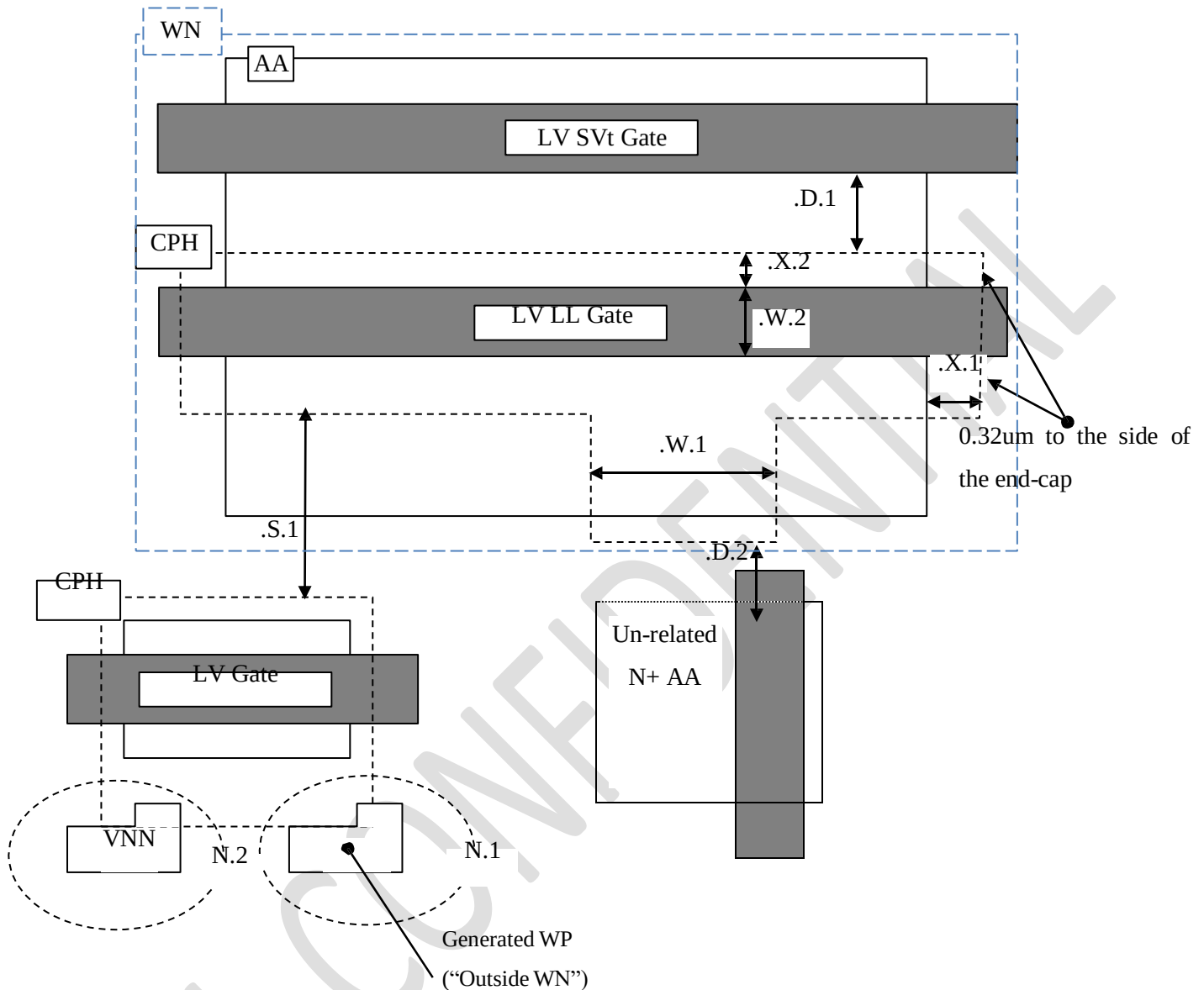


Figure 8-24: CPH (HVt Low Leakage 1.8V PMOS devices) Rules



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

8.2.4 CNL (1.8V LVt nMOS Devices) Rules.

This layer is relevant only for design that includes 1.8 LVt devices. The layer needs to cover the LVt device, only.

8.2.4.1 CNL (1.8V LVt nMOS Devices) Rules

Rule No.	Feature	Action	Drawn
CNL.W.1	Width of CNL	min	0.44
CNL.S.1	Space of CNL layer. Merge if space is under this value	min	0.44
CNL.W.2	Width of LV NMOS GC gate	min	0.18
CNL.X.1	Extension of CNL beyond XN AA region, perpendicular to the transistor width (along the GC end cap). This extension rule must extend 0.32um perpendicular to GC end-cap	min	0.35
CNL.D.1	Distance from CNL to GC over AA , along the S/D direction. AA can be common to CNL transistor or have a separated AA	min	0.32
CNL.D.2	Distance from CNL to un-related GATE	min	0.44
CNL.X.2	Extension of CNL beyond MV gate, perpendicular to the transistor length (along the gate edge).	min	0.32
CNL.N.1	CNL overlap with WN is not allowed		
CNL.N.2	CNL interact with VNN is not allowed		

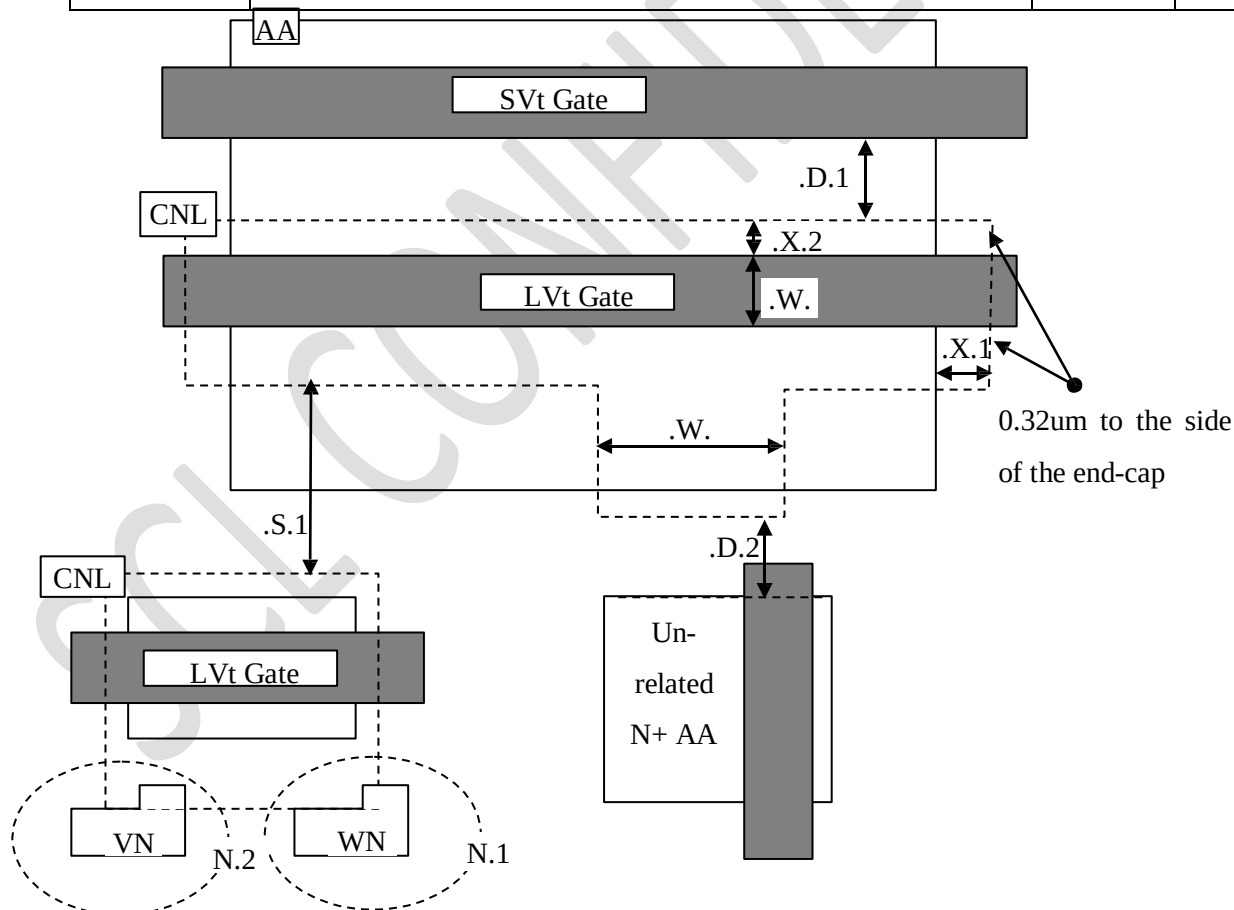


Figure 8-25: CNL (1.8V LVt NMOS Devices) Rules



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

8.2.5 CPL (1.8V LVt pMOS Devices) Rules.

This layer is relevant only for design that includes 1.8 LVt devices. The layer needs to cover the LVt device, only.

8.2.5.1 CPL (1.8V LVt pMOS Devices) Rules

Rule No.	Feature	Action	Drawn
CPL.W.1	Width of CPL	min	0.44
CPL.S.1	Space of CPL layer. Merge if space is under this value	min	0.44
CPL.W.2	Width of LV PMOS GC gate	min	0.18
CPL.X.1	Extension of CPL beyond XP AA region, perpendicular to the transistor width (along the GC end cap). This extension rule must extend 0.32um perpendicular to GC end-cap	min	0.35
CPL.D.1	Distance from CPL to GC over AA, along the S/D direction. AA can be common to CPL transistor or have a separated AA	min	0.32
CPL.D.2	Distance from CPL to un-related GATE	min	0.44
CPL.X.2	Extension of CPL beyond MVt gate, perpendicular to the transistor length (along the gate edge).	min	0.32
CPL.N.1	CPL overlap with WP is not allowed. CPL MUST be inside WN		
CPL.N.2	CPL interact with VNN is not allowed		

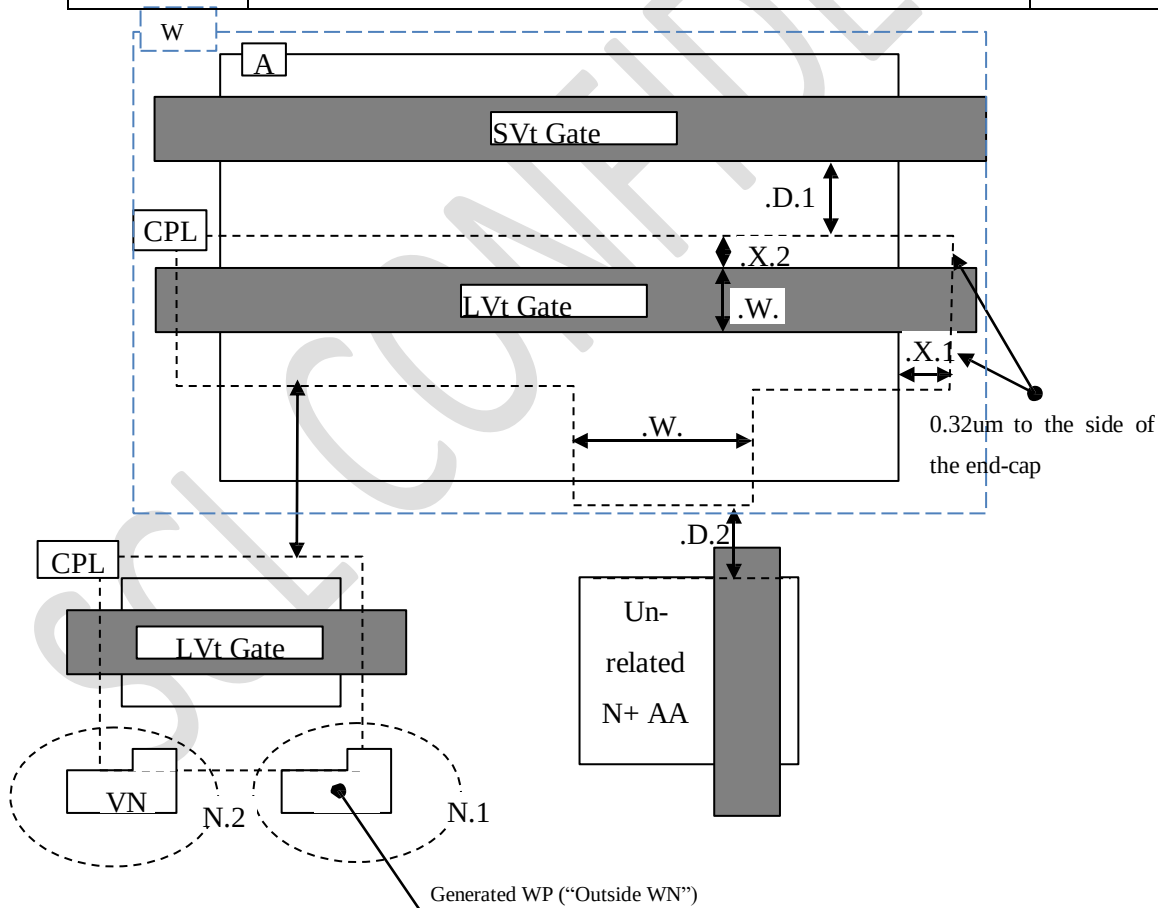


Figure 8-26: CPL (1.8V LVt PMOS Devices) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.5.2 MR (HIPO- High Ohmic P-Type for Poly w/o Salicide RESISTOR) Rules

Two different HIPO resistors are offered: 1kohm/sq and 2kohm/sq, each requires a dedicated implant. Having them both on the same chip is allowed, but requires 2 extra masks: MR for 1kohm/sq, and MR2 for 2kohm/sq. The selection between the resistors is based on marking layers combination.

Table 8-1: Layers that identify the different HIPO resistor types

Nominal resistivity	HIPO drawing layer	HIPO type selection –marking layer
1 kohm/sq	MR (87dt0)	-
2 kohm/sq		MR2 (164dt0)

HIPO Resistor length is defined by XP space. HIPO Resistor width is defined by Poly width. It is recommended to draw DEV_AREA(rs) length - coincident with XP space and width - coincident with poly width.

8.2.5.2.1 MR (HIPO- High Ohmic P-Type for Poly w/o Salicid RESISTOR) Rules		Action	Drawn
MR.N.1	HIPO resistor on AA is not allowed. Resistor must be over STI (field oxide). This rule is checked by rule RES.N.1		
MR.W.1	Width of MR (for photolithography)	min	0.44
MR.S.1	Space between two MR (for photolithography). Merge if space is under this value	min	0.44
MR.R.1	Recommended width GC for HIPO resistor	min	2
MR.D.2	Distance from MR edge to un-related N+ AA inside WP	min	0.26
MR.D.6	Distance from MR edge to non-butt NSUB (XN in N-Well Active Area)	min	0.1
MR.D.3	Distance from MR edge to nMOS Gate	min	0.32
MR.D.4	Distance from MR edge to pMOS Gate	min	0.32
MR.E.3	Enclosure of MR around MR GC resistor	min	0.26
MR.N.4	Overlap of MR on (XN AND AA) is not allowed		
MR.N.5	Overlap of XN and MR (both on GC on field oxide) is not allowed		
MR.A.1	MR island area	min	0.3844um ²
MR.O.1	Overlap of XP and NS (interact MR)	exact	0.3
MR.D.5	Distance from HIPO resistor to un-related implant regions of XN or XP. For DRC purposes, check the distance of un-related XN or XP to (GC AND MR)	min	0.26
MR.RN.2	Contacts to poly of HIPO resistor should be drawn as a single column perpendicular to current flow direction.		
MR.N.3	77dt42 that is NOT covered with NS is not allowed. This is a NON-SALICIDED resistor		
MR.X.1	Extension of XP beyond (GC and MR), and resistor ends	min	0.18
MR.S.2	Space between two XP above (GC and MR)	min	1
MR.N.9	Each type of MR resistor should NOT be mixed with another type. This rule is NOT		



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.5.2.1 MR (HIPO- High Ohmic P-Type for Poly w/o Salicid RESISTOR) Rules		Action	Drawn
	check by DRC		
MR.N.92	MR2 must co-inside MR.		

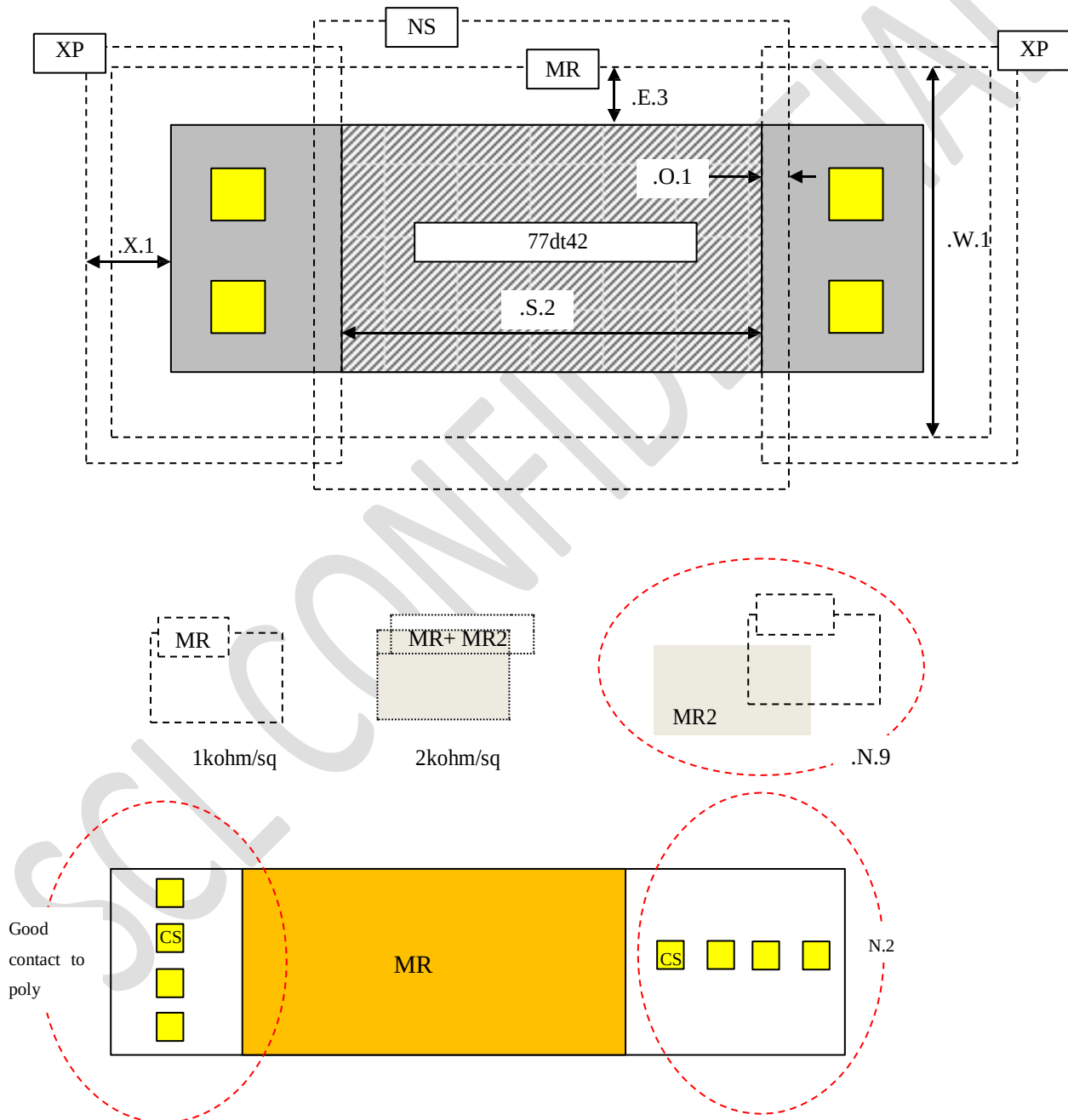


Figure 8-27: MR (HIPO- High Ohmic P-Type Poly w/o salicide RESISTOR) Rules (1/2)

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

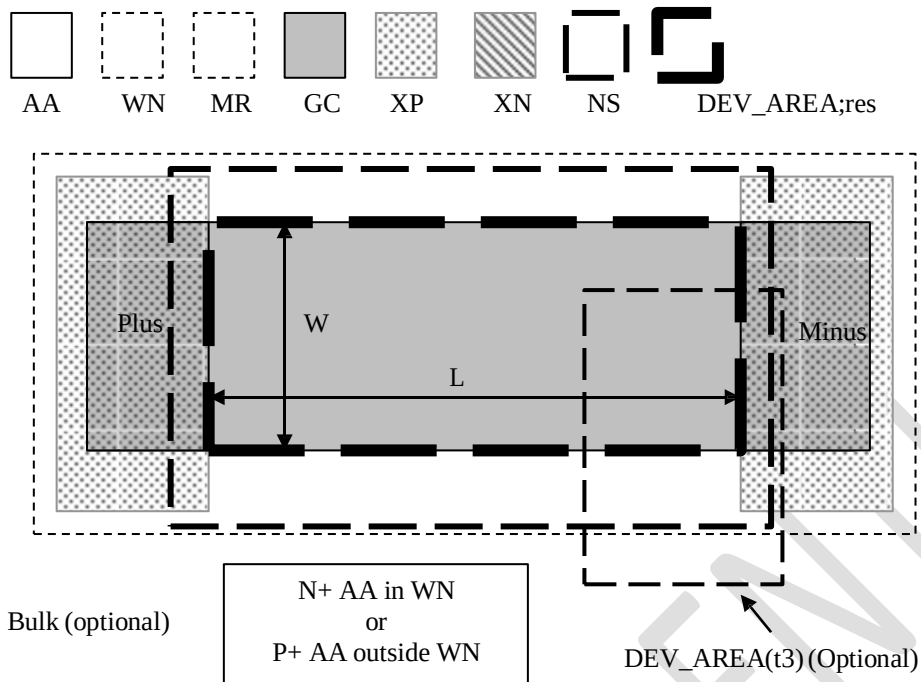


Figure 8-28: rphpoly3t (HIPO–High Ohmic, 1kOhm/sq P Poly resistor, 3 terminals)

SPICE, LVS Parameters: l, w. For mr23t (HIPO–High Ohmic, 2kOhm/sq P Poly resistor) draw MR2 co-incident with MR.

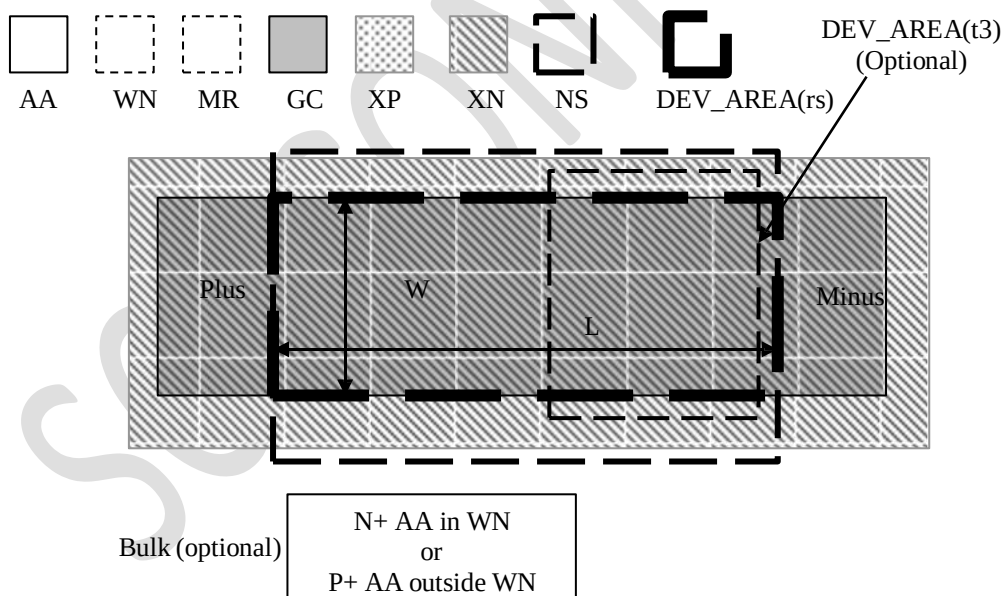


Figure 8-29: rnmppoly3t (NMOPO, Medium Ohmic P Poly w/o salicide resistor).

SPICE, LVS Parameters: w-resistor width (defined by Poly width), l-resistor length (defined by NS layer)



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

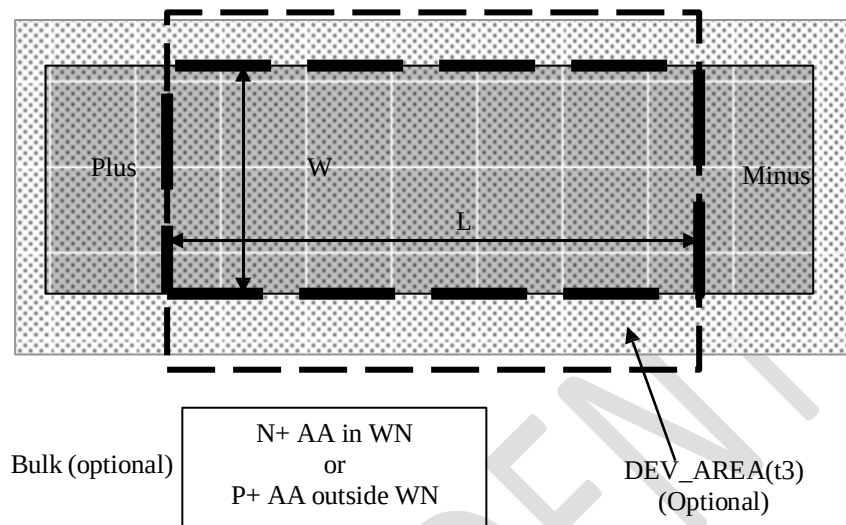
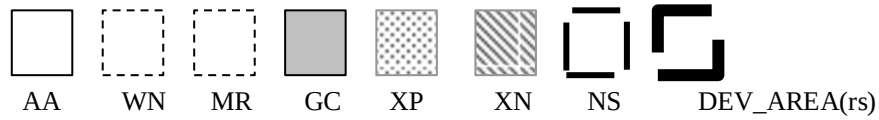


Figure 8-30: rpmpoly3t (PMOPO, Medium Ohmic P Poly w/o salicide resistor)

SPICE, LVS Parameters: w-resistor width (defined by Poly width), l-resistor length (defined by NS layer)



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

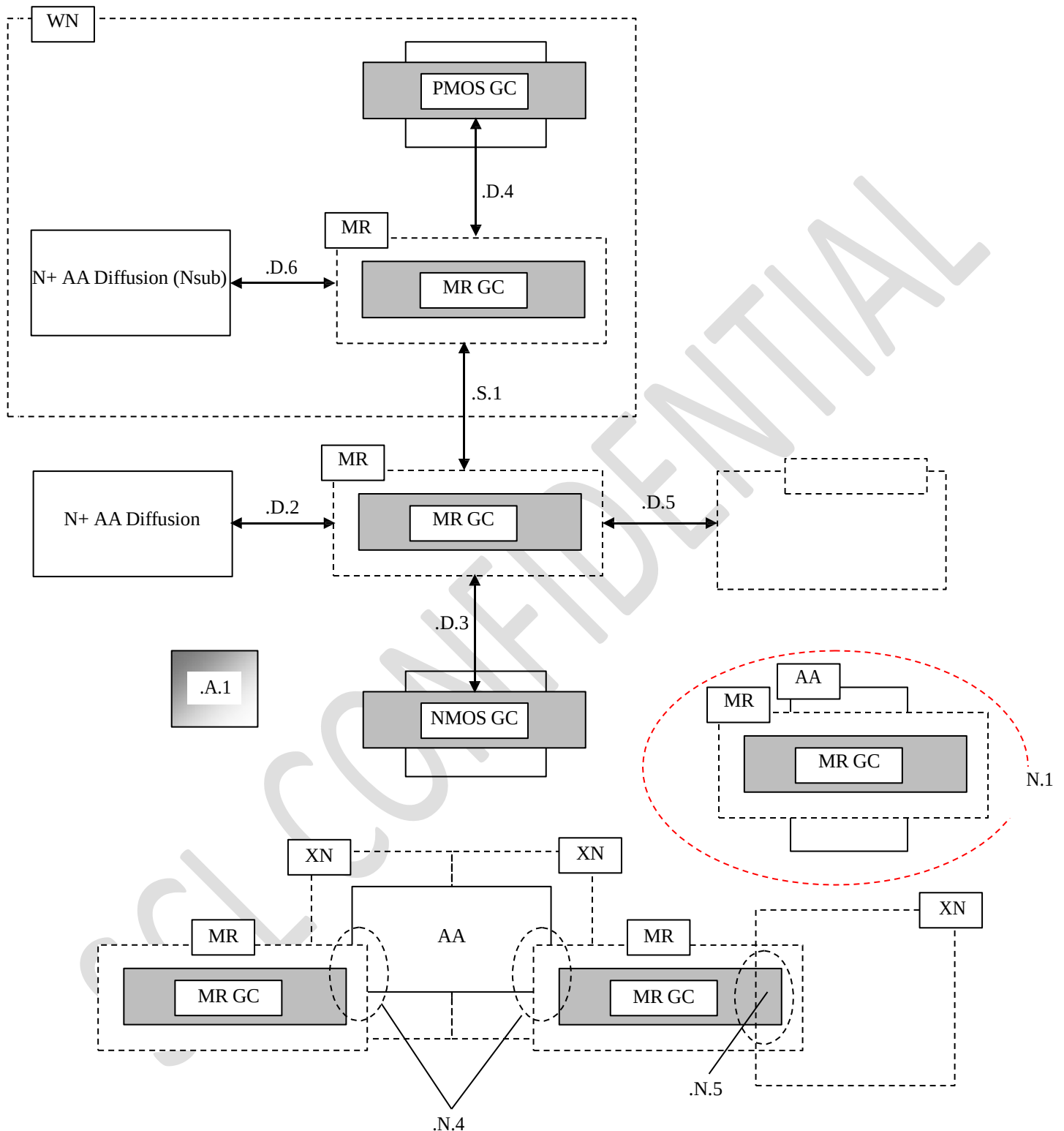


Figure 8-31: MR (HIPO- High Ohmic P-Type Poly w/o salicide RESISTOR) Rules (2/2)



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

8.2.5.3 Medium Ohmic Poly (MOPO) Resistor Rules

MOPO resistors are non salicided poly resistors, implanted with XN or XP.

8.2.5.3.1 NPR (NMOPO, Medium Ohmic N-Type Poly w/o salicided RESISTOR) Rules		Action	Drawn
NPR.E.1	Enclosure of XN around NMOPO resistor	min	0.18
NPR.D.1	Distance from NMOPO resistor to un-related implant regions of MR or XP.	min	0.26
NPR.N.1	NMOPO resistor on AA is not allowed. Resistor must be over STI (field oxide). This rule is checked by rule RES.N.1		
NPR.RN.2	Contacts to poly resistor should be drawn as a single column perpendicular to current flow direction. Refer to rule RER.R.5		
NPR.R.1	Recommended width of GC of NMOPO resistor.	min	2

8.2.5.3.2 PPR (PMOPO, Medium Ohmic P-Type Poly w/o salicided RESISTOR) Rules		Action	Drawn
PPR.E.1	Enclosure of XP around PMOPO resistor	min	0.18
PPR.D.1	Distance from PMOPO resistor to un-related implant regions of MR or XN.	min	0.26
PGR.N.1	PMOPO resistor on AA is not allowed. Resistor must be over STI (field oxide). This rule is checked by rule RES.N.1		
PPR.RN.2	Contacts to poly resistor should be drawn as a single column perpendicular to current flow direction. Refer to rule RER.R.5		
PPR.R.1	Recommend width of GC of PMOPO resistor	min	2

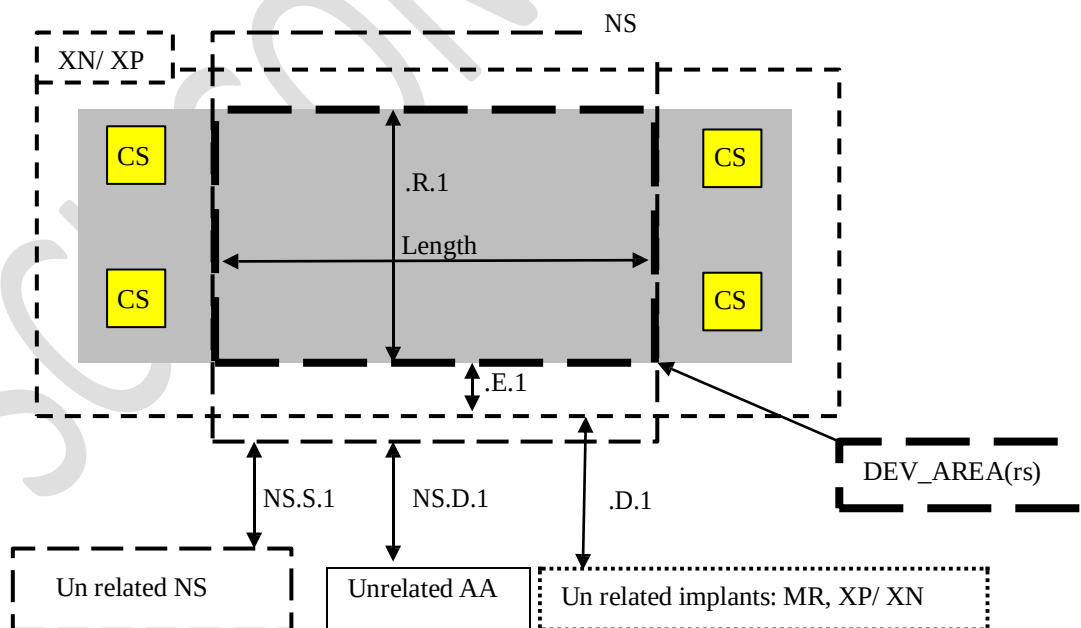


Figure 8-32: NPR/ PPR (NMOPO/PMOPO, Medium Ohmic N/P-Type W/O Salicide Resistor) Rules

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.6 NAR (N+ AA diffusion w/o Salicided Resistor) Rules		Action	Drawn
NAR.R.1	Recommended width of AA of non Salicided N+ Active resistor.	min	2
NAR.N.1	Resistor over WN is not allowed	min	

8.2.7 PAR (P+ AA diffusion w/o Salicided Resistor) Rules		Action	Drawn
PAR.R.1	Recommended width of AA of non Salicided P+ Active resistor	min	2
PAR.N.1	Resistor outside WN is not allowed		

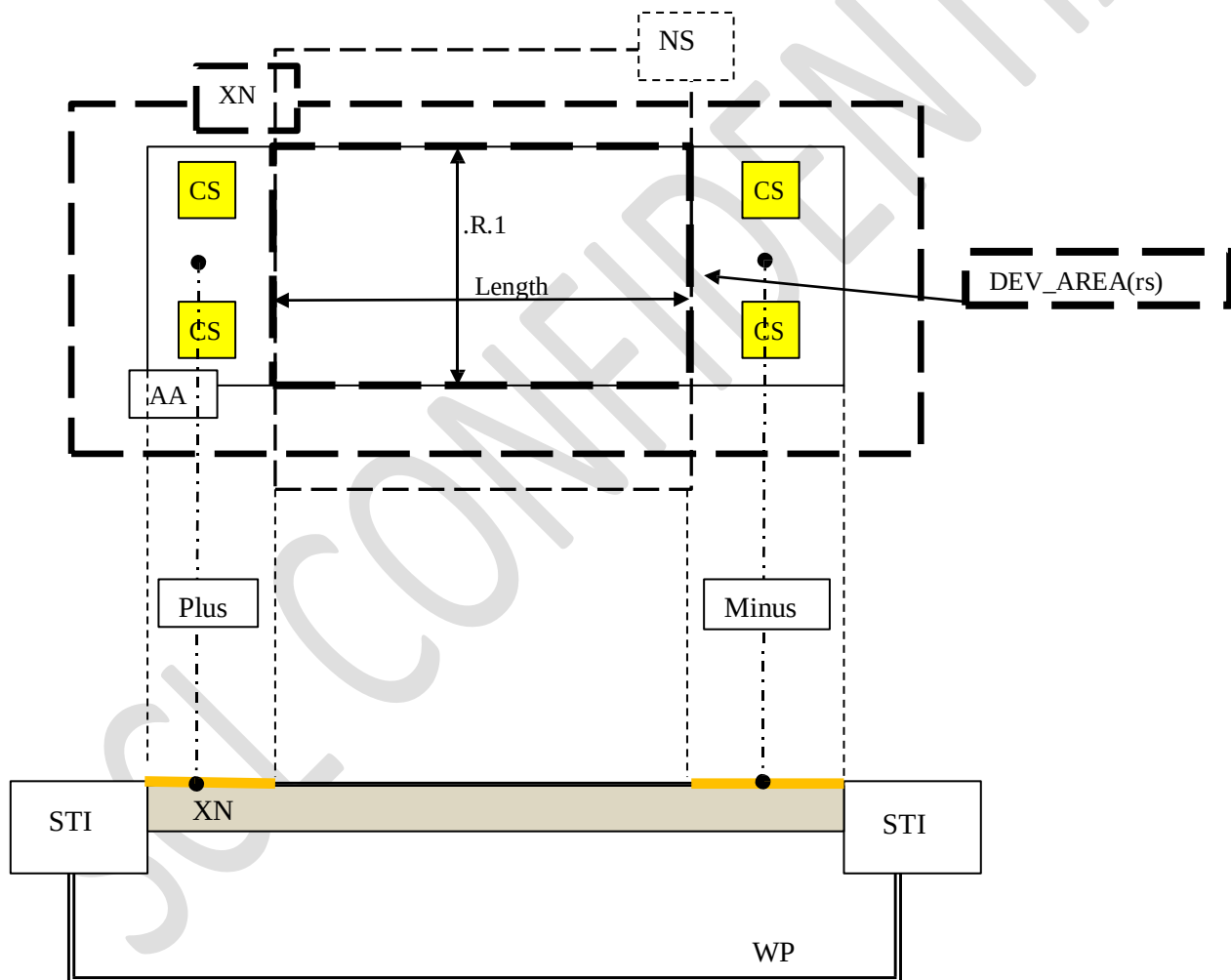


Figure 8-33: NAR (N+ AA diffusion w/o Salicided Resistor) Rules

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

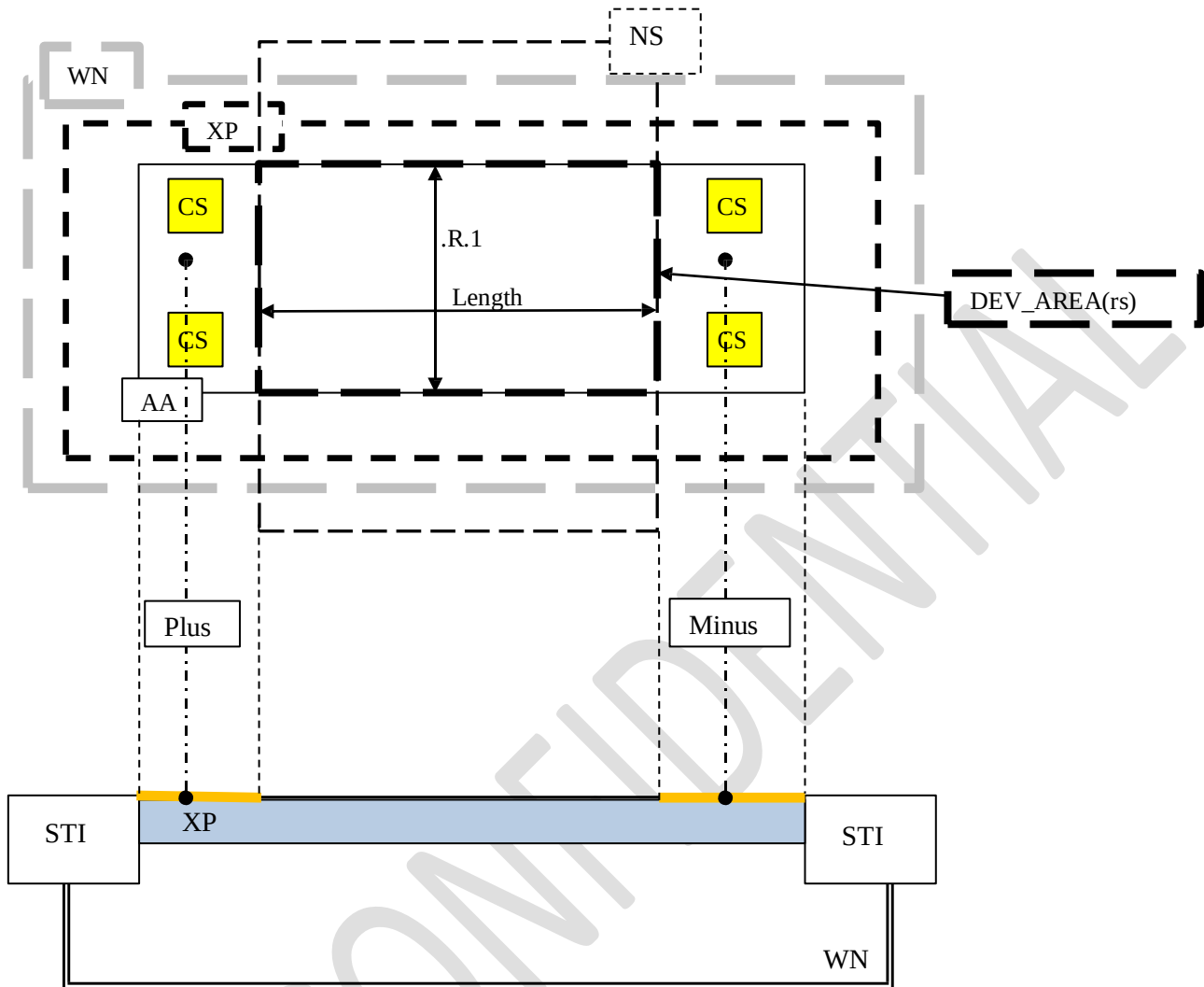


Figure 8-34: PAR (P+ AA diffusion w/o salicide Resistor) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.8 WNSR (N-Well under STI) Resistor Rules		Action	Drawn
WNSR.W.1	Width of WN under STI resistor	min	2.1
WNSR.E.1	Enclosure of N-Well around NSUB. Using for resistors ONLY, replacing rule AA.E.2.	min	0
WNSR.D.2	Distance from dummy AA to Nwell under DEV_AREA(rs)	min	1.2
WNSR.N.1	XP interact DEV_AREA(rs) is not allowed		
WNSR.N.9	Dummy AA in resistor area is not allowed.		

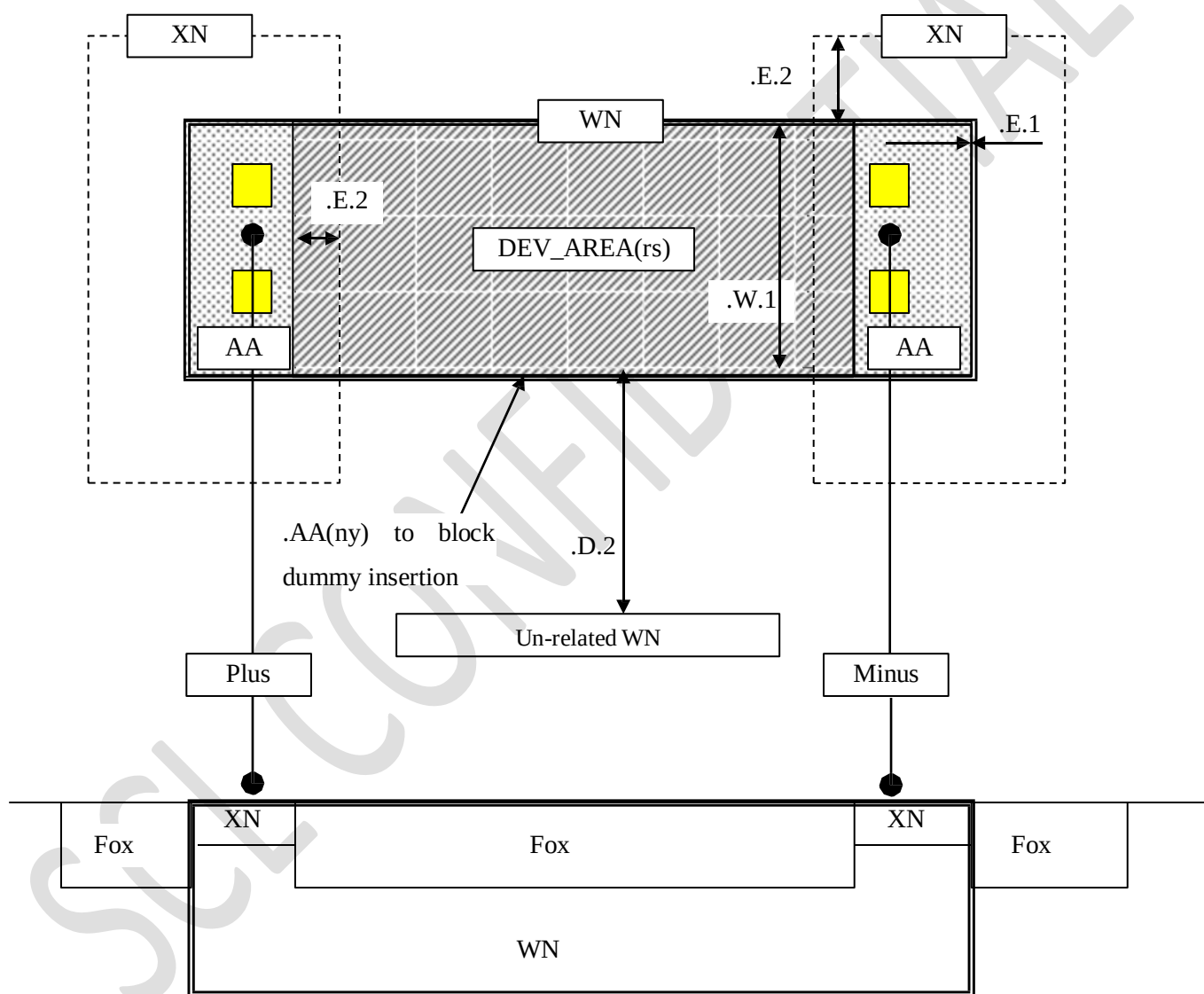


Figure 8-35: WNSR (N-Well under STI) Resistor Rules

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

8.2.8.1 RES (General POLY Resistor) Rules

AA dummy fill insertion run set does not add dummy AA's in and near the resistor area. In order to meet low AA coverage rules (AA.C.1/3/5), keep sufficient GC spaces between the resistors. Dummy Fill run-set will add dummy AA in these spaces. DRC Rule GC.C.4 assists to achieve this, by checking existence of sufficient GC spaces.

In order to keep good matching performance of resistors array, place N+ or P+ ACTIVE bars inside the dense resistors array. This will ensure both analog matching, and assist in meeting minimum AA density rules.

8.2.9 RES (General POLY Resistor) Rules		Action	Drawn
RES.E.1	Enclosure of XN around GC of LOPO resistor	min	0.18
RES.E.2	Enclosure of XP around GC of LOPO resistor	min	0.18
RES.E.3	Enclosure of N-Well on Resistor poly This rule is NOT checked by DRC.	min	0.5
RES.D.1	Distance from XN to P+ GC of LOPO resistor	min	0.18
RES.D.2	Distance from XP to N+ GC of LOPO resistor	min	0.18
RES.D.3	Distance from Resistor poly to N-Well.	min	0.29
RES.N.1	Poly resistor on AA is not allowed. Resistor must be over STI (field oxide). This rule is checked for all kinds of poly resistors: N-LOPO, P-LOPO, NMOPO, PMOPO and HIPO resistors		

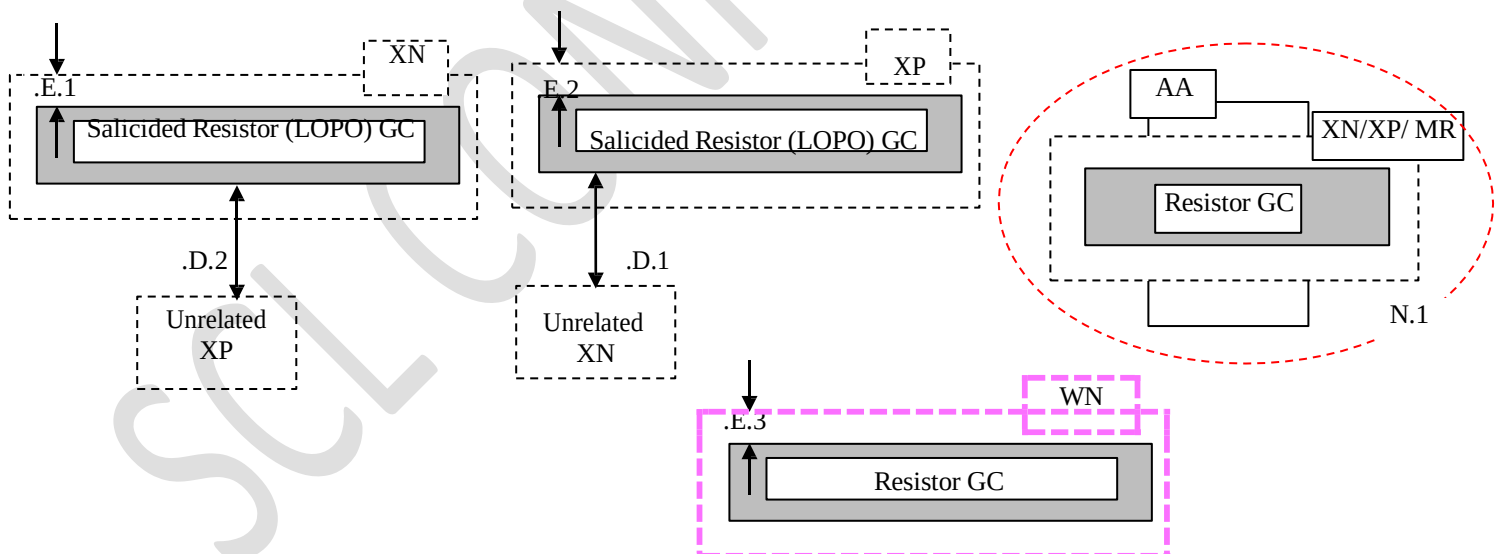


Figure 8-36: RES (General POLY Resistor) Rules

8.2.10 AA and GC coverage in nwell resistor area



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

AA dummy fill insertion run set does not add dummy AA's in and near the resistor area. In order to meet low AA coverage rules (AA.C.1/3/5), design the resistor with sufficient WN spaces. Dummy Fill run-set will add dummy AA in these spaces. DRC Rule GC.C.4 assists to achieve this, by checking existence of sufficient WN spaces.

In order to keep good matching performance of resistors array, place Psubs (P+ ACTIVE) bars inside the dense resistors array. This will ensure both analog matching, and assist in meeting minimum AA density rules.

8.2.10.1 WNAR (N-Well at AA) Resistor Rules		Action	Drawn
WNAR.W.1	Width of WN at Nwell under AA resistor	min	2.1
WNAR.X.1	Extension of AA beyond WN at WNAR resistor	min	1
WNAR.E.1	Enclosure of WN around CS located on salicided AA at WNAR resistor	min	0.3
WNAR.D.3	Distance from NS(inner) to related AA on WN at WNAR resistor	min	0.3
WNAR.E.3	Enclosure of NS(outer) around related AA at WNAR resistor	min	0.22
WNAR.D.1	Distance from NS(inner) to CS at WNAR resistor	min	0.3
WNAR.E.2	Enclosure of XN around (NS(inner)) salicided AA on WN at WNAR resistor	min	0.4
WNAR.N.1	Overlap of DEV_AREA(rs) with XP or XN is not allowed		
WNAR.N.9	Dummy AA in resistor area is NOT allowed For DRC purposes, this rule is covered by WNSR.N.9		
WNAR.D.2	Distance from dummy AA to Nwell under DEV_AREA(rs)	min	1.2



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

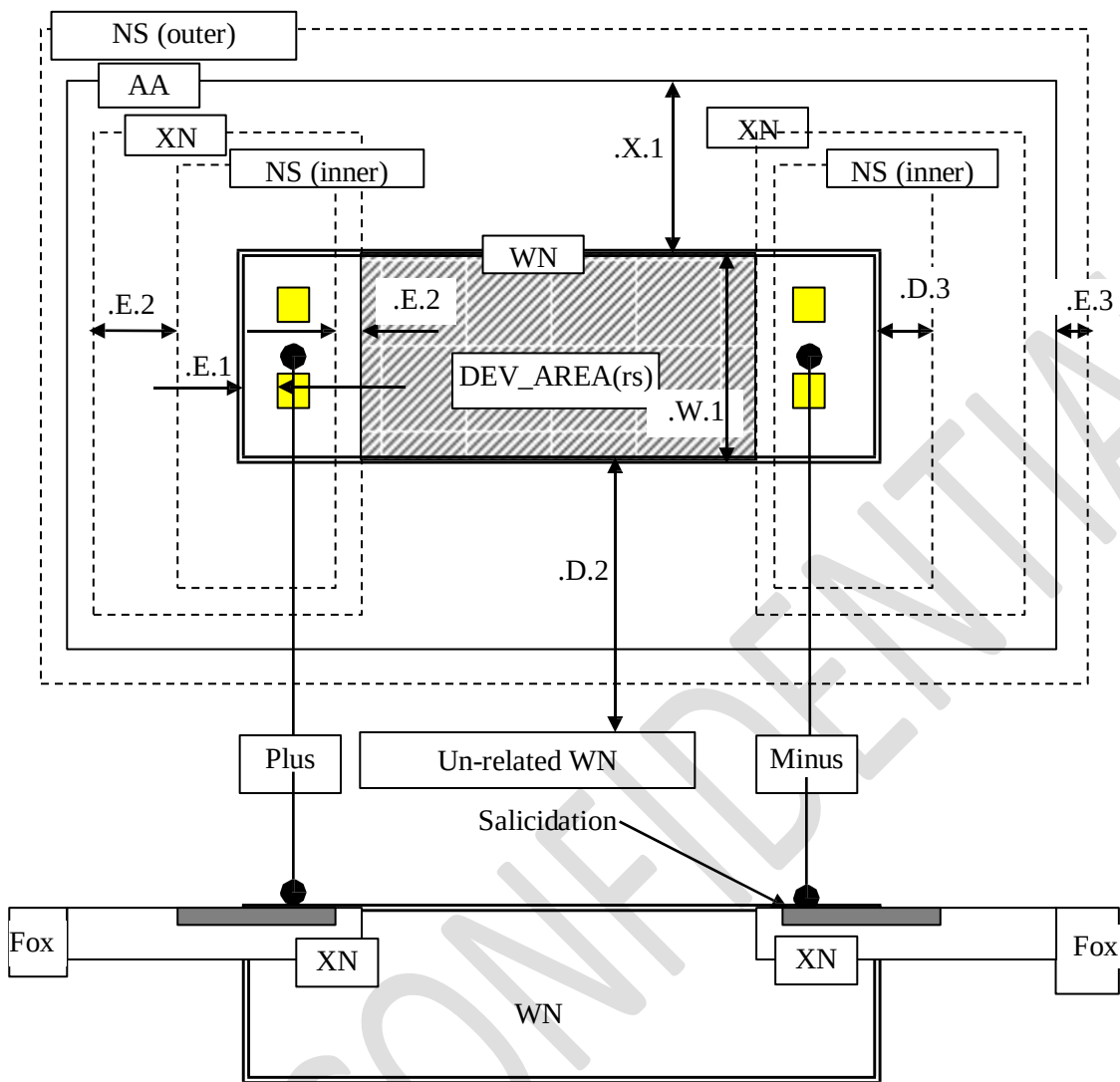


Figure 8-37: WNSR (N-Well at AA) Resistor Rules

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.11 NSDN and NSDP (Non salicided 1.8V P+/WN and 1.8V N+/WP diode) Rules

This section refers to the non salicided *pdio_nosal* and *ndio_nosal* diodes

8.2.11.1 NSDN and NSDP (Non salicided 1.8V P+/WN and 1.8V N+/WP diode) Rules			
NSDN.O.1	Overlap of NS on N+ AA of non Salicided N+/Pwell diode. This rule should be kept along the entire edge of the N+ AA	min	0.22
NSDP.O.1	Overlap of NS on P+ AA of Salicided P+/Nwell diode. This rule should be kept along the entire edge of the P+ AA	min	0.22

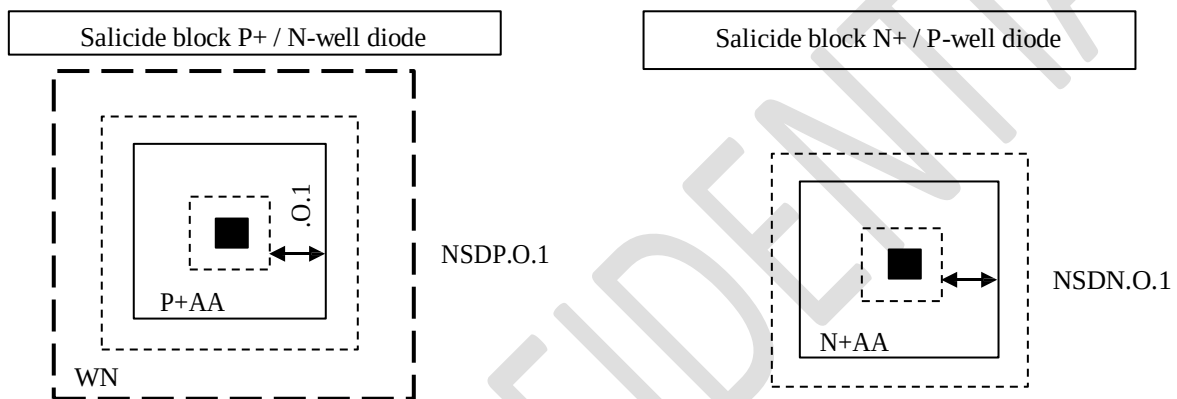


Figure 8-38: NSDN and NSDP (Non salicided 1.8V P+/WN and 1.8V N+/WP diode) Rules

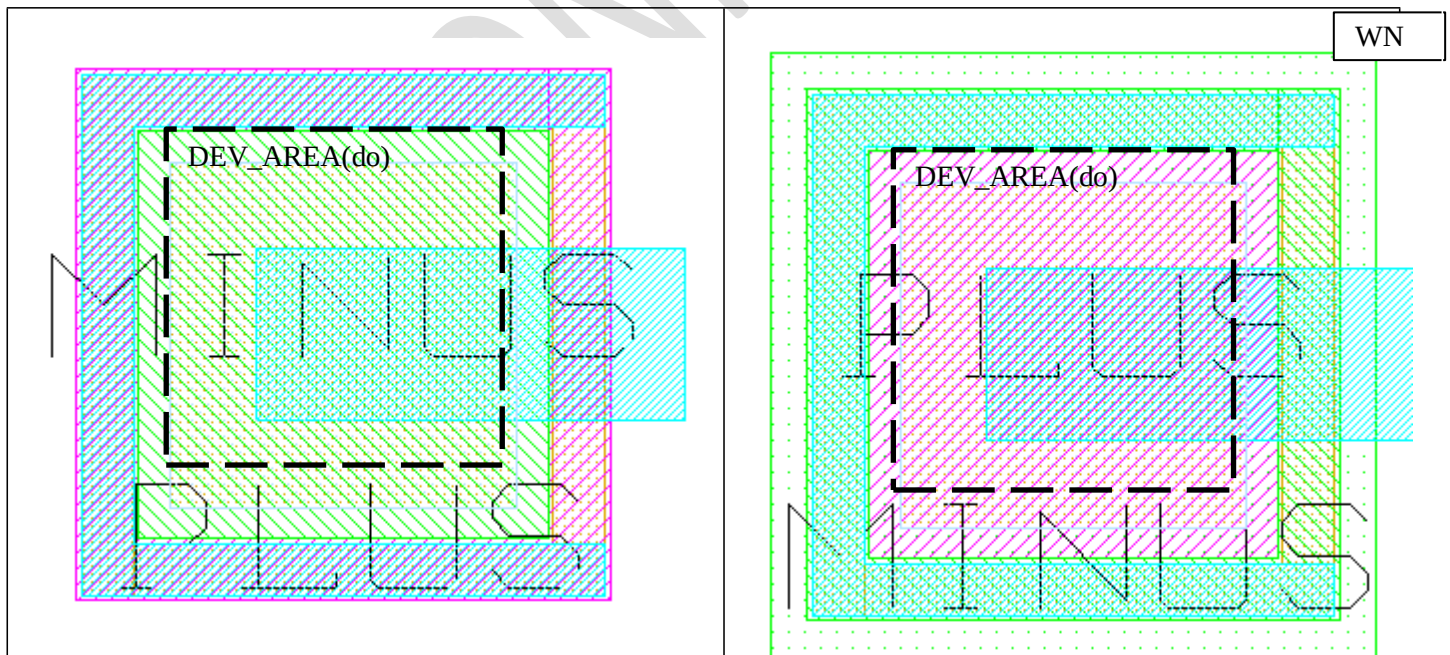


Figure 8-39: DN18 (1.8V N+/WP Diode) SPICE, P+/WN Diode)

Figure 8-40: DP18 (1.8V

8.2.12 Accumulation Capacitor Layout Rules and Recommendations

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

N-Type capacitors are made with top electrode of salicided N-gate Poly over gate oxide (thin or thick) over bottom electrode of WN. P-Type capacitors made with top electrode of salicided P-gate Poly over gate oxide over bottom electrode of WP or Isolated WP.

For optimal model accuracy it is recommended to use square capacitors. For larger capacitance it is better to use multi unit cell, that yield higher accuracy than a single large unit cell.

Take care not to connect capacitors in same N-well in a serial way, since the nwell is common to all of them.

Accumulation capacitors should be marked by DEV_AREA(cp) support layer for DRC and LVS.

8.2.12.1 MOSCAP (Accumulation Capacitors) Rules		Action	Drawn
MOSCAP.N.4	WP accumulation capacitor MUST be over IPW. For DRC purposes, XP AA interact GC gate in WP, outside WB is not allowed		
MOSCAP.N.5	1.8V Three terminal Scalable Accumulation Capacitor (CL3), shares the same N-well polygon with 3.3V N+ Gate Accumulation Capacitor (CH) is NOT allowed		
MOSCAP.N.6	3.3V Three terminal Scalable Accumulation Capacitor (CH3), shares the same N-well polygon with 1.8V N+ Gate Accumulation Capacitor (CL3) is NOT allowed		
MOSCAP.N.7	WN of three terminal Scalable Accumulation Capacitor, CL3 or CH3, interacts with (ACTIVE and XP) in NOT allowed		
MOSCAP.N.8	WN of three terminal Scalable Accumulation Capacitor, CL3 or CH3, interacts with support layer of DEV_AREA(rs) is NOT allowed		

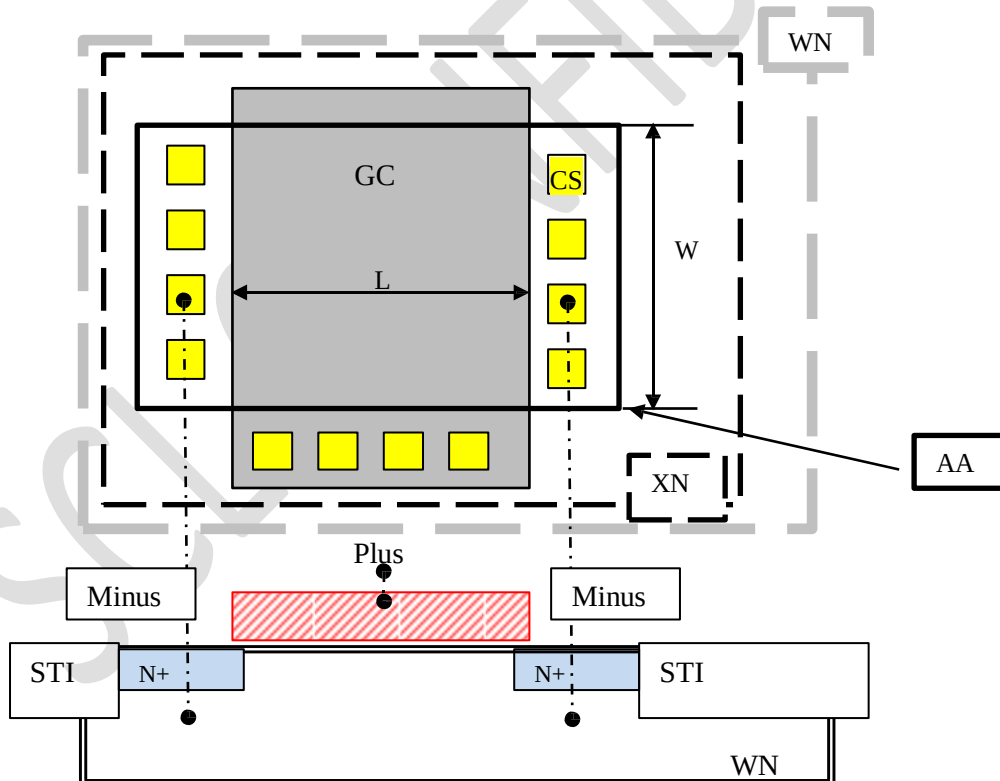


Figure 8-41: MOSCAP (Accumulation Capacitors) Rules

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

8.2.13 MIM (Metal- Insulator- Metal) Capacitors

Metal-Insulator-Metal (MIM) capacitor consists of a metal bottom plate, Dielectric, and Thin conductive top plate. The bottom plate is always the top- most MI (intermediate metal, I=2-5) of the metal combination. The upper electrode is connected to an upper metal layer by the related via.

The different types of MI available at the platform, together with the relevant recognition marking layers are listed at the table below

Table 8-2: Marking layers used to define the different MIM Capacitors, located under TOP_M (ML, MT, MT3) and above MI.

Device name	Description	Design layers	Marking layer	MIM Type layers	Masks
cmim_sq	Single layer, 1fF/um2	CE	77dt56	N/A	CE
cmim_hc	Single layer, High C 1.7fF/um2	CE	77dt56	80dt56	CE
cmim_stk	Stacked layer, High C 3.4fF/um2	CE	77dt56	80dt56; 78dt56	BCE, CE
cmim_uhd	Single layer, UHD 2.8fF/um2	CE	77dt56	78dt43	CE
cmim_stkuhd	Stacked layer, UHD C 5.6fF/um2 (2terminals)	CE	77dt56	78dt43; 78dt56	BCE, CE

Note : UHD is allowed currently in 5M* combination , for other metal combination please ask SCL .

The stacked MIM is comprised of two MIM capacitors, thus yielding twice the capacitance of a single layer for the same layout area.

M(bottom plate): The metal layer that serves as the MIM bottom plate.

M(top plate): The metal layer which directly connected to the MIM top electrode.

M(bottom plate-1) One metal level below the bottom plate.

Top plates of all MIM capacitors are drawn by layer CE. Different MIM marking layers define the location of the MIM capacitor in the stack, and its dielectric type. (see Table 8-2)

The terminal connections (Table 8-2) are located 1-metal level below the MIM. For example, with M3 for 5M1L where the MIM is located on M4. Note that the PDK have an option, to remove this connections, if needed. So for the same example, designer will connect the terminal, manually with M4.

NOTE1	MIM availability depends on the metallization combination.
NOTE2	Combining high C and standard MIM capacitors on the same gds is not allowed. It is checked by DRC rule MIM.N.3 .

8.2.13.1 Methodology for denoting MIM Capacitors in the Design Kit

MIM capacitor top plate is drawn with CE drawing layer. The type of MIM (stacked or single plate), dielectric type (standard or high capacitance), and location in the stack are all manipulated by marking layers. The following table summarizes the methodology used to distinguish between the different types of MIM capacitors.

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Table 8-3: Marking layers Used to Denote Different MIM Capacitors

8.2.13.2 MIM BEOL Dielectric and Conductors (1fF/um2)				
Conductor or Dielectric Layer	Comments	Thickness [nm]	SEM Th	Epsilon /Sheet Rcss
CE (TiN)	Capacitor Electrode of MIM (Upper)	200	N/A	6.5 Ohm/sq
MIM Dielectric (ONO)	MIM Dielectric for 1fF/um2	ONO=11/30/11	N/A	EOT=520A, Ep=5.72
MI	Capacitor Electrode of MIM (Bottom)	540	N/A	80 mOhm/sq

Note: The EOT and equivalent Epsilon, were calculated for modeling purposes. It was calculated as next:
 $EOT = \text{sum of thicknesses}$, $E_p = (\text{sum of thicknesses}) / (\text{Sum of } Th/E_p)$

8.2.13.3 1fF/um² Single plate MIM under TOP_M/ MT/ MT3

By default, MIM capacitors in the design kit, are located one metal level below the thick metal layer of the process. For example, in 4M1L process, the bottom plate of a single MIM capacitor is M3. The upper electrode is connected to TOP_M by TOP_V. Connection to bottom plate is done through the metal below the bottom plate (M(top-2)). demonstrates a cross section of such MIM.

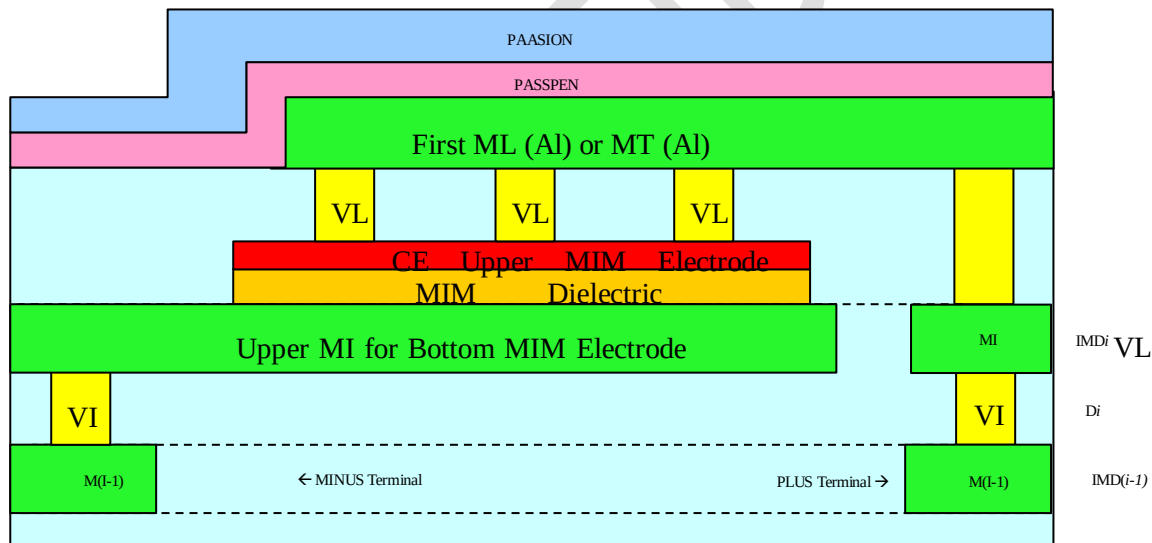


Figure 8-42: Typical Single Plate MIM capacitor, located above most upper MI. Single plate 1.7fF/um² (High Capacitance, cmim_hc) or 2.8fF/um² (Ultra High Density, cmim_uhd) MIM capacitor with a single plate High Capacitance or UHD MIMs with a single plate has the similar layout and cross section as standard MIM capacitor. The only difference is the different dielectric material with higher permittivity.

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.13.4 acked $3.4\text{fF}/\mu\text{m}^2$ (cmim_stk) or $5.6\text{fF}/\mu\text{m}^2$ (cmim_uhdstk) MIM capacitor (2terminals)

Stacked MIM capacitor compose two single plate capacitors, one on top of the other, connected in parallel. See Figure 8-43, Figure 8-44: High C, Stacked MIM Capacitor Cross Section. Note That BCE Mask is Generated

NOTE3	Note that the two capacitors are NOT connected by layout. It is the designer responsibility to connect the MIMs.
NOTE4	In case of having a stacked MIM of $3.4\text{fF}/\mu\text{m}^2$, it is allowed also having MIM of $1.7\text{fF}/\mu\text{m}^2$ at the same chip. In case of having a stacked MIM of $5.6\text{fF}/\mu\text{m}^2$, it is allowed also having MIM of $2.8\text{fF}/\mu\text{m}^2$ at the same chip.

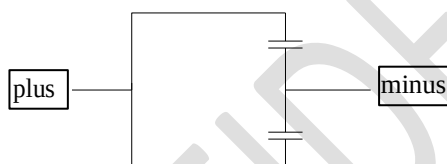


Figure 8-43: High C, Stacked MIM Capacitor Electrical Connection

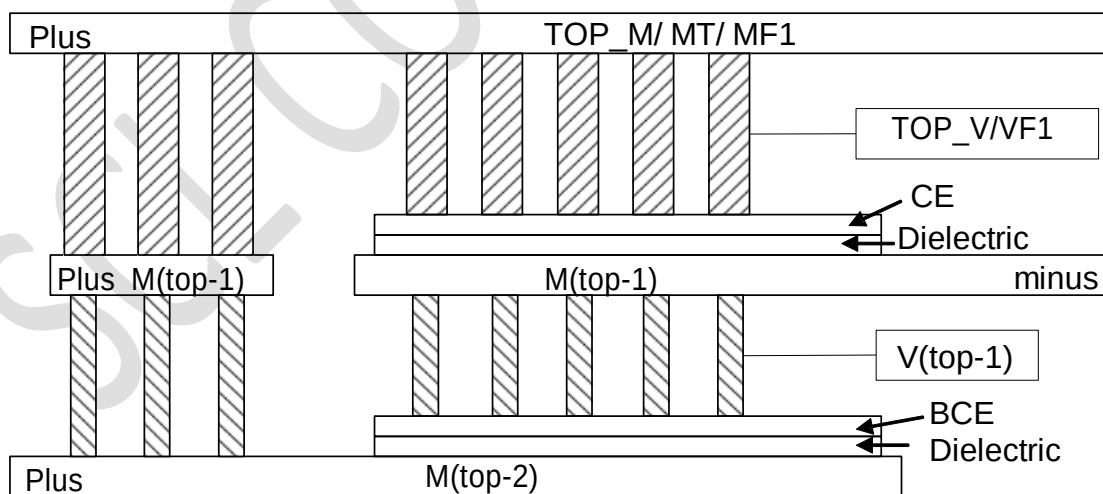


Figure 8-44: High C, Stacked MIM Capacitor Cross Section. Note That BCE Mask is Generated



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

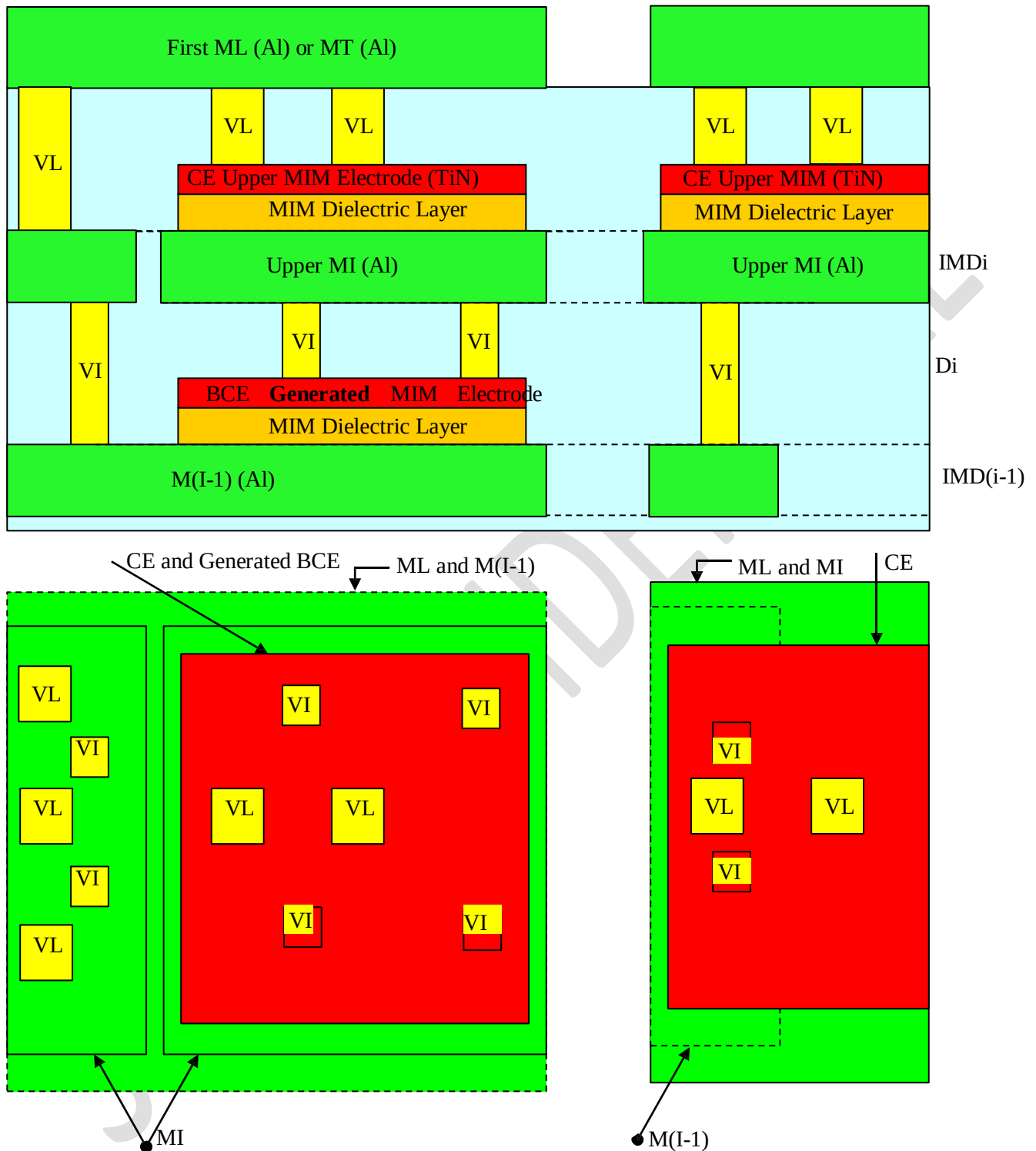


Figure 8-45: Demonstrating High C Single Plates MIMs and Stacked High C MIM on the Same Chip in 4M1L Process



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.13.5 MIM BEOL Dielectric and Conductors (1.7fF/um2)

Conductor or Dielectric Layer	Comments	Thickness [nm]	SEM Th	Epsilon /Sheet R _{ss}
CE (TiN)	Capacitor Electrode of MIM (Upper)	200	N/A	6.5 Ohm/sq
MIM Dielectric (Nitride)	MIM Dielectric for 1.7fF/um2	Nitride=380	N/A	EOT=380Å, E _p =7.8
MI	Capacitor Electrode of MIM (Bottom)	540	N/A	80 mOhm/sq

8.2.13.6 MIM (Metal- Insulator- Metal) Design Rules

Terminology:

Real CE: CE interact with V(top plate).

Non real CE: CE not interact with V(top plate).

CE: real CE or non real CE.

Real Bottom plate: (M(bottom plate) and real CE) Sized by 0.4 per side. “and” is the Boolean formula of and.

Non real Bottom plate: (M(bottom plate) and non real CE) Sized by 0.4 per side.

Bottom Plate: real Bottom plate or non real Bottom plate.

MIM marking layers: M2(mim), DEV_AREA(nm), AREA10(mim), SPC_AREA(mim).

“Recommended” Switch: The next rules: MiM.N.8, MiM.N.9, MiM.N.10 will be checked only when “Recommended” switch is turned on.

8.2.13.7 MIM (Standard, High C single plate, High- C Stacked MIM Capacitor)

Rules		Action	Drawn
MIM.W.1	Width of CE (for photolithography, not a part of a capacitor)	min	0.4
MIM.W.2	Width of CE in MIM capacitor	max	30
MIM.W.3	Width of CE in MIM capacitor	min	4
MIM.W.6	Width of bottom plate metal, that is part of MIM capacitor	max	35
MIM.S.1	Space between two CE where at least one is not a part of MIM capacitor	min	0.8
MIM.S.2	Space between two CE (both CE are MIM capacitors)	min	1.2
MIM.S.3	For single MIM space between two bottom plate metals for single MIM , for MIM array the bottom plate`s may interact (one plate for all array) This rule is not checked in stacked MIM	min	0.8
MIM.S.5	Space between two V(top plate) on CE	min	0.5
MIM.E.1	Enclosure of CE around V(top plate)	min	0.24
MIM.E.2	For stacked MIM: Enclosure of CE around V(top-1) in stacked MIM	min	0.24
MIM.E.3	Enclosure bottom plate metal around CE	min	0.4
MIM.E.4	For stacked MIM: Enclosure of M(top-2) around CE in stacked MIM	min	0.4



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.13.7 MIM (Standard, High C single plate, High- C Stacked MIM Capacitor)		Action	Drawn
Rules			
MIM.D.1	Distance between CE to unrelated MIM marking layer	min	0.4
MIM.D.2	Distance from V(top plate) to CE	min	0.4
MIM.D.3	Distance from V(bottom plate) to CE	min	0.4
MIM.C.1	Coverage of CE area on overall chip area. This rule not applicable for stitching technology	min	0.03
MIM.C.2	For stacked MIM: Coverage of BCE area on overall chip area. Note that BCE is generated from CE and SPC_AREA(mm) in stacked MIM capacitors.	min	0.03
MIM.C.3	Coverage of V(top plate) on CE of MIM capacitor	min	0.01
MIM.C.4	For stacked MIM: Coverage of V(top -1) on CE of stacked MIM capacitor	min	0.006
MIM.A.1	Area of single MIM array. MIM array, having length>250um AND width>250um is NOT allowed. For DRC purposes, 77dt56 having Width>250um AND Length>250um . For several arrays having area close to 62,500um ² , keep a space of >10um (Note-2)	max	62500 um ²
MIM.N.1	CE covered by a different combination of MIM marking layers than the ones listed in Table 8-2 section, is not allowed.		
MIM.N.2	CE interact MIM marking layer should be fully covered, and enclosed by min 0.4um by that layer.		
MIM.N.3	Design having “mix” of different MIM dielectrics is NOT allowed. For example, design having MIM of 1fF/um ² and 1.7fF/um ² or 1fF/um ² and 2.8fF/um ² or 1.7fF/um ² and 2.8fF/um ² , capacitors is NOT allowed.		
MIM.N.5	CE without bottom plate metal is not allowed		
MIM.N.6	For stacked MIM: CE without M(top-2) in stacked MIM is not allowed		
MIM.N.7	V(bottom plate) interact CE is not allowed, except for stacked MIM		
MIM.RN.8	Recommendation: Bottom plate interact GC, GC(dy), or any metal dummy is not allowed. In precision analog or RF design switch can turn on this rule.		
MIM.RN.9	Recommendation: Bottom plate interact nwell resistor or AA resistor is not allowed. In precision analog or RF design switch can turn on this rule.		
MIM.RN.10	Recommendation : CE interact metal layer which is not the MIM bottom plate, or the metal layer connecting to CE is not allowed. For stacked MIM, also allow interaction with M(top-2). In precision analog or RF design switch can turn on this rule.		
MIM.N.11	For stacked MIM: SPC_AREA(mm) not interact with M(top) and M(top-2) is not allowed.		
MIM.N.12	For stacked MIM: At least one polygon of {[M(top-1) not interacting CE] interacting VL) interacting VL -1} should exist in SPC_AREA(mm).		
MIM.N.13	For stacked MIM: [(CE inside SPC_AREA(mm) interact VL) not interact VL-1] is not allowed.		
MIM.N.14	For stacked MIM: [(CE inside SPC_AREA(mm) interact VL-1) not interact VL] is not allowed.		



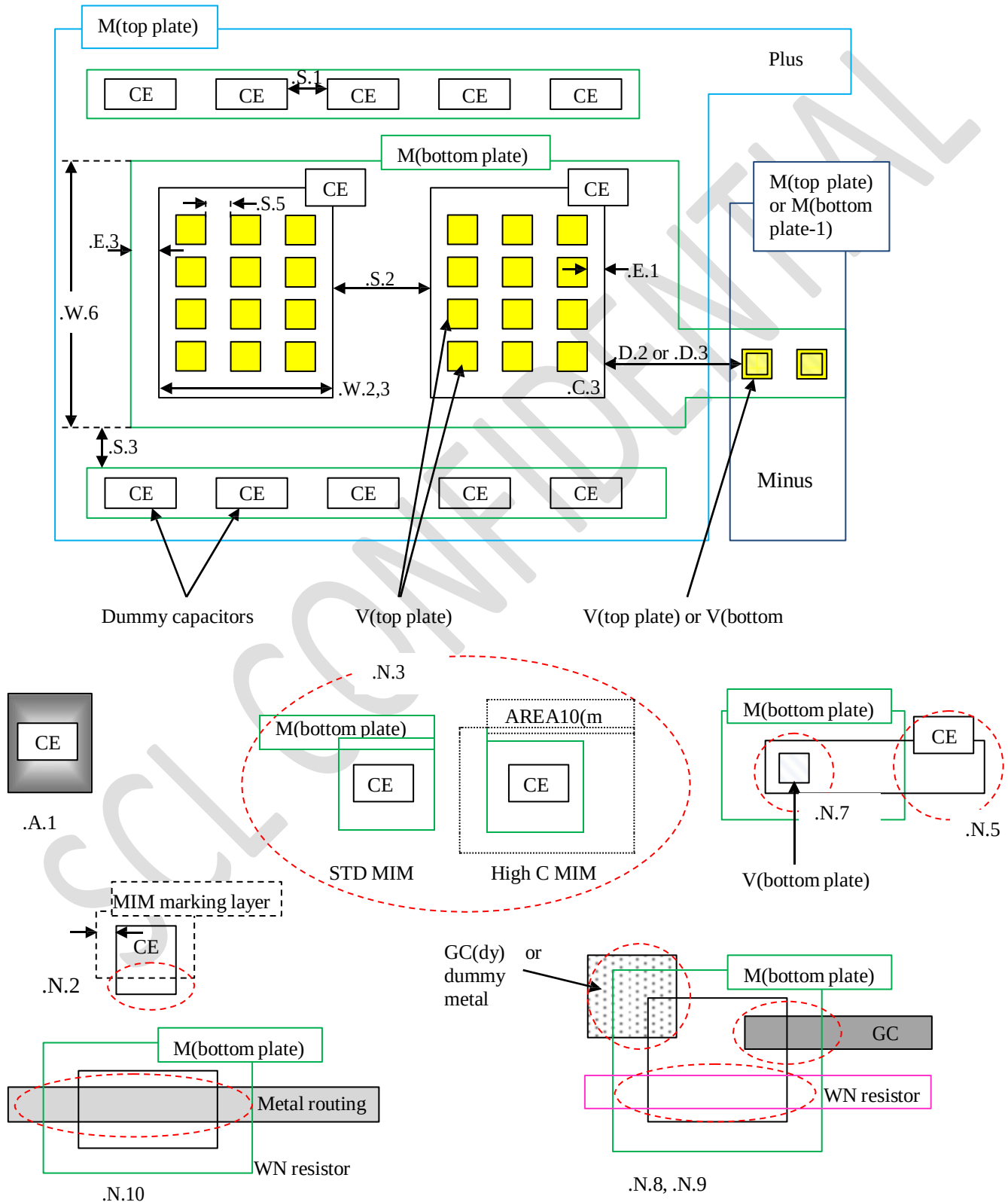
Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Note-1: we recommend not to put devices under the MIM capacitor. This is because the electrical test structures used to develop the MIM modeling did not have such devices under the MIM. However, we can predict that the effect on the MIM performances is only limited.

Note-2: Avoid having on large array of MIM that can introduce some etch bias. Instead, replace with several arrays having maximum area as in MIM.A.1 although there is no "specific rule" demanding a minimum space, for the case of using large array area (close to 250X250um), designer is request to have at least 10um space in between the arrays.



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Figure 8-46: MIM (Standard, High C single plate, High- C Stacked MIM Capacitor) Rules (1/2)

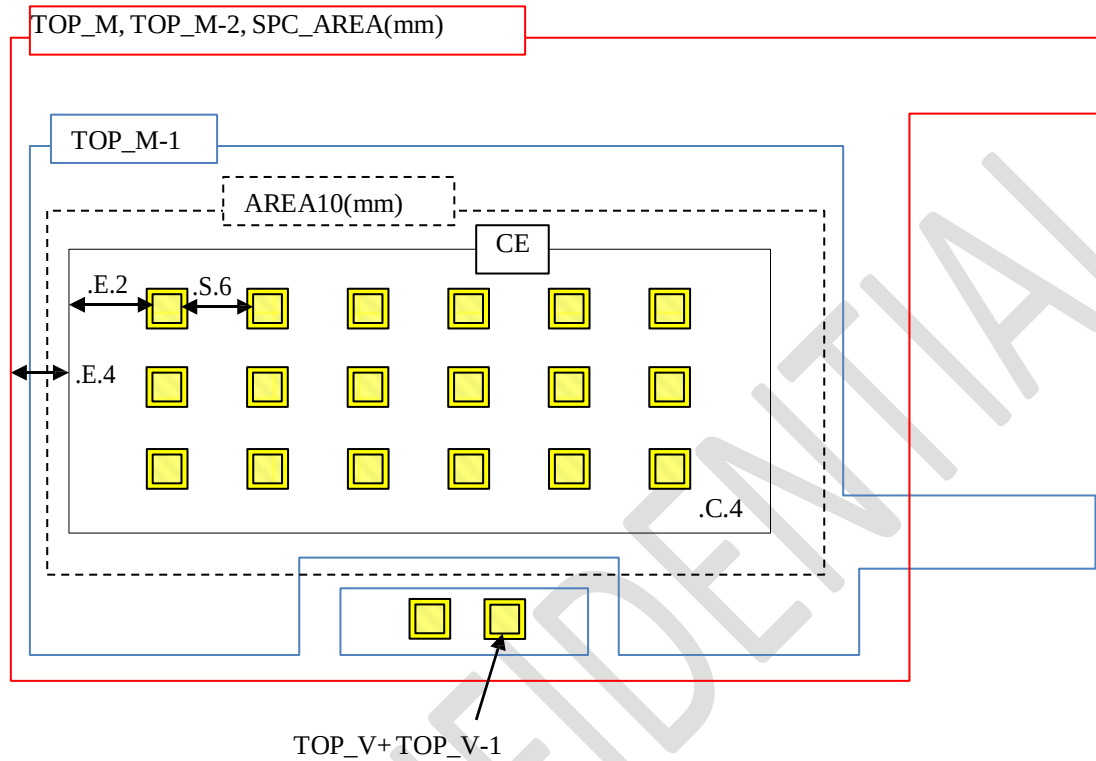


Figure 8-47: MIM (Standard, High C single plate, High- C Stacked MIM Capacitor) Rules (2/2)

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.13.8 xamples for Standard Capacitance MIM Capacitors

8.2.13.8.1 MIM Capacitor Used for Standard Analog Design

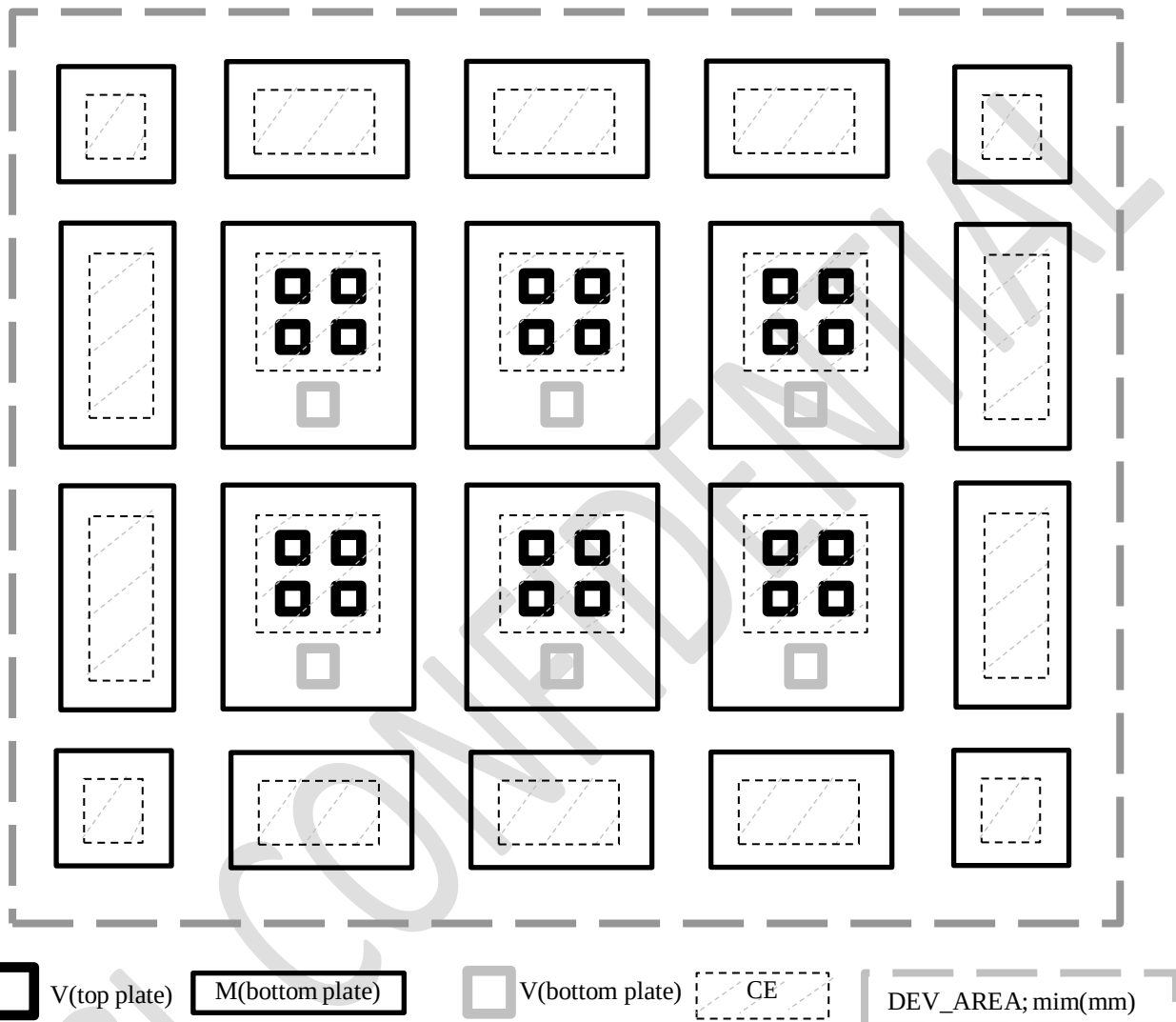


Figure 8-48: Isolated Bottom Plate, Matched MIM Capacitor.

Note the none real CE (floating CE) around the capacitors.

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.13.8.3 Isolated Bottom Plate, Matched MIM Capacitor (Maximum 2x2 Capacitors)

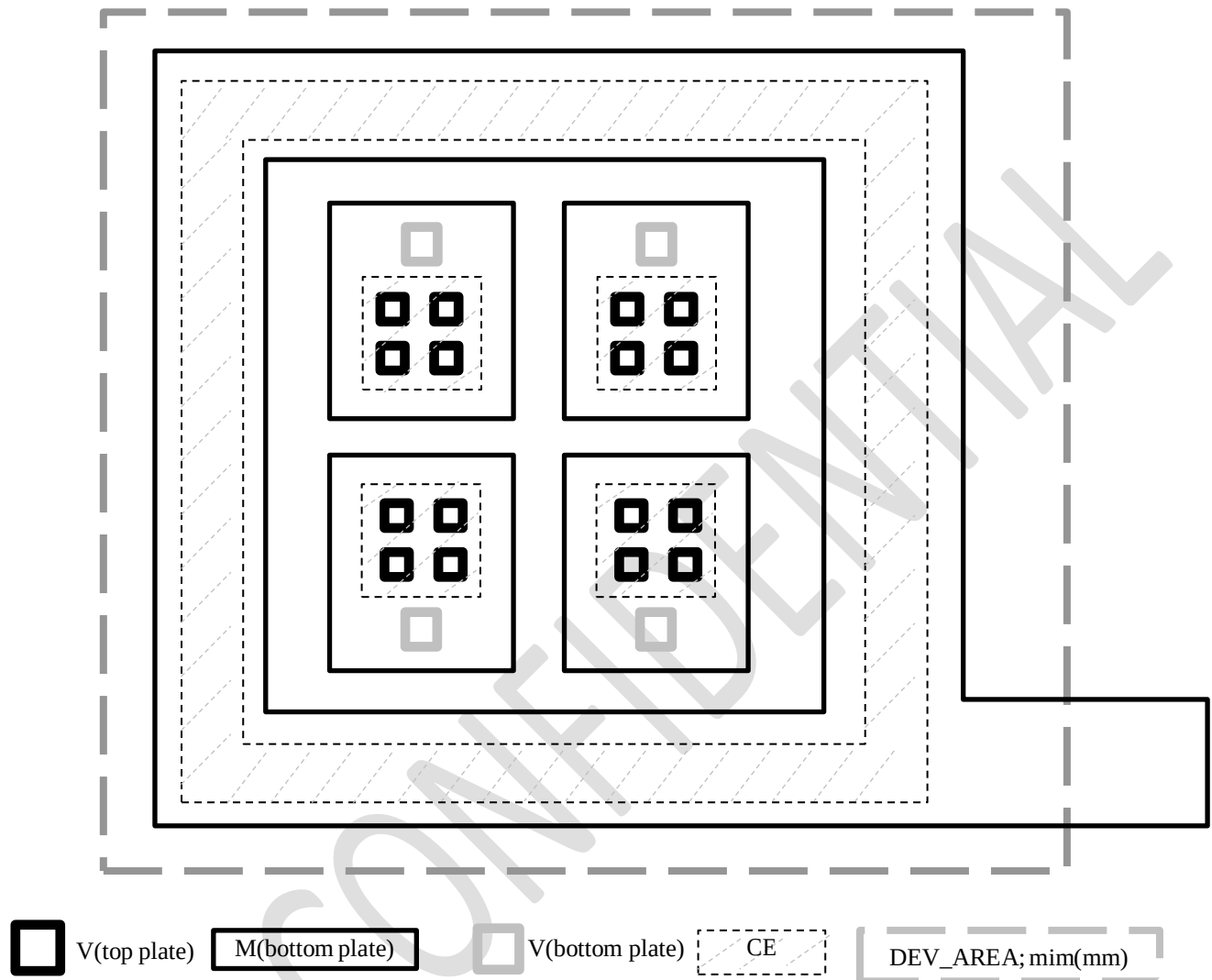


Figure 8-50: Isolated Bottom Plate, Matched MIM Capacitor (Maximum 2x2 Capacitors).

This structure of MIM is not supported by PDK.

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.14 Metal Fringe Capacitor (MFC) Layout Design Rules

The capacitors are made of M1 and MI without last metal any type (see rule MFC.N.9). For example, for 6M1L design, M1/2/3/4/5 will be used for MFC, and ML will not be used. Please refer to Table 7-6. for MFC offering per BEOL metal combination.

The capacitors are made of stacked “T-Shape” metal fingers Figure 8-52. The basic “Unit Cell” is consisting from 4 fingers pairs .The metal fingers width and space dimensions are fixed (exact). The length of the fingers and the number of fingers can be adjusted for achieving the desired capacitance. That is $C=C(l, Nf)$. Details regarding the exact calculation of the capacitance can be found at the related SPICE document. For LVS and capacitance purposes, the capacitor area is calculated using the related 77dt73 support layer.

Substrate terminal need to be considered manually by the physical designer. Depending on the application, the fringe capacitor may be drawn over PWELL (default from the P-cell library) or over an NWELL tied to a quiet Vdd. No additional masks or processing are needed.

The metal fringe capacitor needs to be drawn over STI (field oxide). Note, that SCL dummy fill run-set will introduce some AA dummy’s for process planarization purposes. For very sensitive devices, the dummy AA can be blocked using the related support layer. However, the AA coverage rules (AA.C.*) still need to be satisfied. The affect of this AA dummy’s is <2% change at the parasitic capacitance to substrate.

To reduce the parasitic capacitance of the fringe capacitor, reduce the amount of GC dummy’s under MFC. Note, that SCL dummy fill run-set will NOT introduce some GC dummy’s. For very sensitive devices, the dummy GC can be blocked using the related support layer. However, the GC coverage rules (GC.C.*) still need to be satisfied.

To avoid un-expected parasitic capacitance, it is not allowed having metal wires crossing under or above MFC or dummy’s located under/above the MFC (see rule DM.N.5). Note, that SCL dummy fill insertion run- set will NOT introduce any metal dummy’s. However, the metal coverage rules for all metal layers (M*.C.1/2), should be satisfied. If needed, replace a large area MFC capacitor, with two separated MFC capacitors, and include relevant metal dummy’s in between.

8.2.14.1 MFC (Metal Fringe Capacitor) Rules		Action	Drawn
MFC.W.1	M1 capacitor finger width	exact	0.31
MFC.W.2	MI (I=2, 3, 4, 5) capacitor finger width	exact	0.28
MFC.S.1	M1 capacitor finger space.	exact	0.27
MFC.S.2	MI (I=2, 3, 4, 5) capacitor finger space.	exact	0.3
MFC.S.3	M1 space between finger end and opposite M1 spine.	exact	0.37
MFC.S.4	For MI (I=2,3,4,5): Space between finger end, and opposite spine (of the same level).	exact	0.4
MFC.S.5	Space between capacitor fingers (at any metal level) to un-related metal (at the same level). This is also includes the distance from capacitor fingers (at any metal level) to TOP_M around it.	min	1
MFC.E.1	Enclosure of WN around related DEV_AREA(mc).	min	1
MFC.D.1	Distance from DEV_AREA(mc) to unrelated WN.	min	1
MFC.L.1	Metal finger (M1 or MI) length.	min	1.1



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.14.1 MFC (Metal Fringe Capacitor) Rules		Action	Drawn
MFC.N.1	Capacitors fingers MUST be stacked (alternating terminals). This rule is NOT aplicable for single metal MFC		
MFC.N.2	Number of capacitor fingers pairs (nf), at the same capacitor. For DRC purposes, check under the same DEV_AREA(mc) support layer.	min	4
MFC.N.3	DEV_AREA(mc) MUST coinside the metal edges of the MFC. MFC without related DEV_AREA under is NOT allowed	Not allowed	
MFC.N.4	M(x+1) above MC1x (x=3,4,5,6) is not allowed, this rule not applicable for single metal devices		
MFC.N.6	45 degrees metal under DEV_AREA(mc) is not allowed		
MFC.N.7	Some metal combination, are not allowed to have MFC capacitor. For DRC purposes check that mfc(mc) layer is NOT included at design data, per metal combination		
MFC.N.8	Design having “mix” of different MFC dielectrics (USG and FSG) is NOT allowed.		
MFC.N.9	MFC Device should NOT include any type of the most upper metal layer (ML, MT, MT3). For DRC puroposes, check for not existence of any most upper metal over 77dt73.		

Figure 8-51: MFC (Metal Fringe Capacitor) Rules

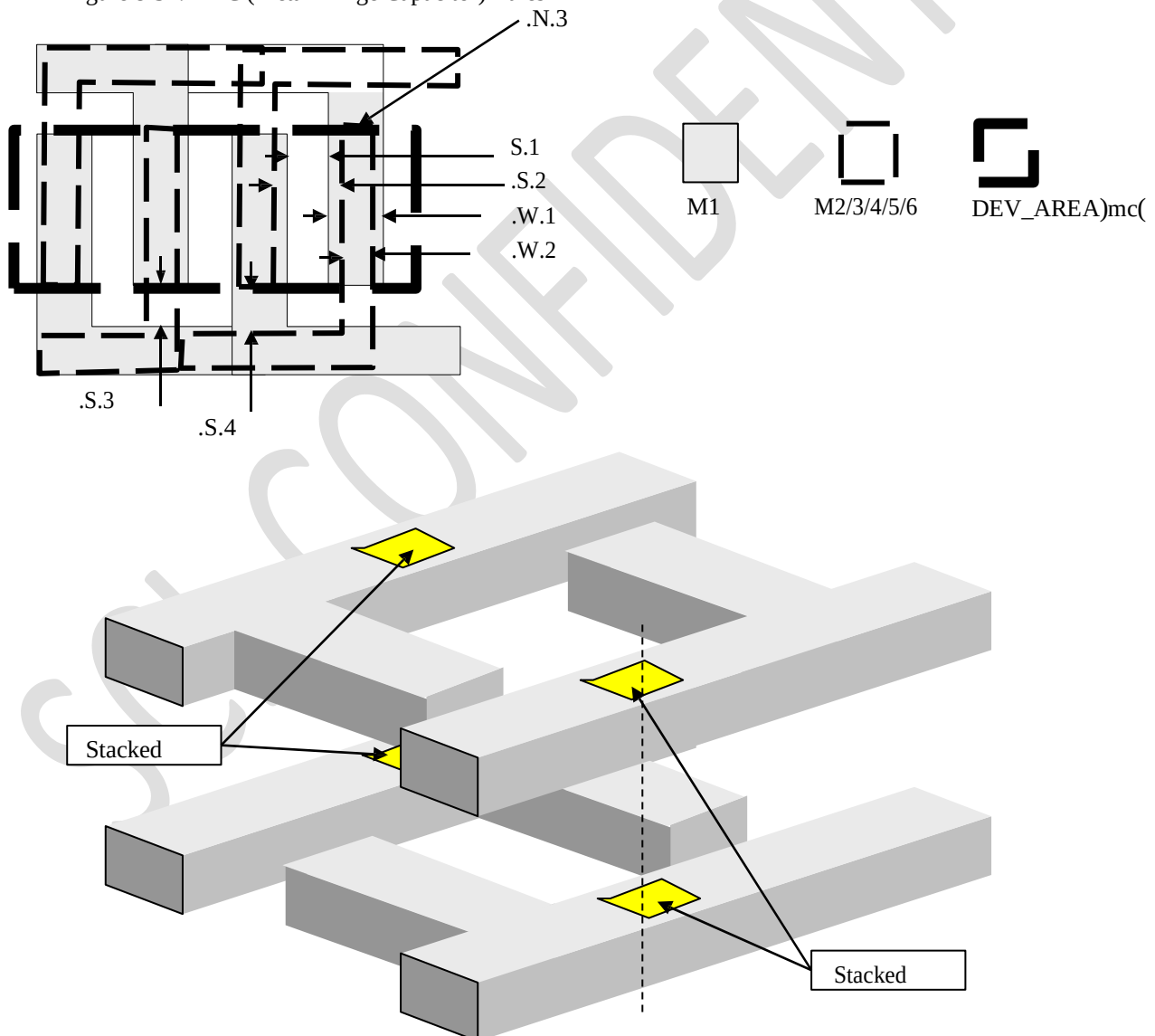


Figure 8-52: Metal Fringe Capacitor (MFC) Schematic Drawing

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

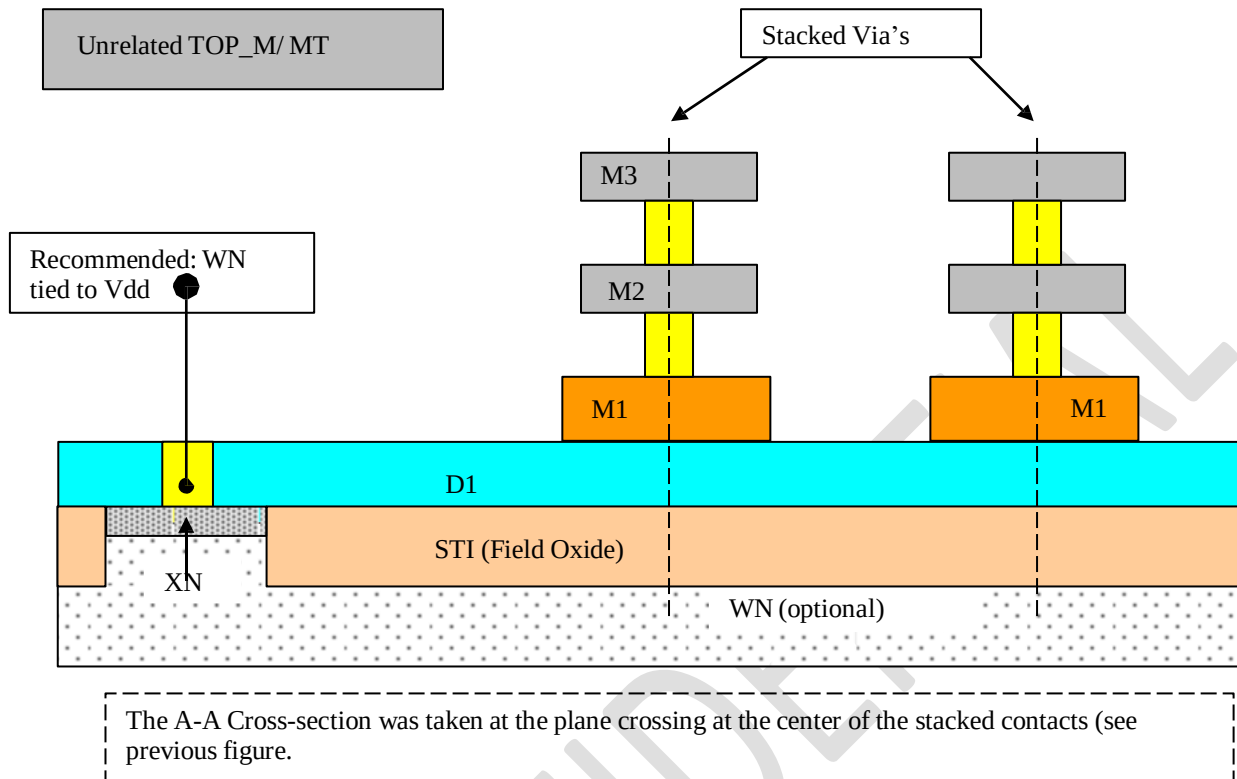


Figure 8-53: Metal Fringe Capacitor (MFC) Schematic Cross-section (example for MFC in 4M1L and 4M1T process)

8.2.15 Design Rules for Electrical Poly Fuse (EPF)

8.2.15.1 Introduction

The ePolyFuse described here, is having a fix dimensions and using the standard CMOS process. The fuse is un-doped poly made over STI, with CoSi₂ on top. No special masks are needed for implementation. We are providing several types of ePF (see Table 8-4). ePF Type-1, used very extensively in production for several years and he is the recommended and the default element to be use. Note, that we provide BurnIC (“wrapper”) for some fo the fuses. Please contact SCL IP team for more details.

Table 8-4: ePolyFuse – types and marking layers used to define the different

Device name	Description	Marking layer	ePF Type layers	Element L/W	Burn with IOs of:
ePF	ePF type-1 (defult fuse to use)	78dt105	N/A	2um/0.2um	3.3V ~ 5V

The ePolyFuse element has a fix length and width (see Table 8-4). The different elements, have different initial resistance (Rinit) and slightly different burned resistance (Rprog).



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Our type-1 characterization showed, for large database of ~1,500 fuses, that the initial resistance (R_{init}) of the fuse is 72 Ohm. After burn, 100% of the fuses showed resistance (R_{prog}) increase by several order of magnitude. For sensing circuit design, there is a need to take into consideration the ratio of R_{prog}/R_{int} that is $>1E+2$. See also section 8.2.15.5.

8.2.15.2 yFuse Layout Rules and Recommendations

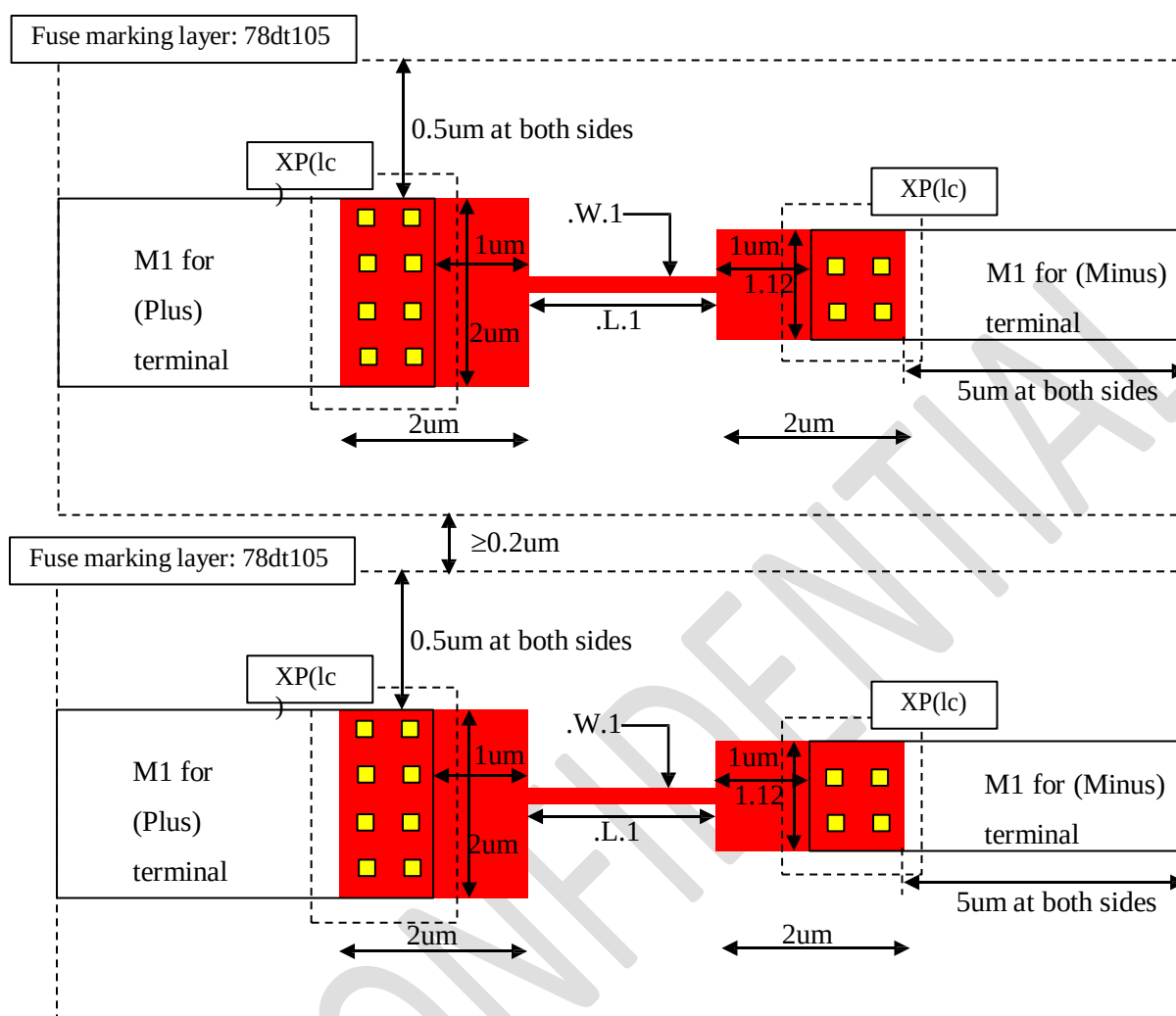
The fuses have a fix dimensions, and basically, design is not allowed to adjust any layer inside the marking layer 78dt105. The following list of rules (8.2.15.3) and the related figure (Figure 8-54) includes the exact dimensions of the fuse.

Important Notes:

1. SCL provide a ready-to-use fuse P-Cell. It is NOT allowed to make any adjustment to the fuse layout, as defined by the ePolyFuse marking layer.
2. Only ONE single fuse is allowed inside each fuse marking layer.
3. The marking layers should be drawn as a simple rectangular having dimension of $3\mu m \times 16\mu m$ with exact enclosure of $0.5\mu m$ (perpendicular to the fuse element) and $3\mu m$ around the poly electrodes (parallel to the fuse element). Minimum distance of $0.2\mu m$ is needed between two fuse marking layers.

8.2.15.3 EPF (ePolyFuse) Rules			Action	Drawn
EPF.W.1	For Type-1: Width of fuse This rule is NOT applicable under (77;108) DEV-AREA blow_up(bp)		exact	0.20
EPF.L.1	For Type-1: Length of fuse This rule is NOT applicable under (77;108) DEV-AREA blow_up(bp)		exact	2
EPF.D.1	Distance between two fuses		min	3
EPF.D.2	M1 is not allowed over the ePolyFuse element. For DRC purposes, check distance of M1(lc) to the ePolyFuse size $0.5\mu m$ /side. Overlap is NOT allowed.			
EPF.N.1	The fuse MUST be consist from Salicided undoped Poly. For DRC purposes, 78dt105 that is not interact with {Poly(lc) NOT NS(lc) NOT AA(lc) NOT DEV_AREA(rs) NOT DEV_AREA(cp)} is NOT allowed			
EPF.N.3	Only one single fuse is allowed inside each fuse marking layer			
EPF.N.2	It is NOT allowed for having the following layers over/under the fuse: • AA(lc), WN(lc), WB(lc), AREA2(lc), XN(lc), NS(lc), • All metals $\leq M2$ (except M1 that used for terminal connections),			

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0



8.2.15.3 EPF (ePolyFuse) Rules		Action	Drawn
	• PD(lc), • Dummy AA, dummy Poly and all $\leq M3$ metal dummies in process (including M1 dummy) For DRC purposes, check that all layers listed above, are NOT interact with 78dt105.		

Figure 8-54: EPF (ePolyFuse) Type-1 Rules

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

8.2.15.4 eFuse Burning Conditions and Guidelines

The platform characterization document, include a detailed description of the experiments and there conditions that were done during the physical qualification of the device. Large set of fuses were analyzed. Note, that this work was done at the device laboratory using pulse generator, and therefore, should be used as complimentary data. The exact conditions and characterizations of each BurnIC, us listed at the IP data sheet and provided by SCL IP team

Tha following data is physical base, for reference purposes, only.

8.2.15.5 EPF (ePolyFuse) burning conditions and Resistance before and after program						
Electrical Performances of fresh and programmed (burned) fuse (at formal Vprog conditions)						
Fuse	Parameter	Unit	Typical	LSL	USL	Evaluation
Type-1	Rinit resistance	Ohm	72	66.6	77.4	1Sigma=1.79 (2.5%). Measured at V=0.1V
	Rprog (burned) resistance	KOhm		7		Burn conditions: See CZ file

8.2.16 Silicon Based Vertical Bipolar Junction (VPNP and VNPN) Transistor Rules

This section refer to silicon-based BJTs, consist from the next types:

- Vertical PNP (P+/WN/Psub): no additional mask is needed. These devices have a fixed size (10x10, 5x5 or 2x2um).
- Vertical NPN (N+/IPW/Deep-WN): one additional mask (WB=Deep WN) is needed. SPICE modeling was done using set of devices with fixed size (10x10, 5x5 or 2x2um).

8.2.16.1 BJT (Vertical Bipolar Transistor) Rules		Action	Drawn
BJT.W.1	Fix size Silicon-based BJTs (both VPNP and VNPN) can emitter area of 10x10, 5x5 or 2x2um. For DRC, check size by min/max rules: 10+/-0.5, 5+/-0.25, or 2+/-0.1um	exact	10x10 5x5 2x2
BJT.N.2	VNPN Emitter must be over isolated P-well (WB not covered by WN)		
BJT.N.3	Each polygon of DEV_AREA(bt) should interact with one polygon of ACTIVE (the BJT emitter) and not touch any other ACTIVE polygon		
BJT.N.4	Scalable BJTs need to be covered by both BJT marking layer and capacitor marking layer, 77dt43. The two layers should be drawn coincidence.		



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

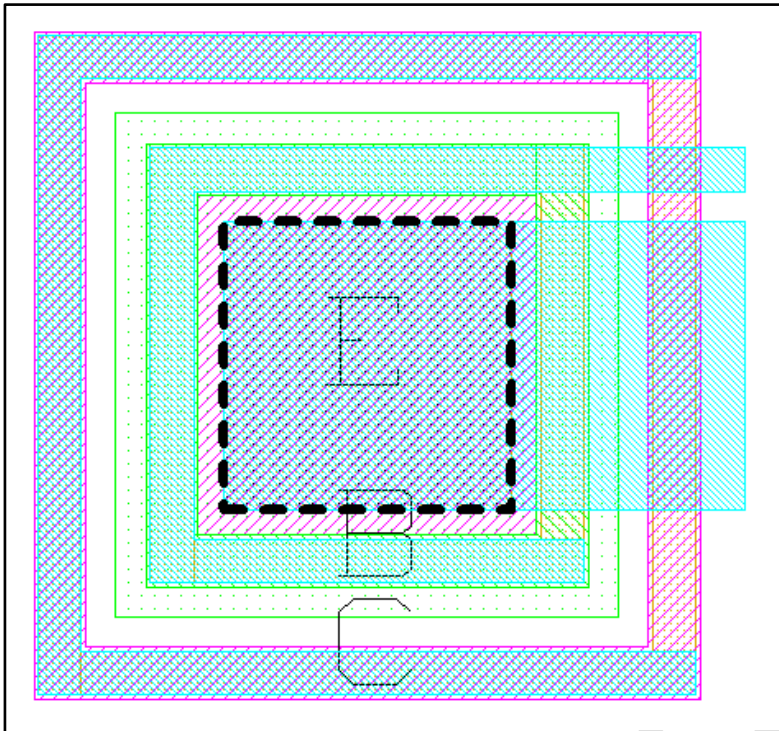


Figure 8-55: VD10/5/2 (VPNP 1.8V 10x10 or 5x5 or 2x2 BJT)

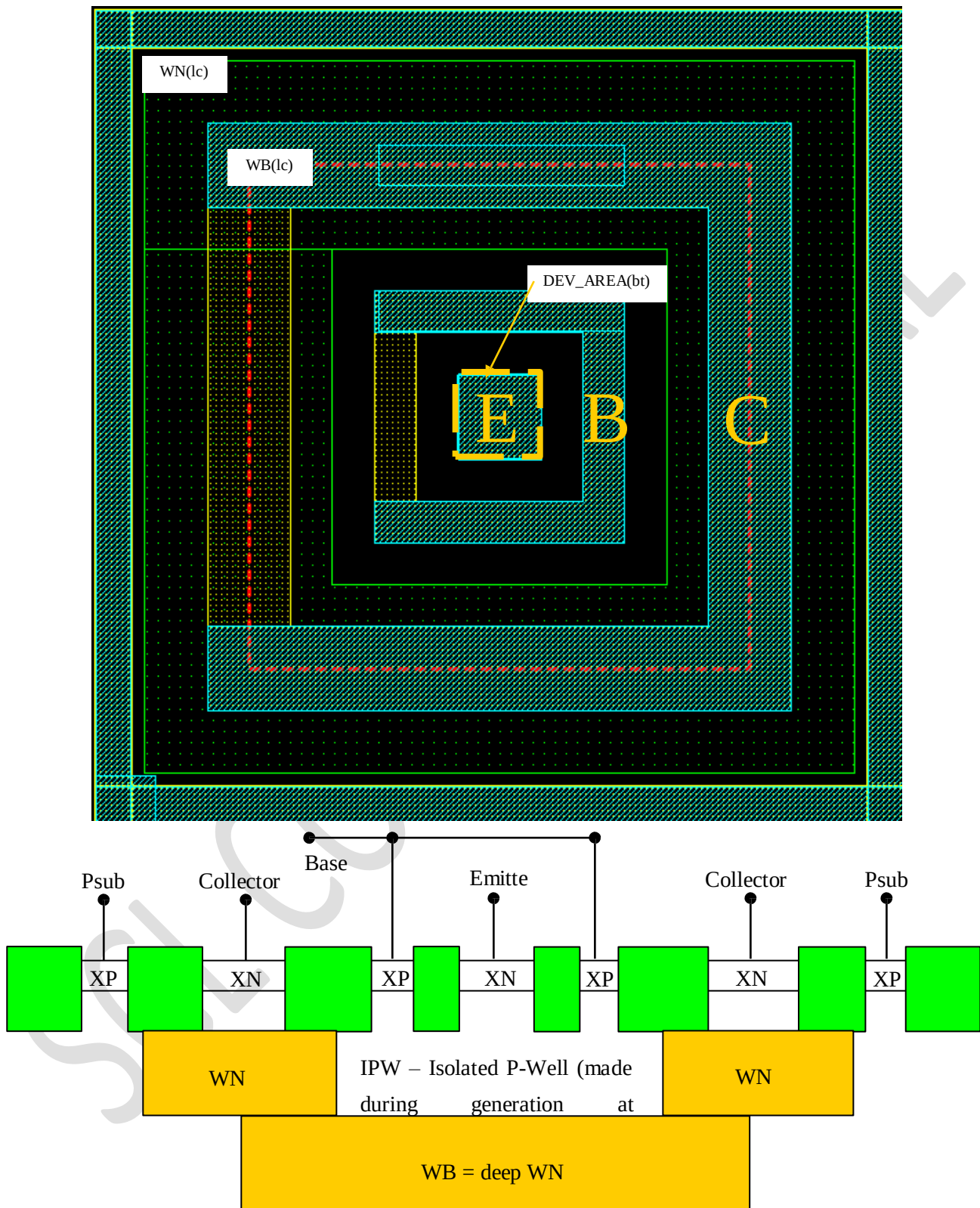


Figure 8-56: VNHV (Vertical NPN 1.8V, 3.3V, 5V 10x10 or 5x5 or 2x2 BJT). SPICE, LVS Parameters: m

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

9 Design Guidelines

9.1 Antenna Rules

To prevent Plasma electrostatic discharge damage to gate oxide during manufacturing process, Follow the antenna rules in this section.

9.1.1 Definitions

Antenna-poly is defined as a poly polygon placed on ACTIVE and on STI area.

Antenna-metal is defined as a single-layer metal that is connected by lower level of metal or poly to some Poly gate.

Antenna-contact/Via is defined as a Contact/Via that is connected to some Poly gate by lower levels of metals or poly.

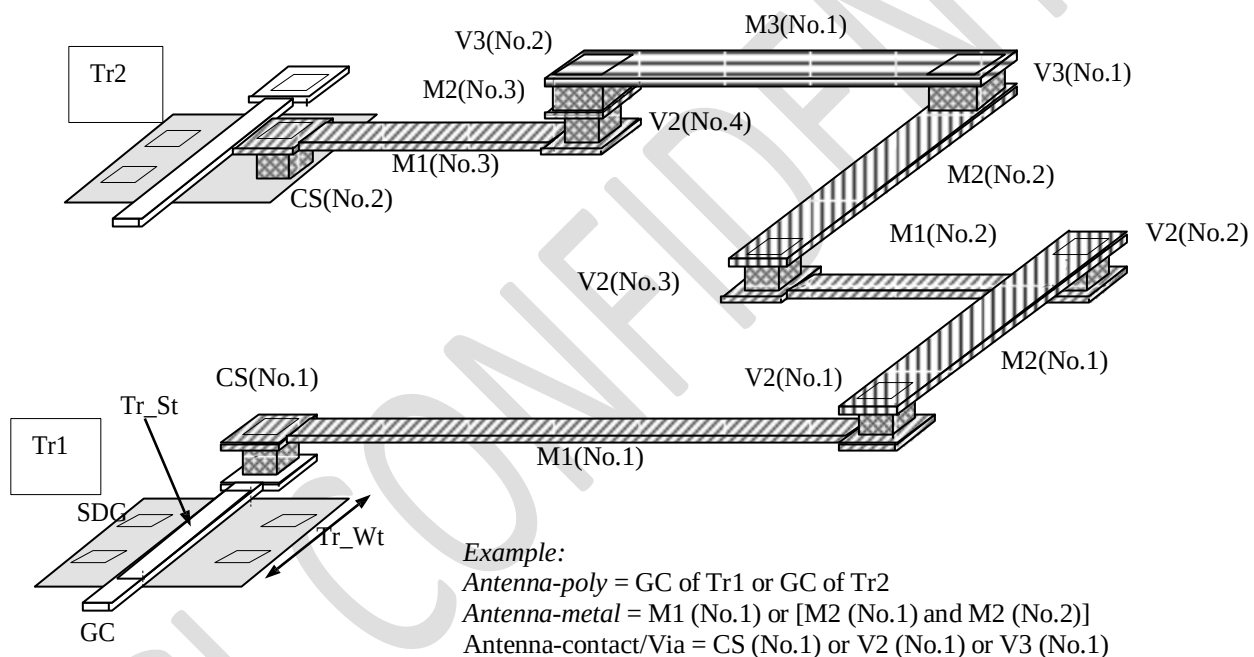


Figure 9-1: Antenna Rules Definitions

9.1.2 Antenna Rules

Any direct connection of an antenna metal to active, by means of lower level metals and Poly, is defined as Antenna Protection Diode. Protection “Diode” can be P+ ACTIVE in N-well or N+ ACTIVE in P-well. Psub or Nsub are also referred to as antenna protection diodes (although not diodes). The area of the “Diode” is defined by the area of the diode’s AA.



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

9.1.2.1 AN (Antenna Ratio) Rules		Action	Ratio
AN.GC.1	Ratio of GC on STI Sidewall exposed area to GC over GOX area. GC on STI Sidewall exposed area = GC thickness x GC perimeter on STI. Physical GC thickness is 0.2um. [GC on STI Sidewall exposed area = 2 (Poly length + Poly width) x 0.2, um ²].	max	200:1
AN.MTL.1	Ratio of a single layer metal perimeter edge area to GC over GOX area (without any protection diode). Metal perimeter edge area (=total Metal sidewall area)=physical Metal thickness x Metal perimeter For M1 to M5: [2*(Metal length + Metal width) x 0.54, um ²] For ML: [2* (Metal length + Metal width) x 0.84, um ²] For MT: [2* (Metal length + Metal width) x 2.0, um ²]		400:1
AN.MTL.2	Ratio of a single layer metal perimeter edge area (see definition in rule AN.MTL.1) to GC over GOX area (when a protection diode with area >0.202um ² is used) Max calculated allowed Ratio= For M1-M5: Diode area*400 + 2200 For ML, MT, MT3: Diode area*8000 + 30000		
AN.CS.1	Max Ratio of overall area of contacts to GC over GOX area		10:1
AN.VIA.1	Max Ratio of overall area of via's to GC over GOX area (without any protection diode). Refer to every via layer separately.		20:1
AN.VIA.2	Max Ratio of overall area of via's to GC over GOX area (with protection diode). Refer to every via layer separately. Ratio: Diode area* 83.33 + 75		



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

9.2 Coverage Guide Lines and Dummy Patterns Insertion / Blocking

In order to pass the coverage rules (*C.*), dummy pattern is needed. Following, are some guidelines for drawing dummy pattern and solving DRC coverage violations:

- Dummies should satisfy all layout rules
- Draw the dummies ONLY using the related dummy layer (see Table 7-1)
- It is recommended, to uniformly distribute the dummy's across the die and across the checking window. Designer should try to avoid having locations (inside the checking window) that will provide most of the coverage, keeping large area (inside the checking window) that are empty. For this platform, we did not specified a DR checking for that.

Note, that fixing the design in one layer, may affect dummies of another layer. For example, consider a case of a design with dummy GC over STI. Later on, any change in AA mask, may cause dummy GC over AA, (which is a violation)!

In order to eliminate unexpected parasitic capacitance, customer may wish to avoid covering of devices with dummies. This can be done by using the relevant dummy blocking layers (*, dt35 or datatype 35, see Table 7-2).

Minimize the use of dummy blocking layer (*, dt35). Carefully cover only desired areas. Blocking large areas for dummy insertion, may lead to low coverage violations.

When using dummy blocking layer, prefer to selectively block dummy of few layers (i.e, use of data type 35) over using NODUMMY layer

On some cases, after embedded of several designed blocks (ex: IP blocks, analog blocks), some DRC AA coverage violations may be found. This is related to the fact that the coverage is checking locally inside a checkbox. To avoid such a problem, make sure the coverage values in both the designed block, and at the core around the block, are at levels of 20~ 60%.

AA or GC dummies w/o XN or XP are allowed.

Block introduction of dummy's on the following structures:

Restriction of AA dummy's, GC dummy's, Metal Dummy's			
Device	Dummy AA	Dummy GC	Dummy Metal
WNSR	Block (WNSR.N.9)		
WNAR	Block (WNAR.N.9)		
MIM	OK	MIM.N.8	Block M(L-2) and ML(DM.D.12) Other Metal layers blocked by MIM.N.8
MFC	OK	OK	Block + 1um
Seal Ring	Block (section 10.3)	Block (section 10.3)	Block (section 10.3)
PAD (Full Metal Stack)	OK	OK	Block (section 10.1)

9.2.1 Maximum and Minimum Active Area (AA) Coverage

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Maximum and Minimum AA coverage is required by the specification of STI (Shallow Trench Isolation) planarization process and in order to minimize variation of transistor width.

9.2.1.1 Max AA Coverage

Both following Condition #1 and Condition #2 must be satisfied.

Condition # 1: Rules (/AA.C.10): Coverage density of *sized* (AA + Dummy AA)

Condition #2: Rule AA.C.8: Coverage density of non *sized* (AA + Dummy AA)

In order to satisfy max AA coverage rule, SCL recommends designing each cell as follows:

Defining AA block pitch (N), create AA spaces with width M according to the following formula:

$$M = N * 0.05 \quad (M \geq 0.94\mu\text{m}, N \leq 180\mu\text{m})$$

$$M = N * 0.02 \quad (M \geq 0.94\mu\text{m}, N \leq 180\mu\text{m}) \text{ to meet rule AA.C.10}$$

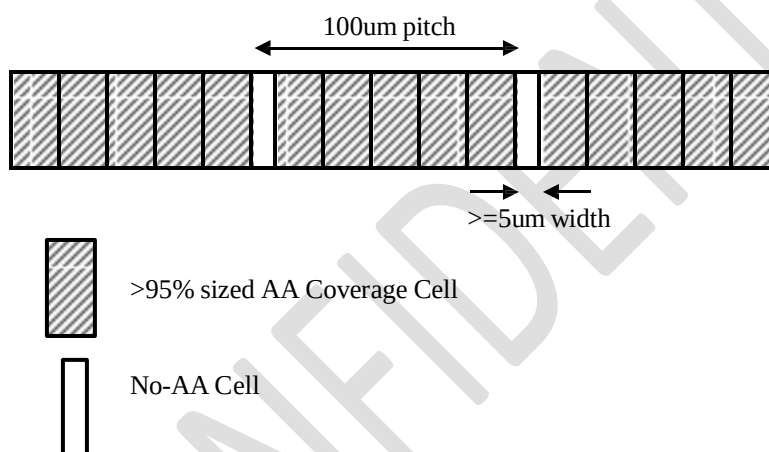


Figure 9-2: Meeting max AA coverage rules

9.2.1.2 Dummy AA Rules

SCL provides customers with Dummy ACTIVE generation run set in order to satisfy minimum coverage density of sized (AA + Dummy AA). See rules AA.C.1/3/5 in section 8.2.1. Dummy AA is floating. For DRC and dummy generation run set procedures, see related document.

The run set maintains wide spaces between dummy AA's. It keeps large distances from Dummy AA to real AA, to GC, and to N-well, minimizing coupling effects.

Two support layers (NODUMMY and AA(ny)) are used for AA dummy blocking. DUMMY FILL generation run set will not generate Dummy ACTIVE inside the polygons drawn with these support layers.

Note that dummy AA rules and AA coverage rules are also checked under external structures (marked with DEV-AREA(et), section 10.4).



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Table 9-1: DAA (Dummy ACTIVE) Rules

DAA (Dummy Active Area) Rules		Action	Drawn
DAA.W.1	Width of dummy AA	min	0.22
DAA.S.1	Space between dummy AA to another dummy AA	min	1.2
DAA.E.1	Enclosure of WN on Dummy ACTIVE	min	0.6
DAA.D.1	Distance of dummy AA to AA. Overlap is not allowed	min	1.2
DAA.D.2	Distance of dummy AA to GC. Overlap is not allowed	min	1.2
DAA.D.3	Distance dummy AA to WN edge.	min	0.6
DAA.D.4	Distance from Dummy ACTIVE to AA(ny) layers	min	1.2
DAA.D.5	Distance of dummy AA to chip edge. Overlap is not allowed	min	10
DAA.A.1	Area of Dummy ACTIVE	min	0.202um ²
DAA.N.1	Extension of Dummy ACTIVE beyond WN is not allowed.		
DAA.N.3	DAA under Fuse marking layer (78dt105) sized by 1um/side is NOT allowed.		

Chip Edge

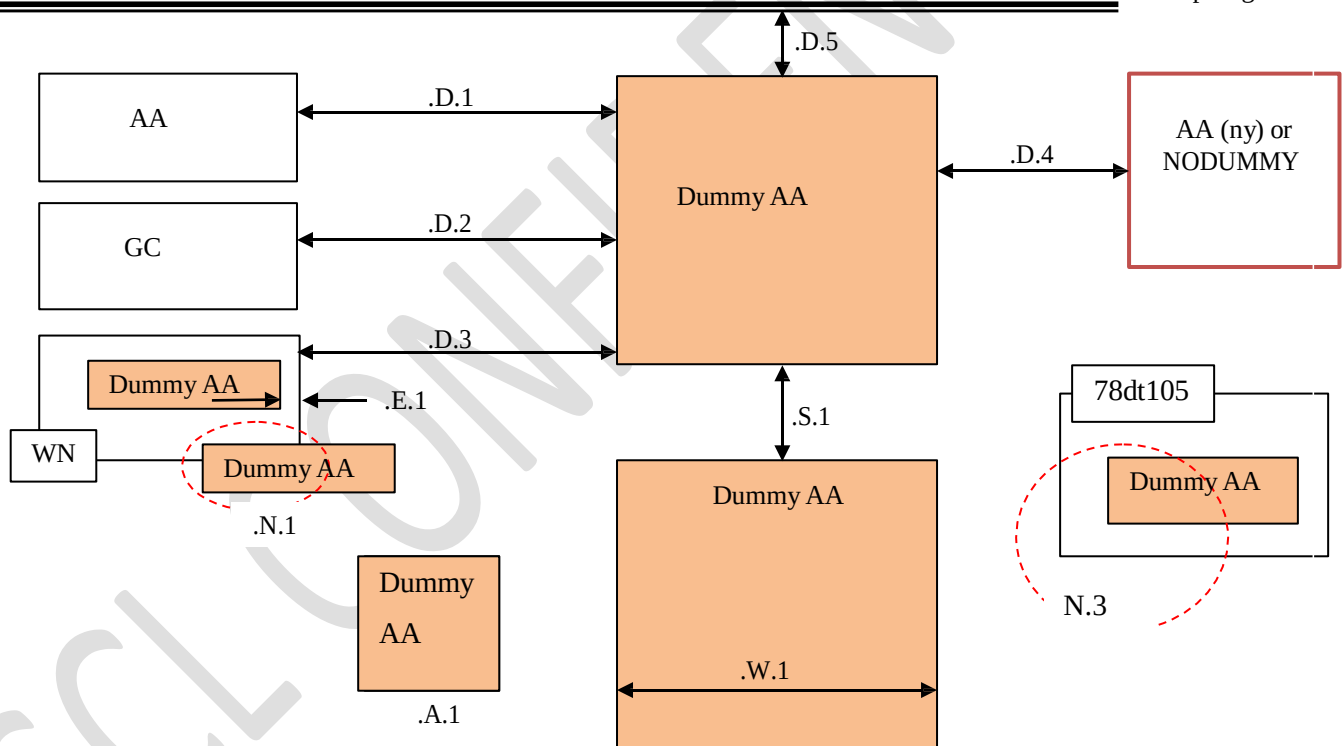


Figure 9-3: DAA (Dummy ACTIVE) Rules

**Design Rules for SL18 Standard Logic Process**

Doc. Name: DR2 0018 SL

Rev. Num: 0

9.2.2 GC/ Metal Coverage

In order to have process margin in RIE (Reactive Ion Etching), min and max coverage rules in GC and all aluminum metal layers are provided.

In order to meet minimum coverage, use the Dummy generation run set described at the utility document. SCL recommends introducing of metal dummies between every two metals which has spacing of more than 10um in between.

9.2.3 DGC (Dummy GC) Rules		Action	Drawn
DGC.W.1	Width of dummy GC	min	0.21
DGC.S.1	Space between two Dummy GCs	min	1.2
DGC.D.1	Distance between dummy GC to GC. Overlap is not allowed	min	1.2
DGC.D.2	Distance of dummy GC to AA	min	1.2
DGC.D.3	Distance between dummy GC to MFC, 77dt73. Overlap is not allowed.	min	1
DGC.D.4	Distance from Dummy GC to GC(ny)	min	1.2
DGC.D.7	Distance of dummy GC to DEV_AREA(et). Overlap is not allowed	min	1.2
DGC.N.1	Dummy GC interact CS or AA is not allowed. DRC also checks that real GC not interacting CS or AA is not allowed		
DGC.N.2	Dummy GC over AA is NOT allowed. Dummy GC MUST be over STI		
DGC.R.1	Recommended size of dummy GC=1x4.	min	1

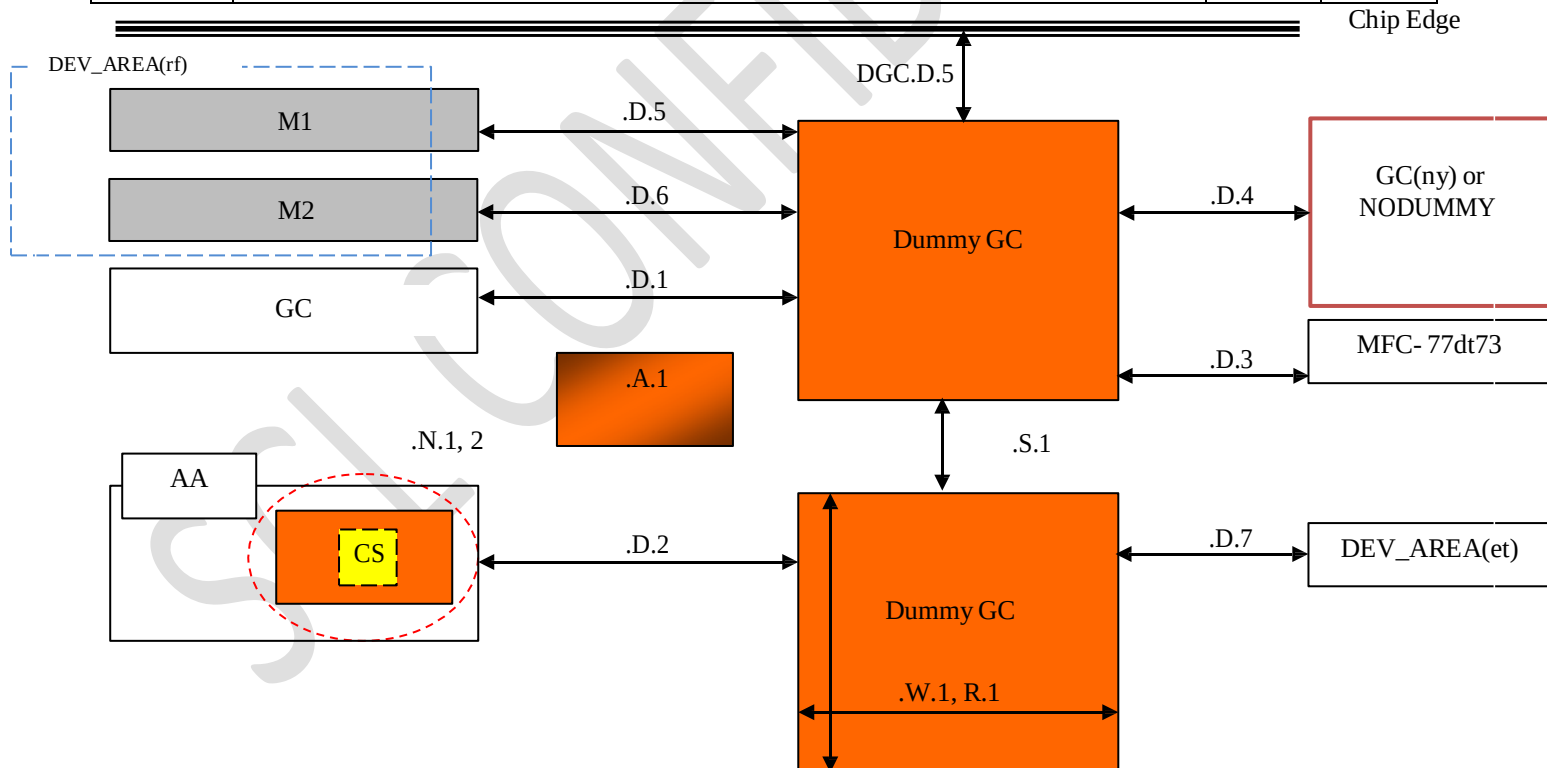


Figure 9-4: DGC (Dummy GC) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

9.2.4 DM (Dummy Metal) Rules		Action	Drawn
DM.W.1	Width of dummy M1,MI,ML	min	2
DM.W.2	Width of dummy MT	min	1.2
DM.W.6	Width of dummy MT3	min	3
DM.S.1	for M1, MI, ML, MT: Space between two metal dummies	min	2
DM.S.3	For MT3: Space between MT3(dy)	min	3
DM.D.1	Distance of dummy Metal to metal(lc) of the same level. Overlap is not allowed. This rule is applicable for all metal layers: M1, MI, ML, MT	min	2
DM.D.2	For MT3: Distance of dummy Metal to metal(lc) of the same level. Overlap is not allowed.	min	3
DM.D.3	Distance between dummy metal to MFC, 77dt73 Overlap is not allowed. This rule is applicable for all metal layers: M1, MI, ML, MT, MT3	min	1
DM.D.6	For M1, MI, ML, and MT: Distance from dummy Metal to M(ny), both at the same metal level. For MT3, refer to rule D.7	min	1.2
DM.D.7	For MT3: MT3(dy) to MT3(ny)	min	2
DM.D.9	Distance of dummy metal to chip edge. Overlap is not allowed This rule is checked only for MT(dy), MT3(dy)	min	2.5
DM.D.10	Distance of dummy metal to DEV_AREA(et). Overlap is not allowed	min	1.2
DM.D.12	Distance between dummy of (MIM bottom plate -1) metal and dummy of MIM top plate metal (MI/ML/MT/MT3) to (MIM). Overlap is not allowed. For DRC purposes, check dummy under { 99dt55 OR CE(lc)}	min	2
DM.A.3	For MT3: Area of dummy MT3	min	9um2
DM.N.5	Dummy metal (any metal) including angled edge is NOT allowed. This rule is checked only for MT(dy)		
DM.R.1	Dummy metal size (M1, MI, ML): 2*5 For MT3 =4*4, space 2.5.		

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

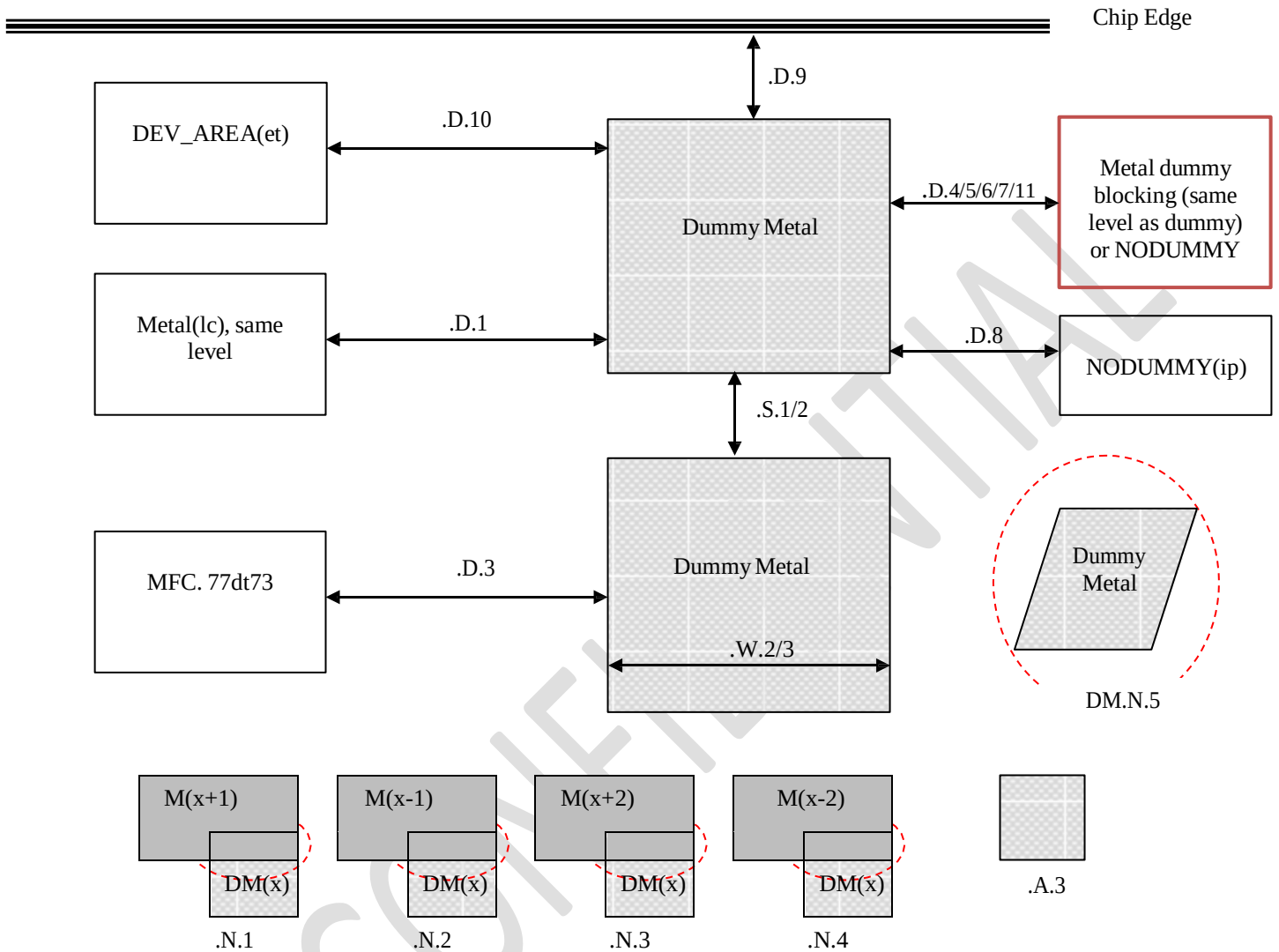


Figure 9-5: DM (Dummy Metal) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

9.3 Metal Slit (for Metal Coverage Reduction and for Stress Relaxation)

Metal slits are needed, in order to reduce the level of stress induced by the metal layer, into the oxides layer under / above it. The rules in this section apply to wide metal lines as defined in rules MS.N.1 and MS.N.4. Slits should be continuous through any change of direction. The inserted slits should be aligned with the long axis parallel to the metal line.

Add support layer mslits (ms) coincident with already existing slit area to enable checking metal slit rules by DRC.

All metal stress relief rules are not applicable under PAD or Dummy PAD. All metal stress relief rules are also applicable for Dummy metals.

9.3.1 MS (Metal Slits for stress relief) Rules		Action	Drawn
MS.W.1	Width of metal slit. This rule is checked for M1(ms), MI(ms), ML(ms), MT(ms).	min	2
MS.W.2	Width of metal slit. This rule is applicable for all metal layers	max	5
MS.W.4	Width and length of MT3 slits	min	3
MS.S.1	Space between slits in same any metal level. This rule is checked for M1(ms), MI(ms), ML(ms), MT(ms).	min	10
MS.S.3	For MT3: space between two MT3 slits	min	3
MS.E.1	Enclosure of Metal around Metal slit in the same level. This rule is checked for M1(ms), MI(ms), ML(ms), MT(ms). For MT3 please refer to MS.E.3,	min	10
MS.E.3	For MT3: Enclosure of MT3 around MT3 slit.	min	3
MS.D.1	Distance between Mx slit to Mx+1 or Mx-1 slit. Overlap is not allowed.	min	2
MS.D.2	Distance between metal slits to Via under This rule is NOT check by DRC	min	0.5
MS.D.13	For MT3: Distance between MT3 slits to MT3 resistor support layer (MT3(r1)). Overlap is not allowed	min	4
MS.D.14	For MT3: Distance between MT3 slits outside pad opening to Pad opening.	min	3
MS.D.15	For MT3: Distance between MT3 slits to DEV_AREA(et). Overlap is not allowed.	min	1.5
MS.D.16	For MT3: Distance between MT3 slits to DEV_AREA(mc). Overlap is not allowed.	min	1.5
MS.D.17	For MT3: Distance between MT3 slits to Metal (any metal) under DEV_AREA(seal). Overlap is not allowed.	min	1.5
MS.A.4	For MT3: MT3 slit area (see also rule MT3.A.2 for hole enclosed by MT3).	min	9



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

9.3.1 MS (Metal Slits for stress relief) Rules		Action	Drawn
MS.N.1	Wide metal without slits is NOT ALLOWED. Wide metal is defined as a metal which both width AND length >35um. For DRC purposes, exclude checking: 1) slits at Pad-Metal,		
MS.N.1	Metal slit layer (any metal) that is not over metal, or that does not exactly overlaps a metal hole (from the same layer) is not allowed. This rule is checked for M1, MI, ML, MT, MT3		
MS.N.3	Length of area in a wide metal at the connection of a narrow metal, in which metal slits are not allowed. _Note: This rule will generate false errors in case the metal or its slits are placed out of grid, or if their edges not in diagonal (45 deg) or in orthogonal (0, 90 deg) angles.	min	22
MS.N.6	Metal slit under pad opening, at the most upper Al metal is NOT allowed. For DRC purposes, check for existence of metal slits (made dt59 or by metal "hole"). This rule is relevant for ML (in 5M1L for example), MT (in 4M1T for example), MT3 (in 4M1T3 for example)		

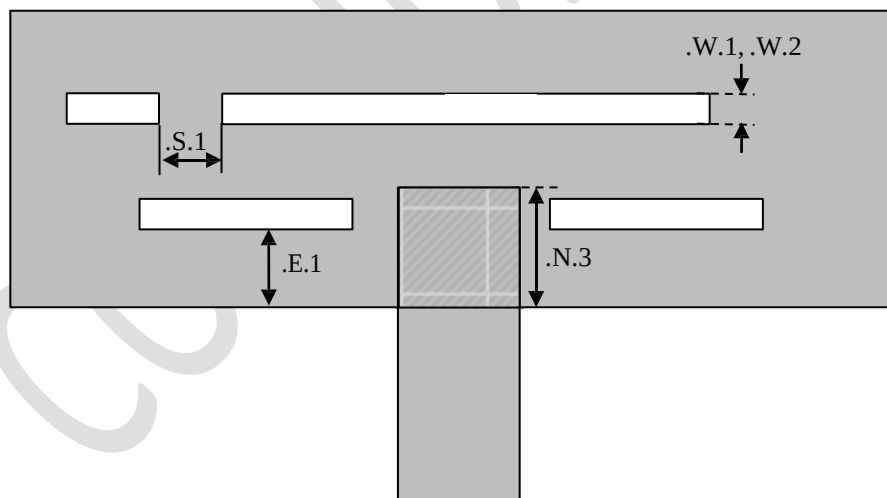


Figure 9-6: MS (Metal Slits for stress relief) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

10 Bonding Pads and External Structures

This section lists design rules for external structures

10.1 Design Rules for Standard Pad Structures (Stacked, Wire-Bond)

The overall performance of bonding pads depends on several parameters: pad type, structure, size, metal thickness of the most upper metal layer as well as the overall number of metals, and more. In addition, the assembly process also affects the bonding integrity and yield. Based on that, designer need to:

Contact the assembly house, verifying that the pad to be used is supported and qualified.

Verify the assembly process and stability for the specific pad to be used.

10.1.1 General Description of the Bond Pad

The following is a description of a “common”, recommended pad structure, applicable to both types of pad arrangements: single in-line and staggered. The structure consists of all-metal stack. The intermediate vias (VI) and top vias (VL’s) are arranged in diamond shape.

10.1.2 Common Pad Rules for Both In-Line Pads and Staggered Pads

The rules in this section, and in the table below, are related to Real PAD: pad in pure aluminum based backed. Some of the rules are not applicable in dummy pads.

Definitions:

Metal: a metal that is both

- Electrically connected to the pad
- Under PAD(lc).

For DRC, that metal is sized by 2um/side.

PAD VIA: VIA which is entirely enclosed by PAD

10.1.3 PD (Pad) Rules		Action	Drawn
PD.W.1/2	Pad opening (min/max) please refer to PD1/2.W.1/2		
PD.S.1	Pad space (min/max) please refer to PD1/2.S.1/2		
PD.S.2	Space between two PAD intermediate VIA’s (at same level). Refer to V2, V3, V4, V5	min	0.45
PD.S.3	Space between two PAD VL (at same level). Refer to TOP_V	min	0.35
PD.E.1	Enclosure of PAD(lc) on GC(dy).	exact	2
PD.D.1	Distance from pad metal to unrelated pad metal or to unrelated routing metal. Unrelated metal is metal that is not connected to the pad metal. Ignore this rule under Pol	min	2.5



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

10.1.3 PD (Pad) Rules		Action	Drawn
PD.D.2	Distance from PAD VIA(x) to PAD VIA(x-1) , where both Via(x) and Via(x-1) are intermediate via. Stacked VIAS of the same type (intermediate or last) is not allowed. Intermediate via refers to V2, V3, V4, V5,	min	0.13
PD.D.3	distance from PAD VIA(x) to PAD VIA(x-1) , where VIA(x) is last via, and Via(x-1) is intermediate via. Ignore this rule under Pol Stacked VIAS of different types (last above intermediate) is not allowed. Intermediate via refers to V2, V3, V4, V5.	min	0.06
PD.RX.1	Extension of all metals beyond edged pad VIA	max	4
PD.C.1	The ratio between via area (at the same level) to pad opening. Refer to V2, V3, V4, V5, TOP_V,	min	5%
PD.C.2	Coverage of PD area on overall chip area.	max	14%
PD.N.1	PD w/o all vias and metals under is not allowed. The pad structure should include all metals. For DRC purposes, ignore pad open at seal area, and POL. The rule is not applicable to M1, V2, M2 and V3 of inner line of staggered pads marked by PAD(dc). Refer to rule PD2.W.2		
PD.N.2	AA, GC, CS, dummy metal interact PD is not allowed		

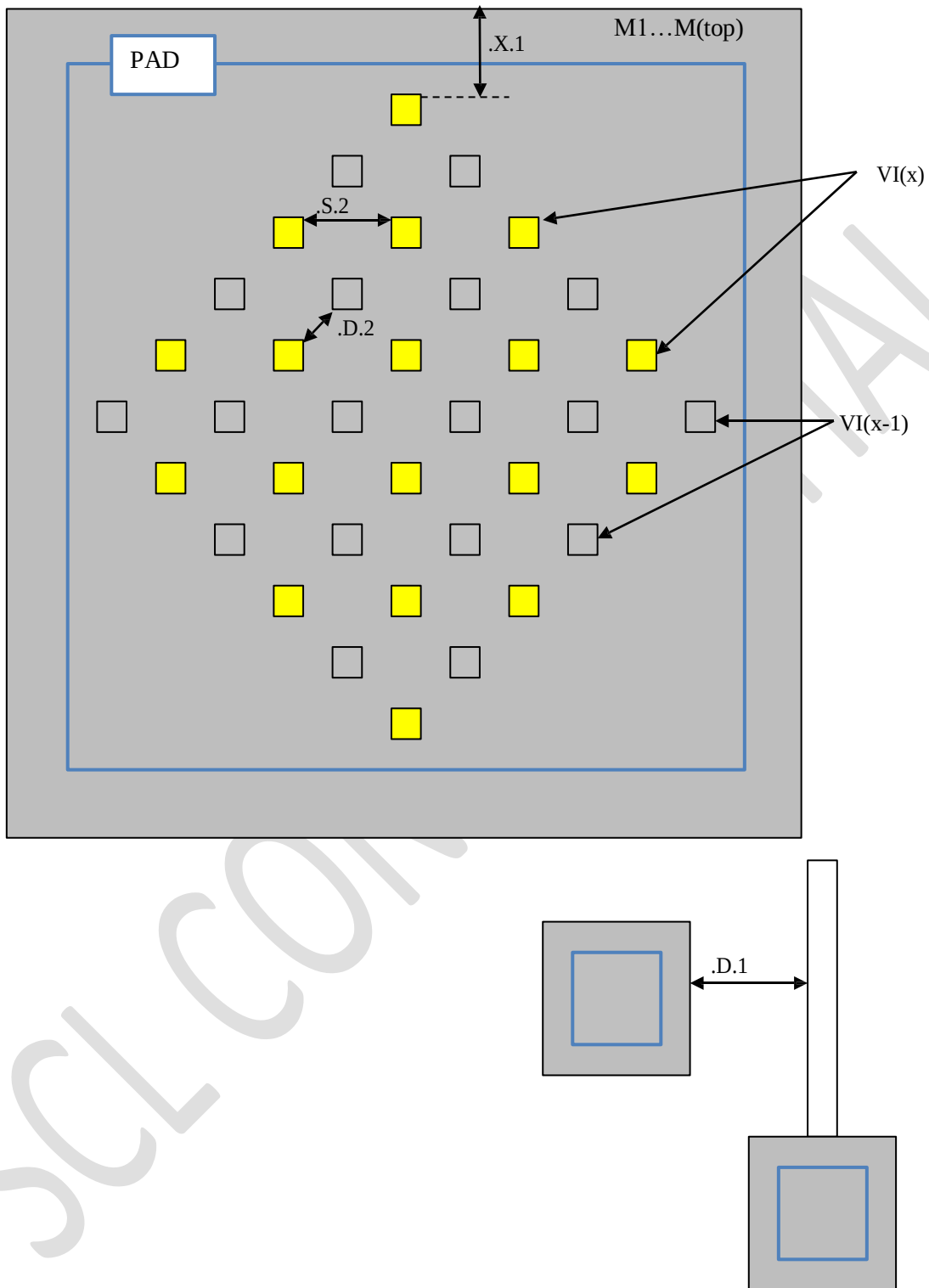


Figure 10-1: General PAD (pad rules)

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

10.1.4 Single In-line Bond Pad Rules

The In-line bond pad arrangement does not require any support layer.

10.1.4.1 PD1 (Single In-Line Bond Pad) Rules		Action	Drawn
PD1.W.1	Pad opening width	min	48x56
PD1.RW.2	Recommended. Pad opening width.	min	75 x 75
PD1.S.1	Space between two pad openings	min	7
PD1.E.1	Enclosure of any metal around pad opening.	min	2
PD1.RD.1	The overall length of the pad (including the space to the next pad), for the first 4 pads, all on the same die edge, closest to the chip corner. That is measured from the center of the 1 st pad to the center of the 2 th pad, 2 nd to 3 rd , 3 rd to 4 th and 4 th to 5 th .	min	80
Note: This rules is for nominal design with pitch of 55um			

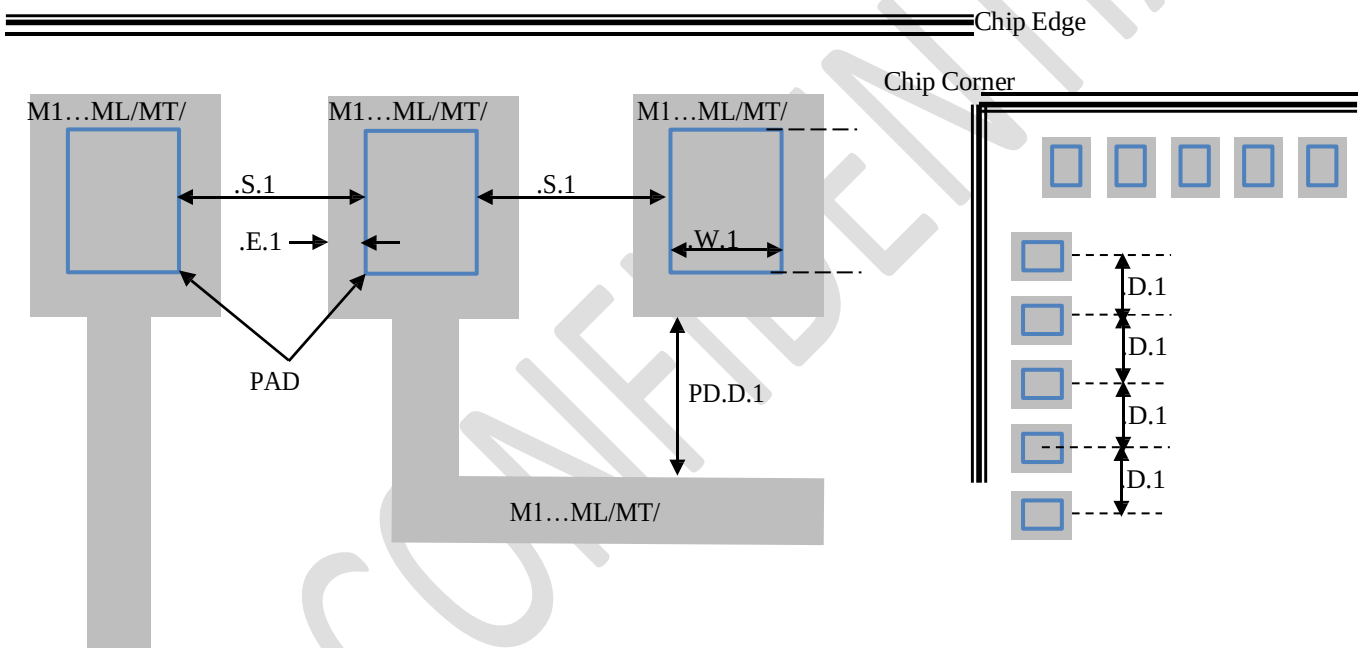


Figure 10-2: PD1 (Single In-Line Bond Pad) Rules

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

10.1.5 Staggered Bond Pad Rules

The Staggered bond pad arrangement requires two support layers for DRC:

1. PD(dc), 68dt27 that is used to identify the inner line of staggered pads for DRC
2. DEV_AREA(pd), 77dt46 that is used to identify the outer line of staggered pads for DRC

10.1.5.1 PD2 (Staggered bond Pad) Rules		Action	Drawn
PD2.W.1	Pad opening width	min	48x56
PD2.W.2	Width of M1 and M2 under the inner pads	min	45
PD2.RW.2	Recommended. Pad opening width.	min	75 x 75
PD2.S.1	Space between two pad openings	min	7
PD2.RD.1	The overall length of the pad (including the space to the next pad), for the first 4 pads, all on the same die edge, closest to the chip corner. That is measured from the center of the 1 st pad to the center of the 2 nd pad, 2 nd to 3 rd , all related to the inner line of pads, and from the center of the 1 st pad to the center of the 2 nd pad, 2 nd to 3 rd , all related to the outer line of pads.	min	80
PD2.RD.2	Distance from the center of the inner pad to center of a neighbor outer pad.	min	31.5
PD2.E.1	Enclosure of any metal directly connected to last metal around pad opening.	min	2
Note: This rules is for nominal design with pitch of 55um.			

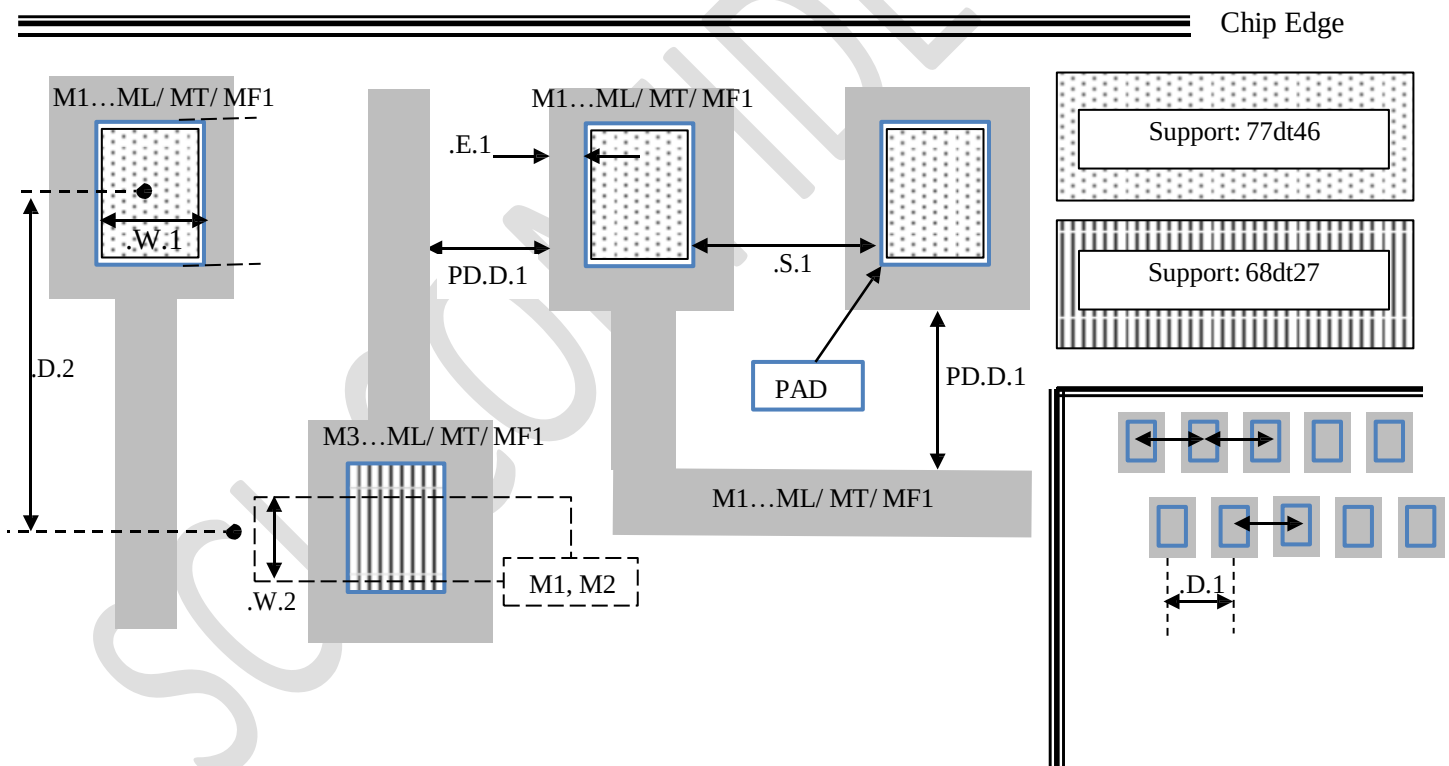


Figure 10-3: PD2 (Staggered bond Pad) Rules.



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

10.1.6 Dummy Pads Rules

In order to reduce the local stress developed at the chip corner, place dummy pad at each corner of the die, at the line end of each corner line. The dummy pads metal should be covered by the PD(dy) support layer for DRC. Note that PAD(dy) does not open a hole in the passivation.

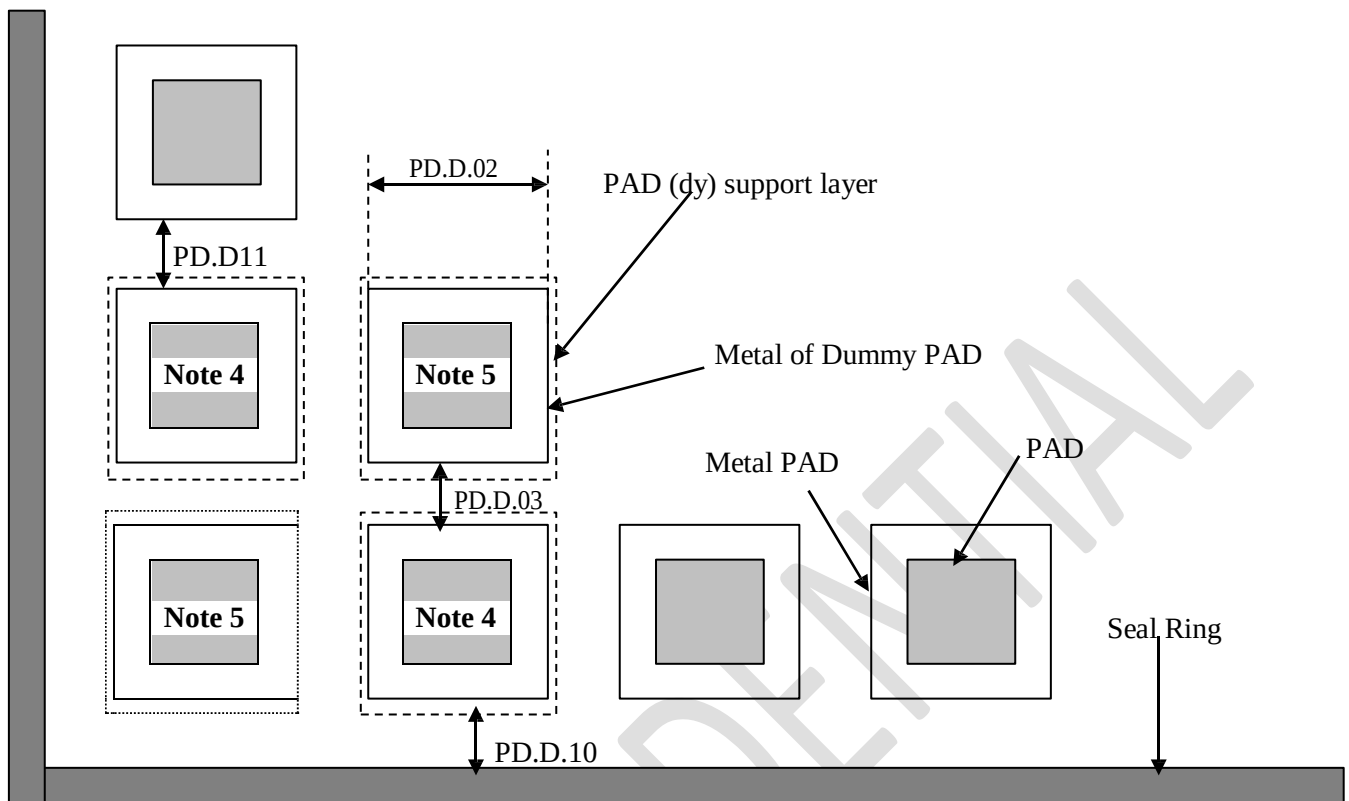
10.1.6.1 DPD (Dummy pads) Rules		Action	Drawn
DPD.W.2	Width of any Metal under layer PAD(dy)	min	40
DPD.W.3	Width of any Metal under layer PAD(dy)	max	80
DPD.S.1	Space between two PD(dy) layers	min	2
DPD.S.2	Space between two PD(dy) intermediate VIA's (at same level). Refer to V2, V3, V4, V5	min	0.58
DPD.S.3	Space between two PD(dy) VL (at same level). Refer to TOP_V	min	0.5
DPD.E.1	Enclosure of PAD(dy) on GC(dy).	exact	2
DPD.E.2	Enclosure of PD(dy) around PD Note that PD under PD(dy) is not mandatory	min	2
DPD.D.1	Distance from PD(dy) to unrelated metal	min	2.5
DPD.D.2	Distance from VIA(x) to VIA(x-1) , where both Via(x) and Via(x-1) are of the same type intermediate. Stacked VIAs of the same type (intermediate or last) is not allowed. Intermediate via refers to V2, V3, V4, V5	min	0.23
DPD.D.3	Distance from VIA(x) to VIA(x-1) , where VIA(x) is last via, and Via(x-1) is intermediate via. Stacked VIAs of different types (last above intermediate) is not allowed. Intermediate via refers to V2, V3, V4, V5. Last via refer to TOP_V,	min	0.16
DPD.D.4	Distance between PD(dy) to DEV_AREA(sl)	min	25
DPD.D.5	Distance between PD(dy) to PAD metal	min	2
DPD.X.1	Extension of any Metal beyond any VIA under PAD(dy)	min	3
DPD.N.1	PD(dy) w/o all vias and metals under is not allowed. The pad structure needs to include all metals.		
DPD.N.2	AA, GC, CS, dummy metal interact PD(dy) is not allowed		
DPD.N.3	Metal layer under PAD(dy), which is not coincide with PAD(dy) is not allowed		



Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0



Note 4: This dummy PAD is **strongly** recommended.

Note 5: This dummy PAD is recommended.

Figure 10-4: DPD (Dummy ads) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

10.2 Die Corner Rules

In order to reduce the impact of structural and/or thermal stress imposed at the die corner during packaging, keep devices and metal corners away from the chip corners. The rules in this section are applicable only for entire chip level. To prevent DRC from checking them on partial design, invoke BLOCK switch in DRC run set.

Table 10-1: DC (Die Corner) Rules

10.2.1 DC (Die Corner) Rules		Action	Drawn
DC.N.1	GC(lc) , N+ AA in pwell, or P+AA in WN are not allowed at die corner area (Table 10-2). This rule is not checked for dummy pads, or under DEV_AREA(et)		
DC.N.2	Metal corner of 90 degrees is not allowed at die corner area (Table 10-2)		

Table 10-2: Definition of Die corner Area for rules DC1.N.1 and DC.N.2

If length of at least one edge of Die $\leq 2\text{mm}$	20um*20um
$< 25\text{mm}^2$, and two edges $> 2\text{mm}$	76um*76um
Die Area	Die corner area for rules DC.N.1 and DC.N.2
$< 100\text{mm}^2$, and two edges $> 2\text{mm}$	110um*110um
$\geq 100\text{mm}^2$, and two edges $> 2\text{mm}$	350um*350um



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

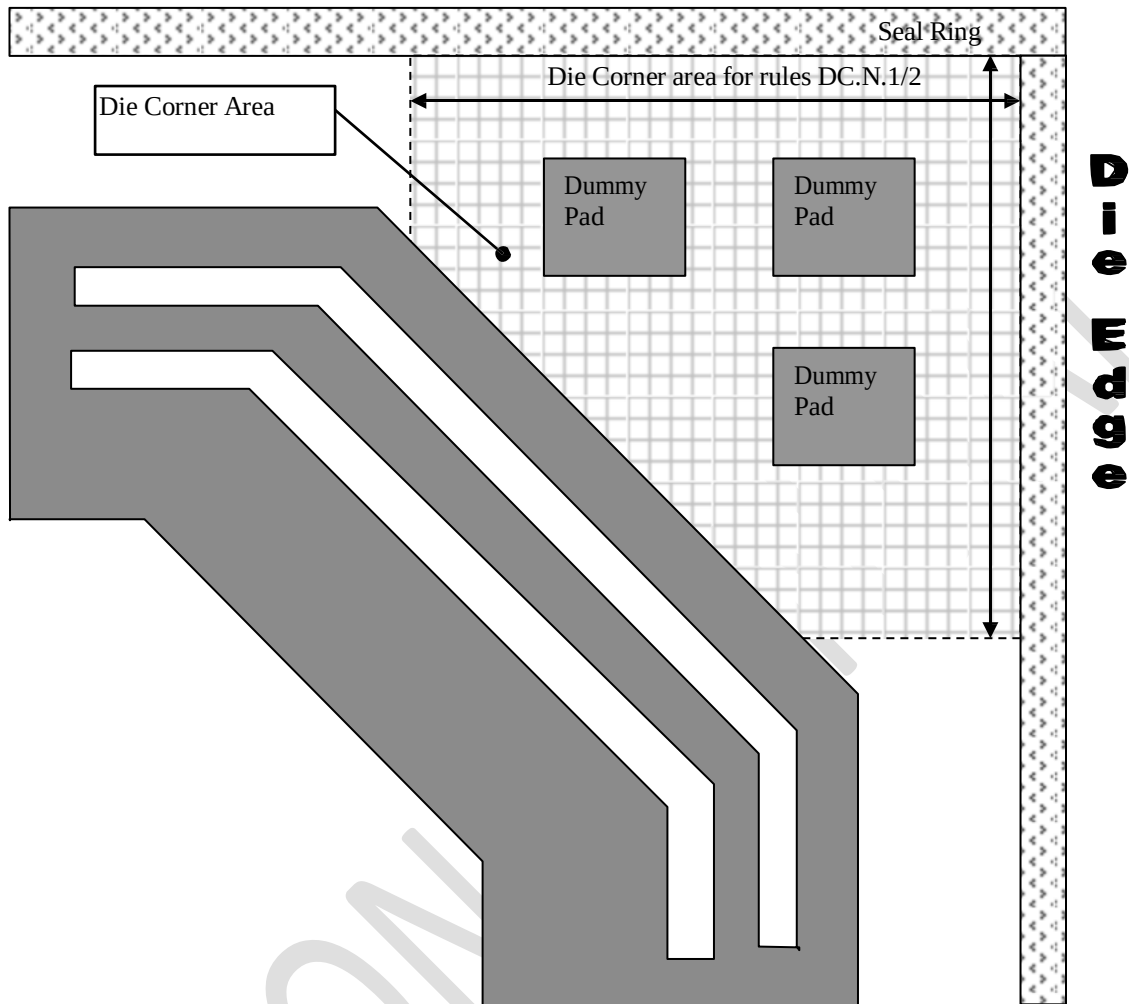


Figure 10-5: DC (Die Corner) Rules

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

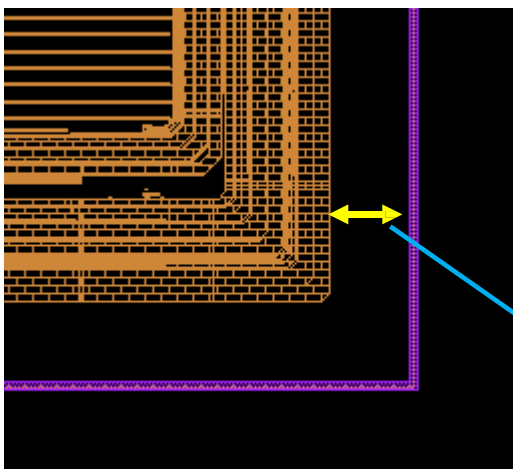
10.3 Seal Ring Layout Rules and Scribe Line

10.3.1 Introduction

The chip seal ring consists of a substrate-contacted metal, providing a low resistance path to substrate ground for surge currents, a physical metal seal against ionic contaminants, and a film de-lamination barrier from environmental exposure during reliability stressing and field usage. The customer has an option to get a well-defined seal ring, which is included in SCL's external structures package, and documented in the technology external structure document. It is strongly recommended to connect the Seal ring metals to Vss metal bus. Since the seal ring consists of all metals, any metal can be used for the connection of the Vss metal bus to the seal ring metals.

10.3.2 Seal Ring Dimensions and Guidelines

The chip ring **MUST** be a complete, un-broken ring around the chip area. A total width of 11um seal ring is described below. The distance from die edge to Seal Ring of 10um is for reference. This distance depends on capability of assembly house see Figure 9-7. For connection between metals, both square and bars holes (Holes: CS and VIA's) are used. The bars should be drawn using dt0 drawn layer. NO special support layer is needed. DRC allows via bars under support layer 77dt45. A customer that likes to use different dimensions, should contact SCL.



The distance from Die edge to Seal Ring should be >10um

Figure 10-6: DC (Die Corner) Rules



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

10.3.3 SR (Seal Ring) Rules		Action	Drawn
SR.W.1	Width of DEV_AREA(sl)	exact	11
SR.W.2	Width of AA	exact	10
SR.W.3	CS width (CS bar)	exact	0.22
SR.W.4	Metal width except for any Metal which is connected to VSS or GND. This rule is applicable for all metal layers	exact	5.9
SR.W.5	Width of V2/3/4/5/6 (via bar)	exact	0.26
SR.W.6	Width of VL	exact	0.36
SR.W.7	Width of PD	exact	2
SR.S.1	Space between CS bars	exact	2.22
SR.S.2	Space between two VI bars	exact	2.18
SR.S.3	Space between two VL	exact	2.08
SR.D.1	Distance of Poly or metal (any metal : M1,Mi,ML,MT,MT3) to internal side of DEV-AREA(sl). See Note below.	Min	10
SR.E.1	Enclosure of DEV_AREA(sl) around AA (inner, far from scribe line)	exact	1
SR.E.2	Enclosure of DEV_AREA(sl) around CS (inner, far from the scribe line)	exact	1.4
SR.E.3	Enclosure of DEV_AREA(sl) around metal (inner, far from the scribe line). This rule is applicable for all metal layers	exact	1
SR.E.4	Enclosure of DEV_AREA(sl) around V2 or V4 or V6 (inner, far from the scribe line)	exact	2.6
SR.E.5	Enclosure of DEV_AREA(sl) around V3 or V5 (inner, far from the scribe line)	exact	1.38
SR.E.6	Enclosure of DEV_AREA(sl) around TOP_V (inner, far from the scribe line)	exact	2.55
SR.O.1	Overlap of WN,XN,NS over DEV-AREA(sl) (77dt45) is NOT allowed		
SR.N.1	Seal ring MUST be one closed ring. For DRC purposes, check DEV_AREA(sl), CS, all metals, all vias and PAD		
SR.N.2	XP should co-inside DEV_AREA(sl)		

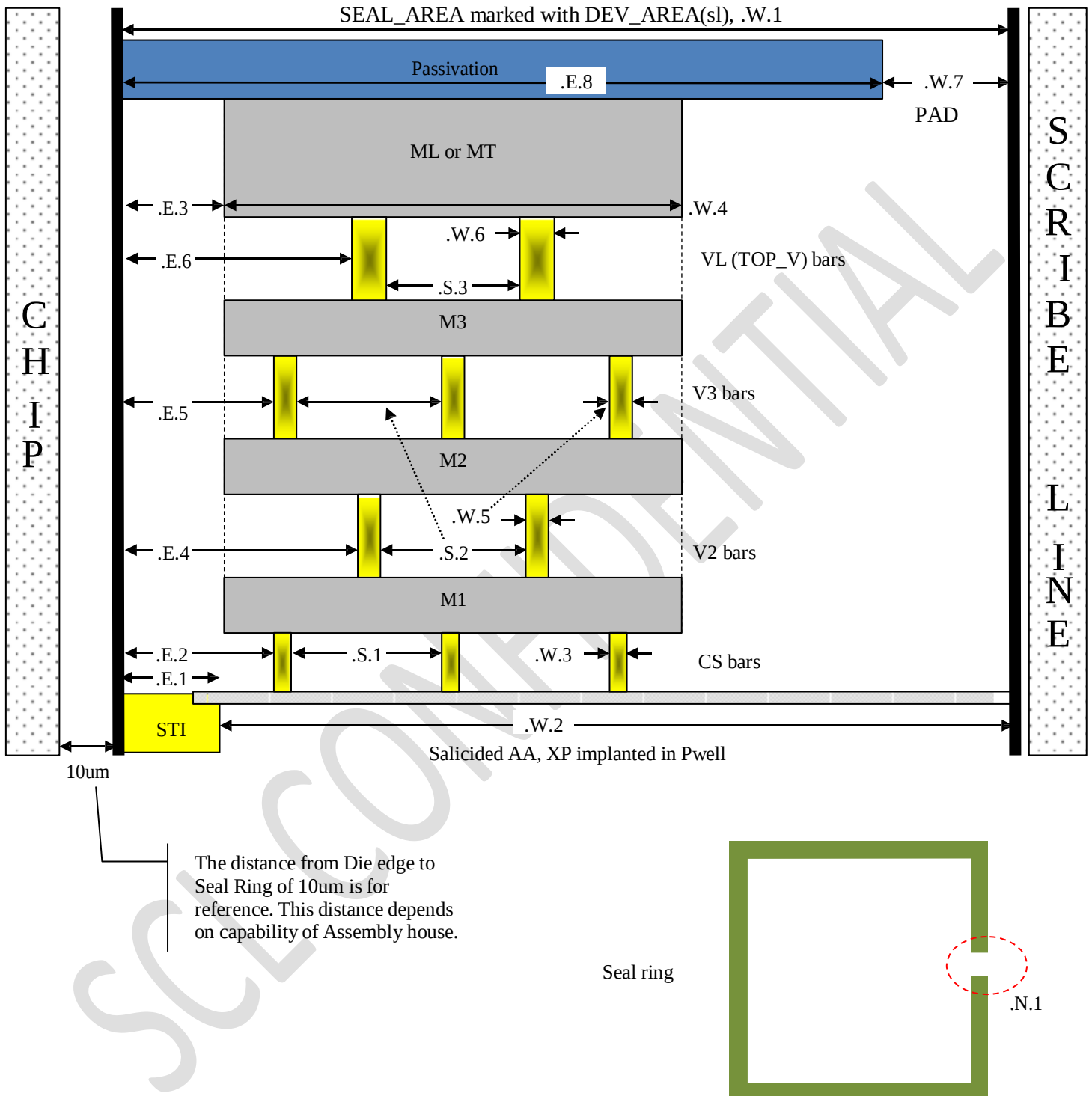


Figure 10-7: SR (Seal Ring) Rules (for example figure for 4M1L or 4M1T).



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

10.4 Instructions for Including External Structures

To avoid DRC violations related to non-electrical external structures, cover them with layer 77dt65-DEV_AREA(et). DRC will not be checked in these areas. It is NOT allowed to use 77dt65 layer for purposes of avoiding DRC checks. Note, that seal ring and PAD's do have an electrical meaning and because of that, should NOT be covered with DEV_AREA(et).

SCL systems follow these guidelines under DEV_AREA(et):

Disable generation of GC(dy) and metal Dummy layers

Disable DRC. DRC will only check two types of rules:

1. AA.C.* of min coverage of sized (real ACTIVE + dummy ACTIVE) described in section 8.2.1 and Dummy ACTIVE rules in section 9.2.1.2.
2. GC.C.1 GC.C.6 of (GC + Dummy GC) coverage, see section 9.2.2 DRC doesn't check distance rules or space rules where only one polygon is covered by DEV_AREA(et).

It is recommended to keep min of 10um distance from external support layer to die edge.

10.4.1 EX (External Structures) Rules		Action	Drawn
EX.A.1	Area of 77dt65	max	40,000um ²
EX.N.2	Any electrical (drawing) data under 77dt65 sized by 1um/side is NOT allowed. For DRC purposes, check existence of GATE, NSUB, PSUB and metal routing		
EX.D.1	Distance from VNN to DEV_AREA(et). VNN cutting DEV_AREA(et) edge is NOT allowed	min	7

SCL provides a GDS file with external structures (*DK2 18SLEXT1*) The files include pads, seal ring, rev letters, CDL's, NIXY's, PCI marks pads and more.



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

11 Parasitic Capacitance Model

The interconnection line-line capacitance simulation results, using TMA RAPHAEL, are listed in this section. At the conduction line, the minimum width of design rule is used. The dielectric thicknesses are the “typical” values.

11.1 Cross Section Technology Parameters

11.1.1 Metal Layers Thicknesses

Conductor Layer	Modeling Thickness	SEM Thickness	SEM Distribution	Min Width	Min Space	Distance between conductor layer and substrate under STI	Recess after etch
-----	[um]	[um]	[%]	[um]	[um]	[um]	[um]
MT3	2.8	~2.83	15%	2.5	2	3.75 (3M1T3) 5.11 (4M1T3)	NA
MT	2.0	~2.05	15%	1.2	1	3.75 (3M1T) 5.11 (4M1T)	NA
ML	0.84	~0.94	15%	0.44	0.46	3.75 (3M1L) 5.11 (4M1L)	NA
M5	0.54	~0.56	15%	0.28	0.28	6.49	NA
M4	0.54	~0.56	15%	0.28	0.28	5.13	NA
M3	0.54	~0.56	15%	0.28	0.28	3.77	NA
M2	0.54	~0.56	15%	0.28	0.28	2.41	NA
M1	0.54	~0.53	15%	0.23	0.23	1.05	NA
Poly	0.2	~0.19	15%	0.18	0.25	0.38	NA
Note: The SEM accuracy for the range of thicknesses is $\pm 5\%$. The “SEM Thickness” is only for validation and customers need to work with the “modeling Thickness”, ONLY.							



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

11.2 Capacitance Modeling BEOL X-Sections

SCL BEOL is having two dielectric option:

1. BEOL with USG (BU)
2. BEOL with FSG (BF)

The following figures include DR cross sections with the effective values for interconnect capacitance. Thus, these cross sections do not present the actual physical layer structure.

Example: for 6M1L, for **BF** (BEOL using **FSG** dielectric)

Dielectric layer	Comments	Modeling Thickness [nm]	SEM Thickness	SEM Distribution	Epsilon
Pass	Passivation nitride	300			7.0
IMD-Last	Metal-last dielectric	1000			3.81
D-Last	Via-last dielectric	800	~795	+/-10%	3.81
IMD5	M 5 dielectric	540			3.77
D5	Via 5 dielectric	820	~795	+/-10%	3.91
IMD4	M4 dielectric	540			3.77
D4	Via 4 dielectric	820	~795	+/-10%	3.91
IMD3	M3 dielectric	540			3.77
D3	Via 3 dielectric	820	~795	+/-10%	3.91
IMD2	M2 dielectric	540			3.77
D2	Via 2 dielectric	820	~795	+/-10%	3.91
IMD1	M1 dielectric	540			3.77
Cap	Cap oxide	-			-
D1'	Contact dielectric on AA	700	652	+/-15%	4.2
D1	Contact dielectric on STI	670			4.2
SIN	Spacer SiNi	--			--
SIN	Spacer Oxide	--			--
STITEOS	STI thickness under Poly	380	350	+/-20%	4.1
Note: The Thickness distribution data is based on SEM monitor data collection.					



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Example: for 6M1L, for BU (BEOL using USG Dielectric)

Dielectric layer	Comments	Modeling Thickness [nm]	SEM Thickness	SEM Distribution	Epsilon
Pass	Passivation nitride	300			7.0
IMD-Last	Metal-last dielectric	1000			3.81
D-Last	Via-last dielectric	800	~795	+/-10%	4.2
IMD5	M 5 dielectric	540			4.2
D5	Via 5 dielectric	820	~795	+/-10%	4.2
IMD4	M4 dielectric	540			4.2
D4	Via 4 dielectric	820	~795	+/-10%	4.2
IMD3	M3 dielectric	540			4.2
D3	Via 3 dielectric	820	~795	+/-10%	4.2
IMD2	M2 dielectric	540			4.2
D2	Via 2 dielectric	820	~795	+/-10%	4.2
IMD1	M1 dielectric	540			4.2
Cap	Cap oxide	-			-
D1'	Contact dielectric on AA	700	652	+/-15%	4.2
D1	Contact dielectric on STI	670			4.2
SIN	Spacer SiNi	--			--
SIN	Spacer Oxide	--			--
STITEOS	STI thickness under Poly	380	350	+/-20%	4.1
Note: The Thickness distribution data is based on 3 years SEM monitor data collection.					

Example: for 6M1T-BF

Dielectric layer	Comments	Thickness [nm]	Thickness Distribution	Epsilon
Pass	Passivation nitride	300		7.0
IMD-Last	Metal-last dielectric	25000		3.81
D-Last	Via-last dielectric	800	+/-20%	3.81
IMD5	M 5 dielectric	540		3.77
D5	Via 5 dielectric	820	+/-20%	3.91

Example: for 6M1T-BU

Dielectric layer	Comments	Thickness [nm]	Thickness Distribution	Epsilon
Pass	Passivation nitride	300		7.0
IMD-Last	Metal-last dielectric	25000		3.81
D-Last	Via-last dielectric	800	+/-20%	4.2
IMD5	M 5 dielectric	540		4.2
D5	Via 5 dielectric	820	+/-20%	4.2

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

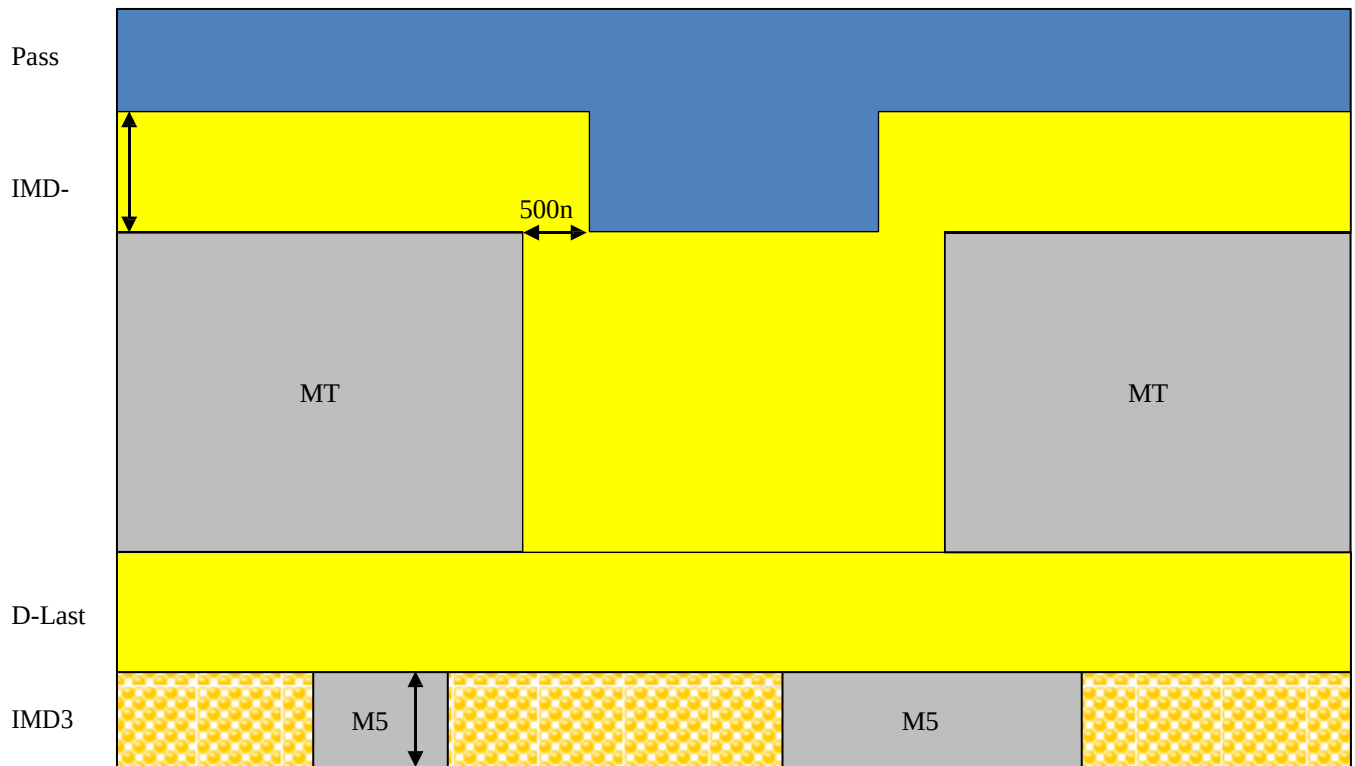


Figure 11-1: Capacitance modeling BEOL X-section of 6M1T. the lower layers (Si –M4) were omitted for simplicity

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

-Example: for 4M1T3-BU

Dielectric layer	Comments	Thickness [nm]	Thickness Distribution	Epsilon
Pass	Passivation nitride	600		7.0
IMD-Last	Metal-last dielectric	200		3.81
D-Last	Via-last dielectric	800	+/-20%	4.2
IMD5	M 5 dielectric	540		4.2
D5	Via 5 dielectric	820	+/-20%	4.2

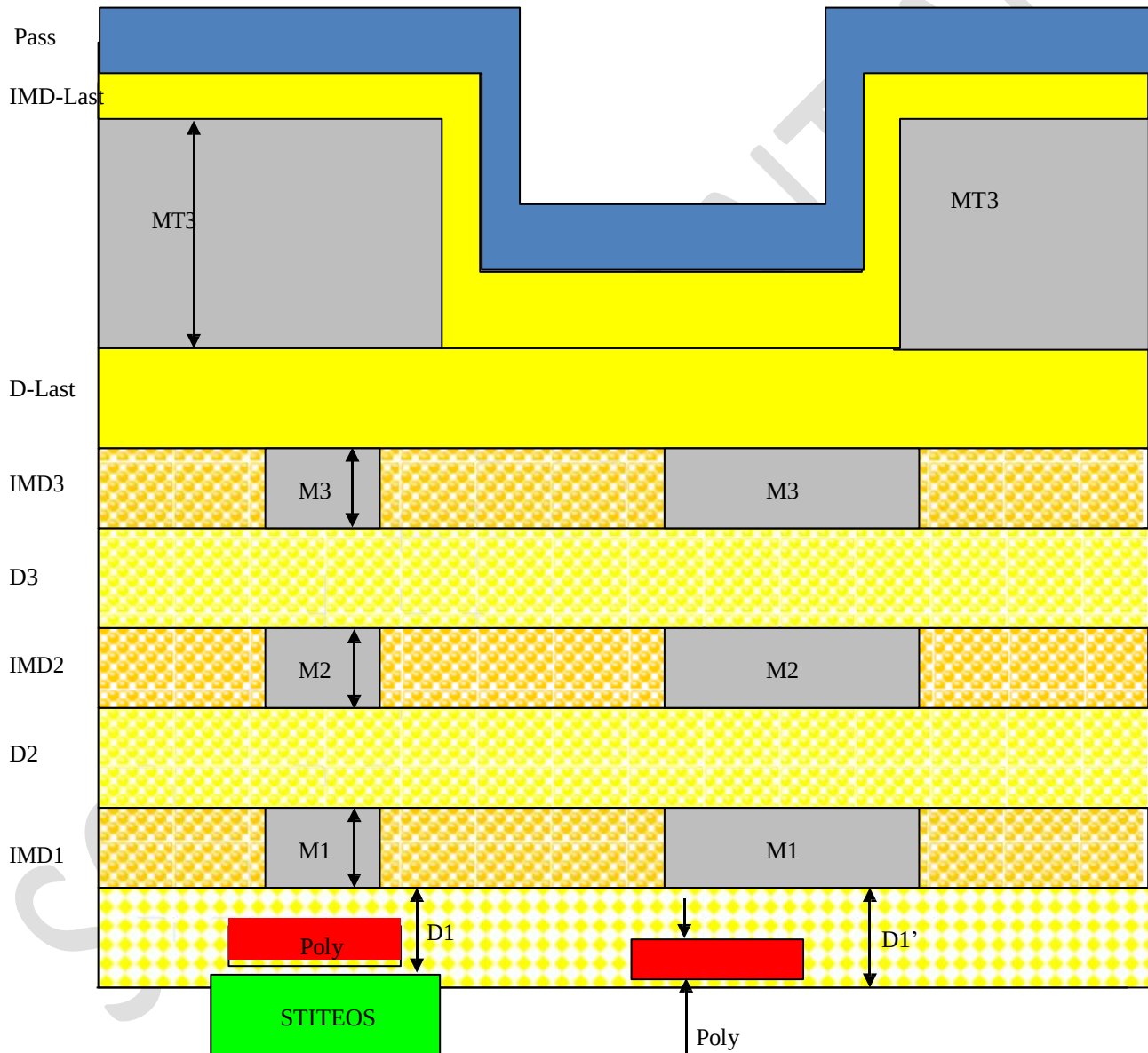


Figure 11-2: Capacitance Modeling BEOL X-section of 4M1T3



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

11.2.1 Definition for Generic Structure Type A

The generic structure type “A” consists on: a top layer that used as a conduction line layer, while the bottom conduction layer is used as an infinite plate.

C_{AREA} [fF/($\mu m * \mu m$)]: Area capacitance.

C_{TOTAL} [fF/ μm]: Total capacitance of the top central line conductor to all others.

C_{BOTTOM} [fF/ μm]: Capacitance between the top center conductor to the bottom ground plane.

C_{COUPLI} [fF/ μm]: Coupling capacitance between two conductors in the top layer.

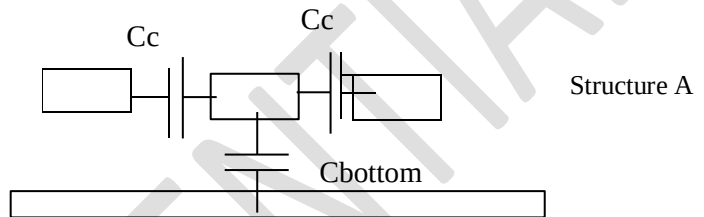
$C_{PERIMETER}$ [fF/ μm]: fringe capacitance per side of top central line to infinite bottom plate.

L_{ATERAL} [fF, fF/ μm]: Lateral coupling capacitance coefficient.

$C_{bottom} = C_a * w + 2 * C_{perimeter}$

$C_{total} = C_{bottom} + 2 * C_{coupli}$

$L_{ateral} = C_{coupli} * s$



11.2.2 Definition for Generic Structure Type B

The generic structure type “B” consists on: a middle layer that used as a conduction line layers between two infinite plate the bottom and the top.

C_{TOTAL} [fF/ μm]: Total capacitance of the central line conductor to all others.

C_{TOP} [fF/ μm]: Capacitance between the middle center conductor to the top ground plane.

C_{BOTTOM} [fF/ μm]: Capacitance between the center conductor to the bottom ground plane.

C_{COUPLI} [fF/ μm]: Coupling capacitance between two conductors in the same layer.

C_{AREA_TOP} [fF/($\mu m * \mu m$)]: Area capacitance to the top layer.

L_{ATERAL} [fF, fF/ μm]: Lateral coupling capacitance coefficient.

C_{AREA_BOT} [fF/($\mu m * \mu m$)]: Area capacitance of the middle layer to bottom layer.

C_{AREA_TOP} [fF/($\mu m * \mu m$)]: Area capacitance of the middle layer to top layer.

C_{SD} [fF/ μm]: Side-wall down capacitance per side of middle central line to infinite bottom plate.

C_{SU} [fF/ μm]: Side-wall up capacitance per side of middle central line to infinite top plate.

$C_{total} = C_{bottom} + C_{TOP} + 2 * C_{coupli}$

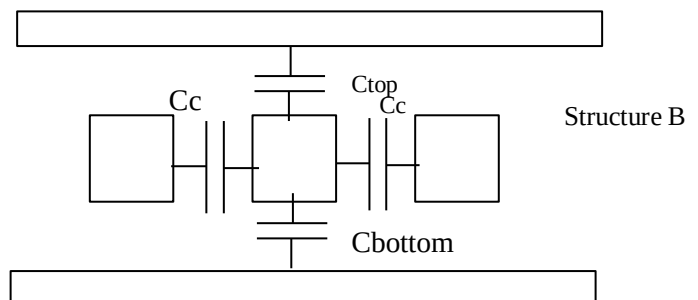
$C_{bottom} = C_{area_bot} * w + 2 * C_{sd}$

$C_{top} = C_{area_top} * w + 2 * C_{su}$

$L_{ateral} = C_{coupli} * s$

w: line width in μm

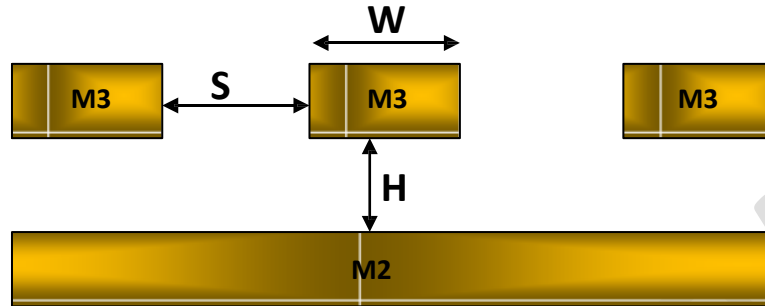
s: line space in μm



Parasitic capacitance modeling for all metal compensations are available in CM2 0018

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

11.2.3 Examples for Parasitic Capacitance Modeling



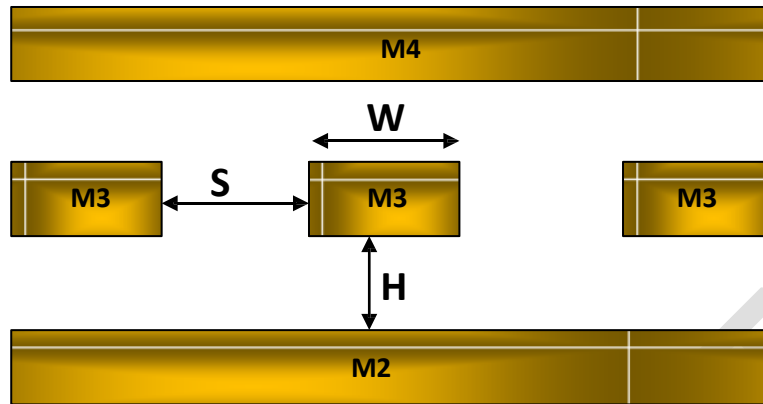
Type	Case	S	W	H	Cbot	Ccap	Ctotal	Cper
		um	um	um	fF/um	fF/um	fF/um	fF/um
A	M3 over M2	0.28	0.28	0.82	3.22E-02	9.74E-02	2.27E-01	1.02E-02
					14.2%	42.9%	100.0%	4.5%
		2	0.28	0.82	7.45E-02	1.44E-02	1.03E-01	3.14E-02
					72.1%	13.9%	100.0%	30.4%

Figure 11-3: Capacitance modeling M3 over M2, using FSG dielectric

Type	Case	S	W	H	Cbot	Ccap	Ctotal	Cper
		um	um	um	fF/um	fF/um	fF/um	fF/um
A	M3 over M2	0.28	0.28	0.82	3.45E-02	1.07E-01	2.49E-01	1.11E-02
					14%	43%	100%	4.4%
		2	0.28	0.82	8.02E-02	1.6E-02	1.12E-01	3.43E-02
					71.4%	14.3%	100%	30.6%

Figure 11-4: Capacitance modeling M3 over M2, using USG dielectric

Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0



Type	Case	S	W	Hup/down	Cbot	Ccap	Ctop	Ctotal	Csd	Csu
		um	um	um	fF/um	fF/um	fF/um	fF/um	fF/um	fF/um
B	M3 between M2/M4	0.28	0.28	0.82	2.45E-02	9.19E-02	2.45E-02	2.33E-01	6.35E-03	6.35E-03
					10.5%	39.5%	10.5%	100.0%	2.7%	2.7%
		2	0.28	0.82	5.79E-02	5.05E-03	5.79E-02	1.26E-01	2.30E-02	2.30E-02
					46.0%	4.0%	46.0%	100.0%	18.3%	18.3%

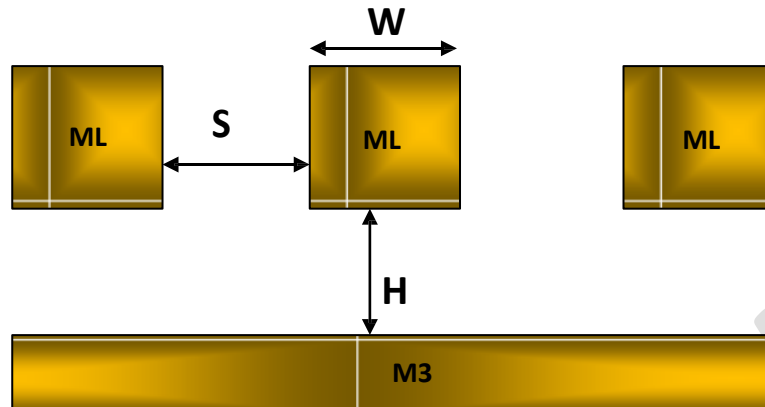
Figure 11-5: Capacitance modeling M3 between M2 and M4, using FSG dielectric.

Type	Case	S	W	Hup/down	Cbot	Ccap	Ctop	Ctotal	Csd	Csu
		um	um	um	fF/um	fF/um	fF/um	fF/um	fF/um	fF/um
B	M3 between M2/M4	0.28	0.28	0.82	2.64E-02	1.01E-01	2.64E-02	2.55E-01	6.83E-03	6.83E-03
					10.4%	39.6%	10.4%	100.0%	2.6%	2.6%
		2	0.28	0.82	6.26E-02	5.63E-03	6.26E-02	1.37E-01	2.50E-02	2.50E-02
					46.0%	4.0%	46.0%	100.0%	18.3%	18.3%

Figure 11-6: Capacitance modeling M3 between M2 and M4, using USG dielectric



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0



Type	Case	S	W	H	Cbot	Ccap	Ctotal	Cper
		um	um	um	fF/um	fF/um	fF/um	fF/um
A	ML over M3	0.46	0.44	0.82	1.53E-02	9.81E-02	2.12E-01	5.51E-03
					7.2%	46.4%	100.0%	2.6%
		2	0.44	0.82	2.67E-02	3.19E-02	9.05E-02	1.12E-02
					29.5%	35.2%	100.0%	12.4%

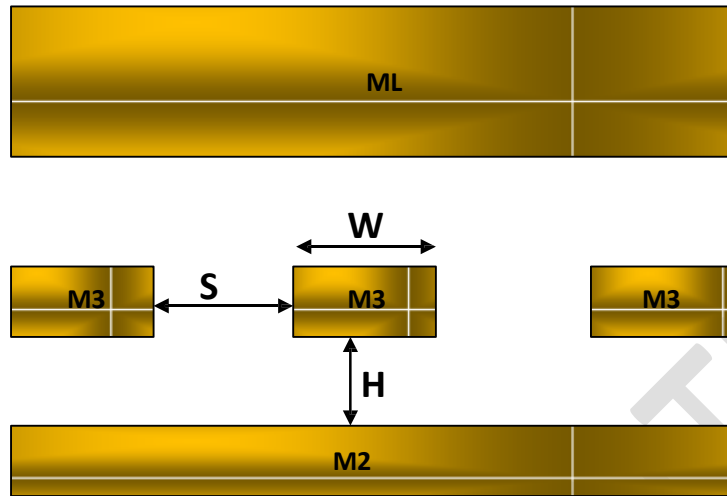
Figure 11-7: Capacitance modeling ML over M3, using FSG dielectric.

Type	Case	S	W	H	Cbot	Ccap	Ctotal	Cper
		um	um	um	fF/um	fF/um	fF/um	fF/um
A	ML over M3	0.46	0.44	0.82	1.66E-02	9.99E-02	2.16E-01	5.81E-03
					7.4%	46.3%	100.0%	2.6%
		2	0.44	0.82	2.88E-02	3.26E-02	9.35E-02	1.21E-02
					30.4%	34.8%	100.0%	12.9%

Figure 11-8: Capacitance modeling ML over M3, using USG dielectric.



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0



Type	Case	S	W	Hup/down	Cbot	Ccap	Ctop	Ctotal	Csd	Csu
		um	um	um	fF/um	fF/um	fF/um	fF/um	fF/um	fF/um
B	M3 between M2/ML	0.28	0.28	0.82	2.45E-02	9.19E-02	2.45E-02	2.33E-01	6.35E-03	6.35E-03
					10.5%	39.5%	10.5%	100.0%	2.7%	2.7%
		2	0.28	0.82	5.79E-02	5.05E-03	5.79E-02	1.26E-01	2.30E-02	2.30E-02
					46.0%	4.0%	46.0%	100.0%	18.3%	18.3%

Figure 11-9: Capacitance modeling M3 between M2 and M4, using FSG dielectric.

Type	Case	S	W	Hup/down	Cbot	Ccap	Ctop	Ctotal	Csd	Csu
		um	um	um	fF/um	fF/um	fF/um	fF/um	fF/um	fF/um
B	M3 between M2/ML	0.28	0.28	0.82	3E-02	1.06E-01	7.79E-03	2.49E-01	8.64E-03	2.42E-03
					12%	42.5%	3.1%	100.0%	3.4%	1%
		2	0.28	0.82	7.05E-02	1.28E-02	1.9E-02	1.15E-01	2.89E-02	8E-03
					61.3%	11.1%	16.5%	100.0%	25%	7%

Figure 11-10: Capacitance modeling M3 between M2 and M4, , using USG dielectric.



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

12 Electro Static Discharge (ESD) Protection

For ESD please refer to EDM SL18 SL1. This is a complementary document includes all ESD-related design rules, ESD device details, ESD protection guidelines, and characterization.

13 Latch Up Rules

13.1 LU (Both Internal Circuit and Periphery Circuit Latch Up) Rules		Action	Drawn
LU0	For both I/O and Internal circuits, distance from any point inside Source and Drain ACTIVE to the nearest Well pick-up ACTIVE in same WN or same WP. This rule is NOT applicable for MOS under marking layer DEV_AREA (ED)	max	30



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

14 Reliability Specification

14.1 Introduction

This chapter describes the reliability guidelines and specification of SCL 0.18um technology platform. The platform technology passed qualification tests as defined by foundry specifications such as JEDEC “foundry process qualification guidelines, (JP001.01) and others. Designer is recommended to check the exact reference list at the platform reliability report, document RAI2 07012. The qualification report also includes reliability performances, under different conditions. For example, at higher temperatures stress of higher voltage stress.

14.2 Electromigration - Introduction

In “Electromigration”, we refer to resistance degradation of interconnects (metal lines as well as contacts and vias), due to aggressive operational conditions of current and temperature. Due to the momentum exchange between the current carrying electrons and the host metal lattice, Aluminum (for Al lines) will drift under the influence of the electron wind. This section provides the relevant information and limits to the designers, in order to achieve the desired operational life time defined below.

Important note: some circuits like ESD protection may work under extreme conditions that are not covered at this section. Designer must refer to the platform ESD manual for more details or contact SCL reliability team.

Based on JEDEC specifications [1], SCL specified the reliability criteria for Electromigration failure as next:

Using failure criteria of 10% resistance change; Provide operational Life-Time (LT) $\geq 100,000$ hr’s with 100% duty cycle, having cumulative failure rate (CumF) of 0.1% under conditions of temperature of 110degC (Junction temperature), and maximum current density as specified below (14.5).

This section covers several topics:

- Electromigration (EM) rules that specify maximum DC current density (J_{DC_max}) through metal lines, per line width
- Electromigration rules, that specify maximum DC current density (J_{DC_max}) through holes – contacts (CS) and Vias, per hole
- Electromigration rules that specify maximum AC current density (J_{AC_max}) based on Average current (J_{AC_avg}) and Root-Mean-Square (RMS) current (J_{RMS}) limits, per line width
- Electromigration rules that specify maximum peak current density (J_{peak}) through metal lines, per line width

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

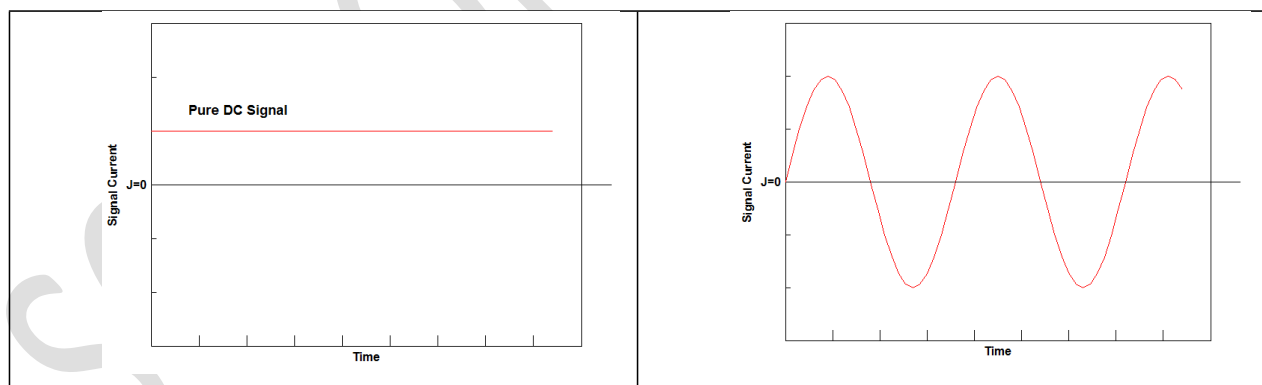
Under AC conditions with 100% duty cycle, it was observed [2], that during the negative half cycle, some of the damages created by the first half cycle were “healed”. That is, under bidirectional current stressing, some of the resistance increase due to current of the first half of the cycle is now reduced at the second half. This is the reason of using “lower average DC value” for calculation, although the nominal AC currents are higher than this average.

Under pulse conditions with lower duty cycle, there is limited self-heating [3], and the life time is inversely proportional to the send power of the time averaged current density. This is the reason referring to the root of the duty cycle in calculations.

Guidelines:

- (1) As a basic guideline, if possible w/o area penalty, it is highly recommended to have more holes (CS and Vias) as needed by the Electromigration calculations. This will reduce holes resistance and resistance variability and process robustness.
- (2) The number of CS located under M1 will affect the maximum current allowed to be pass in between the M1 and the active area. H. Kawasaki *et al.*, [4], investigated the effect of additional contacts placed in-line on the Electromigration failure time of the M1 (Ti/TiN/Al-Cu/TiN stack). The normalized maximum current as function of number of contacts showed, that 2 and 3 contacts increased the maximum current by only +35%, and 69%, respectively. Based on that, a simple reliability rule can be set, that the normalized maximum current allowed to be extracted from a line of n CS, is only $\sqrt{n}$. It is also recommended, if possible, to place the n contacts in parallel and not in line.

14.3 Definitions



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

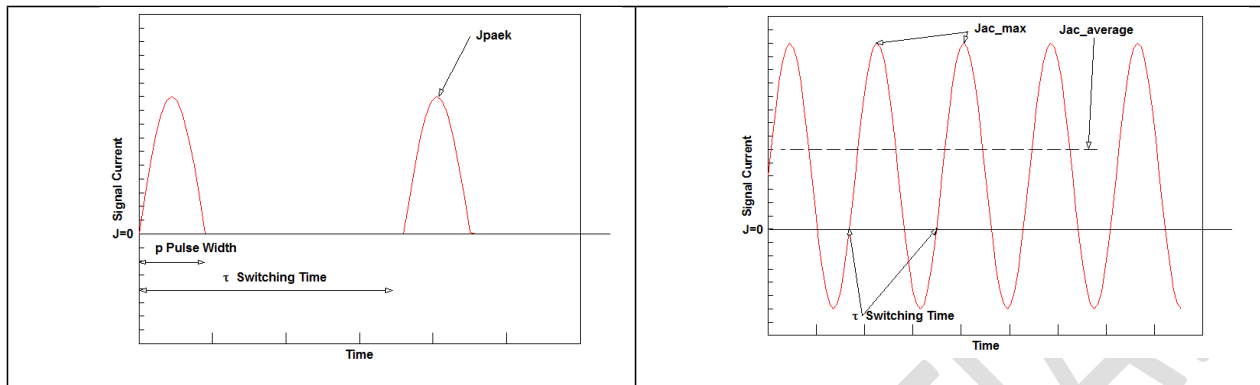


Figure 14-1: (top-left), Pure DC current, (top-right) Terminology for AC operational conditions, having 100% duty cycle, with the relevant parameters used for calculations, (bottom-left) Terminology of non-pure AC operation, with duty cycle that is <100% for J_{peak}, (bottom right) example for non-pure AC condition with 100% duty cycle.

- Under DC conditions, the current applied (J_{DC}) should be below the maximum current limits (J_{DC_max}),
- In AC conditions, τ is the operational switching time, s is switching factor, and f is the current operating frequency. $f = 1/\tau$
- Under non pure AC conditions, at 100% duty cycle, the “equivalent” average DC current (J_{AC_avg}) and can be calculated as:

$$J_{AC_avg} = \frac{s}{\tau} \int_0^{\tau} j(t) dt$$

Note that J_{AC_avg} is NOT relevant for the case of “pure AC”. For Pure AC, we are referring to a fully symmetrical wave around J=0 (that will yield J_{AC_avg}=0). At this case, calculation should be made using RMS.

- Under pure AC conditions and 100% duty cycle, the Root-Mean-Square current density, J_{RMS}, (also known as Joule heating), is calculated as next:

$$J_{RMS} = \sqrt{s f \int_0^{\tau} J^2 dt}$$

- Under AC conditions that do NOT have 100% duty cycle, calculation of the peak current (J_{peak}), is depend on the wave-shape and the duty cycle %. The duty cycle is calculated based on the ratio of pulse width (p) and the overall operational switching time (τ):

$$DutyCycle = \frac{p}{\tau}$$

and J_{peak} is calculated, for all metals using:

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

$$J_{peak}(Al, A/cm^2) = 5 \times 10^6 (A/cm^2) / \sqrt{p/\tau}$$

Note that J_{peak} , is having an upper limit, as listed below (see section 14.7).

- Effective metal line width: in all calculations, use metal effective width, W_{eff} for calculations. Effective line width should be taken as the width after subtraction of slits width:

$$\text{Effective Line Width} = W - W_{slit}$$

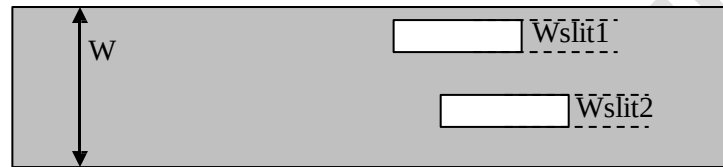


Figure 14-2: Example for the effective metal width. In this example, $W_{eff} = W - W_{slit1} - W_{slit2}$

14.4 Step-by-Step Current Calculation procedure for Metal lines and holes

Calculation should use the limits table listed below:

- For DC:** For power nets for example, compare the DC current to the maximum limits as defined in section 14.5.1.
- For AC (non pure AC, with 100% Duty cycle):** For signal net for example, At this case, calculate the average current (J_{AC_avrg}) and use it to compare to the limits specified for DC (section 14.5.1)
- For AC (Pure AC, with 100% Duty Cycle):** calculate J_{RMS} and compare to the maximum limits as defined in section 14.5.1.
- For AC (non Pure AC, with Duty Cycle <100%):** calculation of J_{peak} is done, for all metals using 14.7

Designer must satisfied the overall Electromigration conditions listed below.

14.5 Current limits for DC (J_{DC_max}) and Average DC (J_{AC_avgr})

14.5.1 Metal Lines max DC currents

The I_{dc} limits for holes and metals lines can be adjusted for different junction temperatures and lifetime criteria. The change is relative to the consumer LT of 100,000hr's provided at temperature of 110degC. The



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

temperature adjustment is not applicable for the J_{RMS} limits since J_{RMS} values given below, are applicable for any temperature.

14.5.1.1 BEOL Metal Lines max DC current density. Units: mA/um									
Temp	75°C	85°C	95°C	105°C	110°C	115°C	125°C	140°C	150°C
M1 and MI	2.92	2.11	1.55	1.16	-	0.88	0.67	0.46	-
TOP_M	4.67	3.36	2.48	1.85	-	1.4	1.08	0.74	-
MT (20kÅ Al Thick top Metal)	17.86	12.06	8.33	5.86	-	4.20	3.06	1.96	-
MT3 (28kÅ Al Thick top Metal)	21.74	14.69	10.14	7.14	-	5.11	3.73	2.39	-

The J_{DC_max} values listed in 14.5.1.1, are the experimental results based on limited amount of data, and used a limited number of test structures configurations. Based on that, design should be made using the J_{DC_max} ONLY. It is NOT allowed having a design, with current densities beyond the numbers listed in 14.5.1.1.

Enhancement of J_{DC_max} for short lines and large widths: Wider metal lines, shows higher EM LT, due to larger grain size. In addition, shorter lines also shows higher EM LT, due to Blech effect [5]. The following table, provide an enhancement factor, that can be used to multiple 110degC J_{DC_max} . This enhancement factor is not available yet and will provided at the next release.

The temperature adjustment factor is applicable for the junction temperature range of 105degC ~ 150degC. For J_{max} at junction temperature lower then 105degC – please use the value for 105degC.

14.5.1.2 Temperature Adjsutment Factor and Time Adjsutment Factor for Metals		
Metal	$J_{DC_max}@110^{\circ}C$ (mA/um)	Other temperature >110°C, Tmax in Kelvin, TAF range
M1 and MI	1.02	Temp. Adjustment factor = Exp [(4885/Tmax) – 12.75], 75degC ~ 150degC Time adjustment factor= (100,000/EOL actual)^0.53 . EOL_actual in hours
TOP_M	1.62	
MT (20kÅ Al Thick top Metal)	4.93	
MT3 (28kÅ Al Thick top Metal)	6	

The **temperature adjustment factor** is applicable only above 110°C. That is, the use of the product below 110°C does not mean that the maximum current can be increased. In the same manner, the time adjustment factor cannot be used to exceed the maximum allowed current values given for 110degC.

The **Time adjustment factor** is applicable for the case less than 100,000 hours are required for reliability



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

14.5.2 Contacts and Vias Max DC and AC-lavg Current

The following table shows the maximum contact and vias current per hole, as a function of the temperature.

14.5.2.1 BEOL Contacts and vias max DC and AC-lavg current. Units: mA/hole								
Temp	75°C	85°C	95°C	105°C	115°C	125°C	140°C	150°C
Contact(CS)	1.55	1.12	0.82	0.61	0.46	0.36	0.25	-
VI	0.82	0.59	0.43	0.32	0.25	0.19	0.13	-
VL	2.22	1.60	1.18	0.88	0.67	0.51	0.35	-

14.5.2.2 Temperature Adjustment Factor for Holes		
Holes	Jdc@110°C (mA/hole)	Other temperature >110°C, Tmax in Kelvin, TAF range
CS	0.55	Temp. Adjustment factor = Exp [(4141/Tmax) – 10.85], 75degC ~ 150degC
VI	0.28	Temp. Adjustment factor = Exp [(4091/Tmax) – 10.68], 75degC ~ 150degC (VF Temp range: 75degC ~ 125degC)
VL	0.78	

14.6 Root-Mean-Square Current (AC-Irms) limit

The RMS current, (JRMS) sets a limiting factor due to Joule heating associated with current flow through the metal lines. JRMS current may exceed JAC_avg, and cause wear out. The actual JRMS value for a given waveform can be calculated as explained above.

It has been found that the current density > 2x10⁶ A/cm² may induce failure due to Joule heating. This value is used for M1, while for other metal levels - a value of 1x10⁶ A/cm² is used.

14.6.1 Root-Mean-Square Current (Irms) limit	
Conductor	Maximum current per μm conductor width
M1	9mA/um
MI	4.5mA/um
ML	9mA/um
MT	20mA/um
MT3	28mA/um

Due to weak temperature dependency of AC-J_{RMS}, the temperature dependency is not provided

14.7 Current limits for Jpeak

Designer should use the lower value from the two steps calculations. Step-

1: Calculation of Jpeak is as listed above.

Step-2: Overall limits for:

- Al lines, for any temperature, Jpeak should be limited to: M1, MI, ML: <18mA/um; MT: <40mA/um, MT3: 56mA/um.
- Contact, for any temperature, Jpeak should be limited to <3mA/hole



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

- Via, for any temperature, J_{peak} should be limited to $<4\text{mA}/\text{hole}$

Based on this calculation, line width can be set.

Calculation example: Assumed duty cycle is 0.1 (10%), based on the *guideline* below, we will use a value of $5\text{E}5\text{A}/\text{cm}^2$: we get: $J_{peak} = 5\text{E}5/0.31 = 1.6\text{E}6\text{ A}/\text{cm}^2 = 0.016\text{A}/\mu\text{m}^2$. For ML, having a thickness of $0.93\mu\text{m}$, it will be $16*0.93=15\text{mA}/\mu\text{m}$ ($<18\text{mA}/\mu\text{m}$ as upper limit), and a value of 15mA need to be use. For the same example, for a duty cycle of 0.02 (2%), we get $J_{peak} = 5\text{E}5/0.14 = 3.5\text{E}6\text{ A}/\text{cm}^2 = 35.4\text{mA}/\mu\text{m}^2$. For ML, having a thickness of $0.93\mu\text{m}$, it will be $35.4*0.93=32.9\text{mA}/\mu\text{m}$ ($>18\text{mA}/\mu\text{m}$ as upper limit), and a value of 18mA need to be use.

Guidelines: Please note that the value of $5\text{E}6\text{ A}/\text{cm}^2$, is the value for sever DAMAGE and Al lines break (open) due to very high temperature (metal is almost not solid). Based on that, it is highly RECOMMENDED using values that are 1/10 (or lower) from the above.

References:

- [1] JEDEC/FSA Joint publication, "Foundry Process Qualification Guidelines (wafer fabrication manufacturing sites)," JP001.01.
- [2] J. Tao, N. W. Cheung and C. Hu, "Metal Electromigration Damage Healing Under Bidirectional Current Stress," *IEEE Electron Device Lett.*, vol. 14, no. 2, pp. 554, 1993.
- [3] B. K. Liew, N. W. Cheung and C. Hu, "Projecting interconnect Electromigration lifetime for arbitrary current waveforms," *IEEE Trans. Electron Device*, vol. 37, no. 5, pp. 1343-1351, 1990.
- [4] H. Kawasaki, C. Lee and F. Pintchovski, "The effect of test structure and stress condition on electromigration failure," Proceeding of the electrochemistry society Symposium on Relability of Metals in Interconnects, Edited by H. S. Rathore, vol. 95-3, 1995, pp. 165-177.
- [5] Milton Ohring, *Reliability and Failure of Electronic Materials and Devices*, Academic Press, New York, 1998.

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

15 Device Tables: Specification of Electrical Parameters

The following table includes the main electrical targets, together with the Upper Spec Limits (USL) and Lower Spec Limits (LSL). These limits represent the most extreme conditions of the process, and may “compensate” for any differences related to other effects like electrical test structure sensitivity, test structure location on the wafer, tester sensitivity, measurement conditions, and more. These target and limits (only) will be used for production control monitoring, SPICE simulation, xRC corner and PDK.

The distribution data presented here, extracted using large amount of production wafers. In general, we used ~ quarter of a million sites (~40kwafers), for calculation. However, the 1Sigma data is for information only. 1Sigma(%) calculated as: $(1Std_Dev/Target)*100\%$. For example, the resistance of WN-Under STI, 1Sigma=31.3hm/sq is $(31.3/1000)*100\%=3.13\%$. Note that this sigma is NOT the matching distribution. For matching data – please refer to platform characterization file.

Designer needs to use the USL and LSL limits, for simulation. Designer also has the possibility, to simulate using more tight distributions, based on the statistical data provided here. Note, that the on-silicon nominal values and distribution are also depend by several parameters like W, L, space, coverage and more.

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

15.1 Resistance of Salicided AA and Poly, CS and Vias, Metals

Item No.	Parameter	Unit	Conditions/Criteria	Typical	Low Limit	High Limit	Evaluation Method / Meas. Conditions
Junctions, and Active Area							
P-SubRS.1	P-Sub Rs	ohm cm	---		1	2	As per silicon vendor specification
NASR.11	N+ AA Rs (Wide Line)	ohm/sq	W>0.38um, STI Periphery.	7	1	15	Kelvin Pattern
NASR.12	N+ AA Rs (Min Line)	ohm/sq	W=0.22um, STI Periphery.	7	1	15	Kelvin Pattern. 1Sigma=0.30 (4.28%)
PASR.11	P+ AA Rs (Wide Line)	ohm/sq	W>0.38um, STI Periphery.	5	1	15	Kelvin Pattern
PASR.12	P+ AA Rs (Min Line)	ohm/sq	W=0.22um, STI Periphery.	5	1	15	Kelvin Pattern. 1Sigma=0.24 (4.8%)
NGSR.11	N+ GC Rs (Wide Line)	ohm/sq	W>1.00um	6	1	20	Kelvin Pattern
NGSR.12	N+ GC Rs (Tr Gate Poly)	ohm/sq	W=0.18um	6	1	20	Kelvin Pattern. 1Sigma=0.31 (5.2%)
PGSR.11	P+ GC Rs (Wide Line)	ohm/sq	W>1.00um	5	1	20	Kelvin Pattern
PGSR.12	P+ GC Rs (Tr Gate Poly)	ohm/sq	W=0.18um	5	1	20	Kelvin Pattern. 1Sigma=0.30 (6.0%)
Backend: contacts and vias' resistance, Metal resistance							
M1RS.11	M1 Rs (first-metal)	mOhm/sq	W=0.23um	80	40	120	Kelvin Pattern
MIRS.11	MI Rs (inter-metal)	mOhm/sq	W=0.28um	80	40	120	Kelvin Pattern
MLRS.11	ML Rs (last-metal)	mOhm/sq	W=0.44um	40	24	56	Kelvin Pattern
MTRS.11	MT Rs (20kÅ Al Thick Top Metal)	mOhm/sq	W=1.2um	15	9	21	Kelvin Pattern
MT3RS.11	MT3 Rs (28kÅ Al Thick Top Metal)	mOhm/sq	W=2.5um	9.3	8.8	11.26	Kelvin Pattern 1sigma=0.21(2%)
NAARC.1	N+ AA Contact Resistance	ohm	Contact=0.22um	10	1	<30	Four-terminal Kelvin cross bridge. 1Sigma=0.65 (6.5%)
PAARC.1	P+ AA Contact Resistance	ohm	Contact=0.22um	10	1	<30	Four-terminal Kelvin cross bridge. 1Sigma=0.56 (5.6%)
NGCRC.1	N+ GC Contact Resistance	ohm	Contact=0.22um	7	1	<30	Four-terminal Kelvin cross bridge. 1Sigma=0.69 (9.8%)
PGCRC.1	P+ GC Contact Resistance	ohm	Contact=0.22um	7	1	<30	Four-terminal Kelvin cross bridge. 1Sigma=0.71 (10%)
VIRC.1	VI (first via) Resistance	ohm	Via=0.26um	6	1	<20	Four-terminal Kelvin cross bridge. 1Sigma=0.69 (11%)
VLRC.1	VL (top-via) Resistance	ohm	Via=0.36um	2.5	1	<13	Four-terminal Kelvin cross bridge. 1Sigma=0.25 (10%)

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

15.2 Transistors: 1.8V, 3.3V, 5V

Item No.	Parameter	Unit	Conditions/Criteria	Typical	Low Limit	High Limit	Evaluation Method / Meas. Conditions
1.8V(Std Vth), Typical Size: L=0.18um, W=10um, nMOSFET, Thin Gate Oxide=40Å, CV measured at inversion.							
N18SVT.VT.1	Vthcc	V	Vd=0.1V; Vb=Vs=0	0.45	0.35	0.55	Constant current method: Vth=Vg @Id=0.1[uA]*(W/L). 1Sigma=0.011 (2.6%)
N18SVT.ID.1	Drive Current	uA/um	Vds=Vgs=1.8V	600	510	690	Transistor measurement. 1Sigma=17.8 (2.97%)
N18SVT.BE.1	Body Effect (Vth Shift)	mV	Vds=0.10V; Vbs=-0.9V	82		<120	Transistor measurement
N18SVT.IS.1	Isup max	nA/um	Vds=2.0V	0.18		<0.30	Transistor measurement
N18SVT.IO.1	Off Leak	pA/um	Vds=2.0V	15		<100	Transistor measurement
N18SVT.BV.1	Surface Breakdown	V	Ids=10 nA/um		3.6		Transistor measurement, W=10um
N18SVT.JB.1	Junction Breakdown	V	Field edge		>10.5		Diode measurement
1.8V(Std Vth), Typical Size: L=0.18um, W=10um, pMOSFET, Thin Gate Oxide=40Å, CV measured at inversion.							
P18SVT.VT.1	Vthcc	V	Vd=0.1V; Vb=Vs=0	-0.48	-0.38	-0.58	Constant current method: Vth=Vg @Id=0.1[uA]*(W/L). 1Sigma=0.013 (2.6%)
P18SVT.ID.1	Drive Current	uA/um	Vds=Vgs=-1.8V	260	220	300	Transistor measurement. 1Sigma=13.6 (5.237%)
P18SVT.BE.1	Body Effect (Vth Shift)	mV	Vds=-0.10V; Vbs=0.9V	180		<220	Transistor measurement
P18SVT.IS.1	Isup max	nA/um	Vds=2.0V	0.0015		<002	Transistor measurement
P18SVT.IO.1	Off Leak	pA/um	Vds=-2.0V	15		<100	Transistor measurement
P18SVT.BV.1	Surface Breakdown	V	Ids=-10nA/um		-3.6		Transistor measurement, W=10um
P18SVT.JB.1	Junction Breakdown	V	Field edge		>10		Diode measurement
1.8V(High Vt), Typical Size: L (Drawn)=0.18um, W=10um, nMOSFET, Tox=40Å, CV measured under inversion.							
N18HVt.VT.1	Vth	V	Vd=0.1V; Vb=Vs=0	0.57	0.46	0.62	Constant current method: Vth=Vg@Id=0.1[uA]*(W/L)
N18HVt.ID.1	Drive Current	uA/um	Vd=Vg=1.8V; Vb=Vs=0	514	453	607	
N18HVt.IO.1	Off Leak	pA/um	Vds=2.0 V	1.5		<10	
1.8V(High Vt), Typical Size: L (Drawn)=0.18um, W=10um, pMOSFET, Tox=40 Å, CV measured under inversion.							
P18HVt.VT.1	Vth (Nominal device)	V	Vd=0.1V; Vb=Vs=0	-0.57	-0.46	-0.67	Constant current method: Vth=Vg@Id=0.1[uA]*(W/L)
P18HVt.ID.1	Drive Current	uA/um	Vd=Vg=1.8V; Vb=Vs=0	226	179	266	
P18HVt.IO.1	Off Leak	pA/um	Vds=2V	1.5		<10	

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Item No.	Parameter	Unit	Conditions/Criteria	Typical	Low Limit	High Limit	Evaluation Method / Meas. Conditions
1.8V(Low Vt), Typical Size: L (Drawn)=0.18um, W=10um, nMOSFET Tox=40Å, CV measured under inversion.							
N18LVt.VT.1	Vth	V	Vd=0.1V; Vb=Vs=0	0.35	0.3	0.4	Constant current method: Vth=Vg@Id=0.1[uA]*(W/L)
N18LVt.ID.1	Drive Current	uA/um	Vd=Vg=1.8V; Vb=Vs=0	700	610	790	
N18LVt.IO.1	Off Leak	pA/um	Vds=2.0 V	155		<5000	
1.8V(Low Vt), Typical Size: L (Drawn)=0.18um, W=10um, pMOSFET, Tox=40 Å, CV measured under inversion.							
P18LVt.VT.1	Vth (Nominal device)	V	Vd=0.1V; Vb=Vs=0	-0.38	-0.32	-0.44	Constant current method: Vth=Vg@Id=0.1[uA]*(W/L)
P18LVt.ID.1	Drive Current	uA/um	Vd=Vg=1.8V; Vb=Vs=0	310	270	350	
P18LVt.IO.1	Off Leak	pA/um	Vds=2V	155		<5000	
1.8V(Native Vth), Typical Size: L=0.50um, W=10um, nMOSFET, Thin Gate Oxide=40A, CV measured at inversion.							
NATLV.VT.1	Vth	V		-0.01	-0.25	0.2	Gm Extrapolation, Not available on EPI wafers
NATLV.ID.1	Drive current	uA/um	Vds=1.8V	620	435	810	Transistor measurement
3.3V(Std Vth), Typical Size: L=0.35um, W=10um, nMOSFET, Thick Gate Oxide=70A, CV measured at inversion.							
NHVSVT.VT.1	Vth	V	Vd=0.1V; Vb=Vs=0	0.72	0.62	0.82	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V). 1Sigma=0.011 (1.5 %)
NHVSVT.ID.1	Drive Current	uA/um	Vds=Vgs=3.3V	600	510	690	Transistor measurement. 1Sigma=0.011 (2.6%)
NHVSVT.BE.1	Body Effect (Vth Shift)	mV	Vds=0.10V; Vbs=-1.65V	130		<200	Transistor measurement
NHVSVT.IS.1	Isub max	nA/um	Vds=3.6V	2.1		<3.0	Transistor measurement
NHVSVT.IO.1	Off Leak	pA/um	Vds=3.6V	0.20		<10	Transistor measurement
NHVSVT.BV.1	Surface Breakdown	V	Ids=50nA/um		6.0		Transistor measurement, W=10um
3.3V(Std Vth), Typical Size: L=0.30um, W=10um, pMOSFET, Thick Gate Oxide=70A, CV measured at inversion.							
PHVSVT.VT.1	Vth	V	Vd=0.1V; Vb=Vs=0	-0.63	-0.53	-0.73	Linear Vth: Gm max extrapolation, -Vds/2 (Vds=0.1V). 1Sigma=0.010 (1.47 %)
PHVSVT.ID.1	Drive Current	uA/um	Vds=Vgs=3.3V	300	255	345	Transistor measurement. 1Sigma=10 (3.33%)
PHVSVT.BE.1	Body Effect (Vth Shift)	mV	Vds=-0.10V; Vbs=1.65V	300		<370	Transistor measurement
PHVSVT.IS.1	Isub max	nA/um	Vds=3.6V	0.22		<0.30	Transistor measurement
PHVSVT.IO.1	Off Leak	pA/um	Vds=3.6V	1.0		<10	Transistor measurement
PHVSVT.BV.1	Surface Breakdown	V	Ids=-1uA/um		-6.0		Transistor measurement, W=10um
3.3V(Native Vth), Typical Size: L=1.20um, W=10um, nMOSFET, Thick Gate Oxide=70A, CV measured at inversion.							
NATHV.VT.1	Vth	V		-0.15	-0.3	0.05	Gm Extrapolation, Not available on EPI wafers

Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

Item No.	Parameter	Unit	Conditions/Criteria	Typical	Low Limit	High Limit	Evaluation Method / Meas. Conditions
NATHV.ID.1	Drive current	uA/um	Vds=3.3V	560	390	730	Transistor measurement
5V (std Vt, 5V Characteristics), Typical Size: L=0.6um, W=10um, nMOSFET, Thick Gate Oxide=109A, CV measured in inversion							
N50SVT.VT.1	Vth	V		0.85	0.75	0.95	Vth = Vg@ Gmax method @ Vds = 0.1V
N50SVT.ID.1	Drive Current	uA/um	Vds=5V	550	470	630	
N50SVT.IO.1	Off Leak	pA/um	Vds=5V	2.4		≤ 30	Formal for SPICE 0.02pA/um, 1Sigma=2.53 (50 %)
N50SVT.BV.1	Breakdown Voltage	V			>7		
5V (std Vt, 5V Characteristics), Typical Size: L=0.5um, W=10um, pMOSFET, Thick Gate Oxide=109A, CV measured in inversion							
P50SVT.VT.1	Vth	V		-0.80	-0.70	-0.90	Vth = Vg@ Gmax method @ Vds = 0.1V
P50SVT.ID.1	Drive Current	uA/um	Vds=5V	300	255	345	
P50SVT.IO.1	Off Leak	pA/um	Vds=5V	1.4		≤ 30	Formal for SPICE 0.4pA/um, 1Sigma=0.52 (40 %)
P50SVT.BV.1	Breakdown Voltage	V			>7		
5V(Native Vth), Typical Size: L=1.50um, W=10um, nMOSFET, Thick Gate Oxide=109A, CV measured at inversion.							
NATH5V.VT.1	V			-0.09	-0.3	0.1	Gm Extrapolation, Not available on EPI wafers
NATH5V.ID.1	uA/um	Vds=5V		520	360	680	Transistor measurement

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

15.3 Device Parameters for Analog Module

Table 15-1: Device Parameters for Analog Resistors, MOSCAP, MFC, MIM

Analog Devices						
Item No.	Parameter	Unit	Typical	Low Limit	High Limit	Evaluation Method / Meas. Conditions
1kohm/sq HIPO resistor	Resistance	Ohm/ sq.	1000	880	1120	W/L=2um/100um. 1Sigma=16.7 (1.67%)
2kohm/sq HIPO resistor	Resistance	Ohm/ sq.	2000	2300	1700	W/L=2um/100um.
NMOPO resistor	Resistance	Ohm/ sq.	400	310	490	W/L=2um/100um. 1Sigma=19.7 (4.93%)
PMOPO resistor	Resistance	Ohm/ sq.	310	260	360	W/L=2um/100um. 1Sigma=6.79 (2.19%)
N+ AA Diffusion resistor	Resistance	Ohm/ sq.	70	50	90	W/L=2um/100um. 1Sigma=1.45 (2.07%)
P+ AA Diffusion resistor	Resistance	Ohm/ sq.	120	80	160	W/L=2um/100um. 1Sigma=2.09 (1.74%)
N-Well resistor under AA	Resistance	Ohm/ sq.	450	350	550	W/L=20um/100um
N-Well resistor under STI	Resistance	Ohm/ sq.	1000	600	1400	W/L=20um/100um. 1Sigma=31.3 (3.13%)
MOSCAP Accumulation Capacitor						
1.8V Nwell	Capacitance	fF / um ²	9.15	8.6	9.70	L=5um, W=5um, in accumulation. 1Sigma=0.09 (0.98%)
3.3V Nwell	Capacitance	fF / um ²	5.10	4.55	5.65	L=5um, W=5um, in accumulation. 1Sigma=0.07 (1.41%)
5V Nwell	Capacitance in accumulation	fF / um ²	3.11	2.77	3.45	L=5um, W=5um, n=400 .Vgb=5.0
MFC (Metal-Fringe-Capacitor)						
CM3 (M13 MFC) using FSG dielectric	Capacitance	fF / um ²	0.50	0.38	0.62	
CM4 (M14 MFC) using FSG dielectric	Capacitance	fF / um ²	0.67	0.51	0.83	
CM5 (M15 MFC) using FSG dielectric	Capacitance	fF / um ²	0.84	0.64	1.04	
USGCM3 (M13 MFC) using USG dielectric	Capacitance	fF / um ²	0.57	0.43	0.7	
USGCM4 (M14 MFC) using USG dielectric	Capacitance	fF / um ²	0.76	0.58	0.94	
USGCM5 (M15 MFC) using USG dielectric	Capacitance	fF / um ²	0.95	0.73	1.18	

Design Rules for SL18 Standard Logic Process

Doc. Name: DR2 0018 SL

Rev. Num: 0

Analog Devices						
Item No.	Parameter	Unit	Typical	Low Limit	High Limit	Evaluation Method / Meas. Conditions
Metal-Insulator-Metal (MIM) Capacitor						
MIM Capacitor (standard capacitance) 1 fF/um2	Area Capacitance	fF / um2	1.0	0.83	1.17	Type B, 25x25um. 1Sigma=0.015 (1.52%)
	Periphery Capacitance	fF / um	0.11			
Single level, high C MIM Capacitor 1.7 fF/um2	Area Capacitance	fF / um ²	1.75	1.45	1.95	Type B, 25x25um. 1Sigma=0.043 (2.5%)
	Periphery Capacitance	fF / um	0.12			
Stacked, high C MIM Capacitor 2x1.7 fF/um2=3.4fF/um2	Area Capacitance	fF / um ²	3.40	2.70	4.10	Stacked Type B, 25x25um. 1Sigma=0.032 (0.93%)
	Periphery Capacitance	fF / um	0.19			
Ultra High Density MIM Capacitor, single plate, 2.8 fF/um2	Area Capacitance	fF / um ²	2.75	2.3	3.2	Type B, 25x25um. 1Sigma=0.05 (1.8%)
	Periphery Capacitance	fF / um	0.054			
Stacked Ultra High Density MIM Capacitor, Stacked 5.6 fF/um2	Area Capacitance	fF / um ²	5.5	4.6	6.4	Stacked Type B, 25x25um. 1Sigma=0.1 (1.8%)
	Periphery Capacitance	fF / um	0.108			

Design Rules for SL18 Standard Logic Process
Doc. Name: DR2 0018 SL
Rev. Num: 0

15.4 BJT's (V_{PNP} and V_{NPN}) for all 0.18um Technologies

Item No.	Parameter	Unit	Typical	Low Limit	High Limit	Evaluation Method / Meas. Conditions
V _{PNP} _1.8V	Current Gain		2.54	1.4	3.6	Emmitter Area = 5um x 5um, 1Sigma=0.1 (4%)
V _{PNP} _1.8V	BV _{CEO}	V	-15.22	-14.13	-16.31	Emmitter Area = 5um x 5um, 1Sigma=0.18 (1.%)
V _{PNP} _3.3V	Current Gain		2.54	1.4	3.6	Emmitter Area = 5um x 5um, 1Sigma=0.1 (4%)
V _{PNP} _3.3V	BV _{CEO}	V	-15.22	-14.13	-16.31	Emmitter Area = 5um x 5um, 1Sigma=0.18 (1.%)
V _{NPN} (for 1.8V, 3.3V, 5V)	Current Gain		5.5	3.5	7.5	Emmitter Area = 5um x 5um 1Sigma=0.33 (6%)
V _{NPN} (for 1.8V, 3.3V, 5V)	BV _{CEO}	V	14.5	>12		Emmitter Area = 5um x 5um, 1Sigma=0.18 (1.2%)



Design Rules for SL18 Standard Logic Process	Doc. Name: DR2 0018 SL
	Rev. Num: 0

16 Appendix A: Mask List

All generation procedures will be done by SCL. Please refer to section 7.8 for the masks needed to be

Mask Name	Description	Data	Generation Flow From Drawn Layers to mask by SCL MDP (sizing values are NOT listed)	GDS # (MDP Output)	mask	OPC	Mask Made
AA	ACTIVE AREA	Dark	AA + dummy AA	9dt0	DUV	No	DRW
Option-WB	Deep N-well (Triple Well)	Clear	WB	4, dt0	MUV	No	DRW
WP	P-WELL	Dark	WN + VNN+ DEV_AREA(hg)	2dt0	MUV	No	GEN
CNS(CP)	complimentary VTN IMPLANT FOR 1.8V STD Tr's	Clear	CN_gen= [(((XN not WN) not AREA2) not VNN) not ((DEV_AREA;res and (GC not AA))size + 0.26)] or (((XN not WN and DEV_AREA;res and (AA not GC) and NS) sized +0.26) and XN) CN=CN_gen or CN(lc)	11dt0	MUV	No	GEN
WN	N-WELL	Clear	WN=WN	3dt0	MUV	No	DRW
CPS (CP)	complimentary VTP IMPLANT FOR 1.8V STD Tr's	Clear	{(((XP and WN) not AREA2) or ((XP and WN and DEV_AREA;res and ((AA not GC) or (GC not AA)) and NS) sized +0.26) and XP)} NOT CP(kt)	12dt0	MUV	No	GEN
Option-CNH	Vt for 1.8V Low Leakage (HVt) NMOS devices	Clear	CNH=CNH1 or CNH2 or CNH_mk CNH_1=(((XN not WN) not AREA2) NOT VNN) NOT DEV_AREA(rs); NOT DEV_AREA(do)); NOT DEV_AREA(bt)); AND (DEV_AREA (lp)) CNH2=CNH(lc) or CNH(shv)	10dt0	MUV	No	GEN/DRW
Option-CPH	Vt for 1.8V Low Leakage (HVt) PMOS devices Or 1.8V Supper HVt Pmos device	Clear	CPH=CPH1 or CPH2 or CPH_mk CPH_1=(((XP and WN) not AREA2) not VNN) not DEV_AREA(rs); not DEV_AREA(do)); not DEV_AREA(bt)); and (DEV_AREA (lp)) CPH2=CPH(lc)or CPH(shv)	65, dt0	MUV	No	GEN/DRW
Option-CNL	1.8V low Vt NMOS Devices	Dark	CNL= [AREA2 or VNN or CNL or WN]	162, dt0	MUV	No	GEN/DRW
Option-CPL	1.8V low Vt PMOS Devices	Dark	CPL=[AREA2 or VNN or CPL] or NOT WN	163, dy0	MUV	No	GEN/DRW
LA	HIGH VOLTAGE BLOCKING	Dark	AREA2	72dt0	MUV	No	DRW
GC	CONTROL GATE POLY	Dark	GC+DUMMY GC	8dt0	DUV	Yes	DRW
LP	1.8V PLDD	Clear	Use CPS Mask	29dt0	MUV	No	GEN
HP	3.3V/ 5V PLDD	Clear	{((XP and WN) AND AREA2) not ((DEV_AREA;res and ((GC not AA) or (AA not GC)) and NS) size + 0.18)} NOT HP(kt)	24dt0	MUV	No	GEN
HN	3.3V/ 5V NLDD	Clear	HN_gen= {(((XN not WN) AND AREA2) not ((DEV_AREA;res and ((GC not AA) or (AA not GC)) and NS) sized +0.26))} NOT HN(kt) HN=HN_gen or HN(lc)	23dt0	MUV	No	GEN
LN	1.8V NLDD	Clear	LN_gen= {(((XN not WN) not AREA2) not ((DEV_AREA;res and NS and (GC not AA)) size + 0.26)) or (((XN not WN and DEV_AREA;res and (AA not GC) and NS) sized +0.26) and XN)} NOT LN(kt) LN=LN_gen or LN(lc)	30dt0	MUV	No	GEN
XN	N+ IMPLANT	Clear	XN	32dt0	MUV	No	DRW
XP	P+ IMPLANT	Clear	XP	31dt0	MUV	No	DRW
Option-MR	HIPO implant for 1kohm/sq HIPO Resistor	Clear	MR_gen=MR not MR2	87dt0	MUV	No	DRW
Option-MR2	HIPO implant for 2kohm/sq HIPO Resistor	Clear	MR2_gen=MR and MR2	164dt0	MUV	No	GEN/DRW
NS	SALICIDE BLOCK	Dark	NS	88dt0	MUV	No	DRW
CS	CONTACT	Clear	CS	34dt0	DUV	No	DRW
M1	METAL 1	Dark	M1 + dummy M1 not M1(ms)	40dt0	DUV	Yes	DRW



Design Rules for SL18 Standard Logic Process	Doc. Name:	DR2 0018 SL
	Rev. Num:	0

V2	VIA 2 (inter Via)	Clear	V2	41dt0	DUV	No	DRW
M2	METAL 2 (inter metal)	Dark	M2 + dummy M2 not M2(ms)	42dt0	DUV	No	DRW
V3	VIA 3 (Inter Via)	Clear	V3	43dt0	DUV	No	DRW
M3	METAL 3 (inter metal)	Dark	M3 + dummy M3 not M3(ms)	44dt0	DUV	No	DRW
V4	VIA 4 (Inter Via)	Clear	V4	45dt0	DUV	No	DRW
M4	METAL 4 (inter metal)	Dark	M4 + dummy M4 not M4(ms)	46dt0	DUV	No	DRW
V5	VIA 5 (inter Via)	Clear	V5	47dt0	DUV	No	DRW
BCE	Top plate of bottom MIM capacitor	Dark	{{(CE and SPC_AREA(mim) } size by 0.05	98dt0	MUV	No	GEN
CE	MIM Capacitor	Dark	CE size by 0.05	99dt0	MUV	No	DRW
M5	METAL 5 (inter metal)	Dark	M5 + dummy M5 not M5(ms)	48dt0	DUV	No	DRW
VL	Top Via	Clear	TOP_V	49dt0	DUV	No	DRW
ML	Top Metal	Dark	TOP_M + dummy TOP_M not TOP_M (ms)	50dt0	DUV	No	DRW
MT	20kÅ Al Thick Top Metal	Dark	MT + dummy MT not MT(ms)	59dt0	MUV	No	DRW
MT3	28kÅ Al Thick Top Metal	Dark	MT3 + dummy MT3 not MT3(ms)	184dt0	DUV	No	DRW
PD	Passivation opening for use with Aluminum top metal	Clear	PAD(dg) NOT PAD(kt)	68dt0	MUV	No	DRW

Table 16-1: Mask Generation Information (By MDP)

17 Revision History

NA



18 Annexure-1: Legal Device List as on 16-10-23

S. No	Device Name	Type	Device Description	SPICE Name	P-cell	LVS
1	MOSFETs	1.8V,Svt	1.8V SVt NMOS Transistor	n18	nmos_18	N
2	MOSFETs	1.8V,Svt	1.8V SVt PMOS Transistor	p18	pmos_18	P
3	MOSFETs	1.8V,Svt	1.8V SVt NMOS Transistor in IPW	n18	nmos_18_iso	N_ISO
4	MOSFETs	1.8V,Hvt-Low Leakage	1.8V HVt NMOS transistor for Low Leakage	n18hvtb3	n18hvt	N18HVT
5	MOSFETs	1.8V,Hvt-Low Leakage	1.8V HVt PMOS transistor for Low leakage	p18hvtb3	p18hvt	P18HVT
6	MOSFETs	1.8V,Hvt-Low Leakage	1.8V HVt NMOS transistor for Low leakage in IPW	n18hvtb3	n18hvt_iso	N18HVT_ISO
7	MOSFETs	1.8V,Native	1.8V NMOS Native Transistor	natlv	nmos_native	NA
8	MOSFETs	3.3V,Svt	3.3V SVt NMOS Transistor	nhv	nmos_33	NH
9	MOSFETs	3.3V,Svt	3.3V SVt PMOS Transistor	phv	pmos_33	PH
10	MOSFETs	3.3V,Svt	3.3V SVt NMOS Transistor in IPW	nhv	nmos_33_iso	NH_ISO
11	MOSFETs	3.3V,Native	3.3V NMOS Native Transistor	nathv	nmos_native_33	NB
12	MOSFETs	5VSvt	5V SVt NMOS transistor	n5v	nmos_5V	N5
13	MOSFETs	5VSvt	5V SVt PMOS transistor	p5v	pmos_5V	P5
14	MOSFETs	5VSvt	5V SVt NMOS transistor in IPW	n5v	nmos_5V_iso	N5_ISO
15	MOSFETs	5V,Native	5V NMOS Native Transistor	nat5v	nmos_native_5	NB5
16	BJT	1.8V, BJT	1.8V Vertical PNP 10um x 10um BJT	vd10	vpnp18_10	VD10
17	BJT	1.8V, BJT	1.8V Vertical PNP 5um x 5um BJT	vd5	vpnp18_5	VD5
18	BJT	1.8V, BJT	1.8V Vertical PNP 2um x 2um BJT	vd2	vpnp18_2	VD
19	BJT	1.8/3.3/5V, BJT	1.8/3.3/5V Vertical NPN 5um x 5um BJT	vdn5	vnnp18_5	VDN5
20	BJT	3.3/5V, BJT	3.3/5 Vertical PNP 10um x 10um BJT	vd10	vpnp33_10	VH10
21	BJT	3.3/5V, BJT	3.3/5 Vertical PNP 5um x 5um BJT	vd5	vpnp33_5	VH5
22	BJT	3.3/5V, BJT	3.3/5 Vertical PNP 2um x 2um BJT	vd2	vpnp33_2	VH
23	Capacitors	3.3V, Acc.Capacitor	1.8V Scalable Accumulation capacitor N+ Gate/ NW, 2 terminals	nwcaph2t	cnwnmos_thn2t	CL
24	Capacitors	1.8V, Acc.Capacitor	1.8V Scalable Accumulation capacitor N+ Gate/NW)3 terminals	nwcaph3t	cnwnmos_thn3t	CL3
25	Capacitors	3.3V, Acc.Capacitor	3.3V Scalable Accumulation capacitor, N+ Gate/NW,2 terminals)	nwcaph2t	cnwnmos_thk2t	CH
26	Capacitors	3.3V, Acc.Capacitor	3.3V Scalable Accumulation capacitor, N+ Gate/NW,3 terminals	nwcaph3t	cnwnmos_thk3t	CH3
27	Capacitors	5V, Acc.Capacitor	5V Accumulation capacitor, N+ Gate/NW.	ch53t	ch53t	CH53T
28	Capacitors	MIM cap	MIM Capacitor, 1fF/um2 capacitor	cmim_sq	cmim_sq	CB
29	Capacitors	MFC in USG	M15 USG 2T MFC	usgcm52t	usgcm52t	usgcm52t
30	Capacitors	MFC in USG	M13 USG 2T MFC	usgcm32t	usgcm32t	usgcm32t
31	Capacitors	MFC in USG	M15 USG 3T MFC	usgcm53t	usgcm53t	usgcm53t
32	Capacitors	MFC in USG	M13 USG 3T MFC	usgcm33t	usgcm33t	usgcm33t
33	Resistors	poly resistor	HIPO-1k High Ohmic P-type Poly w/o salicide 2 terminal resistor	rphpoly2t	rphpoly2t	RC
34	Resistors	poly resistor	HIPO-1k High Ohmic P-type Poly w/o salicide 3 terminal resistor	rphpoly3t	rphpoly3t	RC3
35	Resistors	poly resistor	2k High Ohmic P-type Poly w/o salicide 2 terminal resistor	mr22t	mr22t	MR22T
36	Resistors	poly resistor	2k High Ohmic P-type Poly w/o salicide 3 terminal resistor	mr23t	mr23t	MR23T



S. No	Device Name	Type	Device Description	SPICE Name	P-cell	LVS
37	Resistors	poly resistor	NMOPO–N-type Poly w/o salicide 2 terminals resistor	rnmpoly2t	rnmpoly2t	RE
38	Resistors	poly resistor	NMOPO–N-type Poly w/o salicide 3 terminals resistor	rnmpoly3t	rnmpoly3t	RE3
39	Resistors	poly resistor	PMOPO–P-type Poly w/o salicide 2 terminals resistor	rpmpoly2t	rpmpoly2t	RD
40	Resistors	poly resistor	PMOPO–P-type Poly w/o salicide 3 terminals resistor	rpmpoly3t	rpmpoly3t	RD3
41	Resistors	poly resistor	N-poly salicide 2 terminals resistor	rnlpoly2t	rnlpoly2t	RH
42	Resistors	poly resistor	N-poly salicide 3 terminals resistor	rnlpoly3t	rnlpoly3t	RH3
43	Resistors	poly resistor	P-poly salicide 2 terminals resistor	rplpoly2t	rplpoly2t	RG
44	Resistors	poly resistor	P-poly salicide 3 terminals resistor	rplpoly3t	rplpoly3t	RG3
45	Resistors	diffusion resistor	N+ AA Diffusion w/o salicide 2 terminals resistor	rnplus2t	rnplus2t	RB
46	Resistors	diffusion resistor	N+ AA Diffusion w/o salicide 3 terminals resistor,	rnplus3t	rnplus3t	RB3
47	Resistors	diffusion resistor	P+ AA Diffusion w/o salicide 2 terminals resistor	rpplus2t	rpplus2t	RA
48	Resistors	diffusion resistor	P+ AA Diffusion w/o salicide 3 terminals resistor	rpplus3t	rpplus3t	RA3
49	Resistors	diffusion resistor	N+ AA Diffusion salicide 2 terminals resistor	nasr2t	rnplus_sal2t	RK
50	Resistors	diffusion resistor	N+ AA Diffusion salicide 3 terminals resistor	nasr3t	rnplus_sal3t	RK3
51	Resistors	diffusion resistor	P+ AA Diffusion salicide 2 terminals resistor	pasr2t	rpplus_sal2t	RM
52	Resistors	diffusion resistor	P+ AA Diffusion salicide 3 terminals resistor	pasr3t	rpplus_sal3t	RM3
53	Resistors	well resistor	N-Well under AA 2 terminals resistor	rnwellaa2t	rnwell_AA2t	RF
54	Resistors	well resistor	N-Well under AA 3 terminals resistor	rnwellaa3t	rnwell_AA3t	RF3
55	Resistors	well resistor	N-Well under STI 2 terminals resistor	rnwellsti2t	rnwell_STI2t	RW
56	Resistors	well resistor	N-Well under STI 3 terminals resistor	rnwellsti3t	rnwell_STI3t	RW3
57	Resistors	Metal resistor	M1 resistor	rm1	rm1	R1
58	Resistors	Metal resistor	M2 resistor	rm2	rm2	R2
59	Resistors	Metal resistor	M3 resistor	rm3	rm3	R3
60	Resistors	Metal resistor	M4 resistor	rm4	rm4	R4
61	Resistors	Metal resistor	M5 resistor	rm5	rm5	R5
62	Resistors	Metal resistor	ML resistor	rml	rmL	RL
63	Diodes	1.8V diode	1.8V N+ /Pwell Diode,	dn18	ndio_sal	DN
64	Diodes	1.8V diode	1.8V P+ /Nwell Diode	dp18	pdio_sal	DP
65	Diodes	diode	3.3/5V N+ /Pwell Diode	dnh	ndio_33	N3
66	Diodes	diode	3.3/5V P+ /Nwell Diode	dph	pdio_33	P3
67	Diodes	1.8V diode	1.8V Nwell/Psub Diode	dnwell	nwldio	DW
68	Diodes	diode	3.3/5V Nwell/Psub Diode	dnwell33	nwldio_33	W3
69	Diodes	diode	1.8/3.3/5V Deep-N-Well/Psub Diode	ddwnps18	ddwnps18	DDWNPS18
70	Diodes	diode	1.8/3.3/5V Isolated-P-Well / Deep N-Well diode	ddwnpw18	ddwnpw18	DDWNPW18
71	Diodes	diode	N+ in 5V Pwell diode	dn50	dn50	DN50
72	Diodes	diode	P+ in 5V Nwell diode	dp50	dp50	DP50